

The Three-Dimensional Liquid-Drop Threshold

A Lean-friendly blueprint specification

Autoformalisation specification

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How to read and use this document

What this document is

This is a blueprint of the proof of the results of O. Chodosh and M. Gianocca, *No compromise in the liquid drop model* (arXiv:2608.11517), written in a form that can be turned into a `leanblueprint` project and then autoformalised. Its organising principles are:

- (1) **Every citable claim is its own numbered item.** Mathematical content that would ordinarily live inside prose paragraphs or inside proofs of larger results is extracted into a `lemma`, `proposition`, `theorem` or `definition` with a stable `\label`. Nothing is proved twice; forward dependencies are permitted only when recorded in the acyclic `\uses` graph.
- (2) **Every statement is dependency-explicit.** Its data and hypotheses are either quantified in the statement or imported through a cited named definition or result. There are no unnamed ambient hypotheses. A downstream formaliser should expand a cited setting into Lean parameters while following the corresponding `\uses` edge.
- (3) **Constants are named and their dependencies are declared.** Quantitative statements specify whether constants are absolute or list their allowed parameters. The six high-risk arguments with nested smallness choices additionally carry a numbered `Constants` block recording the order of choice.
- (4) **Dependency edges are machine-readable.** Every nonempty internal dependency set is recorded by `\uses{...}`; dependency-free items omit the macro. Following `leanblueprint` convention, a `\uses on the statement` lists what is needed to *state* the item (definitions, earlier notation); a `\uses on the proof` lists what is needed to *prove* it.
- (5) **The dependency graph is acyclic and machine-readable.** The presentation order is chosen for exposition and contains a small number of forward edges (for example, a theorem may be stated before the technical lemmas that prove it). Formalisation must follow any topological sort of the `\uses` graph, not blindly follow page order. The two most important ordering constraints are called out explicitly (Remarks [13.1](#) and [10.1](#)).
- (6) **Intended Lean module names appear in situ.** Chapter headers list the modules owning their labelled items in a `\module{...}` line. Chapter [41](#) gives the complete assignment, including every definition, convention and constant annotation.

What a downstream tool should do with it

- **To build a blueprint:** delete the `\providecommand` block in the preamble (plasTeX supplies `\uses`, `\lean`, `\leanok`), reuse the existing chapter inputs, and keep the labels verbatim.

The labels are the blueprint’s node names and are referenced from Chapter 41.

- **To autoformalise:** topologically sort the nodes using `\uses` and process them in that order. An item’s `\uses` set is the set of internal items its Lean proof may cite; it need not consist only of earlier pages. If a proof needs an internal result not in its `\uses` set, that is a bug in this document; report it rather than adding a hypothesis.
- **Never discharge an item with `sorry`, `axiom` or an appeal to the literature.** The permitted library base is pinned in Chapter 1. An implementation may refine an item into further local measurability, trace or constant-selection lemmas; it may not add a mathematical hypothesis.

Item-type discipline

- **definition** — introduces notation or a predicate. Expect a Lean `def` or `structure`.
- **lemma** — a step used only inside this document.
- **proposition** — a milestone consumed by a later chapter.
- **theorem** — a headline result, or a result whose statement is quotable outside this development.
- **convention** — a sign, orientation or representative choice. These are the highest-risk items in the whole project: a convention silently flipped in one file and not another is the single most likely source of a false theorem. Each is stated once and referenced by label everywhere.
- **fnote** (Formalisation note) — advice to the formaliser. Carries no mathematical content and generates no blueprint node.
- **test** (Regression test) — a computation the finished Lean development should be able to discharge, used to detect sign errors. Collected in Chapter 39.

Statement of the target

The two results this document exists to prove are stated below informally, for orientation. Their canonical statements — the ones a blueprint should treat as nodes — are Theorem 37.1 in Chapter 37 and Theorem 38.7 in Chapter 38, each stated immediately above its proof.

Main theorem. With $\mathcal{E}$ and V_* as in Definitions 15.2 and 16.5: for $0 < V \leq V_*$ the minimisers of $\mathcal{E}$ among measurable subsets of $\mathbb{R}^3$ of volume V are exactly the translates of a round ball of volume V , up to Lebesgue-null sets; for $V > V_*$ no minimiser exists.

Binding-energy corollary. $e_* := \inf_{0 < |\Omega| < \infty} \mathcal{E}(\Omega)/|\Omega| = 3(9\pi/5)^{1/3}$, the infimum is attained, and every optimiser is a translate of a ball of volume $5/2$, up to a null set.

Remark 0.1 (Volume zero). The statement is deliberately restricted to $V > 0$. A “ball of volume zero” is not a useful formal object, and the $V = 0$ case of the source paper is tacit.

Part I

Ground rules

Chapter 1

The permitted library base

Lean module: (none -- this chapter is a contract, not mathematics)

1.1 The pin

The permitted base is Lean v4.34.0-rc2 and mathlib commit 85e3a25e006c35636f0e53b0e9296caca2685bc0, as recorded in the repository’s `lean-toolchain` and `lake-manifest.json`. Everything in this document that is not on the following list is derived here.

1.2 What may be assumed

- (B1) Real and finite-dimensional linear algebra, Euclidean topology, compactness, smooth partitions of unity (`SmoothPartitionOfUnity`, `SmoothBumpCovering`), and elementary convexity.
- (B2) Completeness of L^p spaces; weak sequential compactness of bounded sequences in a Hilbert space.
- (B3) Lebesgue and Hausdorff measure, Bochner integration, Fubini–Tonelli, monotone and dominated convergence, Radon–Nikodym for *positive* measures, the *positive-functional* Riesz–Markov theorem (`RealRMK.rieszMeasure` together with `RealRMK.integral_rieszMeasure`), and regularity of the resulting positive measures.
- (B4) Banach–Alaoglu (`WeakDual.isCompact_closedBall`) and its sequential form for duals of separable normed spaces (`WeakDual.isSeqCompact_closedBall`).
- (B5) The Besicovitch covering and differentiation theorems for Euclidean balls, including differentiation of one Radon measure with respect to another (`Besicovitch.exists_disjoint_closedBall_covering_ae`, `Besicovitch.ae_tendsto_rnDeriv`).
- (B6) Finite-dimensional Fréchet calculus, the inverse and implicit function theorems (`HasStrictFDerivAt.to_localInverse`), Taylor’s theorem, Rademacher’s theorem (`LipschitzWith.ae_differentiableAt`), and *equidimensional* Jacobian change of variables (`integral_image_eq_integral_abs_det_fderiv_smul`).
- (B7) Picard–Lindelöf local existence and uniqueness, including a locally simultaneous family of solutions Lipschitz in the initial point (`IsPicardLindelof.exists_forall_mem_closedBall_eq_hasDerivWithinAt_lipschitzOnWith`), together with `ODE_solution_unique`.

- (B8) The Fourier transform on Schwartz functions, Fourier inversion, differentiation under the transform, and the L^2 Plancherel isometry. The pinned APIs include `MeasureTheory.Integrable.fourierInv_fourier_eq`, the derivative identities in `Analysis/Fourier/FourierTransformDeriv.lean`, and the L^2 Fourier linear isometry constructed in `Analysis/Fourier/LpSpace.lean`, with norm theorem `MeasureTheory.Lp.norm_fourier_eq`. Real-valued Sobolev functions are sent to the complex theory by the canonical isometric inclusion. The inversion theorem requires integrability of the function and its Fourier transform, and continuity at the evaluation point. In the Sobolev application it is first applied to smooth compactly supported approximants, for which these hypotheses hold, and the result is then extended by the quantitative Sobolev estimate and completeness.

1.3 What may *not* be assumed

The following are *absent* from the base and are derived in this document. Each entry names the chapter that supplies it. The list exists because each of these has previously been mistaken for a library primitive.

Not in the base	Derived in
Signed Riesz representation ((B3) gives positive functionals only; <code>RealRMK.rieszMeasure</code> returns a <code>Measure</code> , not a <code>SignedMeasure</code>)	§3.1
Weak-* compactness stated for signed or vector Radon measures	§3.2
Sets of finite perimeter, reduced boundary, several-variable BV (mathlib’s <code>Analysis/BoundedVariation.lean</code> is one-dimensional)	Ch. 4–8
The <i>lower-dimensional</i> (rectifiable) area formula; (B6) gives the equidimensional one only	§6.2
De Giorgi’s structure theorem; (B5) is not a substitute	Ch. 8
Coarea for BV and for C^1 functions with a weight	§5.1
Any Sobolev, Poincaré, Rellich or isoperimetric inequality	Ch. 7
Elliptic estimates: Campanato, Schauder, maximum and Hopf principles, boundary regularity	Ch. 10
Sard’s theorem in any form	Ch. 11
Differentiability of ODE solutions in the initial point; a global flow generated by a vector field (<code>Dynamics/Flow.lean</code> is an abstract continuous monoid action, never built from an ODE); the flow-box theorem	§14.1
Transversality in any form; tubular neighbourhoods	§14.2, §14.3
Morse theory; Gauss–Bonnet; the degree of a smooth map; classification of compact surfaces	§14.5–§14.8 (degree theory is <i>avoided</i> , not derived — Formalisation note 14.39)
Any regularity theorem for perimeter (quasi)minimisers	Ch. 24

Not in the base	Derived in
Capacitary monotonicity (Agostiniani–Mazzieri)	Ch. 31

Formalisation note 1.1 (The ODE trap). The mathlib ODE results are named by the shape of their conclusion, not by the words “dependence” or “flow”, and the local flow they produce is a bare function rather than a `Flow`. Keyword searches miss them. Grep for conclusion shapes, or read `Mathlib/Analysis/ODE/ExistUnique.lean` directly. In particular [\(B7\)](#) *does* give Lipschitz dependence on the initial point and joint continuity, and *does not* give differentiability in the initial point.

Formalisation note 1.2 (Two false friends). Do not use [\(B6\)](#)’s equidimensional Jacobian theorem where the rectifiable area formula ([§6.2](#)) is required, and do not use [\(B5\)](#)’s differentiation theorem where De Giorgi’s structure theorem (Ch. [8](#)) is required. Both substitutions typecheck as English and neither is valid.

Chapter 2

Global conventions

Lean module: `NoCompromise.BV.Basic`, `NoCompromise.Conventions`,
`NoCompromise.Surface.Geometry`, `NoCompromise.Variation.Straight`

Every item in this chapter is referenced by label from the places that depend on it. A convention silently changed in one place and not another is the most likely way to produce a false theorem in this development, so the conventions are stated once, here, and nowhere else.

2.1 Ambient setting and identification of sets

Convention 2.1 (Ambient space). Work in $\mathbb{R}^3$ with Lebesgue measure. Write $|E|$ for the Lebesgue measure of a measurable $E \subset \mathbb{R}^3$, $B_r(x)$ for the open Euclidean ball, and $B_r := B_r(0)$. B_r^2 denotes the open planar disk of radius r in $\mathbb{R}^2$, $\mathbb{S}^2$ the unit sphere in $\mathbb{R}^3$. All measurability is Lebesgue measurability unless stated otherwise.

Convention 2.2 (Sets modulo null sets). Two measurable sets are identified for variational purposes when their symmetric difference is Lebesgue-null. Every functional in Chapter 15 is invariant under this identification (Lemma 15.4). Uniqueness statements always mean uniqueness modulo translations and null sets. From Chapter 21 onwards a specific representative is fixed; Convention 21.7 records which.

Convention 2.3 (Outward normal and the polar sign). For a set E of finite perimeter in $\mathbb{R}^n$ (with $n = 3$ unless a planar auxiliary result is being stated) the measure-theoretic normal ν_E is always the *outward* one, fixed by the polar decomposition

$$D\mathbf{1}_E = -\nu_E |D\mathbf{1}_E|.$$

For a smooth bounded domain ν is the outward unit normal. This single sign choice propagates into the Gauss–Green formula (Theorem 8.13), the first-variation formulas (§9.2), De Giorgi’s structure theorem (Theorem 8.12), Green’s second identity (§32.1) and the level-set geometry of Chapter 31.

Convention 2.4 (Curvature signs). For a smooth bounded domain E with outward normal ν , let $\mathbb{I}$ be the second fundamental form, κ_1, κ_2 the principal curvatures, and

$$H := \kappa_1 + \kappa_2, \quad \kappa := \kappa_1 \kappa_2,$$

normalised so that *a round sphere of radius R has $H = 2/R > 0$* . Then $|\mathbb{I}|^2 = H^2 - 2\kappa$. In Chapter 31 the surface is a superlevel set $\{u \geq t\}$ with outward normal $\nu = -\nabla u / |\nabla u|$ and $H = \operatorname{div}_{\Sigma_t} \nu$, which is consistent: a round sphere again has $H > 0$.

Convention 2.5 (Sign of the first variation). For a normal variation with scalar speed f ,

$$\frac{d}{dt}\Big|_{t=0} P(E_t) = \int_{\partial E} Hf, \quad \frac{d}{dt}\Big|_{t=0} |E_t| = \int_{\partial E} f.$$

This determines every later sign; it must be fixed before §9.2.

Convention 2.6 (Test-field classes). $C_c^1(U; \mathbb{R}^n)$ denotes continuously differentiable compactly supported vector fields on an open set $U \subset \mathbb{R}^n$; the unmarked ambient dimension is $n = 3$. All suprema defining variation measures are over this class with $|X| \leq 1$ pointwise. Replacing C_c^1 by C_c^∞ changes none of these suprema, by mollification; this is used without comment and is recorded as Lemma 4.3.

2.2 Constant discipline

Convention 2.7 (Constants). Every quantitative item must state the allowed dependence of its constants; the six multi-parameter arguments where quantifier order matters also carry a numbered **Constants** annotation. The rules are:

- (a) “Absolute” means: depends on nothing. Since the ambient dimension is fixed at 3 throughout, what other treatments call a dimensional constant is absolute here.
- (b) A constant written $C(a, b)$ depends only on a and b , monotonically where that is asserted.
- (c) When several smallness parameters occur, the annotation gives the *order of choice*, written as a chain $\gamma \rightsquigarrow \eta \rightsquigarrow \varepsilon$, meaning: γ is chosen first and freely, then $\eta = \eta(\gamma)$, then $\varepsilon = \varepsilon(\gamma, \eta)$. No item may be used with a choice order different from the one it declares.
- (d) Constants are never tracked numerically except in Chapter 16, where exact values are the point.

Formalisation note 2.8 (Why the order of choice is written down). An informal proof can defer the smallness ordering of an ε -regularity argument to a sentence at the end of the proof. In Lean the order *is* the statement: $\forall \gamma \exists \eta \forall \dots$ and $\exists \eta \forall \gamma \dots$ are different theorems, and only one of them is true. Every such chain in this document is therefore explicit at the point of use.

Part II

Foundations: measure theory and BV

Chapter 3

Signed measures from positive ones

Lean module: `NoCompromise.Measure.SignedRiesz`, `NoCompromise.Measure.WeakStar`

The base (B3) represents *positive* functionals only, and (B4) is a compactness statement about duals of normed spaces, not about signed Radon measures. Both gaps are closed here, before any BV object is defined. Everything downstream of Df depends on this chapter.

3.1 Signed Riesz representation

Definition 3.1 (Locally bounded functional). Let $n \geq 1$ and $U \subset \mathbb{R}^n$ be open. A linear map $\Lambda : C_c(U; \mathbb{R}) \rightarrow \mathbb{R}$ is *locally bounded* if for every compact $K \Subset U$ there is $C_K < \infty$ with

$$|\Lambda(\phi)| \leq C_K \|\phi\|_\infty \quad \text{whenever } \text{spt } \phi \subset K. \quad (3.1)$$

Definition 3.2 (Upper variation). For Λ locally bounded and $\phi \in C_c(U)$ with $\phi \geq 0$, set

$$\Lambda^+(\phi) := \sup\{\Lambda(\psi) : \psi \in C_c(U), 0 \leq \psi \leq \phi\}. \quad (3.2)$$

Lemma 3.3 (Λ^+ is finite, homogeneous and additive on the cone). *Let Λ be locally bounded. Then for $\phi, \eta \in C_c(U)$ nonnegative and $t \geq 0$: $\Lambda^+(\phi) < \infty$, $\Lambda^+(t\phi) = t\Lambda^+(\phi)$, and $\Lambda^+(\phi + \eta) = \Lambda^+(\phi) + \Lambda^+(\eta)$.*

Proof. The competing set in (3.2) contains 0 and is bounded above by (3.1), so the supremum is finite and nonnegative; positive homogeneity is immediate. For $\geq$ in additivity, add a competitor for ϕ to one for η . For $\leq$, let $0 \leq \psi \leq \phi + \eta$ and put $\psi_1 := \min\{\psi, \phi\}$, $\psi_2 := \psi - \psi_1$. Then ψ_1, ψ_2 are continuous with compact support, $0 \leq \psi_1 \leq \phi$ and $0 \leq \psi_2 \leq \eta$, whence $\Lambda(\psi) = \Lambda(\psi_1) + \Lambda(\psi_2) \leq \Lambda^+(\phi) + \Lambda^+(\eta)$. $\square$

Theorem 3.4 (Signed Riesz representation). *Let $n \geq 1$, let $U \subset \mathbb{R}^n$ be open, and let $\Lambda : C_c(U; \mathbb{R}) \rightarrow \mathbb{R}$ be linear and locally bounded. Then there are positive, locally finite, regular Borel measures μ^+, μ^- on U with*

$$\Lambda(\phi) = \int_U \phi d\mu^+ - \int_U \phi d\mu^- \quad (\phi \in C_c(U)). \quad (3.3)$$

Proof. Write $g = g_+ - g_-$ for $g \in C_c(U)$ and set $P(g) := \Lambda^+(g_+) - \Lambda^+(g_-)$. Additivity (Lemma 3.3) makes this independent of the chosen nonnegative decomposition: if $g = h - k = h' - k'$ with all four functions nonnegative, then $h + k' = h' + k$, so additivity gives $\Lambda^+(h) - \Lambda^+(k) = \Lambda^+(h') - \Lambda^+(k')$.

Thus P is linear and $P \geq 0$ on the positive cone. Also $Q := P - \Lambda$ is positive, because $P(\phi) = \Lambda^+(\phi) \geq \Lambda(\phi)$ for $\phi \geq 0$. Apply (B3) (`RealRMK.rieszMeasure` and its integral identity) to P and to Q , obtaining μ^+, μ^- and (3.3).

Local finiteness is explicit: for $K \Subset U$ pick $\eta \in C_c(U; [0, 1])$ equal to 1 on K ; then $\mu^+(K) \leq P(\eta) < \infty$ and $\mu^-(K) \leq Q(\eta) < \infty$. $\square$

Lemma 3.5 (Exhaustion and gluing). *Suppose Λ is given only on an exhaustion $U_m \Subset U_{m+1}$, $\bigcup_m U_m = U$, and is locally bounded on each U_m . Performing the construction of Theorem 3.4 on each U_m yields measures which agree on overlaps and glue to a single pair of locally finite measures on U representing Λ .*

Proof. Uniqueness in positive Riesz–Markov, applied to the positive and negative parts produced by (3.2), gives agreement on overlaps; the positive and negative parts are themselves defined by a formula that does not reference m . $\square$

3.2 Weak-* compactness for the signed measures constructed above

Lemma 3.6 ($C_0(U)$ is separable). *For $n \geq 1$ and open $U \subset \mathbb{R}^n$ the space $C_0(U)$ is separable.*

Proof. Compactly supported continuous functions are dense in $C_0(U)$: multiply by the cutoffs ζ_m of Definition 3.7. On a fixed compact support, uniform continuity gives uniform approximation by piecewise-affine functions on rational cubical grids with rational vertex values. The union over m of these countable families is countable and dense. $\square$

Definition 3.7 (Exhaustion cutoffs). *For $n \geq 1$ and open $U \subset \mathbb{R}^n$ fix once and for all compact $K_m \subset \text{int } K_{m+1}$ with $\bigcup_m K_m = U$, and $\zeta_m \in C_c(U; [0, 1])$ with $\zeta_m \equiv 1$ on K_m and $\text{spt } \zeta_m \subset K_{m+1}$.*

Theorem 3.8 (Local weak-* compactness). *Let $n, q \geq 1$, let $U \subset \mathbb{R}^n$ be open, and let λ_j be signed (or $\mathbb{R}^q$ -valued) Radon measures on U which are locally variation-bounded: $\sup_j |\lambda_j|(K) < \infty$ for every compact $K \Subset U$. Then there is a subsequence and a locally finite signed (resp. vector) Radon measure λ with*

$$\int_U \phi d\lambda_j \longrightarrow \int_U \phi d\lambda \quad \text{for every } \phi \in C_c(U).$$

Proof. Multiplication by ζ_m identifies $\zeta_m \lambda_j$ with a uniformly bounded functional on $C_0(U)$, which is separable by Lemma 3.6; (B4) therefore gives a weak-* convergent subsequence for each m . Diagonalise in m . For $L(\phi) := \lim_j \int \phi d\lambda_j$; the limit exists because that integral equals $\int \phi d(\zeta_m \lambda_j)$, and is independent of the larger m chosen because it is the limit of the same scalar sequence. The local variation bounds make L locally bounded in the sense of (3.1), so Theorem 3.4 and Lemma 3.5 represent it. For vector-valued measures, diagonalise over the finitely many components as well. $\square$

Formalisation note 3.9. No compactness theorem stated directly for signed Radon measures has been used, and none may be. Theorem 3.8 is the only compactness statement about measures that later chapters may cite.

Chapter 4

BV functions and sets of finite perimeter

Lean module: NoCompromise.BV.Basic, NoCompromise.BV.Compactness, NoCompromise.BV.Defs, NoCompromise.BV.Rellich, NoCompromise.BV.StrictApprox, NoCompromise.Sobolev.H1Extension, NoCompromise.Sobolev.PlanarDomain, NoCompromise.Sobolev.Rellich

4.1 Density representatives

Definition 4.1 (Density points and essential boundary). For measurable $E \subset \mathbb{R}^n$, $n \geq 1$, put

$$E^{(1)} := \left\{ x : \lim_{r \downarrow 0} \frac{|E \cap B_r(x)|}{|B_r|} = 1 \right\}, \quad E^{(0)} := \left\{ x : \lim_{r \downarrow 0} \frac{|E \cap B_r(x)|}{|B_r|} = 0 \right\},$$

and $\partial^e E := \mathbb{R}^n \setminus (E^{(0)} \cup E^{(1)})$. These are Borel sets: the density limit may equivalently be tested along $r = 1/k$, and for fixed $r > 0$ the map $x \mapsto |E \cap B_r(x)|$ is continuous by local L^1 continuity of translations. This definition belongs to `BV/Defs.lean`; both `coarea` and `traces` import it from there.

4.2 The variation measure

Definition 4.2 (Variation). Let $n \geq 1$. For $f \in L^1_{\text{loc}}(U)$, $U \subset \mathbb{R}^n$ open, set

$$|Df|(U) := \sup \left\{ \int_U f \operatorname{div} X : X \in C^1_c(U; \mathbb{R}^n), |X| \leq 1 \right\}. \quad (4.1)$$

Say $f \in BV(U)$ if $f \in L^1(U)$ and $|Df|(U) < \infty$, and $f \in BV_{\text{loc}}(U)$ if $|Df|(V) < \infty$ for every $V \Subset U$.

Lemma 4.3 (The test class is irrelevant). *The supremum in (4.1) is unchanged if C^1_c is replaced by C^∞_c .*

Proof. $C^\infty_c \subset C^1_c$ gives one inequality. For the other, mollify an admissible X at a radius smaller than $\operatorname{dist}(\operatorname{spt} X, \partial U)$; the mollification is admissible and div of it converges to $\operatorname{div} X$ uniformly. $\square$

Definition 4.4 (Perimeter). For measurable $E \subset \mathbb{R}^n$ and open $A \subset \mathbb{R}^n$, set $P_n(E; A) := |D\mathbf{1}_E|(A)$ and $P_n(E) := P_n(E; \mathbb{R}^n)$. In the ambient dimension $n = 3$ we suppress the subscript and write P . Say $E \subset \mathbb{R}^n$ has *finite perimeter* if $P_n(E) < \infty$, and *locally finite perimeter* if $P_n(E; A) < \infty$ for every $A \in \mathbb{R}^n$.

Theorem 4.5 (Polar decomposition). *Let $n \geq 1$, $U \subset \mathbb{R}^n$ be open, and let $f \in BV_{\text{loc}}(U)$. There is a locally finite positive Radon measure $|Df|$ on U — agreeing on open sets with (4.1) — and a Borel map $\sigma_f : U \rightarrow \mathbb{R}^n$ with $|\sigma_f| = 1$ $|Df|$ -a.e., such that $Df = \sigma_f |Df|$ as $\mathbb{R}^n$ -valued measures. For a set E of locally finite perimeter we write, per Convention 2.3,*

$$D\mathbf{1}_E = -\nu_E |D\mathbf{1}_E|. \quad (4.2)$$

Proof. Apply Theorem 3.4 to $\Lambda_i(\phi) = -\int f \partial_i \phi$; (4.1) supplies (3.1) on every relatively compact subdomain where the variation is finite, so $D_i f = \mu_i^+ - \mu_i^-$ with $\mu_i^\pm$ positive.

Let μ be the sum of the $2n$ measures $\mu_i^\pm$. Positive Radon–Nikodym ((B3)) gives nonnegative densities $a_i^\pm$ with $\mu_i^\pm = a_i^\pm \mu$; the required σ -finiteness holds because U has a compact exhaustion on whose members μ is finite. Put $g_i = a_i^+ - a_i^-$, $g = (g_1, \dots, g_n)$ and $\rho := |g| \mu$. Then $D_i f = g_i \mu$ without invoking signed Radon–Nikodym, and ρ is locally finite since $\int_K |g| d\mu \leq \sum_i \int_K (a_i^+ + a_i^-) d\mu < \infty$.

For every admissible X ,

$$\int_U f \operatorname{div} X = - \int_U X \cdot g d\mu \leq \int_U |X| d\rho. \quad (4.3)$$

Conversely let $O \subset U$ be open, $K \Subset O$, and put $q := -g/|g|$ on $\{|g| > 0\}$ and $q := 0$ elsewhere. Regularity of ρ makes $C_c(O; \mathbb{R}^n)$ dense in $L^1(\rho \llcorner O; \mathbb{R}^n)$, so choose continuous fields converging there to $q\mathbf{1}_K$; compose with the metric projection onto the closed unit ball, which does not increase the error (as $|q\mathbf{1}_K| \leq 1$) and preserves compact support. Approximate uniformly by $Y_j \in C_c^1(O; \mathbb{R}^n)$ with $\|Y_j\|_\infty \leq 1 + \varepsilon_j$ and set $X_j := Y_j/(1 + \varepsilon_j)$, $\varepsilon_j \downarrow 0$. Substituting in the identity preceding (4.3) gives the lower bound $\rho(K)$ for the variation on O . Taking the supremum over $K \Subset O$ and using regularity of ρ shows the supremum in (4.1) equals $\rho(O)$ for every open O . Hence $|Df| = \rho$, and $\sigma_f := g/|g|$ where $|g| > 0$, with an arbitrary unit value on the ρ -null complement. $\square$

Lemma 4.6 (Elementary invariances of the perimeter). *Let $n \geq 1$, let $E, F \subset \mathbb{R}^n$ be measurable, let $A \subset \mathbb{R}^n$ be open, $a \in \mathbb{R}^n$, and $r > 0$. Then*

- (i) $|E \Delta F| = 0 \implies P_n(E; A) = P_n(F; A)$;
- (ii) $P_n(a + E; a + A) = P_n(E; A)$;
- (iii) $P_n(rE; rA) = r^{n-1} P_n(E; A)$;
- (iv) $P_n(\mathbb{R}^n \setminus E; A) = P_n(E; A)$.

The corresponding global identities follow by taking $A = \mathbb{R}^n$.

Proof. (i) is immediate from (4.1), which only sees $\mathbf{1}_E$ as an L^1_{loc} class. For (ii), translation gives a norm-preserving bijection between the admissible test fields on A and $a + A$. For (iii), change variables $y = rx$ and use $\operatorname{div}(X(r \cdot)) = r(\operatorname{div} X)(r \cdot)$; the Jacobian r^n leaves the factor r^{n-1} . Finally (iv) holds because $\int_{E^c} \operatorname{div} X = -\int_E \operatorname{div} X$ for every $X \in C_c^1(A; \mathbb{R}^n)$. $\square$

Lemma 4.7 (Lower semicontinuity). *Let $n \geq 1$ and let $U \subset \mathbb{R}^n$ be open. If $f_j \rightarrow f$ in $L^1_{\text{loc}}(U)$, then $|Df|(U) \leq \liminf_j |Df_j|(U)$. In particular, if $\mathbf{1}_{E_j} \rightarrow \mathbf{1}_E$ in $L^1_{\text{loc}}(U)$, then $P_n(E; U) \leq \liminf_j P_n(E_j; U)$.*

Proof. For each admissible X , $\int f \operatorname{div} X = \lim_j \int f_j \operatorname{div} X \leq \liminf_j |Df_j|(U)$; take the supremum over X . $\square$

Lemma 4.8 (Locality of the perimeter measure). *$P_n(E; \cdot)$ is a Borel measure for every measurable $E \subset \mathbb{R}^n$. Consequently, if $E = F$ a.e. on an open $A \subset \mathbb{R}^n$, then $P_n(E; A) = P_n(F; A)$; and if $E, F \subset \mathbb{R}^n$ have finite perimeter with $\operatorname{dist}(E, F) > 0$, then*

$$P_n(E \cup F) = P_n(E) + P_n(F).$$

Proof. That $P_n(E; \cdot) = |D\mathbf{1}_E|$ is a measure is Theorem 4.5. Agreement a.e. on A makes the distributional derivatives agree on A . For the disjoint union, separate E and F by disjoint open neighbourhoods $A \supset E$, $B \supset F$ with $\operatorname{dist}(A, B) > 0$; then $\mathbf{1}_{E \cup F} = \mathbf{1}_E$ on A and $= \mathbf{1}_F$ on B , and $\mathbf{1}_{E \cup F}$ is a.e. constant on the complement of $A \cup B$, which therefore carries no derivative measure. Additivity of the measure finishes. $\square$

4.3 Translation estimate and local compactness

Lemma 4.9 (Translation estimate). *Let $n \geq 1$, let $Q \Subset U \subset \mathbb{R}^n$ be a cube, and let Q_h be its enlargement by $|h|$. For $f \in BV_{\text{loc}}(U)$ and $|h|$ small enough that $Q_h \Subset U$,*

$$\int_Q |f(x+h) - f(x)| dx \leq |h| |Df|(Q_h). \quad (4.4)$$

Proof. For smooth f this is the fundamental theorem of calculus along the segment from x to $x+h$ followed by Fubini. For general f , mollify: the convolution inequality $\int |\nabla(\rho_\varepsilon * f)| \leq |Df|(Q_h)$ follows directly from (4.1) by translating a test field and applying Fubini, and the left side of (4.4) passes to the limit in L^1 . $\square$

Lemma 4.10 (Piecewise-constant approximation). *Let $n \geq 1$, let $Q \subset \mathbb{R}^n$ be a cube, $\delta > 0$, and let $A_\delta f$ be the function equal on each half-open cube of the side- δ grid to the average of f there. Then, with C_n depending only on n ,*

$$\|f - A_\delta f\|_{L^1(Q)} \leq C_n \delta |Df|(Q_{C_n \delta}). \quad (4.5)$$

Proof. Average $|f(x) - f(y)|$ over pairs x, y in one grid cube and apply (4.4) along coordinate segments. $\square$

Theorem 4.11 (Local BV compactness). *Let $n \geq 1$, let $U \subset \mathbb{R}^n$ be open, and let $f_j \in BV_{\text{loc}}(U)$ satisfy $\sup_j (\|f_j\|_{L^1(V)} + |Df_j|(V)) < \infty$ for every $V \Subset U$. Then a subsequence converges in $L^1_{\text{loc}}(U)$ to some $f \in BV_{\text{loc}}(U)$. If moreover $f_j = \mathbf{1}_{E_j}$ then $f = \mathbf{1}_E$ for a set E of locally finite perimeter in U , that is, $P_n(E; A) < \infty$ for every $A \Subset U$. When $U = \mathbb{R}^n$ this is locally finite perimeter in the sense of Definition 4.4; for general U the hypotheses control no perimeter at ∂U , so no global conclusion is asserted.*

Proof. Fix a cube $Q \Subset U$. For fixed δ the functions $A_\delta f_j$ lie in a finite-dimensional space with coefficient vectors bounded by the uniform L^1 bound, so they have a convergent subsequence. Diagonalise over $\delta = 2^{-m}$; (4.5) makes the resulting subsequence Cauchy in $L^1(Q)$. A second diagonalisation over an exhaustion by cubes gives $L^1_{\text{loc}}(U)$ convergence. Lemma 4.7 gives $f \in BV_{\text{loc}}$. Finally L^1 limits of $\{0, 1\}$ -valued functions are a.e. $\{0, 1\}$ -valued. $\square$

Formalisation note 4.12. No Fréchet–Kolmogorov theorem is used; (4.5) plus finite-dimensionality is the compactness argument. This matters because the base contains no such theorem.

4.4 A two-dimensional Rellich lemma

The harmonic blow-up of §24.7 needs strong L^2 compactness of an H^1 -bounded sequence on a planar disk, and nothing more.

Lemma 4.13 (Planar Gagliardo–Nirenberg). *Let $B \subset \mathbb{R}^2$ be a bounded disk. Then, with C_B depending only on B ,*

$$\|g\|_{L^4(B)} \leq C_B (\|g\|_{L^2(B)} + \|\nabla g\|_{L^2(B)}) \quad (g \in H^1(B)). \quad (4.6)$$

Proof. Apply the one-dimensional fundamental theorem of calculus in each of the two coordinate directions to $|g|^2$ and then Cauchy–Schwarz; this is (7.3) applied to $|g|^2$. Flattening and reflection in finitely many boundary charts (Lemma 4.14) supplies the compactly supported extension the calculation requires. $\square$

Lemma 4.14 (H^1 extension from a Lipschitz domain). *Let $D \subset \mathbb{R}^n$, $n \in \{2, 3\}$, be a bounded Lipschitz domain. There is a bounded linear extension $H^1(D) \rightarrow H^1(\mathbb{R}^n)$ whose image is supported in a fixed bounded neighbourhood of $\overline{D}$.*

Proof. This is the H^1 case of the construction in Lemma 7.5: finitely many Lipschitz graph charts, flatten, reflect evenly across the graph, multiply by a partition of unity, extend the interior piece by zero. $\square$

Lemma 4.15 (L^2 interpolation). $\|h\|_{L^2} \leq \|h\|_{L^1}^{1/3} \|h\|_{L^4}^{2/3}$.

Proof. Hölder with exponents $1/2 = \frac{1}{3} \cdot 1 + \frac{2}{3} \cdot \frac{1}{4}$ applied to $|h|^2 = |h|^{2/3} |h|^{4/3}$. $\square$

Theorem 4.16 (Rellich–Kondrachov on a planar disk). *Let $B \subset \mathbb{R}^2$ be a bounded disk and let g_j be bounded in $H^1(B)$. Then a subsequence converges weakly in $H^1(B)$ and strongly in $L^2(B)$.*

Proof. Weak H^1 convergence along a subsequence is (B2). A bounded sequence in $H^1(B)$ is bounded in $BV(B)$ by Cauchy–Schwarz, so after the extension of Lemma 4.14 the compactness argument of Theorem 4.11 gives a subsequence converging in $L^1(B)$. The differences are uniformly bounded in $L^4(B)$ by Lemma 4.13; Lemma 4.15 upgrades L^1 convergence to L^2 convergence. $\square$

4.5 Strict smooth approximation

Lemma 4.17 (BV product rule with a smooth factor). *For $n \geq 1$, $U \subset \mathbb{R}^n$ open, $f \in BV_{\text{loc}}(U)$ and $\zeta \in C_c^\infty(U)$,*

$$D(\zeta f) = \zeta Df + f \nabla \zeta \, dx.$$

Proof. Immediate for smooth f ; pass to the limit along a mollification, using L^1_{loc} convergence for the second term and weak-* convergence of Df for the first. $\square$

Theorem 4.18 (Strict approximation). *Let $n \geq 1$, let $U \subset \mathbb{R}^n$ be open, and let $f \in BV_{\text{loc}}(U)$. There are $f_j \in C^\infty(U)$ with*

$$f_j \rightarrow f \text{ in } L^1_{\text{loc}}(U), \quad \int_U |\nabla f_j| \rightarrow |Df|(U) \quad (4.7)$$

whenever the right-hand limit is finite. If, moreover, $U = \mathbb{R}^n$ and $f \in BV(\mathbb{R}^n)$, the sequence may be chosen in $C^n_{\text{fty}}(\mathbb{R}^n)$ with $f_j \rightarrow f$ in $L^1(\mathbb{R}^n)$.

Proof. Let $U_k \Subset U_{k+1}$ exhaust U , take a partition of unity ζ_k subordinate to the annuli $U_{k+1} \setminus \overline{U_{k-1}}$, and convolve $\zeta_k f$ at a radius $\varepsilon_{j,k} < \text{dist}(\text{spt } \zeta_k, \partial U)$. Extend each summand by zero before convolution and set

$$f_j := \sum_k \rho_{\varepsilon_{j,k}} * (\zeta_k f).$$

The sum is locally finite and smooth. Choose the radii, one k at a time, so small that

$$\|f_j - f\|_{L^1(U_j)} < j^{-1}, \quad \sum_k \|\rho_{\varepsilon_{j,k}} * (f \nabla \zeta_k) - f \nabla \zeta_k\|_{L^1(U)} < j^{-1};$$

this is possible because each summand is compactly supported and the error budgets may be chosen summable. Lemma 4.17 gives

$$Df_j = \sum_k \rho_{\varepsilon_{j,k}} * (\zeta_k Df) + \sum_k \rho_{\varepsilon_{j,k}} * (f \nabla \zeta_k) dx.$$

For the measure term, convolution and the triangle inequality give $\int |\sum_k \rho_{\varepsilon_{j,k}} * (\zeta_k Df)| \leq \sum_k \int \zeta_k d|Df|$. For the absolutely continuous term, insert $\sum_k f \nabla \zeta_k = 0$ and use the chosen summable commutator errors. Consequently

$$\int_U |\nabla f_j| \leq \sum_k \int_U \zeta_k d|Df| + j^{-1} = |Df|(U) + j^{-1}.$$

Lemma 4.7 supplies the reverse liminf and proves (4.7) whenever $|Df|(U) < \infty$.

For the final whole-space assertion, start instead with $h_j := \rho_{\varepsilon_j} * f$, choosing $\varepsilon_j \downarrow 0$ so that $h_j \rightarrow f$ in $L^1(\mathbb{R}^n)$. Let χ_R be one on B_R , supported in B_{2R} , and satisfy $|\nabla \chi_R| \leq C/R$. Choose $R_j \uparrow \infty$ so that $\chi_{R_j} h_j \rightarrow f$ in L^1 and put $f_j := \chi_{R_j} h_j$. Then

$$\int |\nabla f_j| \leq |Df|(\mathbb{R}^n) + \frac{C}{R_j} \|f\|_{L^1(\mathbb{R}^n)},$$

because convolution does not increase the variation and $\|h_j\|_1 \leq \|f\|_1$. Lower semicontinuity again supplies the reverse liminf. $\square$

Chapter 5

Coarea, slicing and traces

Lean module: NoCompromise.BV.AnnularGluingFull, NoCompromise.BV.BoundaryTraces,
 NoCompromise.BV.Coarea, NoCompromise.BV.CoareaCritical,
 NoCompromise.BV.CoareaLiLebesgue, NoCompromise.BV.CoareaRegular,
 NoCompromise.BV.ExactCuts, NoCompromise.BV.Gluing, NoCompromise.BV.GoodRadii,
 NoCompromise.BV.GoodTruncation, NoCompromise.BV.JumpDisintegration,
 NoCompromise.BV.Slicing, NoCompromise.BV.WeightedCoarea

5.1 Coarea for BV

Theorem 5.1 (Weighted C^1 coarea). *Let $n \geq 2$, let $u \in C^1(U)$ with $U \subset \mathbb{R}^n$ open, and let $g : U \rightarrow [0, \infty]$ be Borel. Then*

$$\int_U g(x) |\nabla u(x)| dx = \int_{\mathbb{R}} \int_{U \cap \{u=t\}} g d\mathcal{H}^{n-1} dt. \quad (5.1)$$

No Sard theorem is used; critical levels are permitted.

Proof. Combine Lemmas 5.2 and 5.3: the first proves (5.1) with $U \setminus Z$ in place of U , where $Z := \{\nabla u = 0\}$; the second shows Z contributes zero to both sides. $\square$

Lemma 5.2 (The regular part). *Let $n \geq 2$, let $u \in C^1(U)$ with $U \subset \mathbb{R}^n$ open, let $g : U \rightarrow [0, \infty]$ be Borel, and put $Z := \{\nabla u = 0\}$. Then*

$$\int_{U \setminus Z} g |\nabla u| dx = \int_{\mathbb{R}} \int_{(U \setminus Z) \cap \{u=t\}} g d\mathcal{H}^{n-1} dt.$$

Proof. Cover $U \setminus Z$ by countably many relatively compact boxes V on each of which one fixed derivative, after permuting coordinates say $\partial_n u$, has constant nonzero sign, subdividing so that each line parallel to e_n meets V in an interval on which u is strictly monotone. Writing $x' = (x_1, \dots, x_{n-1})$, on such a box $\Psi(x', x_n) := (x', u(x))$ is a C^1 diffeomorphism onto its image with Jacobian $|\partial_n u|$, and (B6) gives

$$\int_V g |\nabla u| = \int_{\Psi(V)} g(\Psi^{-1}(y', t)) \frac{|\nabla u|}{|\partial_n u|} (\Psi^{-1}(y', t)) dy' dt. \quad (5.2)$$

For fixed t the inverse graph $x_n = \gamma_t(x')$ satisfies

$$\nabla \gamma_t = -\frac{(\partial_1 u, \dots, \partial_{n-1} u)}{\partial_n u}, \quad \sqrt{1 + |\nabla \gamma_t|^2} = \frac{|\nabla u|}{|\partial_n u|}, \quad (5.3)$$

so the inner integral in (5.2) is the graph definition of $\int_{V \cap \{u=t\}} g d\mathcal{H}^{n-1}$. Disjointify the cover, apply the identity first to simple g , and use monotone convergence. $\square$

Lemma 5.3 (The critical set contributes nothing). *Let $n \geq 2$, let $u \in C^1(U)$ with $U \subset \mathbb{R}^n$ open, and put $Z := \{\nabla u = 0\}$. Then*

$$\mathcal{H}^{n-1}(Z \cap \{u = t\}) = 0 \quad \text{for a.e. } t \in \mathbb{R}.$$

Proof. Fix $K \Subset U$; if $K \cap Z = \emptyset$ there is nothing to prove. Consider the grid cubes of side δ meeting $K \cap Z$, with δ small enough that these lie in U , and put

$$\omega_K(\delta) := \sup\{|\nabla u(x)| : \text{dist}(x, K \cap Z) \leq \sqrt{n}\delta\}.$$

Uniform continuity of ∇u near K and $\nabla u = 0$ on Z give $\omega_K(\delta) \rightarrow 0$. The u -image of each selected cube has length at most $\sqrt{n}\delta\omega_K(\delta)$. For fixed t , the selected cubes whose images contain t cover $K \cap Z \cap \{u = t\}$, so their $(n-1)$ -st powers of the diameters bound its $(n-1)$ -dimensional Hausdorff content at scale $\sqrt{n}\delta$. Integrating in t ,

$$\int_{\mathbb{R}} \mathcal{H}_{\sqrt{n}\delta}^{n-1}(K \cap Z \cap \{u = t\}) dt \leq C_n \omega_K(\delta) \sum_{Q \cap K \cap Z \neq \emptyset} \delta^n \leq C_K \omega_K(\delta). \quad (5.4)$$

As $\delta \downarrow 0$ Hausdorff contents increase to $\mathcal{H}^{n-1}$ while the right side tends to zero; monotone convergence and an exhaustion of Z finish. $\square$

Corollary 5.4 (Coarea with an L^1 weight). *Let $n \geq 2$, let $u \in C^1(U)$ with $U \subset \mathbb{R}^n$ open, let $A \subset U$ be measurable, and let $G \in L^1(A)$. Then*

$$\int_{A \cap \{|\nabla u| > 0\}} G dx = \int_{\mathbb{R}} \int_{A \cap \{u=t\} \cap \{|\nabla u| > 0\}} \frac{G}{|\nabla u|} d\mathcal{H}^{n-1} dt. \quad (5.5)$$

If $|\nabla u| > 0$ throughout A both restrictions may be omitted. This is the form used in Chapter 31.

Proof. Extend G by zero off A and apply Theorem 5.1 to $g := |G|/|\nabla u|$ on $A \cap \{|\nabla u| > 0\}$ and $g := 0$ elsewhere; then split G into positive and negative parts. $\square$

Theorem 5.5 (Coarea for BV). *Let $n \geq 2$, let $U \subset \mathbb{R}^n$ be open, and let $f \in BV_{\text{loc}}(U)$. Then*

$$|Df|(U) = \int_{\mathbb{R}} P_n(\{f > t\}; U) dt. \quad (5.6)$$

Proof. Smooth case. Theorem 5.1 with $g = 1$ gives (5.6) for $f \in C^1$.

$\leq$ for BV. If $|Df|(U) = \infty$, the asserted upper bound for the level-set integral is automatic. Otherwise take f_j as in Theorem 4.18. On each member of a countable relatively compact exhaustion, layer cake and Fubini turn $f_j \rightarrow f$ in L^1 there into, after one diagonal subsequence, $\mathbf{1}_{\{f_j > t\}} \rightarrow \mathbf{1}_{\{f > t\}}$ in L^1_{loc} for a.e. t . Lemma 4.7, Fatou and (4.7) give $\int_{\mathbb{R}} P_n(\{f > t\}; U) dt \leq |Df|(U)$.

$\geq$ for BV. First let $0 \leq f \leq M$. For $X \in C_c^1(U; \mathbb{R}^n)$, Fubini and the layer-cake formula give

$$\int_U f \operatorname{div} X = \int_0^M \int_{\{f > t\}} \operatorname{div} X dx dt \leq \|X\|_{\infty} \int_0^M P_n(\{f > t\}; U) dt;$$

take the supremum over $\|X\|_{\infty} \leq 1$. A bounded signed function is reduced to this case by adding a constant, whose integral against $\operatorname{div} X$ is zero, and shifting the level parameter.

For a general f , put $T_M(f) := \max\{-M, \min\{f, M\}\}$. The truncation chain rule

$$|D(T_M f)|(A) \leq |Df|(A) \quad (A \subset U \text{ open}) \quad (5.7)$$

follows first on every $V \Subset A$ by applying smooth 1-Lipschitz approximations of T_M to a local strict smooth approximation of f , using the ordinary chain rule and Lemma 4.7, and then by exhausting A . Since $T_M f \rightarrow f$ in L^1_{loc} , Lemma 4.7 and (5.7) give $|D(T_M f)|(U) \rightarrow |Df|(U)$ (with extended values allowed). Apply the bounded case to $T_M f$ and let $M \rightarrow \infty$; Tonelli and monotone convergence in the level variable give the reverse inequality for f . $\square$

Corollary 5.6 (Spherical and planar slicing). *Let E have locally finite perimeter. For $x_0 \in \mathbb{R}^3$ and $0 \leq a < b < \infty$,*

$$\int_a^b \mathcal{H}^2(E^{(1)} \cap \partial B_r(x_0)) \, dr = |E \cap (B_b(x_0) \setminus B_a(x_0))|,$$

and, for $\nu \in \mathbb{S}^2$ and $a < b$ in $\mathbb{R}$,

$$\int_a^b \mathcal{H}^2(E^{(1)} \cap \{x : x \cdot \nu = t\}) \, dt = |E \cap \{x : a < x \cdot \nu < b\}|.$$

Proof. The sets E and $E^{(1)}$ agree up to a Lebesgue-null set. Apply Theorem 5.1 with weight $g = \mathbf{1}_{E^{(1)}}$, first with $u(x) = |x - x_0|$ off the null point x_0 and then with $u(x) = x \cdot \nu$. In both cases $|\nabla u| = 1$ on the relevant set, and restriction to the indicated interval gives the two identities. $\square$

Lemma 5.7 (One-dimensional BV slices). *Let $E \subset \mathbb{R}^n$ have locally finite perimeter, $n \geq 2$. For a.e. line ℓ parallel to a fixed direction, $t \mapsto \mathbf{1}_E(x' + te)$ is a one-dimensional BV function; the number of its jumps is finite on compact intervals for a.e. ℓ , and $D_e \mathbf{1}_E$ disintegrates as the one-dimensional jump measures over ℓ .*

Proof. Apply Fubini to (4.1) with test fields of the form $\phi(x')\psi(t)e$; strict approximation (Theorem 4.18) identifies the resulting slice variations with the disintegration of $|D_e \mathbf{1}_E|$. $\square$

5.2 Traces, cut identities and gluing

Theorem 5.8 (Traces on a C^1 domain). *Let $A \subset \mathbb{R}^3$ be a bounded C^1 domain and $f \in BV_{\text{loc}}(\mathbb{R}^3)$. Then f has interior and exterior traces $\text{Tr}_A f, \text{Tr}_{A^c} f \in L^1(\partial A)$, and*

$$D(\mathbf{1}_A f) = \mathbf{1}_A Df - \text{Tr}_A f \, \nu_A \mathcal{H}^2 \llcorner \partial A. \quad (5.8)$$

It also satisfies the exterior formula

$$D(\mathbf{1}_{\mathbb{R}^3 \setminus \bar{A}} f) = \mathbf{1}_{\mathbb{R}^3 \setminus \bar{A}} Df + \text{Tr}_{A^c} f \, \nu_A \mathcal{H}^2 \llcorner \partial A. \quad (5.9)$$

Proof. The bi-Lipschitz BV chain rule (Lemma 7.4, whose proof uses only strict approximation and change of variables) puts f in BV after flattening a C^1 boundary chart. Fubini and one-dimensional BV slicing (Lemma 5.7) then give one-sided limits on a.e. normal line; changing between overlapping charts preserves these limits by the same chain rule and (B6). For (5.8) and (5.9), integrate by parts first for smooth f on the two sides and then along a strict approximation (Theorem 4.18). There is no missing boundary term in the limit: in a graph chart it is exactly the endpoint term in the one-dimensional fundamental theorem of calculus, and $L^1(\partial A)$ convergence of the one-sided traces follows by integrating that one-dimensional identity. $\square$

Definition 5.9 (Good radius). Let E have locally finite perimeter and $x_0 \in \mathbb{R}^3$. Call $r > 0$ a *good radius* for E at x_0 if

$$|D\mathbf{1}_E|(\partial B_r(x_0)) = 0 \quad \text{and} \quad \text{Tr}_{B_r(x_0)} \mathbf{1}_E = \mathbf{1}_{E^{(1)}} \quad \mathcal{H}^2\text{-a.e. on } \partial B_r(x_0).$$

Lemma 5.10 (Almost every radius is good). *For E of locally finite perimeter and any x_0 , a.e. $r > 0$ is a good radius for E at x_0 .*

Proof. The set of r with $|D\mathbf{1}_E|(\partial B_r(x_0)) > 0$ is countable, since $|D\mathbf{1}_E|$ is locally finite. The trace identity holds for a.e. r by Corollary 5.6 together with Theorem 5.8 and Lebesgue's density theorem. $\square$

Proposition 5.11 (Exact cut identities). *Let E have locally finite perimeter and let r be a good radius for E at x_0 . Then*

$$P(E \cap B_r(x_0)) = P(E; B_r(x_0)) + \mathcal{H}^2(E^{(1)} \cap \partial B_r(x_0)), \quad (5.10)$$

$$P(E \setminus B_r(x_0)) = P(E; \mathbb{R}^3 \setminus \overline{B_r(x_0)}) + \mathcal{H}^2(E^{(1)} \cap \partial B_r(x_0)). \quad (5.11)$$

The corresponding identities hold for halfspaces $\{x \cdot \nu < \ell\}$ at a.e. ℓ .

Proof. Apply (5.8) with $A = B_r(x_0)$. At a good radius the two measures on its right side are mutually singular, since $|D\mathbf{1}_E|(\partial B_r) = 0$; taking total variations gives (5.10), and (5.9) gives (5.11); no boundedness of the exterior domain is being assumed. For halfspaces use $x \mapsto x \cdot \nu$ in place of $x \mapsto |x - x_0|$. $\square$

Formalisation note 5.12 (Why exactness matters). Only the *equalities* (5.10)–(5.11), not inequalities, make the perimeter error of a cut equal to twice the section area. Weakening them to inequalities loses a sign in Theorem 19.5 and in Lemma 21.5.

Lemma 5.13 (Good truncation). *Let $E \subset \mathbb{R}^3$ have finite measure and finite perimeter. There are radii $R_n \uparrow \infty$, each a good radius for E at the origin, such that*

$$a_n := \mathcal{H}^2(E^{(1)} \cap \partial B_{R_n}) \longrightarrow 0, \quad |E \setminus B_{R_n}| \longrightarrow 0, \quad P(E \cap B_{R_n}) \longrightarrow P(E). \quad (5.12)$$

Proof. Corollary 5.6 gives

$$\int_n^{n+1} \mathcal{H}^2(E^{(1)} \cap \partial B_r) dr = |E \cap (B_{n+1} \setminus B_n)| \longrightarrow 0,$$

so by Lemma 5.10 choose $R_n \in (n, n+1)$ among the good radii with $a_n \rightarrow 0$; then $|E \setminus B_{R_n}| \rightarrow 0$ because $|E| < \infty$. Since R_n is strictly increasing, $\mathbf{1}_{E \cap B_{R_n}} \uparrow \mathbf{1}_E$ a.e. The trace identity (5.10) gives $P(E \cap B_{R_n}) = P(E; B_{R_n}) + a_n$, hence $\limsup_n P(E \cap B_{R_n}) \leq P(E)$, while global L^1 convergence and Lemma 4.7 give the reverse $\liminf$. $\square$

Formalisation note 5.14 (Shared by two consumers). Lemma 5.13 is the common first step of Lemma 17.1 and Theorem 12.1. It is extracted here so that neither proof re-derives it, and so that the truncation step carries no dependence on the Coulomb energy.

Proposition 5.15 (Gluing formula). *Let $U \subset \mathbb{R}^3$ be open, let $A \Subset U$ be a bounded C^1 domain, and let E, F have locally finite perimeter in U . Put $H := (F \cap A) \cup (E \cap (U \setminus A))$. Then*

$$\begin{aligned} P(H; U) &\leq P(F; A) + P(E; U \setminus \overline{A}) \\ &\quad + \int_{\partial A} |\text{Tr}_A F - \text{Tr}_{A^c} E| d\mathcal{H}^2. \end{aligned} \quad (5.13)$$

For globally finite-perimeter E, F , taking $U = \mathbb{R}^3$ gives the global formula.

Proof. Apply (5.8) on both sides of ∂A and add the distributional derivatives; the boundary measures combine into the trace mismatch term. $\square$

Proposition 5.16 (Annular gluing at a good radius). *Let $A_{a,b} := B_b \setminus \overline{B_a}$, suppose $E_j \rightarrow E$ in $L^1(A_{a,b})$, suppose E_j, E, F have locally finite perimeter, and for a.e. $r \in (a, b)$ put $H_{j,r} := (F \cap B_r) \cup (E_j \setminus B_r)$. Then for every open W with $\overline{B_b} \Subset W$,*

$$P(H_{j,r}; W) \leq P(F; B_r) + P(E_j; W \setminus \overline{B_r}) + \mathcal{H}^2((F^{(1)} \triangle E_j^{(1)}) \cap \partial B_r), \quad (5.14)$$

and

$$\int_a^b \mathcal{H}^2((F^{(1)} \triangle E_j^{(1)}) \cap \partial B_r) dr = |(F \triangle E_j) \cap A_{a,b}|. \quad (5.15)$$

Consequently, if $F = E$ a.e. on the annulus, there are radii $r_j \in (a, b)$ for which the last term of (5.14) tends to zero; a diagonal choice handles expanding annuli. If all three sets have globally finite perimeter, exhaustion by $W \uparrow \mathbb{R}^3$ gives the same formula with global perimeters.

Proof. (5.14) is (5.13) with $A = B_r$; the coarea identity (5.15) is Corollary 5.6 applied to $F \triangle E_j$. The averaging conclusion is Chebyshev in r . $\square$

Formalisation note 5.17. Proposition 5.16 is the precise content of every “interpolate across a slicing annulus” step in this document (Lemma 22.8, Theorem 19.5, Lemma 24.8). No separate gluing theorem may be invoked.

Chapter 6

The lower-dimensional area formula

Lean module: NoCompromise.Area.AlmostLinear, NoCompromise.Area.Cofactor,
 NoCompromise.Area.Formula, NoCompromise.Area.GoodPieces,
 NoCompromise.Area.GoodPiecesCover, NoCompromise.Area.Graph, NoCompromise.Area.Linear,
 NoCompromise.Area.PieceFormula, NoCompromise.Area.RankDeficient,
 NoCompromise.Area.Rectifiable, NoCompromise.Area.Sphere

The base supplies the equidimensional change of variables only (Formalisation note [1.2](#)). This chapter derives the 2-dimensional area formula in $\mathbb{R}^m$ and its consequence for rectifiable subsets of $\mathbb{R}^3$. The decomposition into four elementary lemmas is deliberate: each is separately formalisable and only the last needs Rademacher.

6.1 Linear distortion

Definition 6.1 (2-Jacobian). For a linear $L : \mathbb{R}^2 \rightarrow \mathbb{R}^m$ put $J_2(L) := \sqrt{\det(L^\top L)}$. For $f : \mathbb{R}^2 \rightarrow \mathbb{R}^m$ Lipschitz set $J_2 f(x) := J_2(Df(x))$ at differentiability points and $J_2 f(x) := 0$ elsewhere.

Lemma 6.2 (Linear image measure). *Let $L : \mathbb{R}^2 \rightarrow \mathbb{R}^m$ have rank two with singular values $s_1 \geq s_2 > 0$, and let $B \subset \mathbb{R}^2$ be Borel. Then*

$$\mathcal{H}^2(LB) = s_1 s_2 |B| = J_2(L) |B|. \quad (6.1)$$

Proof. Choose orthonormal singular bases. Isometries preserve Hausdorff measure; the remaining diagonal map is handled first on rectangles and then on Borel sets by outer regularity. $\square$

Constants 6.3. In Lemma [6.4](#) below, C_L may be chosen uniformly over all L with $\|L\| \leq N$ and $s_2(L) \geq 1/N$; it then depends only on N .

Lemma 6.4 (Almost-linear maps). *Let $L : \mathbb{R}^2 \rightarrow \mathbb{R}^m$ have rank two with singular values $s_1 \geq s_2 > 0$, let $B \subset \mathbb{R}^2$ be Borel, and let $h : B \rightarrow \mathbb{R}^m$ satisfy*

$$|h(x) - h(y) - L(x - y)| \leq \varepsilon |x - y| \quad (x, y \in B), \quad \varepsilon < s_2/2. \quad (6.2)$$

Then h is injective, its inverse on $h(B)$ is Lipschitz, and

$$(J_2(L) - C_L \varepsilon) |B| \leq \mathcal{H}^2(h(B)) \leq (J_2(L) + C_L \varepsilon) |B|. \quad (6.3)$$

Proof. Injectivity and the Lipschitz inverse follow from [\(6.2\)](#) and $\varepsilon < s_2/2$. Orthogonal projection onto $L(\mathbb{R}^2)$ followed by L^{-1} differs from the identity by at most ε/s_2 in Lipschitz norm. Apply $\mathcal{H}^2(TC) \leq \text{Lip}(T)^2 \mathcal{H}^2(C)$ to this projection and to h , and combine with [\(6.1\)](#). $\square$

6.2 The area formula in the plane

Definition 6.5 (Uniform differentiability piece). Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}^m$ be Lipschitz. A Borel set G is a *uniform differentiability piece* if there are a compact carrier $G \subset K \subset [-N, N]^2$, an integer $N \geq 1$, and a nondecreasing modulus $\omega : [0, \infty) \rightarrow [0, \infty)$ with $\omega(r) \downarrow 0$ as $r \downarrow 0$ such that f is differentiable at every $x \in K$ and, for all distinct $x, z \in K$,

$$\begin{aligned} \|Df(x)\| &\leq N, & s_2(Df(x)) &\geq 1/N, \\ |f(z) - f(x) - Df(x)(z - x)| &\leq \omega(|z - x|)|z - x|, \\ \|Df(x) - Df(z)\| &\leq \omega(|x - z|). \end{aligned} \tag{6.4}$$

Every Borel subset of such a piece is again a uniform differentiability piece with the same compact carrier, N , and ω .

Lemma 6.6 (The uniform pieces exhaust the rank-two set). *The rank-two differentiability points of a Lipschitz map $f : \mathbb{R}^2 \rightarrow \mathbb{R}^m$ are, up to a Lebesgue-null set, a countable disjoint union of uniform differentiability pieces.*

Proof. Partition first by integer bounds for $|x|$, $\|Df(x)\|$, and $s_2(Df(x))^{-1}$. Lusin's theorem gives countably many compact subsets on which Df is uniformly continuous, up to a null set. At a differentiability point put

$$R_x(r) := \sup_{0 < |y-x| \leq r} \frac{|f(y) - f(x) - Df(x)(y-x)|}{|y-x|}.$$

The supremum may be taken over a countable dense set, so $R_x(r)$ is measurable, and differentiability says $R_x(r) \rightarrow 0$. Egorov on the compact Lusin pieces, followed by a countable exhaustion, makes this convergence uniform. The maximum of the two resulting uniform moduli gives (6.4) on each compact set. Finally replace the countable cover by successive set differences; each resulting Borel piece retains the compact set from which it came as a carrier. $\square$

Lemma 6.7 (Area formula on one uniform piece). *Let G be a uniform differentiability piece, let $A \subset G$ be Borel, and let $q : A \rightarrow [0, \infty]$ be Borel. Then*

$$\int_A q(x) J_2 f(x) dx = \int_{\mathbb{R}^m} \sum_{x \in A \cap f^{-1}(y)} q(x) d\mathcal{H}^2(y). \tag{6.5}$$

Proof. Let K, N, ω be as in Definition 6.5. Partition $[-N, N]^2$ into half-open grid squares of side r . If r is small enough that $4\omega(\sqrt{2}r) < 1/N$, then on the intersection of G with any one square, (6.4) and the last inequality there show that f differs, in the pairwise Lipschitz sense, from any one base derivative $Df(x_Q)$ by at most $2\omega(\sqrt{2}r)$. Lemma 6.4 in fact makes f injective on $K \cap \overline{Q}$ and gives, uniformly for every Borel subset C of $G \cap \overline{Q}$,

$$\left| \mathcal{H}^2(f(C)) - \int_C J_2 f(x) dx \right| \leq C_N \omega(\sqrt{2}r) |C|.$$

Here the replacement of $J_2(Df(x_Q))$ by $J_2(Df(x))$ uses the last line of (6.4) and the uniform local Lipschitz continuity of J_2 on the compact family $\{L : \|L\| \leq N, s_2(L) \geq 1/N\}$. The image $f(C)$ is Borel without a descriptive-set theorem: the continuous injection $f : K \cap \overline{Q} \rightarrow \mathbb{R}^m$ is a homeomorphism onto its compact image, and C is a Borel subset of that compact domain.

Sum this estimate over the grid squares. On the target side the sum of the image indicators is exactly the number of preimages, because the source squares are disjoint and f is injective within each square. The total error is at most $C_N \omega(\sqrt{2}r)|G|$, which tends to zero. This proves (6.5) for $q = 1$. Applying the same argument to Borel subsets of A gives simple weights, and monotone convergence gives every nonnegative Borel q . $\square$

Lemma 6.8 (The rank-deficient part is null in the image). *For $f : \mathbb{R}^2 \rightarrow \mathbb{R}^m$ Lipschitz, $\mathcal{H}^2(f(\{J_2 f = 0\})) = 0$.*

Proof. Let Z be the set of differentiability points at which $J_2 f = 0$. Partition Z by bounds $|x| \leq R$ and $\|Df(x)\| \leq N$. As in the Egorov step of Lemma 6.6, discard a null set and split the remainder into countably many Borel pieces G . With

$$R_x(r) := \sup_{0 < |y-x| \leq r} \frac{|f(y) - f(x) - Df(x)(y-x)|}{|y-x|},$$

these pieces may be chosen so that on each of them

$$\sup_{x \in G} R_x(r) =: \omega_G(r) \rightarrow 0.$$

Fix one such G and cover it by grid squares Q of side r , choosing $x_Q \in G \cap Q$ in each occupied square. For $y \in G \cap Q$, differentiability at x_Q puts $f(y)$ within $C\omega_G(2r)r$ of $f(x_Q) + Df(x_Q)(y-x_Q)$. Since $J_2 f(x_Q) = 0$, the linear image of Q is a line segment of length at most $C_N r$ (or a point). With $\varepsilon_r := \max\{\omega_G(2r), r\} \downarrow 0$, the set $f(G \cap Q)$ is therefore covered by at most C_N/ε_r balls of radius $C\varepsilon_r r$. The sum of their squared diameters is at most $C_N \varepsilon_r r^2$.

There are $O_R(r^{-2})$ occupied squares. Hence the 2-dimensional Hausdorff content of $f(G)$ at scale $C\varepsilon_r r$ is at most $C_{N,R} \varepsilon_r \rightarrow 0$, proving $\mathcal{H}^2(f(G)) = 0$. The discarded nondifferentiability and Egorov sets are Lebesgue-null, and their images are $\mathcal{H}^2$ -null by the elementary Lipschitz inequality $\mathcal{H}^2(f(B)) \leq \text{Lip}(f)^2 \mathcal{H}^2(B)$. Countable union over all pieces, N , and R proves the claim, including the points where $J_2 f$ was defined to be zero because f is not differentiable. $\square$

Theorem 6.9 (Two-dimensional area formula with multiplicity). *Let $A \subset \mathbb{R}^2$ be Borel, $f : \mathbb{R}^2 \rightarrow \mathbb{R}^m$ Lipschitz with $m \geq 2$, and $q : \mathbb{R}^2 \rightarrow [0, \infty]$ Borel. Then*

$$\int_A q(x) J_2 f(x) dx = \int_{\mathbb{R}^m} \sum_{x \in A \cap f^{-1}(y)} q(x) d\mathcal{H}^2(y). \quad (6.6)$$

Proof. Lemma 6.6 partitions the rank-two differentiability set, modulo a null set, into countably many disjoint uniform pieces G_j . Apply Lemma 6.7 to $A \cap G_j$ and sum; on the target side the sums over j combine into the full preimage sum. Everything left in the source is either the null exceptional set discarded in Lemma 6.6, whose image is $\mathcal{H}^2$ -null by the Lipschitz inequality, or lies in $\{J_2 f = 0\}$, whose image is $\mathcal{H}^2$ -null by Lemma 6.8. Both sides therefore ignore it, proving (6.6) with the stated nonnegative Borel weight. $\square$

6.3 Rectifiable sets in $\mathbb{R}^3$

Definition 6.10 (Countably $\mathcal{H}^2$ -rectifiable). $S \subset \mathbb{R}^3$ is *countably $\mathcal{H}^2$ -rectifiable* if it is $\mathcal{H}^2$ -measurable and, up to an $\mathcal{H}^2$ -null set, contained in a countable union of images of Lipschitz maps $\mathbb{R}^2 \rightarrow \mathbb{R}^3$. Its approximate tangent plane $T_x S$ is taken in the *chartwise* sense: fix, once for S , a

countable family of disjoint Borel pieces $A_j \subset \mathbb{R}^2$ and injective Lipschitz maps ψ_j with $D\psi_j$ of rank two on A_j whose images cover S up to an $\mathcal{H}^2$ -null set (such a family exists by Rademacher's theorem), and set $T_x S := \text{range } D\psi_j(z)$ at $x = \psi_j(z)$, $z \in A_j$. Statements about $T_x S$ refer to this fixed family.

Theorem 6.11 (Area formula on a rectifiable set). *Let $S \subset \mathbb{R}^3$ be countably $\mathcal{H}^2$ -rectifiable, $\Phi : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ Lipschitz, and $q : S \rightarrow [0, \infty]$ Borel. Then at $\mathcal{H}^2$ -a.e. $x \in S$ there is a tangential approximate differential $D^S \Phi(x) : T_x S \rightarrow \mathbb{R}^3$, and*

$$\int_S q(x) J_2(D^S \Phi(x)) d\mathcal{H}^2(x) = \int_{\mathbb{R}^3} \sum_{x \in S \cap \Phi^{-1}(y)} q(x) d\mathcal{H}^2(y). \quad (6.7)$$

If Φ is differentiable at x (in particular, if it is C^1 near S), then $D^S \Phi(x) = D\Phi(x)|_{T_x S}$.

Proof. Refine a countable Lipschitz parametrisation of S into disjoint injective pieces $\psi_j : A_j \rightarrow S$, discarding the parameter sets where $J_2 \psi_j = 0$ (null in the image by Lemma 6.8). Apply Rademacher in the parameter plane to ψ_j and $\Phi \circ \psi_j$. At a rank-two point define the unique map $D^S \Phi$ by

$$D^S \Phi(\psi_j(z)) \circ D\psi_j(z) = D(\Phi \circ \psi_j)(z).$$

This is independent of the injective chart: on an overlap, both candidates give the approximate linearisation of Φ along the common approximate tangent plane, and uniqueness follows by testing on two independent tangent density directions. Apply (6.6) to ψ_j and to $\Phi \circ \psi_j$; the a.e. chain rule and cancellation of the common factor $J_2 \psi_j$ give (6.7). Ordinary differentiability of Φ makes the displayed defining identity the usual chain rule and gives the final assertion. $\square$

Lemma 6.12 (Cofactor formula). *For linear $L : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ and a unit vector ν ,*

$$J_2(L|_{\nu^\perp}) = |\text{cof}(L)| \nu. \quad (6.8)$$

Proof. Choose an oriented orthonormal basis a, b of $\nu^\perp$ and use $La \times Lb = \text{cof}(L)(a \times b)$ together with $a \times b = \nu$. $\square$

Corollary 6.13 (Graph area). *For Lipschitz $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ and Borel $G \subset \mathbb{R}^2$,*

$$\mathcal{H}^2(\{(x', f(x')) : x' \in G\}) = \int_G \sqrt{1 + |\nabla f|^2} dx',$$

and for the upward unit normal $\nu = (-\nabla f, 1)/\sqrt{1 + |\nabla f|^2}$ one has $d\mathcal{H}^2 = \sqrt{1 + |\nabla f|^2} dx'$ on the graph.

Proof. Apply (6.6) to $x' \mapsto (x', f(x'))$, whose 2-Jacobian is $\sqrt{1 + |\nabla f|^2}$ by Definition 6.1. $\square$

Lemma 6.14 (Smooth surfaces are rectifiable). *Let $\Sigma \subset \mathbb{R}^3$ be a compact smooth embedded surface. Then Σ is countably $\mathcal{H}^2$ -rectifiable, $\mathcal{H}^2(\Sigma) < \infty$, and for every $x \in \Sigma$ the approximate tangent plane of Definition 6.10 coincides with the classical tangent plane $T_x \Sigma$.*

Proof. Cover Σ by finitely many relatively compact graph subcharts; each is the image of a smooth map whose derivative is bounded on the closure, hence of a Lipschitz map from a bounded planar domain. Then Corollary 6.13 gives its finite area and identifies the tangent plane. $\square$

Lemma 6.15 (Area of a sphere). *For every $R > 0$, $\mathcal{H}^2(\partial B_R) = 4\pi R^2$; in particular $\mathcal{H}^2(\mathbb{S}^2) = 4\pi$.*

Proof. Write ∂B_R as the union of the two graphs $x_3 = \pm\sqrt{R^2 - |x'|^2}$ over B_R^2 , whose common boundary circle is $\mathcal{H}^2$ -null. These graph functions are not Lipschitz up to ∂B_R^2 , so first apply Corollary 6.13 on $B_{(1-1/j)R}^2$, $j \geq 2$, and then let $j \rightarrow \infty$ by monotone convergence. This gives

$$\mathcal{H}^2(\partial B_R) = 2 \int_{B_R^2} \frac{R}{\sqrt{R^2 - |x'|^2}} dx' = 2 \cdot 2\pi \int_0^R \frac{Rr dr}{\sqrt{R^2 - r^2}} = 4\pi R^2.$$

□

Formalisation note 6.16 (A small gap worth closing early). Lemma 6.15 is used by Lemma 16.2 and by Theorem 14.43, and it is the kind of fact that is easy to assume without proof. Note that it must *not* be derived from $P(B_R) = 4\pi R^2$, which is how it would naturally be quoted: that identity is proved *from* this lemma via Corollary 8.14.

Chapter 7

Sobolev, Poincaré and relative isoperimetry

Lean module: NoCompromise.Sobolev.BV, NoCompromise.Sobolev.Extension,
NoCompromise.Sobolev.Maximal, NoCompromise.Sobolev.Poincare,
NoCompromise.Sobolev.RelativeCubes, NoCompromise.Sobolev.RelativeScaling

7.1 The BV Sobolev inequality

Lemma 7.1 (Gagliardo–Nirenberg in $\mathbb{R}^3$, smooth case). *For $f \in C_c^1(\mathbb{R}^3)$,*

$$\|f\|_{L^{3/2}(\mathbb{R}^3)} \leq C \prod_{i=1}^3 \|\partial_i f\|_{L^1}^{1/3} \leq \frac{C}{3} \sum_{i=1}^3 \|\partial_i f\|_{L^1}. \quad (7.1)$$

Proof. Put $A_i(x_1, \dots, \hat{x}_i, \dots, x_3) := \int_{\mathbb{R}} |\partial_i f| dx_i$. The fundamental theorem of calculus gives $|f(x)| \leq A_i$ for each i , hence $|f|^{3/2} \leq (A_1 A_2 A_3)^{1/2}$. Fubini and Cauchy–Schwarz successively in x_1, x_2, x_3 give $\int_{\mathbb{R}^3} (A_1 A_2 A_3)^{1/2} \leq \prod_i \|\partial_i f\|_{L^1}^{1/2}$. Raise to the power $2/3$ and use the arithmetic–geometric mean inequality. $\square$

Theorem 7.2 (BV Sobolev inequality). *For $f \in BV(\mathbb{R}^3)$,*

$$\|f\|_{L^{3/2}(\mathbb{R}^3)} \leq C |Df|(\mathbb{R}^3), \quad (7.2)$$

with C absolute. In particular, for E of finite measure and finite perimeter, $|E|^{2/3} \leq CP(E)$.

Proof. (7.1) plus strict approximation (Theorem 4.18) and Fatou. $\square$

Lemma 7.3 (The planar analogue). *For $f \in C_c^1(\mathbb{R}^2)$,*

$$\|f\|_{L^2(\mathbb{R}^2)}^2 \leq \|\partial_1 f\|_{L^1} \|\partial_2 f\|_{L^1}, \quad \|f\|_{L^2(\mathbb{R}^2)} \leq \frac{1}{2} \sum_{i=1}^2 \|\partial_i f\|_{L^1}, \quad (7.3)$$

For $f \in BV(\mathbb{R}^2)$ the precise measure version is

$$\|f\|_{L^2}^2 \leq |D_1 f|(\mathbb{R}^2) |D_2 f|(\mathbb{R}^2), \quad \|f\|_{L^2} \leq \frac{1}{2} (|D_1 f|(\mathbb{R}^2) + |D_2 f|(\mathbb{R}^2)) \leq |Df|(\mathbb{R}^2).$$

Proof. The identical two-coordinate fundamental-theorem-of-calculus calculation as in Lemma 7.1 proves the two inequalities for $C_c^1(\mathbb{R}^2)$. For $f \in BV(\mathbb{R}^2)$ put $h_\varepsilon := \rho_\varepsilon * f$. The distributional convolution identities give

$$\|\partial_i h_\varepsilon\|_{L^1} \leq |D_i f|(\mathbb{R}^2), \quad i = 1, 2.$$

Apply the smooth inequalities first to $\chi_R h_\varepsilon$ and let $R \rightarrow \infty$; the cutoff error is at most $CR^{-1}\|h_\varepsilon\|_1$ and hence vanishes. Finally $h_\varepsilon \rightarrow f$ in L^1 , so a subsequence converges a.e.; Fatou gives the claimed coordinatewise measure bounds and then the arithmetic–geometric mean bound. This coordinatewise argument is stronger than isotropic strict convergence and does not infer convergence of $|D_i f|$ from Theorem 4.18. $\square$

7.2 Extension and Poincaré

Lemma 7.4 (Bi-Lipschitz chain rule in BV). *Let $U, V \subset \mathbb{R}^n$ be open, let $\Phi : U \rightarrow V$ be a bi-Lipschitz homeomorphism, let $D \subset V$ be open, and let $g \in BV(D)$. Then $g \circ \Phi \in BV(\Phi^{-1}(D))$ and*

$$\|g \circ \Phi\|_{L^1(\Phi^{-1}(D))} + |D(g \circ \Phi)|(\Phi^{-1}(D)) \leq C(\|g\|_{L^1(D)} + |Dg|(D)),$$

where C depends only on n , $\text{Lip}(\Phi)$, and $\text{Lip}(\Phi^{-1})$.

Proof. Take a strict smooth approximation g_j of g . Ordinary change of variables and $\nabla(g_j \circ \Phi) = D\Phi^\top(\nabla g_j) \circ \Phi$ bound both the L^1 norm and the variation of $g_j \circ \Phi$; lower semicontinuity (Lemma 4.7) passes the estimate to $g \circ \Phi$. $\square$

Lemma 7.5 (BV extension). *Let $D \subset \mathbb{R}^n$ ($n = 2, 3$) be a bounded connected Lipschitz domain. There is a bounded linear $\mathcal{E} : BV(D) \rightarrow BV(\mathbb{R}^n)$, with image supported in one fixed bounded neighbourhood of $\overline{D}$, such that*

$$\|\mathcal{E}g\|_{L^1(\mathbb{R}^n)} + |D\mathcal{E}g|(\mathbb{R}^n) \leq C_D(\|g\|_{L^1(D)} + |Dg|(D)). \quad (7.4)$$

Proof. First consider a flat half-cube $Q^+ := (-1, 1)^{n-1} \times (0, 1)$ and a smaller cube $Q_0 \Subset (-1, 1)^n$. For $g \in BV(Q^+)$ define its even reflection by $Rg(x', x_n) := g(x', |x_n|)$. If $X \in C_c^1(Q_0; \mathbb{R}^n)$, split $\int Rg \operatorname{div} X$ over the two half-cubes and change x_n to $-x_n$ in the lower one. The result is the pairing of g with the divergence of the upper test field plus the reflected lower test field, with the normal component reflected with a minus sign. Its pointwise norm is at most $2\|X\|_\infty$. Definition 4.2 therefore gives

$$\|Rg\|_{L^1(Q_0)} \leq 2\|g\|_{L^1(Q^+)}, \quad |D(Rg)|(Q_0) \leq 2|Dg|(Q^+).$$

This direct calculation is the reason no trace theorem for Lipschitz domains is being assumed.

Cover ∂D by finitely many Lipschitz graph charts, flatten each by a bi-Lipschitz map, use the preceding even reflection, multiply by a smooth partition of unity, and extend the piece supported strictly inside D by zero. Lemma 7.4 controls each flattening, Lemma 4.17 controls the cutoffs, and the finite sum of the extended pieces defines the required bounded linear map. Choosing all cutoffs in one fixed bounded collar gives the support assertion. $\square$

Theorem 7.6 (BV–Poincaré). *Let D be a bounded connected Lipschitz domain in $\mathbb{R}^n$, $n = 2, 3$, and $g_D := f_D g$. Then*

$$\|g - g_D\|_{L^1(D)} \leq C_D |Dg|(D), \quad \|g - g_D\|_{L^{n/(n-1)}(D)} \leq C_D |Dg|(D). \quad (7.5)$$

Proof. Suppose the first inequality failed: there would be $f_j \in BV(D)$ with $(f_j)_D = 0$, $\|f_j\|_{L^1(D)} = 1$ and $|Df_j|(D) \rightarrow 0$. By (7.4) the $\mathcal{E}f_j$ are bounded in $BV(\mathbb{R}^n)$ and supported in one fixed compact set; apply the grid compactness argument of Theorem 4.11 on a cube containing that set. A subsequence converges in $L^1(\mathbb{R}^n)$, hence in $L^1(D)$, to some f . Testing distributional derivatives compactly inside D gives $Df = 0$, so f is constant on balls and hence on connected D ; mean zero makes the constant zero, contradicting $\|f\|_{L^1(D)} = 1$. The second inequality follows by applying Theorem 7.2 (or Lemma 7.3 when $n = 2$) to $\mathcal{E}(g - g_D)$ and using the first. $\square$

7.3 Relative isoperimetric inequalities

Remark 7.7 (Three distinct signatures). The cube, ball and planar-disk cases below are *separate statements* with separate constants. The ball case is *not* obtained from the cube case by inclusion — the relative perimeter on the right would be the wrong one. Two of the three are used: (7.6) in Theorem 19.4, and (7.7) in Lemma 21.6 and Lemma 21.8. (7.8) is *not* on the critical path; it is recorded for completeness of the planar toolkit and may be skipped (Formalisation note 40.2).

Proposition 7.8 (Relative isoperimetry, cube). *For a cube $Q \subset \mathbb{R}^3$ of side ℓ and a measurable E with $\mathbf{1}_E \in BV(Q)$,*

$$\min\{|E \cap Q|, |Q \setminus E|\}^{2/3} \leq C_{\text{cube}} P(E; Q), \quad (7.6)$$

with C_{cube} absolute (scale-invariant).

Proof. Apply (7.5) on Q to $g = \mathbf{1}_E$, and put $\theta := |E \cap Q|/|Q|$. If $\theta \leq 1/2$, then $|\mathbf{1}_E - \theta| \geq 1/2$ on $E \cap Q$; if $\theta \geq 1/2$, apply the same argument to $Q \setminus E$. Scaling the unit-cube inequality gives (7.6) with an absolute constant. $\square$

Proposition 7.9 (Relative isoperimetry, ball). *For a ball $B \subset \mathbb{R}^3$ and a measurable E with $\mathbf{1}_E \in BV(B)$,*

$$\min\{|E \cap B|, |B \setminus E|\}^{2/3} \leq C_{\text{ball}} P(E; B), \quad (7.7)$$

with C_{ball} absolute.

Proof. Apply (7.5) on B to $g = \mathbf{1}_E$ and use the same two cases $|E \cap B| \leq |B|/2$ and $|B \setminus E| \leq |B|/2$. Dilation from the unit ball gives (7.7) with an absolute constant. $\square$

Proposition 7.10 (Relative isoperimetry, planar disk). *For a planar disk $D \subset \mathbb{R}^2$ and a measurable $A \subset \mathbb{R}^2$ with $\mathbf{1}_A \in BV(D)$,*

$$\min\{|A \cap D|, |D \setminus A|\}^{1/2} \leq C P_2(A; D). \quad (7.8)$$

Proof. Apply (7.5) in dimension two on D to $g = \mathbf{1}_A$. Splitting according as $|A \cap D| \leq |D|/2$ or $|D \setminus A| \leq |D|/2$, and then scaling the unit-disk inequality, gives (7.8). $\square$

7.4 The weak maximal estimate

Definition 7.11 (Planar maximal density). For a finite Radon measure μ on $\mathbb{R}^2$ put $M\mu(x) := \sup_{r>0} \mu(B_r^2(x))/(\pi r^2)$, and for $R > 0$ put $M_R\mu(x) := \sup_{0<r<R} \mu(B_r^2(x))/(\pi r^2)$.

Lemma 7.12 (Weak maximal estimate). *For a finite Radon measure μ on $\mathbb{R}^2$ and $t > 0$,*

$$|\{M\mu > t\}| \leq \frac{25}{t} \mu(\mathbb{R}^2), \quad (7.9)$$

and the same bound holds for $M_R\mu$.

Proof. Let $K \subset \{M\mu > t\}$ be compact and choose for each $x \in K$ a ball $B_x = B_{r_x}^2(x)$ with $\mu(B_x) > t\pi r_x^2$. The five-covering lemma (a consequence of (B5)) gives a disjoint subfamily B_i whose fivefold dilates cover K , so $|K| \leq 25 \sum_i \pi r_i^2 < \frac{25}{t} \sum_i \mu(B_i) \leq \frac{25}{t} \mu(\mathbb{R}^2)$. Inner regularity removes the compactness assumption. Restricting radii gives the M_R version. $\square$

Chapter 8

De Giorgi's structure theorem

Lean module: NoCompromise.DeGiorgi.Blowup, NoCompromise.DeGiorgi.ConeConcentration,
 NoCompromise.DeGiorgi.ExactDensity, NoCompromise.DeGiorgi.NormalFlux,
 NoCompromise.DeGiorgi.Rectifiability, NoCompromise.DeGiorgi.Reduced,
 NoCompromise.DeGiorgi.ReducedDensity, NoCompromise.DeGiorgi.ReducedPerimeter,
 NoCompromise.DeGiorgi.SmoothBoundary, NoCompromise.DeGiorgi.Structure

No perimeter minimality is used anywhere in this chapter. The output is Theorem 8.12 and its corollary Theorem 8.13, which are the interface every later chapter uses.

Notation 8.1. Throughout the chapter $E \subset \mathbb{R}^3$ has locally finite perimeter, $\mu_E := |D\mathbf{1}_E|$, and $D\mathbf{1}_E = -\nu_E \mu_E$ per Convention 2.3.

8.1 The reduced boundary

Definition 8.2 (Reduced boundary). Put

$$\partial^* E := \left\{ x \in \text{spt } \mu_E : \lim_{r \downarrow 0} \frac{D\mathbf{1}_E(B_r(x))}{\mu_E(B_r(x))} \text{ exists and has norm } 1 \right\}, \quad (8.1)$$

and at such x define $\nu_E(x)$ to be the *negative* of that limit, consistently with Convention 2.3. The set $\partial^* E$ is Borel: $x \mapsto \mu_E(B_r(x))$ and $x \mapsto D\mathbf{1}_E(B_r(x))$ are Borel for fixed r , the limit can be tested along rational radii, and $\text{spt } \mu_E$ is closed.

Lemma 8.3 (Polar differentiation). *For μ_E -a.e. x ,*

$$\lim_{r \downarrow 0} \frac{1}{\mu_E(B_r(x))} \int_{B_r(x)} |\nu_E(y) - \nu_E(x)| d\mu_E(y) = 0. \quad (8.2)$$

Consequently

$$\mu_E(\mathbb{R}^3 \setminus \partial^* E) = 0. \quad (8.3)$$

Proof. Write $\nu_E = (\nu_1, \nu_2, \nu_3)$ and decompose each bounded component as $\nu_i = \nu_i^+ - \nu_i^-$ with $\nu_i^\pm \geq 0$. Each $\nu_i^\pm \mu_E$ is a finite positive Radon measure on compact sets, so **Besicovitch.ae_tendsto_rnDeriv** ((B5)) applies to it against μ_E and identifies its ball averages with its density $\nu_i^\pm$ at μ_E -a.e. point. Summing the six statements and using the triangle inequality gives (8.2). Discarding a μ_E -null set puts such points in $\text{spt } \mu_E$, whence (8.3). $\square$

Formalisation note 8.4 (Which declaration to use). The statement needed is a Lebesgue-point property of an $L^1(\mu_E)$ vector field, not a ratio of two measures. The componentwise route above uses only `Besicovitch.ae_tendsto_rnDeriv`, which is in the base. A packaged alternative is `VitaliFamily.ae_tendsto_average_norm_sub` at `Besicovitch.vitaliFamily`, converting balls to the Vitali filter by `Besicovitch.tendsto_filterAt`; either is acceptable.

8.2 Density and perimeter bounds at a reduced point

Lemma 8.5 (Normal-flux comparison). *Let $0 \in \partial^* E$ with $\nu_E(0) = e_3$, and write $\mu := \mu_E$. Then for all sufficiently small $r > 0$,*

$$\mu(B_r) \leq -2 D\mathbf{1}_E(B_r) \cdot e_3, \quad (8.4)$$

and for a.e. $r > 0$,

$$D\mathbf{1}_E(B_r) \cdot e_3 = \int_{E^{(1)} \cap \partial B_r} \frac{x_3}{r} d\mathcal{H}^2, \quad |D\mathbf{1}_E(B_r) \cdot e_3| \leq \mathcal{H}^2(E^{(1)} \cap \partial B_r). \quad (8.5)$$

Proof. (8.4) is (8.2) at $x = 0$. For (8.5), no Gauss–Green theorem is available yet. Instead test the defining distributional identity for $D\mathbf{1}_E$ with $X(x) = \psi(|x|)e_3$, where $\psi \in C_c^1((0, \infty))$. Radial coarea gives

$$\int \psi(|x|)e_3 \cdot dD\mathbf{1}_E(x) = - \int_0^\infty \psi'(r) \int_{E^{(1)} \cap \partial B_r} \frac{x_3}{r} d\mathcal{H}^2 dr.$$

Thus the pushforward of $e_3 \cdot D\mathbf{1}_E$ by $x \mapsto |x|$ is the distributional derivative of the locally integrable inner integral. Taking cumulative masses from 0 to r yields $D\mathbf{1}_E(B_r) \cdot e_3 = \int_{E^{(1)} \cap \partial B_r} x_3/r d\mathcal{H}^2$ for a.e. r . Lemma 5.7 also gives $D\mathbf{1}_E(\{0\}) = 0$: each coordinate derivative disintegrates over parallel lines, and the single transverse base point has measure zero. The possible atom on ∂B_r is absent for a.e. r , so the cumulative mass is exactly the mass of B_r . The final inequality is immediate. $\square$

Lemma 8.6 (Two-sided volume density at a reduced point). *For every $x \in \partial^* E$,*

$$\liminf_{r \downarrow 0} \frac{|E \cap B_r(x)|}{r^3} > 0, \quad \liminf_{r \downarrow 0} \frac{|B_r(x) \setminus E|}{r^3} > 0. \quad (8.6)$$

Proof. Normalise $x = 0$, $\nu_E(0) = e_3$, and put $f(r) := |E \cap B_r|$, so $f'(r) = \mathcal{H}^2(E^{(1)} \cap \partial B_r)$ for a.e. r by Corollary 5.6. Apply Theorem 7.2 to $E \cap B_r$, use (5.10), and combine with (8.4)–(8.5):

$$f(r)^{2/3} \leq C(\mu(B_r) + f'(r)) \leq C f'(r). \quad (8.7)$$

Since $f(r) \downarrow 0$ as $r \downarrow 0$, integrating $(f^{1/3})' \geq c$ from 0 to r gives $f(r) \geq cr^3$. Apply the same argument to $\mathbb{R}^3 \setminus E$, whose outward normal at 0 is $-e_3$ (Lemma 4.6(iv)). $\square$

Lemma 8.7 (Perimeter upper bound at a reduced point). *For $x \in \partial^* E$ there is $C_x < \infty$ with $\mu_E(B_r(x)) \leq C_x r^2$ for all small r .*

Proof. Combine (8.4)–(8.5) with $\mathcal{H}^2(E^{(1)} \cap \partial B_r) \leq 4\pi r^2$ at a.e. sufficiently small radius. For an arbitrary small r , choose such radii $r_k \downarrow r$ and use $\mu_E(B_r) \leq \liminf_k \mu_E(B_{r_k})$. $\square$

8.3 Halfspace blow-up and the exact density

Theorem 8.8 (Halfspace blow-up). *For every $x \in \partial^* E$,*

$$\mathbf{1}_{(E-x)/r} \longrightarrow \mathbf{1}_{\{y: y \cdot \nu_E(x) < 0\}} \quad \text{in } L^1_{\text{loc}}(\mathbb{R}^3) \text{ as } r \downarrow 0. \quad (8.8)$$

Proof. Normalise $x = 0$, $\nu_E(0) = e_3$. For any $r_j \downarrow 0$ put $E_j := E/r_j$. Lemmas 8.6 and 8.7 plus Theorem 4.11 give a subsequence with $\mathbf{1}_{E_j} \rightarrow \mathbf{1}_F$ in L^1_{loc} , F nontrivial. Introduce $\mu_j(A) := r_j^{-2} \mu_E(r_j A)$; by Lemma 8.7 the $\mu_j(B_R)$ are bounded, so Theorem 3.8 gives $\mu_j \rightarrow \lambda$ after a further subsequence. Moreover (8.2) and the same bound give

$$\begin{aligned} |D\mathbf{1}_{E_j} + e_3 \mu_j|(B_R) &\leq r_j^{-2} \int_{B_{Rr_j}} |\nu_E - e_3| d\mu_E \\ &= \frac{\mu_E(B_{Rr_j})}{r_j^2} \cdot \frac{\int_{B_{Rr_j}} |\nu_E - e_3| d\mu_E}{\mu_E(B_{Rr_j})} \longrightarrow 0. \end{aligned} \quad (8.9)$$

The distributional convergence from $\mathbf{1}_{E_j} \rightarrow \mathbf{1}_F$ therefore identifies

$$D\mathbf{1}_F = -e_3 \lambda, \quad |D\mathbf{1}_F| = \lambda, \quad (8.10)$$

so $D\mathbf{1}_F = -e_3 |D\mathbf{1}_F|$. Convolving $\mathbf{1}_F$: the first two derivatives vanish and the third is nonpositive, so every convolution depends only on x_3 and is non-increasing. Letting the convolution scale tend to zero and using (8.6) at every fixed rescaled radius — which shows both phases meet every ball about the origin — gives $F = \{x_3 < 0\}$ modulo null sets. Every subsequence has the same limit, so the full limit (8.8) holds. $\square$

Theorem 8.9 (Exact perimeter density). *For every $x \in \partial^* E$,*

$$\lim_{r \downarrow 0} \frac{\mu_E(B_r(x))}{\pi r^2} = 1. \quad (8.11)$$

Proof. Lemma 4.7 applied to (8.8) gives $\liminf r^{-2} \mu_E(B_r) \geq \pi$. For the reverse, choose $r_j \downarrow 0$ along which $r^{-2} \mu_E(B_r)$ approaches its limsup. Coarea in the radial variable (Corollary 5.6) and (8.8) give, after a subsequence, trace convergence

$$\mathbf{1}_{(E/r_j)^{(1)}} \longrightarrow \mathbf{1}_{\{x_3 < 0\}} \quad \text{in } L^1(\partial B_\rho) \quad (8.12)$$

for a.e. ρ in any fixed compact interval of positive radii: the ρ -integral of the left side tends to zero, so a subsequence converges for a.e. ρ . Fix such a $\rho > 1$, as close to 1 as desired, chosen also so that each ρr_j is a good radius (Lemma 5.10 discards at most a null set of ρ).

The polar error has the exact scalar form

$$\mu_E(B_{\rho r_j}) + D\mathbf{1}_E(B_{\rho r_j}) \cdot e_3 = \int_{B_{\rho r_j}} (1 - \nu_E \cdot e_3) d\mu_E = o(\mu_E(B_{\rho r_j})), \quad (8.13)$$

and Lemma 8.7 makes its value divided by r_j^2 tend to zero. The already proved radial flux identity (8.5) gives

$$-D\mathbf{1}_E(B_{\rho r_j}) \cdot e_3 = - \int_{E^{(1)} \cap \partial B_{\rho r_j}} \frac{x_3}{\rho r_j} d\mathcal{H}^2,$$

and after scaling (8.12) sends the right side divided by r_j^2 to $-\int_{\{x_3 < 0\} \cap \partial B_\rho} (x_3/\rho) d\mathcal{H}^2 = \pi \rho^2$. Hence $\mu_E(B_{\rho r_j})/r_j^2 \rightarrow \pi \rho^2$, and $B_{r_j} \subset B_{\rho r_j}$ gives $\limsup_j \mu_E(B_{r_j})/r_j^2 \leq \pi \rho^2$. Let $\rho \downarrow 1$. $\square$

Lemma 8.10 (Cone concentration). *Let $x \in \partial^* E$, $T := \nu_E(x)^\perp$, and $\mu_r(A) := r^{-2}\mu_E(x + rA)$. For every $\varepsilon > 0$, with*

$$V_\varepsilon := \{y \in B_1 : \text{dist}(y, T) > \varepsilon|y|\},$$

one has $\lim_{r \downarrow 0} \mu_r(V_\varepsilon) = 0$. Equivalently, writing $X(x, T, \varepsilon) := \{y : \text{dist}(y - x, T) < \varepsilon|y - x|\}$,

$$\lim_{r \downarrow 0} \frac{\mu_E(B_r(x) \setminus X(x, T, \varepsilon))}{\mu_E(B_r(x))} = 0. \quad (8.14)$$

Proof. The concentration does *not* follow from the scalar density limit alone; it follows from a mass balance. Put $W := \{y \in B_1 : \text{dist}(y, T) < (\varepsilon/2)|y|\}$, an open set containing $(T \cap B_1) \setminus \{0\}$, whose omitted point is $\mathcal{H}^2$ -null. Lower semicontinuity under (8.8) gives $\liminf_r \mu_r(W) \geq \mathcal{H}^2(T \cap B_1) = \pi$. Since V_ε and W are disjoint while $\mu_r(B_1) \rightarrow \pi$ by (8.11), $\limsup_r \mu_r(V_\varepsilon) = 0$. Dividing by $\mu_r(B_1) \rightarrow \pi$ gives (8.14). $\square$

8.4 A rectifiability criterion

Lemma 8.11 (Rectifiability from density and cone concentration). *Let μ be a Radon measure on $\mathbb{R}^3$ such that for μ -a.e. x there is a plane T_x with*

$$0 < \lim_{r \downarrow 0} \frac{\mu(B_r(x))}{\pi r^2} < \infty, \quad \lim_{r \downarrow 0} \frac{\mu(B_r(x) \setminus X(x, T_x, \varepsilon))}{\mu(B_r(x))} = 0 \quad (8.15)$$

for every $\varepsilon > 0$, where $X(x, T, \varepsilon) = \{y : \text{dist}(y - x, T) < \varepsilon|y - x|\}$. Then μ is carried by countably many Lipschitz graphs.

Proof. Fix $0 < \varepsilon < 10^{-2}$ and a countable ε -net $\{T_j\}$ in the compact Grassmannian of two-planes. Up to a null set, partition the good points into countably many sets G with fixed parameters c, C, m, j such that for $x \in G$ and $0 < r < 1/m$,

$$cr^2 \leq \mu(B_r(x)) \leq Cr^2, \quad \mu(B_r(x) \setminus X(x, T_x, 2\varepsilon)) \leq \delta r^2, \quad \angle(T_x, T_j) < \varepsilon. \quad (8.16)$$

Intersect with a countable grid of cubes of diameter $< 1/(10m)$. On one resulting piece take distinct x, y , put $d := |x - y|$, and suppose $y - x \notin X(x, T_x, 4\varepsilon)$. With $\rho := \varepsilon d/20$, elementary triangle estimates give

$$B_\rho(y) \subset B_{2d}(x) \setminus X(x, T_x, 2\varepsilon), \quad (8.17)$$

whence $c\varepsilon^2 d^2/400 \leq \mu(B_\rho(y)) \leq \delta(2d)^2$. Choosing $\delta < c\varepsilon^2/1600$ is a contradiction. Hence every difference of two points of G lies within 5ε of the fixed plane T_j , the orthogonal projection π_j onto T_j is injective on G , and

$$|\pi_{T_j^\perp}(x - y)| \leq \frac{5\varepsilon}{\sqrt{1 - 25\varepsilon^2}} |\pi_j(x - y)|. \quad (8.18)$$

Thus G lies in a Lipschitz graph after extending the height from $\pi_j(G)$ by the McShane formula. All indices, and a sequence $\varepsilon \downarrow 0$, are countable. $\square$

8.5 The structure theorem and Gauss–Green

Theorem 8.12 (De Giorgi structure theorem). *Let $E \subset \mathbb{R}^3$ have locally finite perimeter. Then $\partial^* E$ is countably $\mathcal{H}^2$ -rectifiable and*

$$D\mathbf{1}_E = -\nu_E \mathcal{H}^2 \llcorner \partial^* E, \quad \mu_E = \mathcal{H}^2 \llcorner \partial^* E, \quad P(E; A) = \mathcal{H}^2(A \cap \partial^* E). \quad (8.19)$$

Moreover (8.8) and (8.11) hold at every point of $\partial^ E$, hence at μ_E -a.e. point.*

Proof. At a reduced point, (8.11) supplies the density hypothesis and Lemma 8.10 the cone hypothesis of (8.15) with $T_x = \nu_E(x)^\perp$. With (8.3), Lemma 8.11 gives countably many Lipschitz graphs whose union G carries μ_E . This is initially a statement about μ_E , so one more step is required for rectifiability in the $\mathcal{H}^2$ sense of Definition 6.10. Split ∂^*E into the countably many sets on which

$$\mu_E(B_r(x)) \geq (\pi/2)r^2 \quad (0 < r < 1/m).$$

They cover ∂^*E by (8.11). On each such set the five-covering lemma, followed by outer regularity of μ_E , gives $\mathcal{H}^2(A) \leq C\mu_E(A)$ for every Borel subset A . Apply this to $A = \partial^*E \setminus G$, whose μ_E -measure is zero by (8.3). Hence $\mathcal{H}^2(\partial^*E \setminus G) = 0$, and ∂^*E is countably $\mathcal{H}^2$ -rectifiable.

Intersect the graph cover with ∂^*E and refine it into disjoint Borel pieces G_k . On subsets where the density bounds for μ_E are uniform, the five-covering lemma applied in both directions shows $\mu_E \llcorner G_k$ and $\mathcal{H}^2 \llcorner G_k$ are mutually absolutely continuous. Rademacher and the graph area formula (Corollary 6.13) give $\mathcal{H}^2(G_k \cap B_r(x))/(\pi r^2) \rightarrow 1$ for $\mathcal{H}^2$ -a.e. $x \in G_k$; here one also uses the planar Lebesgue-density theorem for the parameter subset of each graph chart. The Besicovitch differentiation theorem, first applied to $\mathbf{1}_{G_k} \in L^1_{\text{loc}}(\mu_E)$, also gives $\mu_E(B_r(x) \setminus G_k)/\mu_E(B_r(x)) \rightarrow 0$ for μ_E -a.e. $x \in G_k$. Thus the numerator below may equivalently be restricted to G_k . Differentiation of one measure with respect to the other, followed by (8.11), therefore gives

$$\frac{d\mu_E}{d(\mathcal{H}^2 \llcorner G_k)}(x) = \lim_{r \downarrow 0} \frac{\mu_E(B_r(x))}{\mathcal{H}^2(G_k \cap B_r(x))} = 1$$

a.e. on each G_k . Since (8.3) says μ_E has no mass off their union, $\mu_E = \mathcal{H}^2 \llcorner \partial^*E$; combining with the outward polar convention gives (8.19). $\square$

Theorem 8.13 (Gauss–Green). *For E of locally finite perimeter and $X \in C^1_c(\mathbb{R}^3; \mathbb{R}^3)$,*

$$\int_E \operatorname{div} X \, dx = \int_{\partial^*E} X \cdot \nu_E \, d\mathcal{H}^2. \quad (8.20)$$

Proof. By definition of the distributional derivative, $\int_E \operatorname{div} X = - \int X \cdot D\mathbf{1}_E$; substitute (8.19). $\square$

Corollary 8.14 (Smooth boundaries). *If E is a bounded domain with C^1 boundary then $\partial^*E = \partial E$ up to $\mathcal{H}^2$ -null sets, ν_E is the classical outward normal, and $P(E) = \mathcal{H}^2(\partial E)$.*

Proof. In a C^1 graph chart, Fubini and the one-dimensional fundamental theorem of calculus in the vertical variable give, for every compactly supported smooth test field in the chart,

$$\int_E \operatorname{div} X = \int_{\partial E} X \cdot \nu \, d\mathcal{H}^2;$$

Corollary 6.13 supplies exactly the graph Jacobian and the classical outward normal in this calculation. A partition of unity over finitely many charts shows $D\mathbf{1}_E = -\nu \mathcal{H}^2 \llcorner \partial E$. The defining ball averages then put every boundary point in ∂^*E with this normal, while points off ∂E lie in the topological interior of E or its complement and therefore outside $\operatorname{spt} |D\mathbf{1}_E|$. Substitute in (8.19). $\square$

Remark 8.15 (Dependency audit for this chapter). Every ingredient used above — polar differentiation (Lemma 8.3), BV compactness (Theorem 4.11), slicing (Corollary 5.6), weighted coarea (Theorem 5.1), the five-covering lemma (Lemma 7.12), the BV Sobolev inequality (Theorem 7.2) and the rectifiable area formula (Theorem 6.11) — is proved earlier in this Part or is in the declared base. In particular the phase-preserving cylindrical deformation of Chapter 23 is *not* used here, so there is no cycle between that construction and this chapter.

Chapter 9

Transport of perimeter and first-variation formulas

Lean module: NoCompromise.Variation.ConstantTranslation,
 NoCompromise.Variation.FieldStability, NoCompromise.Variation.OneDimensionalTranslation,
 NoCompromise.Variation.Perimeter, NoCompromise.Variation.Piola,
 NoCompromise.Variation.StraightCofactor, NoCompromise.Variation.StraightDiffeo,
 NoCompromise.Variation.SymmetricDifferenceFull, NoCompromise.Variation.TransportC1,
 NoCompromise.Variation.VolumeBoundary

9.1 Bi-Lipschitz transport

Theorem 9.1 (Transport of perimeter). *Let $\Phi : U \rightarrow V$ be a C^1 diffeomorphism with Φ, Φ^{-1} locally Lipschitz, and let $E \subset U$ have locally finite perimeter in U . Then $\Phi(E)$ has locally finite perimeter in V , at $\mathcal{H}^2$ -a.e. $x \in \partial^* E \cap U$*

$$\nu_{\Phi(E)}(\Phi(x)) = \text{sign}(\det D\Phi(x)) \frac{\text{cof } D\Phi(x) \nu_E(x)}{|\text{cof } D\Phi(x) \nu_E(x)|}, \quad (9.1)$$

and for every Borel $A \subset U$

$$P(\Phi(E); \Phi(A)) = \int_{\partial^* E \cap A} |\text{cof } D\Phi \nu_E| d\mathcal{H}^2. \quad (9.2)$$

Proof. First work on a connected component of U , on which $s := \text{sign}(\det D\Phi) \in \{-1, 1\}$ is constant. C^2 case. For $X \in C_c^1(V; \mathbb{R}^3)$, change variables and use the Piola identity (Lemma 9.2):

$$\int_{\Phi(E)} \text{div } X = \int_E (\text{div } X)(\Phi) |\det D\Phi| = s \int_E \text{div}((\text{cof } D\Phi)^T X(\Phi)). \quad (9.3)$$

Insert Gauss–Green (8.20) and then the rectifiable area formula (6.7) with Φ ; the cofactor formula (6.8) identifies the resulting Jacobian, giving (9.1)–(9.2).

C^1 case. Convolve the coordinate functions of Φ on compact subsets. The cofactor fields converge uniformly, so the C^2 Piola identities pass to the distributional Piola identity for Φ , and (B6) gives (9.3) directly. An exhaustion gives the local statement. Apply the argument on each connected component of U and combine the resulting locally finite measures. $\square$

Lemma 9.2 (Piola identity). *For $\Phi \in C^2(U; \mathbb{R}^3)$, each row of $\text{cof } D\Phi$ is divergence-free, $\sum_j \partial_j (\text{cof } D\Phi)_{ij} = 0$ for every i (equivalently, the columns of $(\text{cof } D\Phi)^\top$ are divergence-free), and $D\Phi (\text{cof } D\Phi)^\top = (\det D\Phi)I$.*

Proof. Expand the 3×3 cofactors: all terms containing a second derivative cancel in pairs. The second identity is the cofactor expansion of the determinant. $\square$

Formalisation note 9.3 (Piecewise-chart proof schema). There is no separate “piecewise transport theorem” in the formal target. Whenever finitely many C^1 charts are used, apply Theorem 9.1 on each explicitly specified open piece and then apply Proposition 5.15 one interface at a time. Each interface contributes the L^1 difference of its two BV traces in (5.13); an orientation-reversing chart uses the factor $\text{sign}(\det D\Phi)$ in (9.1). Thus a consumer must name its pieces, charts and traces instead of invoking an unquantified piecewise result.

9.2 Straight perturbations

Definition 9.4 (Straight perturbation). For $X \in C_c^1(\mathbb{R}^3; \mathbb{R}^3)$ and $t \in \mathbb{R}$ put $F_t^X(x) := x + tX(x)$.

Lemma 9.5 (F_t^X is a global C^1 diffeomorphism). *Let $L := \|DX\|_\infty$ and $|t|L < 1$. Then*

$$|F_t^X(x) - F_t^X(y)| \geq (1 - |t|L)|x - y|, \quad (9.4)$$

F_t^X is a bijection of $\mathbb{R}^3$, and it is a C^1 diffeomorphism with $DF_t^X = I + tDX$ invertible.

Proof. (9.4) follows from the mean value theorem, giving injectivity; $I + tDX$ is invertible for $|t|L < 1$, so F_t^X is a local C^1 diffeomorphism. It equals the identity off a compact set, hence is proper, so its image is open and closed in $\mathbb{R}^3$ and therefore all of $\mathbb{R}^3$. The inverse function theorem gives a global C^1 inverse. $\square$

Lemma 9.6 (Cofactor expansion of a straight perturbation). *Uniformly on $\text{spt } X$, as $t \rightarrow 0$,*

$$\text{cof } DF_t^X = I + t((\text{div } X)I - DX^\top) + O(t^2), \quad \det DF_t^X = 1 + t \text{div } X + O(t^2).$$

Consequently, for a two-plane $T = \nu^\perp$,

$$J_2(DF_t^X|_T) = 1 + t(\text{div } X - \nu \cdot DX \nu) + o(t) = 1 + t \text{div}_T X + o(t), \quad (9.5)$$

with the error uniform in x and ν .

Proof. Polynomial expansion of the cofactor matrix, then (6.8). $\square$

Theorem 9.7 (First variation of perimeter). *Let E have locally finite perimeter and $X \in C_c^1(\mathbb{R}^3; \mathbb{R}^3)$. If $A \subset \mathbb{R}^3$ is bounded and open with $\text{spt } X \Subset A$, then, for all sufficiently small $|t|$, $F_t^X(A) = A$ and*

$$\left. \frac{d}{dt} \right|_{t=0} P(F_t^X(E); A) = \int_{\partial^* E \cap A} \text{div}_\tau X \, d\mathcal{H}^2 = \int_{\partial^* E} \text{div}_\tau X \, d\mathcal{H}^2. \quad (9.6)$$

If E has finite perimeter globally, the same identity holds with $P(F_t^X(E))$ on the left.

Proof. For small $|t|$, the distance from $\text{spt } X$ to ∂A and (9.4) give $F_t^X(A) = A$. Theorems 9.1 and 6.11 therefore give

$$P(F_t^X(E); A) = \int_{\partial^* E \cap A} J_2(DF_t^X|_{T_x \partial^* E}) d\mathcal{H}^2.$$

Subtract $P(E; A)$ and insert (9.5). The integrand vanishes off the compact support of X , the error there is uniform, and local finite perimeter makes its $\mathcal{H}^2$ measure finite, so dominated convergence applies. If E has global finite perimeter, the identical exterior contributions give the global assertion. $\square$

Theorem 9.8 (First variation of volume). *Let $E \subset \mathbb{R}^3$ have locally finite perimeter and finite measure, and let $X \in C_c^1(\mathbb{R}^3; \mathbb{R}^3)$. Then*

$$\left. \frac{d}{dt} \right|_{t=0} |F_t^X(E)| = \int_E \text{div } X = \int_{\partial^* E} X \cdot \nu_E d\mathcal{H}^2. \quad (9.7)$$

Proof. Change variables using $\det DF_t^X = 1 + t \text{div } X + O(t^2)$, then apply (8.20). $\square$

9.3 The symmetric-difference formula

The following is proved without any flow or straightening theorem, so that it is available before Chapter 14.

Lemma 9.9 (Weighted one-dimensional translation identity). *Let $q \in BV_{\text{loc}}(\mathbb{R}; \{0, 1\})$ and $\zeta \in C_c(\mathbb{R}; [0, \infty))$. Then*

$$\lim_{t \rightarrow 0} \frac{1}{|t|} \int_{\mathbb{R}} \zeta(s) |q(s+t) - q(s)| ds = \int_{\mathbb{R}} \zeta d|Dq|. \quad (9.8)$$

Proof. On every compact interval containing $\text{spt } \zeta$ and its sufficiently small translates, a $\{0, 1\}$ -valued BV function has only finitely many essential jumps: each alternation contributes one unit of variation, so infinitely many would contradict local finite variation. After changing q on a null set it is therefore the indicator of a finite union of intervals there. For $|t|$ smaller than the separation of the finitely many jump points, the symmetric difference consists of disjoint intervals of length $|t|$, one at each jump. Uniform continuity of ζ makes the contribution of the interval at a jump s_j , divided by $|t|$, tend to $\zeta(s_j)$. Summing gives $\sum_j \zeta(s_j) = \int \zeta d|Dq|$. $\square$

Lemma 9.10 (Constant fields). *For a constant vector a , E of locally finite perimeter, and $\zeta \in C_c(U; [0, \infty))$,*

$$\lim_{t \rightarrow 0} \frac{1}{|t|} \int_U \zeta(x) |\mathbf{1}_E(x - ta) - \mathbf{1}_E(x)| dx = \int_{\partial^* E} \zeta |a \cdot \nu_E| d\mathcal{H}^2. \quad (9.9)$$

Proof. The assertion is immediate if $a = 0$. Otherwise write $e := a/|a|$ and slice along the lines $x' + \mathbb{R}e$. On each slice apply (9.8) with shift $t|a|$ and weight $s \mapsto \zeta(x' + se)$. Integrate using Fubini and BV slicing (Lemma 5.7). The passage of the limit through the transverse integral is dominated by the one-dimensional BV estimate

$$\frac{1}{|\tau|} \int_K |q(s + \tau) - q(s)| ds \leq |Dq|(K_{|\tau|}),$$

obtained by integrating the variation of q over the intervals between s and $s + \tau$. The factor $|a|$ from the shift and the sliced variation combine to $|a \cdot \nu_E|$, and Theorem 8.12 identifies the resulting measure. $\square$

Lemma 9.11 (Stability in the field). *Let $X, Y \in C_c^1(\mathbb{R}^3; \mathbb{R}^3)$ be supported in a fixed compact set with $|t| \max(\|DX\|_\infty, \|DY\|_\infty) < 1/2$, let $K \Subset U$ and let $K_0 \Subset K' \Subset U$ be fixed open neighbourhoods such that K_0 contains all intermediate inverse images of K . Then for $u \in BV(U)$,*

$$\int_K |u((F_t^X)^{-1}z) - u((F_t^Y)^{-1}z)| dz \leq C|t| \|X - Y\|_{L^\infty(K')} |Du|(K'). \quad (9.10)$$

Proof. First take u smooth. Join F_t^X to F_t^Y by $F_t^{Y+s(X-Y)}$, differentiate $u((F_t^{Y+s(X-Y)})^{-1}z)$ in s , use the inverse-derivative bound from (9.4), integrate in s and z , and change variables; only $\int_{K_0} |\nabla u|$ occurs. For general u , mollify on a neighbourhood of K' at radii smaller than $\text{dist}(\overline{K_0}, \mathbb{R}^3 \setminus K')$. The convolution estimate gives $\int_{K_0} |\nabla u_\varepsilon| \leq |Du|(K')$, while local L^1 convergence and the uniform bi-Lipschitz change of variables make both composed functions converge in $L^1(K)$. Let $\varepsilon \downarrow 0$. $\square$

Theorem 9.12 (Symmetric-difference formula). *For E of locally finite perimeter and $X \in C_c^1(\mathbb{R}^3; \mathbb{R}^3)$,*

$$\lim_{t \rightarrow 0} \frac{|E \triangle F_t^X(E)|}{|t|} = \int_{\partial^* E} |X \cdot \nu_E| d\mathcal{H}^2. \quad (9.11)$$

Proof. Take a finite partition of unity $\{\zeta_j\}$ on a fixed compact neighbourhood containing $E \triangle F_t^X(E)$ for every sufficiently small $|t|$, by functions supported in cubes of diameter δ , with slightly larger cubes of diameter at most 3δ . Choose $a_j := X(x_j)$ in each cube and a compactly supported field Y_j equal to a_j on the larger cube. For sufficiently small t the integrand weighted by ζ_j only sees the constant part of Y_j , so (9.9) applies. On the larger cube $\|X - Y_j\|_\infty \leq \omega_X(3\delta)$, so (9.10) bounds the sum of all replacement errors by $C\omega_X(3\delta) |D\mathbf{1}_E|(K')$, and the same replacement on the boundary integral has error at most $\omega_X(3\delta) |D\mathbf{1}_E|(K')$. The finite cover may be chosen with dimension-dependent overlap, so C does not depend on δ . Let $t \rightarrow 0$ first, then $\delta \rightarrow 0$; outside the chosen fixed neighbourhood both sides vanish. $\square$

Chapter 10

Elliptic and potential-theoretic estimates

Lean module: NoCompromise.Elliptic.Boundary, NoCompromise.Elliptic.Caccioppoli, NoCompromise.Elliptic.CampanatoComparison, NoCompromise.Elliptic.CampanatoGrowth, NoCompromise.Elliptic.CampanatoHolder, NoCompromise.Elliptic.CampanatoHolderDatum, NoCompromise.Elliptic.CampanatoIteration, NoCompromise.Elliptic.ClassicalAdmissible, NoCompromise.Elliptic.ClassicalKernelFlux, NoCompromise.Elliptic.FrozenDecay, NoCompromise.Elliptic.HarmonicDerivative, NoCompromise.Elliptic.HarmonicMeanValue, NoCompromise.Elliptic.HolderInterpolation, NoCompromise.Elliptic.Hopf, NoCompromise.Elliptic.HopfExterior, NoCompromise.Elliptic.InteriorH2, NoCompromise.Elliptic.KelvinRemovable, NoCompromise.Elliptic.Newtonian, NoCompromise.Elliptic.NewtonianSchauder, NoCompromise.Elliptic.NondivSchauder, NoCompromise.Elliptic.Quasilinear, NoCompromise.Elliptic.Schauder, NoCompromise.Elliptic.SobolevChain, NoCompromise.Elliptic.StrongMaximumHarmonic, NoCompromise.Elliptic.TraceL1, NoCompromise.Elliptic.WeakDirichlet, NoCompromise.Elliptic.WeakMaximum, NoCompromise.Elliptic.WeakSolutions, NoCompromise.Sobolev.PoincareTrace

Throughout this chapter, $0 < \alpha < 1$.

Remark 10.1 (Ordering). Sections 10.2 and 10.1 are pure chart calculus and do not use De Giorgi’s theorem or transport of perimeter. They are therefore an independent low-cost entry point to the elliptic layer. Their later consumers include the mean-value formula, the Coulomb first variation, exterior flux identities, Appendix K, and Green’s identity. The machine build order is determined by the `\uses` DAG, as explained in the opening chapter; no backward edge into Chapters 8 or 9 remains.

Convention 10.2 (Domain classes). Below, “admissible domain” means: a cube, a ball, a smooth annulus, or a bounded connected C^1 domain (in particular a $C^{1,1/2}$ or $C^{2,\alpha}$ domain). These are exactly the classes used in Theorem 30.1, in (13.1) and in (10.5).

10.1 Trace inequalities

Proposition 10.3 (Poincaré and L^2 trace). *Let G be an admissible domain, let $z \in H^1(G)$, and put $z_G := \int_G z$. Then*

$$\|z - z_G\|_{L^2(G)} \leq C_G \|\nabla z\|_{L^2(G)}, \quad (10.1)$$

$$\|\operatorname{Tr} z\|_{L^2(\partial G)} \leq C_G (\|z\|_{L^2(G)} + \|\nabla z\|_{L^2(G)}). \quad (10.2)$$

Proof. For the trace bound, cover ∂G by finitely many graph charts, flatten, and apply the one-dimensional fundamental theorem of calculus along the normal coordinate; an interior partition of unity completes the estimate. Lemma 4.14 gives the bounded H^1 extension to a fixed cube. If (10.1) failed, normalize a mean-zero counterexample z_j by $\|z_j\|_2 = 1$ and $\|\nabla z_j\|_2 \rightarrow 0$. Theorem 4.11, applied to the extensions, gives strong L^1 convergence on their common compact support. The extensions are uniformly bounded in L^6 : apply (7.2) to a smooth approximation of $|z_j|^4$ and use $\int |z_j|^3 |\nabla z_j| \leq \|z_j\|_6^3 \|\nabla z_j\|_2$ (with truncation before cancellation if the L^6 norm is not yet known). Interpolation $\|h\|_2 \leq \|h\|_1^{2/5} \|h\|_6^{3/5}$ then gives strong L^2 convergence. The limit has zero distributional gradient on G , so connectedness makes it constant and mean zero makes it zero, contradicting its L^2 norm one. Balls and annuli follow by specialization, and dilation records the scaled constants. $\square$

Proposition 10.4 (L^1 trace). *For $Z \in W^{1,1}(G; \mathbb{R}^3)$ with G admissible,*

$$\|\operatorname{Tr} Z\|_{L^1(\partial G)} \leq C_G (\|Z\|_{L^1(G)} + \|\nabla Z\|_{L^1(G)}). \quad (10.3)$$

Proof. Cover ∂G by finitely many graph charts, flatten each chart, and apply the one-dimensional fundamental theorem of calculus in the normal coordinate to each component of Z . Integrating tangentially and summing a partition of unity gives (10.3); the derivatives of the cutoffs are absorbed by the $L^1(G)$ term. $\square$

10.2 The $W^{1,1}$ divergence theorem

Theorem 10.5 ($W^{1,1}$ Gauss–Green on a C^1 domain). *Let G be an admissible domain and $Z \in W^{1,1}(G; \mathbb{R}^3)$. Then*

$$\int_G \operatorname{div} Z \, dx = \int_{\partial G} \operatorname{Tr} Z \cdot \nu_G \, d\mathcal{H}^2. \quad (10.4)$$

Proof. For smooth Z supported in one flattened graph chart this is the one-dimensional fundamental theorem of calculus in the normal variable followed by Fubini; differentiating the graph change of variables produces exactly the Euclidean normal and surface Jacobian. Summing over a partition of unity, the terms in which the derivative hits a cutoff cancel because the cutoffs sum to one. Approximate a general $W^{1,1}$ field in each chart by convolution: (10.3) makes the traces converge in $L^1(\partial G)$ and the volume terms converge in $L^1(G)$. $\square$

Corollary 10.6 (Two consequences used later). *(i) For a bounded C^1 domain Ω of volume V ,*

$$\int_{\partial \Omega} x \cdot \nu \, d\mathcal{H}^2 = 3V.$$

(ii) For $y \in \Omega$ fixed and $X \in C_c^\infty(\mathbb{R}^3; \mathbb{R}^3)$, the field $W_y(x) := X(x)/|x - y|$ lies in $W^{1,1}(\Omega; \mathbb{R}^3)$ and

$$\int_{\Omega} \operatorname{div} W_y \, dx = \int_{\partial \Omega} \frac{X \cdot \nu}{|x - y|} \, d\mathcal{H}^2. \quad (10.5)$$

Proof. (i) is (10.4) with $Z(x) = x$. For (ii), $|x - y|^{-1}$ and $|\nabla_x |x - y|^{-1}|$ are both in $L_{\text{loc}}^1(\mathbb{R}^3)$, so $W_y \in W^{1,1}$; apply (10.4). $\square$

Formalisation note 10.7. Corollary 10.6(ii) is the correct justification of the last line of the nonlocal first variation (28.1). It replaces an invalid claim of a small-sphere estimate uniform in points y approaching the boundary.

10.3 Weak Dirichlet and Neumann problems

Proposition 10.8 (Weak Dirichlet problem). *Let G be admissible, $f \in L^2(G)$, and let g admit one $H^1(G)$ extension. Then $\mathcal{J}_D(z) := \frac{1}{2} \int_G |\nabla z|^2 + \int_G f z$ has a unique minimiser on $g + H_0^1(G)$, characterised by*

$$\int_G \nabla z \cdot \nabla \phi = - \int_G f \phi \quad (\phi \in H_0^1(G)). \quad (10.6)$$

Proof. Poincaré on H_0^1 makes a minimising sequence bounded; (B2), weak continuity of the affine constraint and lower semicontinuity give a minimiser; differentiating gives (10.6); testing the difference of two solutions with itself gives uniqueness. $\square$

Proposition 10.9 (Weak Neumann problem). *Let G be admissible, $f \in L^2(G)$, $h \in L^2(\partial G)$ with the compatibility condition*

$$\int_G f = \int_{\partial G} h. \quad (10.7)$$

Then $\mathcal{J}_N(z) := \frac{1}{2} \int_G |\nabla z|^2 + \int_G f z - \int_{\partial G} h \operatorname{Tr} z$ has a minimiser on $\{z \in H^1(G) : z_G = 0\}$, unique modulo constants, and

$$\int_G \nabla z \cdot \nabla \phi = - \int_G f \phi + \int_{\partial G} h \operatorname{Tr} \phi \quad (10.8)$$

for every $\phi \in H^1(G)$.

Proof. (10.1)–(10.2) give coercivity; the direct method gives a minimiser and (10.8) first for mean-zero ϕ , then for all ϕ by (10.7). Testing differences gives uniqueness modulo constants. $\square$

10.4 Caccioppoli, maximum and Hopf principles

Lemma 10.10 (Caccioppoli). *Let $0 < \lambda \leq \Lambda < \infty$ and let $A \in L^\infty(U; \mathbb{R}^{n \times n})$ satisfy $\lambda |\xi|^2 \leq \xi \cdot A(x) \xi$ and $|A(x)| \leq \Lambda$ a.e., and let $u \in H_{\text{loc}}^1(U)$ solve $\operatorname{div}(A \nabla u) = \operatorname{div} G$ weakly with $G \in L_{\text{loc}}^2(U; \mathbb{R}^n)$. For every $\eta \in C_c^1(U)$,*

$$\int \eta^2 |\nabla u|^2 \leq C(\lambda, \Lambda) \left(\int u^2 |\nabla \eta|^2 + \int \eta^2 |G|^2 \right). \quad (10.9)$$

In particular this applies to weakly harmonic functions with $A = I$ and $G = 0$.

Proof. Test with $\eta^2 u$ and use ellipticity and Young's inequality to absorb both terms containing $\eta |\nabla u|$ into the left side. $\square$

Lemma 10.11 (Weak maximum principle). *Let G be an admissible domain and $u \in H^1(G)$. If $\Delta u \geq 0$ weakly on G and $u \leq 0$ in the trace sense on ∂G , then $u \leq 0$ a.e. in G .*

Proof. Test with u_+ to get $\nabla u_+ = 0$. $\square$

Lemma 10.12 (Mean value property and smoothness). *Let $n \in \{2, 3\}$ and $U \subset \mathbb{R}^n$ be open. A weakly harmonic function on U is smooth and satisfies the mean value property on every ball $B \Subset U$.*

Proof. On concentric compactly contained balls, apply Lemma 10.17 first with right side zero and then to every weak derivative already obtained. Induction gives H_{loc}^m for every m , and Lemma 10.18 gives a smooth representative. For a smooth harmonic function h , differentiation under the integral and the divergence theorem on a Euclidean ball give

$$\frac{d}{dr} \int_{\partial B_r(x)} h = \frac{1}{|\partial B_r|} \int_{\partial B_r(x)} \partial_\nu h = \frac{1}{|\partial B_r|} \int_{B_r(x)} \Delta h = 0.$$

The spherical average therefore equals its limit $h(x)$ at radius zero. For completeness, the smooth divergence identity used in this line is obtained by covering the sphere by finitely many graph patches, applying the one-dimensional fundamental theorem of calculus in the normal coordinate, and summing a partition of unity; the cutoff-derivative terms cancel because the partition sums to one. This is the same chart calculation recorded in the earlier divergence-theorem section, with the identical argument in dimension two. Thus the proof works in both dimensions used later and does not invoke the three-dimensional trace theorem. $\square$

Proposition 10.13 (Strong maximum principle). *A harmonic function on a connected open G attaining an interior maximum is constant.*

Proof. Equality of the spherical average forces equality a.e. on a ball; continuity and a chain of overlapping balls propagate constancy. $\square$

Lemma 10.14 (Harmonic derivative estimates). *Let $n \in \{2, 3\}$, let h be harmonic on $B_R \subset \mathbb{R}^n$, and let $r < R$. For every integer $k \geq 0$,*

$$\begin{aligned} \|D^k h\|_{L^\infty(B_r)} &\leq C_{n,k,r,R} \|h\|_{L^\infty(B_R)}, \\ \|D^k h\|_{L^\infty(B_r)} &\leq C_{n,k,r,R} \|h\|_{L^2(B_R)}. \end{aligned} \tag{10.10}$$

If $k \geq 1$, the last norm may instead be $\|\nabla h\|_{L^2(B_R)}$, with a possibly different constant.

Proof. Choose $r < r_1 < r_2 < R$. Differentiating the mean-value formula on balls of a fixed radius smaller than $r_2 - r_1$ gives the first estimate. The mean-value property and Cauchy–Schwarz first give $\|h\|_{L^\infty(B_{r_2})} \leq C \|h\|_{L^2(B_R)}$, which gives the second. For the last assertion apply the same L^2 -to- L^∞ estimate to each harmonic derivative $\partial_i h$ on B_{r_1} , and then differentiate its mean-value formula $k - 1$ more times. Nested radii keep every averaging ball inside B_R . $\square$

Proposition 10.15 (Hopf boundary lemma with sign). *Let $G \subset \mathbb{R}^3$ be open, let $u \in C(G \cup \{p\})$ be weakly harmonic and strictly positive in G , and let $u(p) = 0$ at $p \in \partial G$. Suppose an interior tangent ball $B_R(q) \subset G$ satisfies $p \in \partial B_R(q)$. Then*

$$\liminf_{s \downarrow 0} \frac{u(p + s(q - p)/R) - u(p)}{s} > 0.$$

If ∂G is differentiable at p , so that its outward unit normal is $\nu_G(p) = (p - q)/R$, and u is differentiable at p , this is equivalently

$$\partial_{\nu_G} u(p) < 0 \tag{10.11}$$

for the outward normal ν_G of G .

Proof. On $B_R(q) \setminus \overline{B_{R/2}(q)}$ use the harmonic barrier

$$b(x) := \frac{R}{|x - q|} - 1, \quad (10.12)$$

zero at $|x - q| = R$ and positive inside. Proposition 10.13 gives a positive lower bound for u on the inner sphere, so Lemma 10.11 gives $u \geq \varepsilon b$ on the annulus for some $\varepsilon > 0$. Take the inward difference quotient at p . $\square$

Corollary 10.16 (The exterior sign). *Let $K = \overline{\text{int } K} \subset \mathbb{R}^3$ be compact with C^2 boundary, and let $u \in C^1(\mathbb{R}^3 \setminus K)$ be harmonic on $\mathbb{R}^3 \setminus K$, with $u = 1$ on ∂K and $0 < u < 1$ outside K . Then*

$$\partial_{\nu_K} u < 0 \quad \text{on } \partial K.$$

Proof. At every point of the C^2 boundary there is an interior tangent ball for $G = \mathbb{R}^3 \setminus K$. Apply Proposition 10.15 to $1 - u$ on $G = \mathbb{R}^3 \setminus K$. The inner-boundary outward normal of G is $-\nu_K$, so (10.11) gives $\partial_{\nu_K}(1 - u) > 0$. $\square$

10.5 Interior regularity

Lemma 10.17 (Interior H^2 estimate). *Let $n \in \{2, 3\}$ and let $\Delta u = f$ weakly on $B_R \subset \mathbb{R}^n$ with $f \in L^2(B_R)$. Then for $r < R$*

$$\|u\|_{H^2(B_r)} \leq C_{r,R} (\|u\|_{L^2(B_R)} + \|f\|_{L^2(B_R)}). \quad (10.13)$$

Proof. The difference quotient $\delta_h u$ satisfies $\int \nabla(\delta_h u) \cdot \nabla \phi = - \int (\delta_h f) \phi$ on a slightly smaller domain. Test with $\phi = \eta^2 \delta_h u$. Although $\delta_h f$ need not be uniformly bounded in L^2 , discrete integration by parts gives $|\int (\delta_h f) \phi| = |\int f \delta_{-h} \phi| \leq \|f\|_2 \|\nabla \phi\|_2$. With (10.9) this gives an H^1 bound independent of h ; weak compactness and the characterisation of weak derivatives by difference quotients give (10.13). $\square$

Lemma 10.18 (Sobolev chain to C^k). *For $u \in H^1(\mathbb{R}^3)$,*

$$\|u\|_{L^6(\mathbb{R}^3)} \leq C \|\nabla u\|_{L^2(\mathbb{R}^3)}. \quad (10.14)$$

More generally, for $n \in \{2, 3\}$, integers $m, k \geq 0$, open $V \Subset U \subset \mathbb{R}^n$, and $m > k + n/2$,

$$\|u\|_{C^k(\overline{V})} \leq C_{U,V,m,k} \|u\|_{H^m(U)} \quad (10.15)$$

whenever the right side is finite (or, locally, after inserting a cutoff equal to one on V). Consequently, in either dimension, iterating (10.13) for $u \in H_{\text{loc}}^1(U)$ satisfying $\Delta u = f$ with $f \in H_{\text{loc}}^k(U)$ gives $u \in H_{\text{loc}}^{k+2}(U)$; in particular weakly harmonic functions are smooth.

Proof. For smooth compactly supported u , apply (7.2) to $|u|^4$ and use Hölder:

$$\|u\|_6^4 = \| |u|^4 \|_{3/2} \leq C \int |u|^3 |\nabla u| \leq C \|u\|_6^3 \|\nabla u\|_2.$$

Cancel $\|u\|_6^3$ (the zero case being immediate) and use H^1 density to obtain (10.14); the same estimate applies after multiplying by a compactly supported cutoff. For (10.15), multiply by a smooth cutoff equal to one near $\overline{V}$ and supported compactly in U , then extend by zero. First perform

the calculation for smooth compactly supported approximants. They and their Fourier transforms are integrable, and they are continuous, so `MeasureTheory.Integrable.fourierInv_fourier_eq` applies at every point. Use the unitary Fourier convention $\hat{u}(\xi) = (2\pi)^{-n/2} \int e^{-ix \cdot \xi} u(x) dx$; it is obtained from Mathlib's $e^{-2\pi i x \cdot \xi}$ convention by a frequency dilation, with the corresponding change of variables in Plancherel. For $|\alpha| \leq k$,

$$|D^\alpha u(x)| \leq (2\pi)^{-n/2} \int_{\mathbb{R}^n} |\xi|^{|\alpha|} |\hat{u}(\xi)| d\xi \leq C \|u\|_{H^m} \left(\int_{\mathbb{R}^n} \frac{|\xi|^{2|\alpha|}}{(1 + |\xi|^2)^m} d\xi \right)^{1/2}.$$

The last integral is finite exactly when $m > |\alpha| + n/2$; dominated convergence in the Fourier integral gives continuity of $D^\alpha u$. Convolution approximates the localized function in H^m . Applying the displayed estimate to differences makes the approximants Cauchy in C^k ; their L^2 limit identifies the resulting C^k representative with u . Thus no continuity of an arbitrary H^m function is assumed when invoking inversion. Cutoffs and exhaustion give the local statement. Finally apply Lemma 10.17 successively to weak derivatives, using nested balls, to obtain the asserted elliptic bootstrap. $\square$

10.6 Schauder estimates

Theorem 10.19 (Newtonian Schauder estimate). *Let $N(x) := -1/(4\pi|x|)$ and let $\Delta z = f$ weakly on B_1 with $f \in C^{0,\alpha}(B_1)$, $0 < \alpha < 1$. Then*

$$\|z\|_{C^{2,\alpha}(B_{1/2})} \leq C_\alpha (\|z\|_{L^\infty(B_1)} + \|f\|_{C^{0,\alpha}(B_1)}). \quad (10.16)$$

Proof. Choose a cutoff $\eta \equiv 1$ on $B_{3/4}$ and write $z = N * (\eta f) + h$ with h harmonic on $B_{3/4}$. In $D_{ij}(N * (\eta f))(x)$, differentiate first outside a ball of radius ε and integrate the boundary term explicitly; the limit is a principal-value integral plus a fixed multiple of $\delta_{ij}f(x)$. In the principal-value term subtract $f(x)$; the spherical mean of $D_{ij}N$ away from the origin vanishes, and $|f(y) - f(x)| \leq [f]_{C^{0,\alpha}}|x - y|^\alpha$. Split at $|x - y| = |x - z|$ and use $|D^2N(\xi)| \leq C|\xi|^{-3}$, $|D^3N(\xi)| \leq C|\xi|^{-4}$; near and far integrals each give $C|x - z|^\alpha$. Lemma 10.14 controls h . The calculation is first made for smooth f ; uniform estimates and convolution of a Hölder f pass to the limit. $\square$

Lemma 10.20 (Hölder interpolation). *For $z \in C^{2,\alpha}(B_\rho)$ and $\varepsilon > 0$,*

$$\|Dz\|_{C^{0,\alpha}(B_\rho)} \leq \varepsilon \|D^2z\|_{C^{0,\alpha}(B_\rho)} + C_{\varepsilon,\rho} \|z\|_{L^\infty(B_\rho)}. \quad (10.17)$$

Proof. Bound the first difference quotient at scale s once by $s\|D^2z\|_\infty$ and once by $2s^{-1}\|z\|_\infty$ and optimise s ; the Hölder seminorm follows from the same two-scale split. $\square$

Theorem 10.21 (Nondivergence Schauder estimate). *Let $a_{ij} \in C^{1,\alpha}(B_1)$ be uniformly elliptic with constants λ, Λ , let $b_i \in C^{0,\alpha}(B_1)$, and let $z \in C^{1,\alpha}(B_1)$ satisfy, in the sense of distributions,*

$$a_{ij}D_{ij}z + b_iD_iz = f$$

with $f \in C^{0,\alpha}(B_1)$. Then

$$\|z\|_{C^{2,\alpha}(B_{1/2})} \leq C(\|z\|_{C^{1,\alpha}(B_1)} + \|f\|_{C^{0,\alpha}(B_1)}), \quad (10.18)$$

with $C = C(\alpha, \lambda, \Lambda, \|a\|_{C^{1,\alpha}}, \|b\|_{C^{0,\alpha}})$. Moreover the dependence on the a-priori $C^{1,\alpha}$ norm can be removed:

$$\|z\|_{C^{2,\alpha}(B_{1/2})} \leq C(\|z\|_{L^\infty(B_1)} + \|f\|_{C^{0,\alpha}(B_1)}). \quad (10.19)$$

Proof. Since $a \in C^{1,\alpha}$ and $z \in C^{1,\alpha}$, the displayed equation is equivalent distributionally to

$$\partial_i(a_{ij}\partial_j z) = g := f + (\partial_i a_{ij} - b_j)\partial_j z, \quad g \in C^{0,\alpha}. \quad (10.20)$$

For a coordinate direction e_k , the difference quotient $w_h := (z(\cdot + he_k) - z)/h$ satisfies on a smaller ball

$$\operatorname{div}(a(\cdot + he_k)\nabla w_h) = \delta_h g - \operatorname{div}((\delta_h a)\nabla z).$$

Lemma 10.28 writes $\delta_h g$ as the divergence of a vector field uniformly bounded in $C^{0,\alpha}$, while $\delta_h a$ and ∇z are uniformly $C^{0,\alpha}$. Theorem 10.27, applied on nested balls, therefore gives a uniform interior $C^{1,\alpha}$ bound for w_h : its L^2 gradient term is uniform by the Caccioppoli estimate for the displayed difference-quotient equation, because $\|w_h\|_{L^2} \leq C\|\nabla z\|_\infty$ and its vector datum is uniformly $C^{0,\alpha}$. Passing to the limit gives $D_k z \in C^{1,\alpha}$ for every k , hence $z \in C^{2,\alpha}$ and (10.18).

To prove (10.19), now that $z \in C^{2,\alpha}$ is known, fix a ball $B_\rho(x_0) \Subset B_{3/4}$, freeze a at x_0 , and make a fixed linear change of coordinates which turns $a(x_0)_{ij}D_{ij}$ into the Laplacian. The scaled form of Theorem 10.19 applied on this ball has right side

$$f - b_i D_i z - (a_{ij} - a_{ij}(x_0))D_{ij} z.$$

Choose ρ so that the $C^{0,\alpha}$ coefficient-oscillation multiple of the top $C^{2,\alpha}$ seminorm is less than one half and absorb it. Then use Lemma 10.20, with its parameter chosen after ρ , to absorb the $C^{0,\alpha}$ norm of the drift term $b_i D_i z$. A finite cover of $B_{1/2}$ and the usual overlap comparison of the local $C^{2,\alpha}$ norms give (10.19). This argument only estimates the already-constructed solution and uses no boundary solvability theorem. $\square$

Constants 10.22. The order of choice in the proof is $\rho \rightsquigarrow \varepsilon$: the radius is fixed first from $[a]_{C^{0,\alpha}}$, then the interpolation parameter from ρ . Reversing the order gives a false statement.

10.7 The divergence-form Campanato estimate

Lemma 10.23 (Comparison with the frozen solution). *Let $n \in \{2, 3\}$, let $A \in C^{0,\alpha}(B_1; \mathbb{R}^{n \times n})$ be uniformly elliptic, let $G \in C^{0,\alpha}(B_1; \mathbb{R}^n)$, and let $w \in H^1(B_1)$ solve*

$$\operatorname{div}(A(x)\nabla w) = \operatorname{div} G \quad (10.21)$$

weakly in B_1 . Fix $B_r(x_0) \Subset B_1$ and let h solve $\operatorname{div}(A(x_0)\nabla h) = 0$ with $h - w \in H_0^1(B_r(x_0))$. Then

$$\int_{B_r} |\nabla(w - h)|^2 \leq C \int_{B_r} |A - A(x_0)|^2 |\nabla w|^2 + C \int_{B_r} |G - G(x_0)|^2. \quad (10.22)$$

Proof. Test the difference of the two equations with $w - h$. $\square$

Lemma 10.24 (Oscillation decay for the frozen solution). *With h as in Lemma 10.23 and $\theta \in (0, 1)$,*

$$\int_{B_{\theta r}} |\nabla h - (\nabla h)_{B_{\theta r}}|^2 \leq C\theta^{n+2} \int_{B_r} |\nabla h - (\nabla h)_{B_r}|^2, \quad (10.23)$$

with $n + 2 = 5$ in dimension three and $n + 2 = 4$ for the two-dimensional graph equation. One also has the non-centred energy decay

$$\int_{B_{\theta r}} |\nabla h|^2 \leq C\theta^n \int_{B_r} |\nabla h|^2. \quad (10.24)$$

Proof. After a linear change of variables the derivatives of h are harmonic. Their mean-value formula and interior L^2 -to- L^∞ estimate give (10.24); applying the derivative estimate to $\nabla h - (\nabla h)_{B_r}$ gives (10.23). $\square$

Lemma 10.25 (Subcritical gradient-energy growth). *Under the hypotheses of Lemma 10.23, for every $\beta \in (0, n)$ there is C_β such that, whenever $B_{2r}(x) \subset B_{3/4}$,*

$$\int_{B_r(x)} |\nabla w|^2 \leq C_\beta r^\beta. \quad (10.25)$$

The constant depends only on β , α , the ellipticity and $C^{0,\alpha}$ bounds of A, G , and $\|\nabla w\|_{L^2(B_1)}$.

Proof. Put $\Psi(r) := \int_{B_r(x)} |\nabla w|^2$ and compare w with the frozen solution h on $B_r(x)$. The constant-coefficient estimate for h (obtained from Lemma 10.24 after retaining its mean gradient) and (10.22) give

$$\Psi(\theta r) \leq C(\theta^n + r^{2\alpha})\Psi(r) + Cr^{n+2\alpha}.$$

Given $\beta < n$, first choose θ so that $C\theta^n \leq \frac{1}{4}\theta^\beta$, then choose the starting radius so that $Cr^{2\alpha} \leq \frac{1}{4}\theta^\beta$. Iteration on the radii $\theta^j r$ (the forcing exponent $n + 2\alpha$ is strictly larger than β) proves (10.25); larger radii are covered by the original L^2 bound. $\square$

Lemma 10.26 (Campanato iteration). *Let $\Phi : (0, r_0] \rightarrow [0, \infty)$ be nondecreasing and satisfy*

$$\Phi(\theta r) \leq \frac{1}{2}\theta^{n+2\alpha}\Phi(r) + Cr^{n+2\alpha} \quad (10.26)$$

for some fixed $\theta \in (0, 1)$ and all $r \leq r_0$. Then $\Phi(\rho) \leq C'\rho^{n+2\alpha}$ for $\rho \leq r_0$.

Proof. Induct on j at $r = \theta^j r_0$, summing the geometric series with ratio $1/2$, then compare a general ρ with the nearest such radius. $\square$

Theorem 10.27 (Campanato/ $C^{1,\alpha}$ estimate, divergence form). *Under the hypotheses of Lemma 10.23, w has a representative in $C^{1,\alpha}(B_{1/2})$ with, for every $B_\rho(x_0) \subset B_{1/2}$ and every $x, y \in B_{1/2}$,*

$$\int_{B_\rho(x_0)} |\nabla w - (\nabla w)_{B_\rho(x_0)}|^2 \leq C\rho^{2\alpha}, \quad |\nabla w(x) - \nabla w(y)| \leq C|x - y|^\alpha, \quad (10.27)$$

where C depends on α , the ellipticity constants, $[A]_{C^{0,\alpha}}$, $[G]_{C^{0,\alpha}}$ and $\|\nabla w\|_{L^2(B_1)}$.

Proof. Put $\Phi(r) := \int_{B_r} |\nabla w - (\nabla w)_{B_r}|^2$. Comparison with the frozen solution gives, on balls compactly contained in $B_{3/4}$,

$$\Phi(\theta r) \leq C\theta^{n+2}\Phi(r) + Cr^{2\alpha} \int_{B_r} |\nabla w|^2 + Cr^{n+2\alpha}. \quad (10.28)$$

Choose $\beta \in (\max\{0, n - 2\alpha\}, n)$ and put $\gamma := (\beta + 2\alpha - n)/2 \in (0, \alpha)$. Lemma 10.25 changes the middle term in (10.28) into $Cr^{n+2\gamma}$. Choosing θ and applying Lemma 10.26 with γ in place of α gives $\nabla w \in C^{0,\gamma}$ on $B_{2/3}$, hence ∇w is bounded there. Consequently $\int_{B_r} |\nabla w|^2 \leq Cr^n$ on smaller balls. Insert this sharper bound back into (10.28) and apply the same iteration with exponent α ; this proves the first estimate in (10.27). For the second, that estimate implies $|(\nabla w)_{B_{2^{-j}r}(x)} - (\nabla w)_{B_{2^{-j-1}r}(x)}| \leq C(2^{-j}r)^\alpha$, so the averages converge at every Lebesgue point; comparing the two chains from x and y at the first radius larger than $|x - y|$ gives the Hölder estimate. $\square$

Lemma 10.28 ($C^{-1,\alpha}$ data from a Hölder function). *Let $U \subset \mathbb{R}^n$ be open and $g \in C^{0,\alpha}(U)$. For each $h \neq 0$ and coordinate direction e_k , on $U_h := \{x \in U : x + te_k \in U \text{ for every } t \text{ between } 0 \text{ and } h\}$,*

$$\frac{g(x + he_k) - g(x)}{h} = \partial_k \left(\frac{1}{h} \int_0^h g(x + te_k) dt \right),$$

so the left side equals $\operatorname{div} G_h$ for

$$G_h(x) := e_k \frac{1}{h} \int_0^h g(x + te_k) dt, \quad \|G_h\|_{C^{0,\alpha}(U_h)} \leq \|g\|_{C^{0,\alpha}(U)}.$$

Proof. Direct computation. □

10.8 The quasilinear Schauder lemma

Theorem 10.29 (Quasilinear Schauder). *Let $A \in C^2(\mathbb{R}^2; \mathbb{R}^2)$ with $DA(p)$ uniformly elliptic for $|p| \leq M$, and suppose $\operatorname{div} A(\nabla f) = g$ weakly in B_1^2 with $f \in C^{1,\alpha}$, $g \in C^{0,\alpha}$, $\|\nabla f\|_\infty \leq M$. Then $f \in C^{2,\alpha}(B_{1/2}^2)$, and its derivatives satisfy*

$$\|\nabla f\|_{C^{1,\alpha}(B_{1/2}^2)} \leq C, \quad \|f\|_{C^{2,\alpha}(B_{1/2}^2)} \leq \|f\|_{L^\infty(B_1^2)} + C,$$

using the sum norm for the full $C^{2,\alpha}$ norm. Here C depends only on α , M , the ellipticity constants of DA on $\{|p| \leq M\}$, $\|A\|_{C^2(\{|p| \leq M\})}$, $[\nabla f]_{C^{0,\alpha}}$ and $\|g\|_{C^{0,\alpha}}$. The derivative estimate is unchanged by adding a constant to f ; the full norm includes the displayed zeroth-order term.

Proof. Fix a coordinate direction and form $w_h(x) := (f(x + he_k) - f(x))/h$. Subtracting the two weak equations and using the fundamental theorem of calculus in the gradient variable, on $B_{1-|h|}$

$$\operatorname{div}(A_h(x) \nabla w_h) = g_h, \quad A_h(x) = \int_0^1 DA((1-t)\nabla f(x) + t\nabla f(x + he_k)) dt. \quad (10.29)$$

The A_h are uniformly elliptic and uniformly $C^{0,\alpha}$, and by Lemma 10.28 the data are $g_h = \operatorname{div} G_h$ with G_h uniformly $C^{0,\alpha}$. Moreover $\|w_h\|_{L^2}$ is bounded by the L^2 norm of $D_k f$ on a slightly larger disk; applying Lemma 10.10 to (10.29) with nested cutoffs therefore gives a uniform interior L^2 bound for ∇w_h . Theorem 10.27 now gives a uniform $C^{1,\alpha}$ bound for w_h on $B_{1/2}$; letting $h \rightarrow 0$ gives $D_k f \in C^{1,\alpha}$, hence $f \in C^{2,\alpha}$. Summing the derivative bounds and adding $\|f\|_\infty$ gives the stated full-norm estimate. Only weak equations are used; second derivatives are never assumed in advance. □

10.9 Boundary estimates

Notation 10.30. $B_r^+ := B_r \cap \{x_3 > 0\}$ and $\Gamma_r := B_r \cap \{x_3 = 0\}$.

Theorem 10.31 (Boundary $C^{1,\alpha}$, Dirichlet). *Let $w \in H^1(B_1^+)$ solve $\operatorname{div}(A \nabla w) = \operatorname{div} G$ weakly in B_1^+ , where $A \in C^{0,\alpha}(\overline{B_1^+}; \mathbb{R}^{3 \times 3})$ is uniformly elliptic, $G \in C^{0,\alpha}(\overline{B_1^+}; \mathbb{R}^3)$, and $w = 0$ in the trace sense on Γ_1 . Then*

$$\oint_{B_\rho^+} |\nabla w - (\nabla w)_{B_\rho^+}|^2 \leq C \rho^{2\alpha}, \quad (10.30)$$

so $w \in C^{1,\alpha}$ up to $\Gamma_{1/2}$. The constant depends only on α , ellipticity, the displayed coefficient and datum norms, and $\|\nabla w\|_{L^2(B_1^+)}.$

Proof. Freeze the coefficients on a half-ball and solve the constant-coefficient comparison problem with the same curved boundary data and zero data on Γ_r . A linear shear followed by a tangential linear change turns the frozen operator into the Laplacian without moving $\{x_3 = 0\}$; odd reflection of the comparison solution is harmonic, so its gradient satisfies (10.23) on half-balls. Testing the difference with itself gives (10.22) with B_r replaced by B_r^+ , the flat-boundary term vanishing because the difference has zero trace. The resulting non-centred recurrence is

$$\int_{B_{\theta r}^+} |\nabla w|^2 \leq C(\theta^3 + r^{2\alpha}) \int_{B_r^+} |\nabla w|^2 + Cr^{3+2\alpha}.$$

The proof of Lemma 10.25, unchanged for half-balls, first gives every subcritical growth exponent $\beta < 3$. Insert this in the centred oscillation recurrence to obtain $C^{0,\gamma}$ regularity of ∇w for some $\gamma > 0$, hence a local gradient bound; reinsert that bound and apply Lemma 10.26 a second time to obtain the full exponent α in (10.30). $\square$

Theorem 10.32 (Boundary $C^{2,\alpha}$, Dirichlet, divergence form). *Let $w \in H^1(B_1^+)$ solve $\operatorname{div}(A\nabla w) = \operatorname{div} G$, where $A \in C^{1,\alpha}(\overline{B_1^+}; \mathbb{R}^{3 \times 3})$ is uniformly elliptic, $G \in C^{1,\alpha}(\overline{B_1^+}; \mathbb{R}^3)$, and the trace $w|_{\Gamma_1}$ is $C^{2,\alpha}$. Then $w \in C^{2,\alpha}(\overline{B_{1/2}^+})$, with an estimate depending only on ellipticity, the displayed norms and $\|\nabla w\|_{L^2(B_1^+)}$ (the energy cannot be dropped: multiples of a solution with zero data are solutions). For a $C^{2,\alpha}$ curved boundary the same holds after flattening. Iterating with smooth coefficients, data, and boundary gives all higher regularity.*

Proof. Subtract a $C^{2,\alpha}$ extension of the boundary value. Tangential difference quotients of the resulting flattened equation have zero Dirichlet trace and satisfy a divergence equation whose right side consists of the divergence of a $C^{0,\alpha}$ vector field plus a $C^{0,\alpha}$ scalar g . Absorb g as the divergence of $(0, 0, \int_0^{x_3} g(x_1, x_2, s) ds)$. Applying (10.30) to those quotients and passing to the limit controls every second derivative with at least one tangential index; the equation, whose a_{33} is bounded below, solves algebraically for $\partial_{33}w$. $\square$

Theorem 10.33 (Boundary $C^{2,\alpha}$, nondivergence form). *Let $a \in C^{1,\alpha}(\overline{B_1^+})$ be uniformly elliptic, let $f \in C^{0,\alpha}(\overline{B_1^+})$, and let $z \in C^{1,\alpha}(\overline{B_1^+})$ satisfy $a_{ij}D_{ij}z = f$ distributionally in B_1^+ . If the trace of z on Γ_1 is $C^{2,\alpha}$, then $z \in C^{2,\alpha}(\overline{B_{1/2}^+})$, with an estimate depending on the displayed norms, ellipticity, and the $C^{2,\alpha}$ boundary trace.*

Proof. Subtract a $C^{2,\alpha}$ extension of the boundary trace, so that $z = 0$ on Γ_1 . The distributional product rule rewrites the equation as

$$\operatorname{div}(a\nabla z) = g, \quad g := f + (\partial_i a_{ij})\partial_j z \in C^{0,\alpha}.$$

For a tangential direction e_k , the difference quotient $w_h = \delta_h z$ still has zero trace and satisfies

$$\operatorname{div}(a(\cdot + he_k)\nabla w_h) = \delta_h g - \operatorname{div}((\delta_h a)\nabla z).$$

Lemma 10.28 expresses the first term as a divergence with a uniform $C^{0,\alpha}$ coefficient, and the second vector field is uniformly $C^{0,\alpha}$. The L^2 norms of the tangential difference quotients are uniformly bounded by $\|\nabla z\|_2$, and a boundary Caccioppoli estimate with nested half-ball cutoffs gives the uniform L^2 bounds for their gradients required by Theorem 10.31. That theorem gives uniform $C^{1,\alpha}$ estimates for the tangential difference quotients. Thus all second derivatives with at least one tangential index are $C^{0,\alpha}$. Finally uniform ellipticity gives $a_{33} \geq \lambda > 0$, and the original equation solves algebraically for $D_{33}z$. $\square$

Formalisation note 10.34. Theorem 10.33 is a standalone boundary regularity result. The interior proof of Theorem 10.21 does not use boundary solvability or a method-of-continuity construction; this keeps the formal dependency graph literal and acyclic.

Theorem 10.35 (Boundary regularity, Neumann, in signed-distance coordinates). *Let G be a bounded $C^{3,\alpha}$ domain, $0 < \alpha < 1$, and in signed-distance coordinates (in which the coefficient matrix A has one derivative less than the boundary) let*

$$\partial_i(A_{ij}\partial_j z) = f, \quad A_{3\alpha} = A_{\alpha 3} = 0 \text{ on } \{x_3 = 0\}, \quad (10.31)$$

with conormal condition $A_{3j}\partial_j z = h$, $A \in C^{1,\alpha}$, $f \in C^{0,\alpha}$ and $h \in C^{1,\alpha}$. Require constants $0 < \lambda \leq \Lambda < \infty$, uniform over the finite boundary atlas, such that

$$\xi \cdot A(x)\xi \geq \lambda|\xi|^2, \quad \|A(x)\|_{\text{op}} \leq \Lambda \quad \text{for every } x \text{ in a chart and every } \xi \in \mathbb{R}^3.$$

In particular $A_{33} \geq \lambda > 0$ on the flat boundary, hence $A_{33} \geq \lambda/2$ on a neighbourhood of it by continuity; the proofs use this neighbourhood form. More precisely, assume $z \in H^1(G)$ and interpret the equation and conormal datum weakly; on each flattened half-ball, for test functions supported away from the curved face,

$$\int_{B_1^+} A \nabla z \cdot \nabla \phi = - \int_{B_1^+} f \phi - \int_{\Gamma_1} h \phi. \quad (10.32)$$

Here the domain is $x_3 > 0$, so $h = (A \nabla z)_3$ is the inward-coordinate conormal; the outward conormal on the flat face is $-h$. Thus (10.32) has exactly the same sign as the usual formula $\int_G A \nabla z \cdot \nabla \phi = - \int_G f \phi + \int_{\partial G} g \phi$ when g denotes the outward conormal. Then $z \in C^{1,\alpha}$ up to the boundary. If $A \in C^{2,\alpha}$, $f \in C^{1,\alpha}$, and $h \in C^{2,\alpha}$ (as is the case when G is $C^{4,\alpha}$), then $z \in C^{2,\alpha}$ up to the boundary, with a uniform estimate over a finite boundary atlas. The estimates depend on λ, Λ , the coefficient and datum norms at the stated orders, $\|z\|_{H^1(G)}$, and the fixed chart and cutoff bounds. Iterating with smooth coefficients and data (still with $0 < \alpha < 1$ and $\lambda > 0$) gives smoothness.

Proof. For the first assertion, the explicit function $q(x) := \chi(x_3) x_3 h(x_1, x_2) / A_{33}(x_1, x_2, 0)$, where χ is a smooth cutoff equal to one near 0, has conormal derivative h on the flat boundary; put $w := z - q$, which has homogeneous conormal data. The right side can be put in the precise form required for reflection. Set

$$P(x) := e_3 \int_0^{x_3} f(x_1, x_2, s) ds, \quad H := P + e_3 h(x_1, x_2) - A \nabla q.$$

Then $H \in C^{0,\alpha}$, $\text{div } H = f - \text{div}(A \nabla q)$ distributionally, and $H_3 = 0$ on Γ_1 : here $P_3 = 0$ and $(A \nabla q)_3 = h$ on the flat face. Thus (10.32) for w is exactly

$$\int_{B_1^+} A \nabla w \cdot \nabla \phi = \int_{B_1^+} H \cdot \nabla \phi.$$

Reflect w evenly; reflect $A_{\alpha\beta}$, A_{33} and H_α evenly, and $A_{\alpha 3}$, $A_{3\alpha}$ and H_3 oddly. Vanishing of the cross coefficients and of H_3 on the boundary makes all reflected coefficients and data $C^{0,\alpha}$. On the reflected side the matrix is $RA(x', -x_3)R$, where $R = \text{diag}(1, 1, -1)$ is orthogonal, so the same ellipticity and operator-norm bounds hold there. The homogeneous conormal condition cancels

the boundary term, so the reflected function solves a divergence-form equation on a full ball. Theorem 10.27 gives w , and hence $z = w + q$, in $C^{1,\alpha}$ up to the boundary.

Under the stronger hypotheses, $q \in C^{2,\alpha}$ and $H \in C^{1,\alpha}$. Tangential difference quotients of the homogeneous-conormal equation satisfy a divergence equation whose vector datum is $\delta_h H - (\delta_h A) \nabla w$, uniformly in $C^{0,\alpha}$; the shifted cross coefficients and the normal component of this datum still vanish on the flat face. Indeed $\delta_h H_3 = 0$ and $\delta_h A_{3\alpha} = 0$ there, while homogeneous conormal data and $A_{3\alpha} = 0$ give $A_{33} \partial_3 w = 0$ and hence $(\delta_h A_{33}) \partial_3 w = 0$. Repeat the reflected gradient estimate (using Lemma 10.28 if a scalar divergence datum is retained); this controls tangential–tangential and tangential–normal second derivatives. Expanding

$$A_{33} \partial_{33} z = f - \sum_{(i,j) \neq (3,3)} A_{ij} \partial_{ij} z - (\partial_i A_{ij}) \partial_j z$$

controls the remaining normal derivative. $\square$

10.10 Newtonian potential and Kelvin removability

Proposition 10.36 (Newtonian equation for bounded densities). *For f bounded, measurable and compactly supported, put $v(x) := \int f(y) |x - y|^{-1} dy$. Then $-\Delta v = 4\pi f$ distributionally.*

Proof. For a test function ϕ , truncate the kernel at radius ε , integrate by parts away from the pole, and use $-\int_{\partial B_\varepsilon(y)} \partial_r |x - y|^{-1} dA_x = 4\pi$. The remaining volume and small-sphere errors tend to zero by dominated convergence. $\square$

Proposition 10.37 (Kelvin removability). *Let q be bounded and harmonic on a punctured ball $B_1 \setminus \{0\}$. Then q extends harmonically, hence smoothly, across the origin.*

Proof. Choose a radial cutoff η_ε which is zero on B_ε , one outside $B_{2\varepsilon}$, with $|\nabla \eta_\varepsilon| \leq C/\varepsilon$. Caccioppoli on $B_{3\varepsilon} \setminus B_{\varepsilon/2}$ gives $\int_{B_{2\varepsilon} \setminus B_\varepsilon} |\nabla q|^2 \leq C\varepsilon \|q\|_\infty^2$. Summing this estimate on dyadic annuli first shows that $q \in H^1(B_{1/2})$ after assigning any value at the origin. For a test function ϕ , use $\eta_\varepsilon \phi$ in the punctured weak equation; the term with $\nabla \eta_\varepsilon$ is bounded by Cauchy–Schwarz and $\int |\nabla \eta_\varepsilon|^2 \leq C\varepsilon$, hence tends to zero. So $\int \nabla q \cdot \nabla \phi = 0$ across the point, and Lemma 10.18 gives smoothness. $\square$

Chapter 11

Sard-type theorems

Lean module: NoCompromise.Sard.Equidimensional, NoCompromise.Sard.FourToThree,
NoCompromise.Sard.OneDimensional, NoCompromise.Sard.Scalar,
NoCompromise.Sard.ThreeDimensional

Exactly three Sard statements are used in this development, and each is proved from one-dimensional calculus. No general Sard theorem is needed or permitted.

Lemma 11.1 (Equidimensional critical values). *Let $n \in \{2, 3\}$ and $F : U \subset \mathbb{R}^n \rightarrow \mathbb{R}^n$ be C^1 . Then*

$$|F(\{x : \det DF(x) = 0\})| = 0.$$

Proof. Work on a compact cube $Q \Subset U$ where $\|DF\| \leq M$. Given $\varepsilon \in (0, 1)$, uniform continuity of DF gives $r_0 > 0$ with

$$\|DF(x) - DF(y)\| < \varepsilon \quad \text{whenever } x, y \in Q, |x - y| < 2\sqrt{n}r_0. \quad (11.1)$$

Partition Q into cubes of side $r < r_0$. If a cube meets the critical set, choose a critical point x_0 in it; Taylor's formula along line segments gives

$$F(x) = F(x_0) + DF(x_0)(x - x_0) + R(x), \quad |R(x)| \leq C_n \varepsilon r. \quad (11.2)$$

The linear image in (11.2) lies in a parallelepiped of rank at most $n - 1$ and diameter $\leq C_n M r$; its $C_n \varepsilon r$ -neighbourhood is covered by at most $C_n(1 + M/\varepsilon)^{n-1}$ cubes of side $C_n \varepsilon r$, hence has n -volume at most $C_{n,M} \varepsilon r^n$. There are $O(r^{-n})$ source cubes, so the critical image in Q has outer measure $\leq C_{n,M,Q} \varepsilon$. Let $\varepsilon \downarrow 0$ and exhaust U by compact cubes. $\square$

Corollary 11.2 (Chart version). *Let M, N be second-countable C^1 manifolds of the same dimension $n \in \{2, 3\}$ and let $F : M \rightarrow N$ be C^1 . The image under F of its critical set has measure zero in every target chart. In particular it is null for any smooth volume measure on N .*

Proof. Cover M by countably many relatively compact coordinate charts. On each chart Lemma 11.1 makes the coordinate image of the critical set Lebesgue-null. Target coordinate changes are locally C^1 , hence locally Lipschitz on compact subsets, and so preserve null sets. A countable union is null. A smooth volume measure has a locally bounded density with respect to Lebesgue measure in every target chart. $\square$

Lemma 11.3 (One-dimensional critical values). *Let $g \in C^1([a, b])$ and $Z := \{x : g'(x) = 0\}$. Then $|g(Z)| = 0$. The same holds on an open interval by exhaustion.*

Proof. Given $\varepsilon > 0$, continuity of g' gives an open $O \supset Z$ on which $|g'| < \varepsilon$; write $O = \bigsqcup_j I_j$. The mean value theorem gives $\text{diam } g(I_j) \leq \varepsilon |I_j|$, so $|g(Z)| \leq \varepsilon \sum_j |I_j| \leq \varepsilon(b-a)$. $\square$

Lemma 11.4 (Two-dimensional scalar Sard). *Let $U \subset \mathbb{R}^2$ be open, $g \in C^2(U)$, $C(g) := \{Dg = 0\}$. Then $|g(C(g))| = 0$.*

Proof. Work on a compact square $Q \Subset U$ and put $C_1 := \{Dg = 0\}$, $C_2 := \{Dg = 0, D^2g = 0\}$.

For $x \in C_1 \setminus C_2$ some $h := \partial_i g$ has $Dh(x) \neq 0$; the elementary implicit-function argument (contraction mapping in one coordinate) gives a neighbourhood W_x in which $\{h = 0\}$ is a C^1 curve containing $C_1 \cap W_x$. Parametrise each compact subarc by a C^1 map γ ; at a parameter whose image lies in C_1 , $(g \circ \gamma)' = Dg(\gamma) \cdot \gamma' = 0$, so Lemma 11.3 makes $g(C_1 \cap W_x)$ null. Second countability gives a countable such family.

For $C_2 \cap Q$, subdivide into squares of side r ; Taylor with integral remainder and uniform continuity of D^2g give $\text{osc}_{Q_r} g \leq \omega(r)r^2$ with $\omega(r) \rightarrow 0$. There are $O(r^{-2})$ squares, so the total image length is $O(\omega(r)) \rightarrow 0$. $\square$

Theorem 11.5 (Three-dimensional scalar Sard). *Let $U \subset \mathbb{R}^3$ be open and $f \in C^3(U)$. Then $f(\{Df = 0\})$ has one-dimensional Lebesgue measure zero. In particular, almost every $t \in \mathbb{R}$ is a regular value of f .*

Proof. Work on a compact cube $Q \Subset U$ and put $C_1 := \{Df = 0\}$, $C_2 := C_1 \cap \{D^2f = 0\}$, $C_3 := C_2 \cap \{D^3f = 0\}$.

For $x \in C_1 \setminus C_2$, some $h := \partial_i f$ has $Dh(x) \neq 0$; near x the zero set of h is a C^2 surface, parametrised by $\Phi : W \subset \mathbb{R}^2 \rightarrow U$. Then $f \circ \Phi$ is C^2 and every preimage of a point of C_1 is a critical point of it, so Lemma 11.4 makes $f(C_1 \cap \Phi(W))$ null; a countable chart cover finishes.

For $x \in C_2 \setminus C_3$, some $h := \partial_{ij} f$ has $Dh(x) \neq 0$, so C_2 lies locally on a C^1 surface. On a compact surface patch, the graph description and a square grid give a cover by $O(r^{-2})$ balls of radius r ; if such a ball meets C_2 , Taylor at a point of C_2 gives $\text{osc } f \leq Cr^3$, so the image intervals have total length $O(r) \rightarrow 0$.

Finally cover $C_3 \cap Q$ by the $O(r^{-3})$ grid cubes of side r ; at a point of C_3 all Taylor coefficients through order three vanish, and uniform continuity of D^3f gives $\text{osc}_{Q_r} f \leq \omega(r)r^3$ with $\omega(r) \rightarrow 0$. The total image length is $O(\omega(r))$. $\square$

Theorem 11.6 (The $4 \rightarrow 3$ statement). *Let $F : U \subset \mathbb{R}^4 \rightarrow \mathbb{R}^3$ be C^2 and $C(F) := \{x \in U : \text{rank } DF(x) < 3\}$. Then $|F(C(F))|_3 = 0$.*

Proof. Exhaust U by compact cubes and split $C(F)$ by rank.

Rank two. Some 2×2 minor is nonzero, so the inverse function theorem in source and target coordinates puts F locally in the form

$$F(s, t, y_1, y_2) = (s, t, h(s, t, y_1, y_2)), \quad (11.3)$$

in which the critical points are exactly those with $D_y h = 0$. For fixed (s, t) , Lemma 11.4 makes $|\{h(s, t, y) : D_y h(s, t, y) = 0\}|_1 = 0$; Fubini makes the three-dimensional measure of the critical image zero. The coordinate changes are locally bi-Lipschitz, hence preserve null sets; a countable chart cover finishes.

Rank one. A nonzero first derivative gives local coordinates

$$F(s, y) = (s, H(s, y)), \quad y \in \mathbb{R}^3, \quad H \in \mathbb{R}^2, \quad (11.4)$$

in which every point of the stratum satisfies $D_y H = 0$. Fix s , cover a compact y -cube by $O(r^{-3})$ cubes of side r ; if a cube meets $\{D_y H = 0\}$, Taylor about such a point puts its $H(s, \cdot)$ -image in a disk of radius Cr^2 . The total area is $O(r^{-3})O(r^4) = O(r) \rightarrow 0$. Fubini in s and a countable chart cover finish.

Rank zero. Cover by $O(r^{-4})$ four-dimensional grid cubes; Taylor about a point with $DF = 0$ puts the image in a ball of radius Cr^2 , total volume $O(r^{-4})O(r^6) = O(r^2) \rightarrow 0$. $\square$

Remark 11.7. Corollary 11.2 is used for path transversality and for choosing a regular value of a Gauss map; Theorem 11.5 for regular level values of a scalar function; Theorem 11.6 for disk transversality. These are the only three uses.

Chapter 12

Smooth approximation of finite-perimeter sets

Lean module: `NoCompromise.BV.SmoothApprox`

Theorem 12.1 (Smooth bounded approximation with exact volume). *Let $E \subset \mathbb{R}^3$ have finite positive volume and finite perimeter. There are bounded open sets $\tilde{E}_j$ with smooth boundary such that*

$$|\tilde{E}_j| = |E|, \quad \mathbf{1}_{\tilde{E}_j} \rightarrow \mathbf{1}_E \text{ in } L^1(\mathbb{R}^3), \quad P(\tilde{E}_j) \rightarrow P(E).$$

Proof. Step 1: truncate. Lemma 5.13 supplies good radii $R_j \uparrow \infty$ with $|E \setminus B_{R_j}| \rightarrow 0$ and $P(E \cap B_{R_j}) \rightarrow P(E)$. Put $F_j := E \cap B_{R_j}$.

Step 2: mollify and choose a level. Put $u_j := \rho_{\varepsilon_j} * \mathbf{1}_{F_j}$. For each j , choose $\varepsilon_j > 0$ so small that convolution changes $\mathbf{1}_{F_j}$ in L^1 by at most j^{-1} and changes its total variation by at most j^{-1} . The strict-approximation argument (Theorem 4.18), together with the convergence of F_j , then gives

$$u_j \rightarrow \mathbf{1}_E \text{ in } L^1, \quad \int |\nabla u_j| \rightarrow P(E). \quad (12.1)$$

For fixed $0 < a < b < 1$, layer cake and Fubini give $\int_a^b |\{u_j > t\} \Delta E| dt \rightarrow 0$, while Theorem 5.5 gives $\int_a^b P(\{u_j > t\}) dt \leq \int |\nabla u_j|$. Choose $a_j \downarrow 0$, $b_j \uparrow 1$ and then a common $t_j \in (a_j, b_j)$ for which the symmetric difference tends to zero and the perimeter limsup is at most $P(E)$; Lemma 4.7 makes the perimeters converge. By Theorem 11.5, t_j may simultaneously be chosen a regular value of u_j , so $E_j := \{u_j > t_j\}$ has smooth boundary.

Step 3: correct the volume. Put $\lambda_j := (|E|/|E_j|)^{1/3} \rightarrow 1$ and $\tilde{E}_j := \lambda_j E_j$. Then $|\tilde{E}_j| = |E|$ and $P(\tilde{E}_j) = \lambda_j^2 P(E_j) \rightarrow P(E)$ by Lemma 4.6(iii). The required L^1 convergence after dilation is not inferred from boundedness of the individual E_j : explicitly,

$$\|\mathbf{1}_{\lambda_j E_j} - \mathbf{1}_E\|_1 \leq \lambda_j^3 \|\mathbf{1}_{E_j} - \mathbf{1}_E\|_1 + \|\mathbf{1}_E(\cdot/\lambda_j) - \mathbf{1}_E\|_1 \rightarrow 0,$$

where the last term tends to zero by strong continuity of dilations on $L^1(\mathbb{R}^3)$, proved first for compactly supported continuous functions and then by density. $\square$

Corollary 12.2 (Energy convergence). *With $\tilde{E}_j$ as in Theorem 12.1, $D(\tilde{E}_j) \rightarrow D(E)$ and hence $\mathcal{E}(\tilde{E}_j) \rightarrow \mathcal{E}(E)$.*

Proof. The L^1 -Lipschitz estimate (15.4) and the Coulomb scaling (15.7). $\square$

Off the critical path: Theorem [13.7](#), the only consumer of Theorem [12.1](#), needs volume and perimeter convergence only. The preceding corollary is recorded because it is the natural companion of Lemma [17.1](#), which proves the same energy convergence for the bounded truncation. A formalisation may skip it; see Formalisation note [40.2](#).

Chapter 13

The sharp isoperimetric inequality

Lean module: NoCompromise.Isoperimetric.ABP

Remark 13.1 (Acyclicity). The *inequality* (13.3) is proved here from the ABP contact-set argument and depends on nothing from Part IV. The *equality case* needs the regularity theory for ω -minimal sets and is therefore deferred to Theorem 27.1. Chapter 20 onwards uses only the inequality. This ordering is what keeps the dependency graph acyclic, and it must be preserved: if the regularity chapters are ever allowed to cite the equality case, the development becomes circular.

Lemma 13.2 (The ABP auxiliary Neumann problem). *Let $G \subset \mathbb{R}^3$ be a bounded connected smooth domain. There is a smooth $z : \overline{G} \rightarrow \mathbb{R}$ with*

$$\Delta z = \frac{P(G)}{|G|} \text{ in } G, \quad \partial_\nu z = 1 \text{ on } \partial G. \quad (13.1)$$

Proof. Compatibility (10.7) holds because $\int_G P(G)/|G| = P(G) = \int_{\partial G} 1$. Apply Proposition 10.9 with $f = P(G)/|G|$, $h = 1$ and the sign convention $\partial_\nu z = h$. In an upper-half-ball signed-distance chart, the inward-coordinate conormal appearing in Theorem 10.35 is the negative of this outward datum (with the smooth chart density absorbed into A, f, h). To check ellipticity, write the chart as $x = \Psi(y)$, set $J = |\det D\Psi|$, and use the pulled-back Laplacian coefficient

$$A = J(D\Psi)^{-1}(D\Psi)^{-\top}, \quad \xi \cdot A\xi = J|(D\Psi)^{-\top}\xi|^2.$$

On finitely many smaller chart closures, J is bounded above and below by positive constants, and $D\Psi$ and its inverse are bounded. This gives uniform $0 < \lambda \leq \Lambda < \infty$. Signed-distance coordinates make the normal direction orthogonal to the tangential ones on the flat face, giving $A_{3\alpha} = A_{\alpha 3} = 0$ there. The smooth boundary gives coefficients and data of every required order, so Theorem 10.35 applies and gives smoothness up to the boundary. $\square$

Lemma 13.3 (The contact set covers the unit ball of gradients). *Let z be as in Lemma 13.2 and let*

$$\Gamma := \{x \in G : z(y) \geq z(x) + \nabla z(x) \cdot (y - x) \quad \forall y \in \overline{G}\}$$

be its lower contact set. Then $B_1 \subset \nabla z(\Gamma)$, and $D^2 z \geq 0$ on Γ .

Proof. For $p \in B_1$ the function $z - p \cdot x$ attains its minimum on $\overline{G}$. A boundary minimum would force $\partial_\nu(z - p \cdot x) \leq 0$, but $\partial_\nu(z - p \cdot x) = 1 - p \cdot \nu > 0$. So the minimum is interior, where $\nabla z = p$ and $D^2 z \geq 0$; such a point lies in Γ by convexity of the supporting-plane condition at an interior minimum. $\square$

Proposition 13.4 (Smooth connected case). *For a bounded connected smooth domain $G \subset \mathbb{R}^3$,*

$$P(G)^3 \geq 36\pi |G|^2. \quad (13.2)$$

Proof. The critical values of ∇z are null by Lemma 11.1. Cover the regular set by countably many inverse function neighbourhoods and disjointify them; on each resulting Borel piece the equidimensional change-of-variables theorem from (B6) applies injectively. Summing gives $|\nabla z(\Gamma)| \leq \int_{\Gamma} |\det D^2 z|$. On Γ the Hessian is positive semidefinite, so Lemma 13.3 and the arithmetic–geometric mean inequality give

$$|B_1| \leq \int_{\Gamma} \det D^2 z \leq \int_{\Gamma} \left(\frac{\Delta z}{3}\right)^3 \leq |G| \left(\frac{P(G)}{3|G|}\right)^3.$$

Since $|B_1| = 4\pi/3$, this is (13.2). $\square$

Formalisation note. The argument uses only that ∂G is C^3 : the Neumann solution z of Lemma 13.2 is then C^2 in G and C^1 up to the boundary with $\partial_{\nu} z = 1$, which is all the contact-set argument needs. The formalisation proves the proposition, Corollary 13.6 and Theorem 13.7 in this C^3 form (the smooth case being a special case), and Step 2 of Theorem 27.1 applies it to a $C^{3,\beta}$ representative.

Lemma 13.5 (Strict superadditivity of $t^{2/3}$). *For $a, b > 0$, $a^{2/3} + b^{2/3} > (a + b)^{2/3}$.*

Proof. Strict concavity of $t \mapsto t^{2/3}$ on $(0, \infty)$ with value 0 at 0. $\square$

Corollary 13.6 (Smooth, possibly disconnected). *For a bounded smooth domain G with components G_i ,*

$$P(G) \geq (36\pi)^{1/3} \sum_i |G_i|^{2/3} \geq (36\pi)^{1/3} |G|^{2/3},$$

and the second inequality is strict unless there is exactly one component.

Proof. Apply (13.2) to each component and then Lemma 13.5. $\square$

Theorem 13.7 (Sharp isoperimetric inequality). *For every measurable $E \subset \mathbb{R}^3$ of finite measure and finite perimeter,*

$$P(E) \geq c_I |E|^{2/3}, \quad c_I := (36\pi)^{1/3}. \quad (13.3)$$

Proof. Apply Corollary 13.6 to the smooth exact-volume approximants of Theorem 12.1 and pass to the limit. $\square$

Remark 13.8 (The equality case is not proved here). Equality in (13.3) holds exactly for balls up to null sets; that is Theorem 27.1, proved in Chapter 27 after the regularity theory.

Chapter 14

Flows, transversality and the total curvature bound

Lean module: NoCompromise.Flow.FlowBox, NoCompromise.Flow.ODE,
NoCompromise.Surface.Geometry, NoCompromise.Surface.Morse,
NoCompromise.Surface.MorseCount, NoCompromise.Surface.Shape,
NoCompromise.Surface.TotalCurvature, NoCompromise.Topology.Parity,
NoCompromise.Topology.Transversality, NoCompromise.Topology.Tubular

This chapter supplies the differential-topological layer. Its output is used in exactly three places: Theorem 14.14 by Lemma 29.3 and Lemma 30.9, Theorem 14.12 by Lemma 32.3, and the total curvature bound (14.28) by Lemma 31.17.

It is the most expensive part of the project (Formalisation note 14.8). Formalisation note 14.45 records what the present design already avoids, and Formalisation note 14.46 what might still be avoided.

14.1 From Picard–Lindelöf to a global smooth flow

Theorem 14.1 (Global flow). *Let $X : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be C^k , $1 \leq k \leq \infty$, globally Lipschitz with constant L and bounded by M . Then there is a unique $\Phi : \mathbb{R} \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ with*

$$\Phi_t(x) = x + \int_0^t X(\Phi_s(x)) ds. \quad (14.1)$$

In particular a compactly supported C^k field satisfies these hypotheses, with $M = \|X\|_\infty$ and $L = \|DX\|_\infty$.

Proof. The local Picard–Lindelöf family (B7) applies on every closed ball: continuity supplies M , the mean value theorem supplies the spatial Lipschitz hypothesis with constant L . Its remaining hypothesis is the interval condition $M \cdot \max(t_{\max} - t_0, t_0 - t_{\min}) \leq a - r$ relating the time interval to the ball radii $r < a$ on which the family is posed. It holds on an arbitrarily long interval: given $T > 0$, take $r = a/2$ with $a \geq 2MT$, so that $a - r = a/2 \geq MT$. *This is the only place global boundedness is used, and it is what upgrades a local statement to a flow defined for all time.*

A local solution obeys $|\Phi_t(x) - x| \leq M|t|$, so on a prescribed bounded time interval its graph stays in a fixed compact ball. The quantitative local theorem supplies a time step depending only on the bounds of X on that ball; finitely many consecutive steps cover the interval, and uniqueness makes consecutive solutions agree and extensions from different intervals compatible. $\square$

Lemma 14.2 (Group law). $\Phi_0 = \text{Id}$, $\Phi_{t+s} = \Phi_t \circ \Phi_s$, $\Phi_t^{-1} = \Phi_{-t}$.

Proof. Uniqueness in (14.1) after a time translation. $\square$

Lemma 14.3 (Grönwall and bi-Lipschitz dependence). $|\Phi_t(x) - \Phi_t(y)| \leq e^{L|t|}|x - y|$; hence each Φ_t is a bi-Lipschitz homeomorphism.

Proof. Subtract (14.1) at x and y and apply the elementary Grönwall inequality, proved by differentiating $e^{-Lt} \int_0^t L|\Phi_s(x) - \Phi_s(y)| ds$, first for $t \geq 0$ and then backwards in time. $\square$

Lemma 14.4 (The variational equation). For fixed x the linear matrix integral equation

$$A(t, x) = I + \int_0^t DX(\Phi_s(x))A(s, x) ds \quad (14.2)$$

has a unique solution, and $(t, x) \mapsto A(t, x)$ is continuous.

Proof. Picard iteration in the finite-dimensional matrix space converges uniformly on compact time intervals: the difference of the m -th and $(m+1)$ -st iterates is bounded by $C(L|t|)^m/m!$. The same estimate and continuity of $DX \circ \Phi$ give joint continuity. $\square$

Theorem 14.5 (C^k dependence on the initial point). $D_x \Phi_t(x) = A(t, x)$, continuously in (t, x) ; each Φ_t is a C^1 diffeomorphism with $(D\Phi_t)^{-1} = D\Phi_{-t}(\Phi_t(x))$; and if $X \in C^k$ then $(t, x) \mapsto \Phi_t(x)$ is C^k . In particular a smooth vector field has a smooth flow.

Proof. For an increment z , subtract (14.1) at $x+z$ and x , then subtract (14.2) applied to z . By Lemma 14.3 all points lie in one compact set and $|X(p+w) - X(p) - DX(p)w| \leq \omega(|w|)|w|$ with $\omega(r) \downarrow 0$ a modulus of continuity for DX there. A second Grönwall estimate gives, uniformly for t in a fixed compact interval,

$$\frac{|\Phi_t(x+z) - \Phi_t(x) - A(t, x)z|}{|z|} \leq C_t \omega(e^{L|t|}|z|) \rightarrow 0. \quad (14.3)$$

Invertibility follows from Lemma 14.2 and the chain rule. For $X \in C^k$, differentiate (14.2) inductively: at order j the highest derivative of Φ occurs linearly with coefficient $DX(\Phi)$, while the forcing contains only derivatives of X of order $\leq j$ and already-constructed lower-order derivatives of Φ . The same integral-equation and Grönwall argument gives C^j dependence; induct to $j = k$. The ODE $\partial_t \Phi = X \circ \Phi$ supplies the time and mixed derivatives. $\square$

Theorem 14.6 (Flow box). Let $X \in C^k$ with $X(p) \neq 0$. Choose an affine $(n-1)$ -disk S through p whose tangent space is complementary to $\mathbb{R}X(p)$, and set

$$\Psi : S \times (-\varepsilon, \varepsilon) \rightarrow \mathbb{R}^n, \quad \Psi(y, t) := \Phi_t(y). \quad (14.4)$$

After shrinking the domain, Ψ is a C^k coordinate chart, and in these coordinates X becomes ∂_t .

Proof. At $(p, 0)$, $D\Psi(v, s) = v + sX(p)$, an isomorphism; apply the inverse function theorem (B6), and differentiate in t . $\square$

Corollary 14.7 (Flows on a manifold). Let M be a smooth manifold without boundary, let $U \subset M$ be open, and let X be a C^k vector field on U , $1 \leq k \leq \infty$. Through every $p \in U$ there is a unique maximal integral curve, the resulting local flow is C^k and satisfies the group law wherever both sides are defined, and the flow-box conclusion holds wherever $X \neq 0$. A maximal curve cannot have a finite endpoint while remaining in a compact subset of U . Consequently a C^k vector field on a compact manifold is complete.

Proof. Perform the Euclidean local construction in charts. Coordinate representatives agree on overlaps by ODE uniqueness, so they glue; the difference-quotient proof of Theorem 14.5 is invariant under smooth coordinate changes and gives C^k dependence. The group law follows from uniqueness, and Theorem 14.6 applies in a chart. If a curve is confined to a compact subset of U , finitely many relatively compact existence charts cover that set and furnish a common positive continuation time after taking the minimum of their local times. Thus a finite maximal endpoint would admit an extension, a contradiction. Take $U = M$ when M is compact. $\square$

Formalisation note 14.8 (Cost). §14.1 is a mathlib-scale contribution on its own: global flow from Picard–Lindelöf, group law, Grönwall, the variational equation solved by Picard iteration in matrix space, C^k by induction, and straightening. Budget accordingly. It is the single largest item in this chapter, and Formalisation note 14.46(a) records the one route that might remove it.

14.2 Relative transversality

Theorem 14.9 (Relative transversality for paths and disks). *Let N be a compact path (a 1-manifold with boundary) or a compact disk, let $\Sigma \subset \mathbb{R}^3$ be a compact smooth embedded surface, and let $f : N \rightarrow \mathbb{R}^3$ be smooth, already transverse to Σ on a neighbourhood of ∂N , and with $f|_{\partial N}$ transverse to Σ (for a path this means $f(\partial N) \cap \Sigma = \emptyset$). Then for every $\epsilon > 0$ there is a smooth $g : N \rightarrow \mathbb{R}^3$ with $\|g - f\|_{C^1} < \epsilon$, $g = f$ on a neighbourhood of ∂N , g transverse to Σ , and $g|_{\partial N}$ transverse to Σ .*

Proof. Choose boundary collars $U_0 \Subset U_1$ whose closures lie in the neighbourhood where f is transverse, and a smooth $\chi : N \rightarrow [0, 1]$ vanishing on $\overline{U_0}$ and positive on $N \setminus \overline{U_0}$ (obtained from a smooth collar coordinate by a cutoff flat at the inner edge of U_0). For $v \in \mathbb{R}^3$ set

$$f_v(p) := f(p) + \chi(p)v. \quad (14.5)$$

On $\{\chi > 0\}$ the evaluation map $\mathcal{F}(p, v) := f_v(p)$ is a submersion, since its v -derivative is $\chi(p)I$. Hence $Z := \mathcal{F}^{-1}(\Sigma) \cap (\{\chi > 0\} \times \mathbb{R}^3)$ is a smooth manifold of dimension $\dim N + 2$. Let $\pi : Z \rightarrow \mathbb{R}^3$ be the parameter projection. Directly from the tangent-space equation

$$D_p f_v(\xi) + \chi(p)w \in T_{f_v(p)}\Sigma \quad (14.6)$$

one sees that $D\pi$ is onto at (p, v) exactly when $D_p f_v(T_p N) + T_{f_v(p)}\Sigma = \mathbb{R}^3$. So a regular value v of π makes f_v transverse wherever $\chi > 0$.

For a path π has dimensions $3 \rightarrow 3$ and Corollary 11.2 makes its critical values null; for a disk it has dimensions $4 \rightarrow 3$ and Theorem 11.6 does the same. So every ball about 0 contains a regular v . On $\overline{U_0}$ the map is unchanged; elsewhere transversality follows from regularity of v . $\square$

Corollary 14.10 (Preimage structure). *Let $g : N \rightarrow \mathbb{R}^3$ be transverse to Σ with $g|_{\partial N}$ transverse to Σ , as produced by Theorem 14.9. A transverse path has a compact zero-dimensional inverse image of Σ , hence finitely many intersections. A transverse disk has a compact one-dimensional inverse image, a compact 1-manifold with boundary whose boundary is exactly its intersections on ∂N . (Without transversality of $g|_{\partial N}$ the disk statement fails: the inclusion of the unit disk into $\{z = 0\}$ is transverse to the unit sphere centred at $(2, 0, 0)$, yet the preimage is the single boundary point $(1, 0)$.)*

Proof. The preimage theorem, i.e. the implicit function theorem (B6) in a defining-function chart for Σ . $\square$

Lemma 14.11 (Even cardinality of a compact 1-manifold boundary). *The boundary of a compact one-dimensional manifold with boundary has even cardinality.*

Proof. Choose a finite cover by interval and half-interval charts and subdivide at the finitely many chart endpoints. The resulting finite graph has degree two at every interior vertex and degree one exactly at the manifold boundary points; the handshake identity $\sum_v \deg v = 2\#\{\text{edges}\}$ makes the number of degree-one vertices even. $\square$

14.3 Uniform tubular coordinates

Theorem 14.12 (Tubular neighbourhood). *Let $\Sigma \subset \mathbb{R}^3$ be a compact smooth embedded surface with a smooth unit normal n . Then there is $\varepsilon > 0$ such that*

$$\Theta : \Sigma \times (-\varepsilon, \varepsilon) \rightarrow \mathbb{R}^3, \quad \Theta(p, t) := p + t n(p) \quad (14.7)$$

is a diffeomorphism onto an open neighbourhood of Σ .

Proof. $D\Theta$ is invertible at every $(p, 0)$, so the inverse function theorem gives injective normal coordinates near each such point; compactness gives finitely many neighbourhoods and a uniform local radius. If no smaller uniform radius made Θ globally injective, there would be $p_j + t_j n(p_j) = q_j + s_j n(q_j)$ with $(p_j, t_j) \neq (q_j, s_j)$ and $|t_j| + |s_j| \rightarrow 0$; after taking limits $p_j, q_j \rightarrow p$, and for large j both pairs lie in the same injective normal chart at p , a contradiction. $\square$

14.4 Orientation by parity

Proposition 14.13 (Parity, orientation and component count). *Let $\Sigma \subset \mathbb{R}^3$ be a compact connected smooth embedded closed surface. Then:*

- (i) *the parity $I(x) \in \mathbb{Z}/2$ of the number of transverse intersections with Σ of a path from a fixed far base point z to $x \notin \Sigma$ is independent of the path;*
- (ii) *the two parity classes select compatible local sides, hence a continuous — and by smoothness of Σ , smooth — global unit normal n ; in particular Σ is orientable;*
- (iii) *$\mathbb{R}^3 \setminus \Sigma$ has exactly two connected components.*

Proof. (i): A piecewise smooth path can be perturbed with endpoints fixed, by Theorem 14.9, to meet Σ transversely in finitely many points. Concatenating two paths gives a closed loop γ ; smooth its finitely many corners in small balls disjoint from the other intersection points. Choose $a \in \mathbb{R}^3$ and smooth $\rho : [0, 1] \rightarrow [0, 1]$ zero near 0 and one near 1; the explicit map

$$F(re^{i\theta}) := (1 - \rho(r))a + \rho(r)\gamma(e^{i\theta}) \quad (14.8)$$

is a smooth disk spanning γ , constant near the centre and agreeing with γ on a boundary collar. Apply Theorem 14.9 in its disk case to perturb the interior while fixing that collar. By Corollary 14.10 the inverse image of Σ is a compact 1-manifold whose boundary is exactly the intersections of γ with Σ , and Lemma 14.11 makes that set even.

(ii): I is locally constant on the complement. In a defining-function chart around a point of Σ the two local sides have opposite parity, so both values occur; two local positive sides with the

same parity agree on their common connected collar. Only after this gluing may Theorem 14.12 be invoked.

(iii): Let x, y have the same parity and take a transverse path between them, with an even number of intersections. Pair consecutive intersections; between the points of one pair the path lies on one local side of Σ . Since a connected smooth surface is path connected and remains so after deleting finitely many points — cover a connecting path by coordinate disks and, whenever it passes through a forbidden point, replace the corresponding diameter segment by a small semicircle in that disk — join the two intersection points by a path on Σ avoiding the other intersections. Push it a small distance into the same globally labelled side inside the tubular neighbourhood and join it to the old path by two small collars. This removes exactly two intersections and stays in the complement; repeat. $\square$

Theorem 14.14 (Component count for a disjoint union of surfaces). *Let $\Sigma = \Sigma_1 \dot{\cup} \cdots \dot{\cup} \Sigma_k \subset \mathbb{R}^3$ be a finite disjoint union of compact connected smooth embedded closed surfaces. Then $\mathbb{R}^3 \setminus \Sigma$ has exactly $k + 1$ connected components.*

Proof. Induct on k ; the case $k = 1$ is Proposition 14.13(iii). Suppose the result for $\Sigma_1, \dots, \Sigma_{k-1}$. The connected surface Σ_k lies in one connected component U of their complement; U is open and connected in Euclidean space, hence path connected, and its tubular neighbourhood about Σ_k has positive distance from the other surfaces. Apply the parity and path-surgery argument of Proposition 14.13 to paths lying in U . The spanning disk used only to prove parity independence may cross the other surfaces; that is irrelevant, since transversality is taken only with respect to Σ_k , and the path surgery itself takes place in U . Thus $U \setminus \Sigma_k$ has exactly two components and all others are unchanged. $\square$

14.5 The shape operator and height functions

Everything in this section is pointwise linear algebra on a smooth surface. No Morse theory, no flow, and no moving frame is used.

Definition 14.15 (Shape operator). For a smooth embedded surface $\Sigma \subset \mathbb{R}^3$ with unit normal n , the *shape operator* at p is $A_p X := D_X n$ for $X \in T_p \Sigma$, the second fundamental form is $\Pi_p(X, Y) := \langle A_p X, Y \rangle$, and

$$H(p) := \operatorname{tr} A_p, \quad \kappa(p) := \det A_p,$$

consistently with Convention 2.4. A_p is symmetric, so Π_p is a symmetric bilinear form on the two-dimensional space $T_p \Sigma$.

Lemma 14.16 (Hessian of a height function). *Let $\Sigma \subset \mathbb{R}^3$ be a smooth embedded surface with unit normal n , $v \in \mathbb{S}^2$, and $h_v := \langle \cdot, v \rangle$. Then p is a critical point of $h_v|_\Sigma$ if and only if $n(p) = \pm v$, and at such a point*

$$\operatorname{Hess}_p(h_v|_\Sigma) = -\langle n(p), v \rangle \Pi_p = \mp \Pi_p, \quad (14.9)$$

where the upper displayed sign corresponds to $\langle n(p), v \rangle = +1$.

Proof. $d_p h_v(X) = \langle X, v \rangle$, which vanishes for all $X \in T_p \Sigma$ iff $v \perp T_p \Sigma$ iff $n(p) = \pm v$. At such a p , for X, Y tangent,

$$\operatorname{Hess}_p h_v(X, Y) = \langle D_X Y, v \rangle = \langle n(p), v \rangle \langle D_X Y, n(p) \rangle = -\langle n(p), v \rangle \Pi_p(X, Y),$$

where the middle step uses $v = \langle n(p), v \rangle n(p)$ (valid because $v = \pm n(p)$), while the last follows by differentiating $\langle Y, n \rangle = 0$ and using Definition 14.15. $\square$

Lemma 14.17 (The Gauss map Jacobian is the Gauss curvature). *Let $\Sigma \subset \mathbb{R}^3$ be a smooth embedded surface with unit normal n . Then $T_{n(p)}\mathbb{S}^2 = T_p\Sigma$, the differential $d_p n : T_p\Sigma \rightarrow T_{n(p)}\mathbb{S}^2$ equals A_p , and*

$$\det(d_p n) = \det A_p = \kappa(p), \quad J_2(d_p n) = |\kappa(p)|. \quad (14.10)$$

Proof. $T_{n(p)}\mathbb{S}^2 = n(p)^\perp = T_p\Sigma$ because $n(p) \perp T_p\Sigma$. Then $d_p n(X) = D_X n = A_p X$ by Definition 14.15. For a linear map between two-dimensional inner-product spaces Definition 6.1 gives $J_2 = |\det|$. $\square$

Convention 14.18 (Orientations for the Gauss map). Orient Σ by the unit normal n of Proposition 14.13(ii) and orient $\mathbb{S}^2$ by its outward normal, which at the point $n(p)$ is $n(p)$. Under the canonical identification $T_p\Sigma = T_{n(p)}\mathbb{S}^2$ of Lemma 14.17 these two orientations agree, so

$$\varepsilon_n(p) := \text{sign } \det(d_p n) = \text{sign } \kappa(p) \quad (14.11)$$

with no further sign. *This is the one place where a stray sign would invert the conclusion of §14.8; it is recorded as a convention so that it is fixed once.*

Lemma 14.19 (Nondegeneracy and the sign of the curvature). *Let $v \in \mathbb{S}^2$ and let p be a critical point of $h_v|_\Sigma$. Then p is nondegenerate if and only if $\kappa(p) \neq 0$, and in that case*

$$\text{sign } \kappa(p) = (-1)^{\text{ind}_p(h_v)}. \quad (14.12)$$

Proof. By (14.9), $\det \text{Hess}_p h_v = \det(\pm \text{II}_p) = \det \text{II}_p = \kappa(p)$, the sign disappearing because the form is two-dimensional. So p is nondegenerate iff $\kappa(p) \neq 0$. For a nondegenerate critical point of a function on a surface, $\text{sign } \det \text{Hess} = (-1)^{\text{ind}}$: indices 0 and 2 give two eigenvalues of the same sign, index 1 two of opposite signs. $\square$

Lemma 14.20 (Critical points split by normal direction). *Let $\Sigma \subset \mathbb{R}^3$ be closed and embedded with unit normal n and let $v \in \mathbb{S}^2$. Then*

$$\text{Crit}(h_v|_\Sigma) = n^{-1}(v) \sqcup n^{-1}(-v), \quad (14.13)$$

and $h_v|_\Sigma$ is a Morse function if and only if both v and $-v$ are regular values of n .

Proof. (14.13) is the first assertion of Lemma 14.16, the union being disjoint since $v \neq -v$. By Lemma 14.17, y is a regular value of n iff $\kappa \neq 0$ on $n^{-1}(y)$, and by Lemma 14.19 that is exactly nondegeneracy of the corresponding critical points. $\square$

14.6 Morse functions and regular bands

Lemma 14.21 (A Morse height function exists). *Let Σ be a compact connected smooth embedded closed surface with the unit normal n supplied by Proposition 14.13(ii). Then for almost every $v \in \mathbb{S}^2$ both v and $-v$ are regular values of n , and for every such v the height function h_v is Morse with finitely many critical points, all lying in $n^{-1}(\{v, -v\})$.*

Proof. Corollary 11.2 applied to n , a C^1 map between two-surfaces, makes the set of critical values $\mathcal{H}^2$ -null; the union of that set with its antipodal image is still null. For v outside it, Lemma 14.20 makes h_v Morse, and its critical set is then discrete and compact, hence finite. $\square$

Lemma 14.22 (Distinct critical values). *Let h be Morse on the compact surface Σ . There is a Morse function $\tilde{h}$ on Σ with the same critical points, the same Hessians there — hence the same index counts c_0, c_1, c_2 — and with pairwise distinct critical values.*

Proof. Write the critical points as $p_1, \dots, p_k$, choose disjoint neighbourhoods $W_i \Subset V_i$ and smooth b_i supported in V_i with $b_i \equiv 1$ on W_i , and shrink the W_i so that $m := \min_{\Sigma \setminus \bigcup_i W_i} |\nabla_\Sigma h| > 0$. Choose arbitrarily small $c = (c_1, \dots, c_k)$ avoiding the finitely many affine hyperplanes $h(p_i) + c_i = h(p_j) + c_j$ and small enough that

$$\sum_i |c_i| \|\nabla_\Sigma b_i\|_\infty < m/2. \quad (14.14)$$

Put $\tilde{h} := h + \sum_i c_i b_i$. On each W_i it differs from h by the additive constant c_i , so its critical points and Hessians there are those of h ; (14.14) prevents a critical point on $\Sigma \setminus \bigcup_i W_i$, so none is created; and the critical values $h(p_i) + c_i$ are distinct. $\square$

Formalisation note 14.23 (Why “same Hessians” is in the statement). Proposition 14.42 reads the index counts c_0, c_1, c_2 off h_v itself, while Proposition 14.35 needs distinct critical values. The bridge is that Lemma 14.22 changes neither the critical points nor their Hessians, hence neither the indices. That is automatic from the construction, but it has to be part of the statement rather than left inside the proof.

Lemma 14.24 (Local Morse coordinates). *Let g be smooth near $0 \in \mathbb{R}^2$ with $g(0) = 0$, $Dg(0) = 0$ and $D^2g(0)$ nonsingular of index λ . There are smooth local coordinates in which*

$$g = \begin{cases} x_1^2 + x_2^2, & \lambda = 0, \\ x_1^2 - x_2^2, & \lambda = 1, \\ -x_1^2 - x_2^2, & \lambda = 2. \end{cases} \quad (14.15)$$

Proof. Taylor’s integral formula writes

$$g(x) = \frac{1}{2} x^\top B(x) x, \quad B(x) = 2 \int_0^1 (1-t) D^2g(tx) dt, \quad (14.16)$$

with $B(0) = D^2g(0)$. After a fixed linear change a diagonal entry of $B(0)$ is nonzero; complete its square, use the resulting coordinate change (whose Jacobian at zero is invertible), and repeat on the Schur complement. $\square$

Lemma 14.25 (Classification of a one-manifold with a tangent field). *Let M be a connected Hausdorff, second-countable smooth one-manifold without boundary and suppose a smooth nowhere-zero tangent vector field on M is specified. Then M is diffeomorphic to S^1 or to $\mathbb{R}$. Consequently a noncompact such M has exactly two ends. In particular this conclusion applies to every connected component of a regular level set of a smooth function on an oriented surface.*

Proof. Normalize the specified field to unit length and use Corollary 14.7 to take its maximal integral curve through a base point. Every orbit is open by the flow-box theorem, so the chosen orbit and the union of all other orbits are disjoint open sets; connectedness makes the chosen orbit all of M . If its orbit map is injective, it is a bijective local diffeomorphism from its maximal open interval onto M , hence a diffeomorphism; every open interval is diffeomorphic to $\mathbb{R}$. Otherwise the curve returns to its base point. ODE uniqueness then makes it periodic and extends it for all time. Its period group is closed and, by a flow box, contains no nonzero sequence tending to zero; it is

therefore $\tau\mathbb{Z}$ for a least $\tau > 0$, and the orbit map descends to a diffeomorphism $\mathbb{R}/\tau\mathbb{Z} \simeq S^1 \rightarrow M$. For a regular level set on an oriented surface, rotation by $\pi/2$ of the nonzero tangential gradient supplies the required nowhere-zero tangent field. $\square$

Lemma 14.26 (Regular bands are smooth products). *Let h be smooth on the compact surface Σ and let $a < b$ be such that $h^{-1}([a, b])$ contains no critical point. Then*

$$\Theta : h^{-1}(a) \times [0, b - a] \rightarrow h^{-1}([a, b]), \quad \Theta(p, t) := \Phi_t(p) \quad (14.17)$$

is a diffeomorphism, where Φ is the flow of

$$X := \frac{\nabla_\Sigma h}{|\nabla_\Sigma h|^2}. \quad (14.18)$$

Proof. X is smooth on an open neighbourhood of the band, and

$$\frac{d}{dt}h(\Phi_t(p)) = dh(X) = 1. \quad (14.19)$$

A trajectory starting on $h^{-1}(a)$ and run for $0 \leq t \leq b - a$ remains in the compact band, so the finite-step continuation of Theorem 14.1 rules out a shorter maximal interval. Hence (14.17) is a smooth bijection, with explicit smooth inverse

$$q \mapsto (\Phi_{a-h(q)}(q), h(q) - a). \quad (14.20)$$

$\square$

Corollary 14.27 (Sublevel retraction across a critical-value-free interval). *Let h be Morse on the compact surface Σ and let $a < b$.*

- (i) *If h has no critical value in $(a, b]$, then $\{h \leq a\}$ is a deformation retract of $\{h \leq b\}$.*
- (ii) *If h has no critical value in $[a, b)$, then $\{h \leq a\}$ is a deformation retract of $\{h < b\}$.*

In either case the two sets have the same number of connected components.

Proof. In case (i), $h^{-1}((a, b])$ contains no critical point, so the flow Φ of (14.18) is defined there and, by (14.19), a trajectory started at q with $a < h(q) \leq b$ exists backwards until its height tends to a . If its limiting point is regular, the ordinary local flow extends it to time $a - h(q)$. Near a critical point p of level a , Morse coordinates and equivalence of the coordinate and Riemannian norms give, on a sufficiently small coordinate ball,

$$|\nabla_\Sigma h(x)| \geq c|x|, \quad |h(x) - a| \leq C|x|^2, \quad |X(x)| \leq \frac{C'}{\sqrt{h(x) - a}} \quad (h(x) > a). \quad (14.21)$$

Along a descending trajectory, height itself is the time variable. Thus a segment descending from height $a + \delta$ has length at most

$$\int_0^\delta \frac{C'}{\sqrt{\tau}} d\tau = 2C'\sqrt{\delta}. \quad (14.22)$$

A bootstrap with two concentric coordinate balls shows that a trajectory starting sufficiently near p cannot leave the larger ball before reaching height a : otherwise its length would have a fixed positive lower bound, contrary to (14.22). The same estimate makes it Cauchy at the finite

terminal time and makes its endpoint tend to p as $q \rightarrow p$. Consequently every trajectory has a well-defined endpoint on level a , and

$$r(q) := \begin{cases} q, & h(q) \leq a, \\ \Phi_{a-h(q)}(q), & a < h(q) \leq b \end{cases} \quad (14.23)$$

is a well-defined retraction of $\{h \leq b\}$ onto $\{h \leq a\}$. It is continuous at regular points of $h^{-1}(a)$ because $\Phi_0 = \text{Id}$, and (14.22) proves continuity at a critical point of that level. The identical length estimate, applied to partial trajectory segments, shows jointly in (q, s) that

$$H_s(q) := \begin{cases} q, & h(q) \leq a, \\ \Phi_{s(a-h(q))}(q), & a < h(q) \leq b, \end{cases} \quad 0 \leq s \leq 1,$$

is continuous and stays in $\{h \leq b\}$; it is a deformation from the identity to r . Thus every component meets $\{h \leq a\}$ and two such components cannot be joined upstairs without being joined after applying r , so the component counts agree. Case (ii) uses the same formulas with $h(q) < b$. $\square$

Lemma 14.28 (Local structure at a critical point). *Let h be Morse on the surface Σ , let p be a critical point of index λ and $c := h(p)$, and let D be a Morse coordinate ball about p as in Lemma 14.24. Put $L := h^{-1}(c)$.*

- (i) $\lambda = 0$: $\{h < c\} \cap D = \emptyset$ and $\{h > c\} \cap D = D \setminus \{p\}$, and $L \cap D = \{p\}$.
- (ii) $\lambda = 2$: $\{h > c\} \cap D = \emptyset$ and $\{h < c\} \cap D = D \setminus \{p\}$, and $L \cap D = \{p\}$.
- (iii) $\lambda = 1$: $L \cap D$ is the union of four smooth arcs (rays) r_1, r_2, r_3, r_4 from p to ∂D , meeting only at p ; $\{h < c\} \cap D$ and $\{h > c\} \cap D$ each have exactly two connected components (sectors); and in cyclic order around p the rays and sectors alternate as

$$r_1, S_{12}^-, r_2, S_{23}^+, r_3, S_{34}^-, r_4, S_{41}^+,$$

where $S_{\bullet}^- \subset \{h < c\}$ and $S_{\bullet}^+ \subset \{h > c\}$. In particular the two sectors flanking r_1 are S_{41}^+ and S_{12}^- , those flanking r_2 are S_{12}^- and S_{23}^+ , those flanking r_3 are S_{23}^+ and S_{34}^- , and those flanking r_4 are S_{34}^- and S_{41}^+ .

Proof. Read (14.15) off in the chart. For $\lambda = 1$ the set $\{x_1^2 = x_2^2\}$ is the union of the four rays $x_2 = \pm x_1, \pm x_1 > 0$; $\{x_1^2 < x_2^2\}$ is the pair of vertical sectors and $\{x_1^2 > x_2^2\}$ the pair of horizontal ones; and the cyclic alternation is immediate. $\square$

Formalisation note 14.29 (No handle attachment is used). Lemma 14.28 is purely local: it says nothing relating $h^{-1}(c - \varepsilon)$ to $h^{-1}(c + \varepsilon)$. An earlier draft of this section asserted such a relation — that the level set is obtained by deleting two arcs and inserting two others — and used it to run a surgery argument. That assertion is true but its proof is the handle-attachment theorem, which requires constructing a modified function with a bump near p and is a substantial development in its own right. §14.7 is written so that it is never needed: every comparison there is between $\{h < c\}$, $\{h \leq c\}$ and a sublevel set at a nearby regular value, all handled by Lemma 14.28 and Corollary 14.27.

14.7 The Morse count is at most two

Definition 14.30 (Merging saddle). Let h be Morse on the compact connected surface Σ and let p be a critical point of index 1 with $c := h(p)$. With the notation of Lemma 14.28(iii), write $A(S)$ for the connected component of $\{h < c\}$ containing a sector S , and $B(S)$ for the component of $\{h > c\}$ containing S . Say p is

- *sub-merging* if $A(S_{12}^-) \neq A(S_{34}^-)$;
- *super-merging* if $B(S_{23}^+) \neq B(S_{41}^+)$.

Lemma 14.31 (Adjacent components are constant along a regular level arc). *Let h be smooth on Σ , $c \in \mathbb{R}$, $L := h^{-1}(c)$, and let $q \in L$ be a regular point of h . Then q has a neighbourhood U with $U \cap \{h < c\}$ and $U \cap \{h > c\}$ both connected and nonempty; writing $\phi^-(q)$ and $\phi^+(q)$ for the components of $\{h < c\}$ and $\{h > c\}$ containing them, the maps $\phi^\pm$ are locally constant on the set of regular points of L , hence constant on each connected subset of it.*

Proof. By the implicit function theorem (B6), in suitable coordinates $h - c$ is the first coordinate on a ball U ; then $U \cap \{h < c\}$ and $U \cap \{h > c\}$ are the two open half-balls, each connected and nonempty. For $q' \in U \cap L$ the same half-balls witness $\phi^\pm(q') = \phi^\pm(q)$. $\square$

Lemma 14.32 (Closing up a sublevel set). *Let h be Morse with distinct critical values on the compact connected surface Σ , let c be a critical value with unique critical point p , and let λ be its index. Then*

$$\#\text{comp}\{h \leq c\} - \#\text{comp}\{h < c\} = \begin{cases} +1, & \lambda = 0, \\ -1, & \lambda = 1 \text{ and } p \text{ sub-merging}, \\ 0, & \text{otherwise.} \end{cases}$$

Proof. Write $L := h^{-1}(c)$, so $\{h \leq c\} = \{h < c\} \sqcup L$. A component of $\{h \leq c\}$ is a union of components of $\{h < c\}$ together with part of L , so it suffices to determine which components of $\{h < c\}$ get identified and whether any part of L forms a component by itself.

Away from a smaller Morse coordinate ball around p , the compact regular level set is a compact one-manifold with boundary and therefore has finitely many components. Inside the ball it consists of the pieces listed in Lemma 14.28. Thus L is a finite graph whose only possible nonmanifold vertex is p . By Lemma 14.31, ϕ^- is constant on each edge in the regular part $L \setminus \{p\}$. Consequently the closure of an edge attaches along all of its regular points to one and the same component of $\{h < c\}$; distinct such components can be identified by adjoining L only at the vertex p . A closed regular-level component not meeting p likewise attaches to exactly one sublevel component and identifies nothing.

At p : if $\lambda = 0$ then by Lemma 14.28(i) a neighbourhood of p meets $\{h < c\}$ not at all and meets L only in $\{p\}$, so $\{p\}$ is an isolated point of $\{h \leq c\}$ and contributes one new component. If $\lambda = 2$ then by (ii) the punctured ball lies in $\{h < c\}$, so p is adjacent to exactly one component and nothing changes. If $\lambda = 1$ then by (iii) p is adjacent to exactly the two sectors S_{12}^- and S_{34}^- , so it identifies $A(S_{12}^-)$ with $A(S_{34}^-)$: the count drops by one precisely when these differ, i.e. precisely when p is sub-merging. No component of $\{h < c\}$ is created or destroyed otherwise. $\square$

Lemma 14.33 (Counting merging saddles). *Let h be Morse with distinct critical values on the compact connected surface Σ , with c_0 minima and c_2 maxima. Then the number of sub-merging saddles is exactly $c_0 - 1$, and the number of super-merging saddles is exactly $c_2 - 1$.*

Proof. List the critical values $c^{(1)} < \dots < c^{(k)}$ and pick regular values $a_0 < c^{(1)} < a_1 < c^{(2)} < \dots < c^{(k)} < a_k$. Put $N_j := \#\text{comp}\{h \leq a_j\}$. Then $N_0 = 0$ since $\{h \leq a_0\} = \emptyset$, and $N_k = 1$ since $\{h \leq a_k\} = \Sigma$ is connected.

Fix j and write $c := c^{(j)}$. Corollary 14.27(ii) applied on $[a_{j-1}, c]$ gives $\#\text{comp}\{h < c\} = N_{j-1}$, and Corollary 14.27(i) applied on $(c, a_j]$ gives $N_j = \#\text{comp}\{h \leq c\}$. So Lemma 14.32 computes $N_j - N_{j-1}$ as $+1$ at a minimum, -1 at a sub-merging saddle, and 0 otherwise.

Summing over j , $1 - 0 = c_0 - \#\{\text{sub-merging}\}$. Apply the same argument to $-h$: its minima are the maxima of h , its index-1 points are those of h because $\text{ind}_p(-h) = 2 - \text{ind}_p(h)$, and sub-merging for $-h$ is super-merging for h because $\{-h < -c\} = \{h > c\}$. $\square$

Lemma 14.34 (No saddle merges on both sides). *Let h be Morse with pairwise distinct critical values on the compact connected surface Σ . Then no saddle of h is both sub-merging and super-merging.*

Proof. Let p be a saddle, $c := h(p)$, $L := h^{-1}(c)$, and adopt the notation of Lemma 14.28(iii). The distinct-critical-value hypothesis makes p the only critical point at level c ; hence $L \setminus \{p\}$ is a smooth one-manifold without boundary, by the implicit function theorem (B6).

Let σ be the connected component of $L \setminus \{p\}$ containing $r_1 \setminus \{p\}$. It is not compact, since r_1 accumulates at $p \notin \sigma$, and it carries the nowhere-zero tangent field obtained by rotating $\nabla_\Sigma h$ by $\pi/2$ using the orientation of Σ . Hence Lemma 14.25 makes it diffeomorphic to $\mathbb{R}$ with exactly two ends. Both ends accumulate only at p : L is compact and $L \setminus \{p\}$ is a manifold, so any limit point of σ outside σ lies in L and is not a manifold point, hence equals p . Near p the set L is the union of the four rays (Lemma 14.28(iii)), and each ray accounts for exactly one end; the two ends of σ therefore lie along two *distinct* rays, one of which is r_1 . Write r_j for the other, $j \in \{2, 3, 4\}$.

By Lemma 14.31, ϕ^- and ϕ^+ are constant on the connected set σ . Along the r_1 end, the flanking sectors are S_{41}^+ and S_{12}^- , so $\phi^-(\sigma) = A(S_{12}^-)$ and $\phi^+(\sigma) = B(S_{41}^+)$. Reading the flanking sectors of r_j off Lemma 14.28(iii):

- $j = 2$: flanking sectors S_{12}^-, S_{23}^+ , so $B(S_{41}^+) = \phi^+(\sigma) = B(S_{23}^+)$ and p is not super-merging.
- $j = 4$: flanking sectors S_{34}^-, S_{41}^+ , so $A(S_{12}^-) = \phi^-(\sigma) = A(S_{34}^-)$ and p is not sub-merging.
- $j = 3$: flanking sectors S_{23}^+, S_{34}^- , so both equalities hold and p is neither.

In every case at least one of the two merging properties fails. $\square$

Formalisation note. The formal proof follows the integral curve of $n \times \nabla_\Sigma h$ through a point of r_1 directly: it is injective, stays in $L \setminus \{p\}$, and tends to p at both ends along two distinct rays, which gives the two-ends statement used above without invoking Lemma 14.25. That lemma is formalised independently, in its abstract form (second countability is not needed) and in the embedded form for level components.

Proposition 14.35 (The Morse count is at most two). *Let h be Morse on a compact connected surface Σ , with c_λ critical points of index λ . Then*

$$c_0 - c_1 + c_2 \leq 2. \quad (14.24)$$

Proof. By Lemma 14.22 we may assume the critical values are distinct, since the perturbation changes no c_λ . Then Lemma 14.33 produces $c_0 - 1$ sub-merging and $c_2 - 1$ super-merging saddles, and Lemma 14.34 makes these two sets disjoint, so $c_1 \geq (c_0 - 1) + (c_2 - 1)$. $\square$

Remark 14.36 (What is *not* claimed). Proposition 14.35 is an inequality, and that is all this development needs. No Euler characteristic is defined, no homology is computed, no cell structure is built, no handle is attached, and the classification of compact surfaces is not used. In particular the identity $c_0 - c_1 + c_2 = 2 - 2g$ is deliberately *not* proved, and neither is independence of h .

14.8 The total curvature bound

Lemma 14.37 (Regular fibres are finite). *Let $f : \Sigma \rightarrow \Sigma'$ be a C^1 map between compact surfaces, Σ without boundary, and let y be a regular value. Then $f^{-1}(y)$ is finite.*

Proof. The inverse function theorem (B6) makes $f^{-1}(y)$ discrete; it is closed in the compact Σ , hence finite. $\square$

Definition 14.38 (Local index sum). Let Σ, Σ' be smooth surfaces with Σ compact, without boundary and oriented, Σ' oriented, and let $f : \Sigma \rightarrow \Sigma'$ be C^1 . For a regular value $y \in \Sigma'$ put $\varepsilon_f(p) := \text{sign det } d_p f$ using the orientations, and

$$\iota_y(f) := \sum_{p \in f^{-1}(y)} \varepsilon_f(p) \in \mathbb{Z}, \quad (14.25)$$

a finite sum by Lemma 14.37. At a nonregular value define $\iota_y(f) := 0$. Thus $y \mapsto \iota_y(f)$ is an everywhere-defined integer-valued function.

Proof. The critical set is closed in the compact source, so its image is compact; hence the regular-value set is open. At a regular value the fibre is finite. Choose pairwise disjoint inverse-function neighbourhoods of its points; by compactness, after shrinking the common target neighbourhood, its full preimage lies in their union. The signs of the local Jacobians are constant there, so $\iota_y(f)$ is locally constant on the regular-value set. Its extension by zero over the closed critical-value set is therefore Borel. $\square$

Formalisation note 14.39 (This is not a degree). $\iota_y(f)$ is *not* asserted to be independent of y , and no such statement is proved or used. Proving independence would require a transversality statement for a path in Σ' against f , which Theorem 14.9 does *not* supply — it is about maps into $\mathbb{R}^3$ transverse to a surface in $\mathbb{R}^3$. The antipodal averaging in Theorem 14.43 is what makes independence unnecessary; see Formalisation note 14.44.

Lemma 14.40 (The Gauss map extends). *Let $\Sigma \subset \mathbb{R}^3$ be a compact smooth embedded closed surface with unit normal n . There is a Lipschitz $N : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ with $N|_{\Sigma} = n$ and $DN(p)|_{T_p \Sigma} = d_p n$ for every $p \in \Sigma$.*

Proof. Theorem 14.12 gives a tubular neighbourhood $\Theta(p, t) = p + tn(p)$ and hence a smooth nearest-point projection π onto Σ there. Put $N := \chi \cdot (n \circ \pi)$ with χ a smooth cutoff equal to one near Σ and supported in the tubular neighbourhood. Then $N|_{\Sigma} = n$, and since $\pi|_{\Sigma} = \text{id}$ and $d_p \pi|_{T_p \Sigma} = \text{id}$, the tangential differential is $d_p n$. $\square$

Theorem 14.41 (Total curvature as an integral of index sums). *Let $\Sigma \subset \mathbb{R}^3$ be a compact connected smooth embedded closed surface with the unit normal n of Proposition 14.13(ii). Then*

$$\int_{\Sigma} \kappa d\mathcal{H}^2 = \int_{\mathbb{S}^2} \iota_y(n) d\mathcal{H}^2(y), \quad (14.26)$$

both sides being finite.

Proof. By Lemma 6.14 and Lemma 14.40 the area formula (6.7) applies to n on the rectifiable set Σ , with $J_2(d_p n) = |\kappa(p)|$ by (14.10). Apply it twice, with the nonnegative Borel weights

$$q_{\pm} := \max\{\pm \text{sign } \kappa, 0\},$$

and subtract; both integrals are finite because κ is continuous on the compact Σ . Since $q_{\pm}J_2(dn) = \kappa^{\pm}$, the left side becomes $\int_{\Sigma} \kappa$, and the right side becomes

$$\int_{\mathbb{S}^2} \left(\sum_{p \in n^{-1}(y), \kappa(p) > 0} 1 - \sum_{p \in n^{-1}(y), \kappa(p) < 0} 1 \right) d\mathcal{H}^2(y).$$

By Corollary 11.2 almost every y is a regular value of n , and at a regular value every $p \in n^{-1}(y)$ has $\kappa(p) \neq 0$ with $\varepsilon_n(p) = \text{sign } \kappa(p)$ by (14.11); so for almost every y the bracket is $\iota_y(n)$. The right-hand integral is supported on $\mathbb{S}^2$, whose total $\mathcal{H}^2$ -measure is 4π by Lemma 6.15. At the exceptional critical values no pointwise identification with $\iota_y(n)$ is needed, because that target set is $\mathcal{H}^2$ -null. $\square$

Formalisation note. The identity is formalised for every compact smooth embedded surface and every unit normal field; connectedness and the particular normal of Proposition 14.13 are not needed.

Proposition 14.42 (Antipodal index sums add up to the Morse count). *Let $\Sigma \subset \mathbb{R}^3$ be a compact connected smooth embedded closed surface, let n be the unit normal supplied by Proposition 14.13, and let $v \in \mathbb{S}^2$ be such that both v and $-v$ are regular values of n — almost every v , by Lemma 14.21. Let c_{λ} be the number of critical points of h_v of index λ . Then*

$$\iota_v(n) + \iota_{-v}(n) = c_0 - c_1 + c_2. \quad (14.27)$$

Proof. By (14.11), $\varepsilon_n(p) = \text{sign } \kappa(p)$, and by Lemma 14.19 this equals $(-1)^{\text{ind}_p(h_v)}$ at every critical point of h_v . Hence

$$\iota_v(n) = \sum_{p \in n^{-1}(v)} (-1)^{\text{ind}_p(h_v)}, \quad \iota_{-v}(n) = \sum_{p \in n^{-1}(-v)} (-1)^{\text{ind}_p(h_v)}.$$

Adding these and using the disjoint splitting (14.13) of Lemma 14.20 gives (14.27). $\square$

Theorem 14.43 (Total curvature bound). *Let $\Sigma \subset \mathbb{R}^3$ be a compact connected smooth embedded closed surface. Then*

$$\int_{\Sigma} \kappa d\mathcal{H}^2 \leq 4\pi. \quad (14.28)$$

Proof. Orient Σ and fix n by Proposition 14.13(ii). The antipodal map is an isometry of $\mathbb{S}^2$, so it preserves $\mathcal{H}^2$ and

$$\int_{\mathbb{S}^2} \iota_y(n) d\mathcal{H}^2(y) = \frac{1}{2} \int_{\mathbb{S}^2} (\iota_y(n) + \iota_{-y}(n)) d\mathcal{H}^2(y).$$

For almost every y both y and $-y$ are regular values of n (Lemma 14.21), and for such y Proposition 14.42 and Proposition 14.35 applied to the Morse function h_y give

$$\iota_y(n) + \iota_{-y}(n) = c_0 - c_1 + c_2 \leq 2.$$

Hence, with (14.26) and $\mathcal{H}^2(\mathbb{S}^2) = 4\pi$,

$$\int_{\Sigma} \kappa d\mathcal{H}^2 = \int_{\mathbb{S}^2} \iota_y(n) d\mathcal{H}^2(y) \leq \frac{1}{2} \cdot 2 \cdot 4\pi = 4\pi.$$

$\square$

Formalisation note 14.44 (Why averaging replaces degree theory). The classical route asserts $\iota_y(n) = \deg(n)$ for every regular y and then bounds $\deg(n) \leq 1$. Establishing y -independence costs a transversality statement this document does not have (Formalisation note 14.39). It is also unnecessary: the quantity actually needed is the *average* of $\iota_y(n)$ over $\mathbb{S}^2$, and the pointwise bound $\iota_y(n) + \iota_{-y}(n) \leq 2$ from Proposition 14.42 bounds that average by one directly. This removes an entire section from the chapter and, with it, the only place where orientations of one-manifold cobordisms would have had to be tracked.

Formalisation note 14.45 (Why this route). An earlier design proved (14.28) from a finite smooth triangulation, a local Gauss–Bonnet formula on curved triangles, and $\chi = b_0 - b_1 + b_2 \leq 2$ via the cellular chain complex. That route required, in addition to everything above: abstract simplicial complexes with geometric realisation, a four-part simplicial gluing lemma with a quotient-complex argument, a moving-frame connection and structure equation, Stokes on a curved triangle with corners, the turning-number calculation (polar decomposition and angle lifting over the disk), and the $b_2 = 1$ dual-graph argument. The pinned mathlib supplies none of it: `AlgebraicTopology/Simplicial*` is category-theoretic simplicial sets and `Analysis/Convex/SimplicialComplex` is convex geometry, so all of it would have been built from zero.

What replaces it is §14.5 (pointwise linear algebra), §14.7 (component counting plus Lemma 14.34) and §14.8 (the area formula plus antipodal averaging). The exchange is not free: the sharp *identity* $\int \kappa = 2\pi\chi$ is lost, and only the inequality (14.28) survives (Remark 14.36). Since Lemma 31.17 is the sole consumer and needs only the inequality, that costs nothing here. It would matter if this development were reused as a general-purpose surface-theory library, which it is not intended to be.

Formalisation note 14.46 (The remaining cost centre). After the exchange described in Formalisation note 14.45, the dominant cost of this chapter is §14.1 together with §14.6: the global flow is needed only for Lemma 14.26 and Lemma 14.25, which feed only Corollary 14.27 and Lemma 14.34, which in turn feed §14.7. The shape-operator calculations and the index-integral formula can be proved independently of that flow construction. The final total-curvature bound uses Proposition 14.35, so the complete `Surface/TotalCurvature.lean` file depends transitively on `Flow/`. Two observations for anyone trying to reduce that dependency further.

- (a) Corollary 14.27 is used only for the *count* of connected components, never for the retraction itself. Its (ii) half already has a flow-free proof of surjectivity — a component V of $\{h < b\}$ has $\min_{\overline{V}} h$ attained at an interior point, hence at a critical point of value $< b$ — and only injectivity uses the flow. If injectivity can be obtained without it, §14.1 would be needed only for Lemma 14.25, whose sole use is to know that a connected one-manifold has two ends. I have verified neither step.
- (b) On a level set $\Sigma_t = \{u = t\}$ of the capacity potential the Gauss map is $-\nabla u/|\nabla u|$, which extends to the whole exterior region, and by (30.14) it has degree one on a large sphere. A degree comparison between Σ_t and that sphere would give $\deg(n) \leq 1$ directly, retiring §14.6 and §14.1. The obstruction is that the critical set of a harmonic function need not be discrete — precisely the difficulty Chapter 31 is designed to route around — so the comparison would have to be run with that set excised. Whether that can be done is open; do not plan on it.

§14.2, §14.3 and Theorem 14.14 cannot be retired under any of these variants: Lemma 29.3 and Lemma 30.9 use them independently of this chapter.

Part III

The liquid-drop functional

Chapter 15

The functional and its elementary properties

Lean module: NoCompromise.Energy.Coulomb, NoCompromise.Energy.CoulombDefs,
NoCompromise.Energy.Defs, NoCompromise.Energy.NullInvariance,
NoCompromise.Energy.Scaling

15.1 Definitions

Definition 15.1 (Coulomb potential and energy). For measurable $E \subset \mathbb{R}^3$ of finite measure put

$$v_E(x) := \int_E \frac{dy}{|x-y|}, \quad D(E) := \frac{1}{2} \iint_{E \times E} \frac{dx dy}{|x-y|} = \frac{1}{2} \int_E v_E(x) dx,$$

and for disjoint measurable E, F put $I(E, F) := \iint_{E \times F} \frac{dx dy}{|x-y|}$.

Definition 15.2 (Energy). $\mathcal{E}(E) := P(E) + D(E)$.

Definition 15.3 (Fixed-volume minimiser). A *fixed-volume minimiser at volume $V > 0$* is a measurable $\Omega \subset \mathbb{R}^3$ with $|\Omega| = V$, $P(\Omega) < \infty$, and $\mathcal{E}(\Omega) \leq \mathcal{E}(F)$ for every measurable F with $|F| = V$.

Lemma 15.4 (Null-set invariance). *If $|E \triangle F| = 0$ then $|E| = |F|$, $P(E) = P(F)$, $D(E) = D(F)$ and $\mathcal{E}(E) = \mathcal{E}(F)$.*

Proof. Lemma 4.6(i) for the perimeter; for D , the double integral only sees $\mathbf{1}_E$ as an L^1 class. $\square$

15.2 Quantitative Coulomb estimates

Notation 15.5 (The Coulomb constant). $C_0 := 2\pi + 1$.

Lemma 15.6 (Uniform potential bound). *Let $|E| = M < \infty$. Then for every $x \in \mathbb{R}^3$ and $r > 0$,*

$$v_E(x) \leq \int_{B_r(x)} \frac{dy}{|x-y|} + \frac{M}{r} = 2\pi r^2 + \frac{M}{r}, \quad (15.1)$$

and consequently

$$\|v_E\|_{L^\infty} \leq C_0 M^{2/3}, \quad D(E) \leq \frac{1}{2} C_0 M^{5/3}. \quad (15.2)$$

In particular $D(E) < \infty$ for every set of finite measure.

Proof. (15.1) splits the integral at radius r and uses $|x - y|^{-1} \leq r^{-1}$ outside $B_r(x)$; the interior integral is exactly $2\pi r^2$. If $M > 0$ choose $r = M^{1/3}$; if $M = 0$ both assertions are trivial. No rearrangement inequality is needed. $\square$

Lemma 15.7 (Bilinear identity). *For measurable E, F of finite measure,*

$$D(E) - D(F) = \frac{1}{2} \int_{\mathbb{R}^3} (\mathbf{1}_E - \mathbf{1}_F)(v_E + v_F) dx. \quad (15.3)$$

Proof. Expand both sides for the truncated kernels $\min\{|x - y|^{-1}, k\}$, where every integral is finite and the algebraic identity is a direct computation; then let $k \rightarrow \infty$. Lemma 15.6 makes $(\mathbf{1}_E - \mathbf{1}_F)(v_E + v_F)$ absolutely integrable, so positive and negative parts may be passed separately by monotone convergence. $\square$

Lemma 15.8 (L^1 -Lipschitz estimate). *If $|E|, |F| \leq M$ then*

$$|D(E) - D(F)| \leq C_0 M^{2/3} |E \Delta F|. \quad (15.4)$$

In particular D is continuous under global L^1 convergence with uniformly bounded volumes.

Proof. Bound the integrand of (15.3) using (15.2); the factor $\mathbf{1}_E - \mathbf{1}_F$ is supported on $E \Delta F$ and bounded by 1. $\square$

Formalisation note 15.9 (Where (15.4) is used). Three times, and it is essential in all three: Theorem 19.5 (Coulomb splitting), Lemma 20.2 (volume penalisation), and Lemma 20.4 (perimeter quasiminimality).

Lemma 15.10 (Disjoint-union formula). *For disjoint measurable E, F of finite measure,*

$$D(E \cup F) = D(E) + D(F) + I(E, F), \quad (15.5)$$

and $I(E, F) > 0$ whenever both sets have positive measure.

Proof. Expand the double integral over $(E \cup F) \times (E \cup F)$; strict positivity holds because the kernel is strictly positive off the diagonal, and the diagonal has product measure zero. $\square$

Lemma 15.11 (Cross-interaction decays under translation). *Let E, F be bounded, contained in B_R , and let $e \in \mathbb{S}^2$. Then for $t > 2R$,*

$$I(E, F + te) \leq \frac{|E| |F|}{t - 2R}, \quad (15.6)$$

so $I(E, F + te) \rightarrow 0$ as $t \rightarrow \infty$.

Proof. For $x \in E$, $y \in F + te$ one has $|x - y| \geq t - 2R$. $\square$

15.3 Scaling

Lemma 15.12 (Translation and dilation). *For $a \in \mathbb{R}^3$, $r > 0$ and measurable E of finite measure,*

$$\begin{aligned} |a + E| &= |E|, & P(a + E) &= P(E), & D(a + E) &= D(E), \\ |rE| &= r^3 |E|, & P(rE) &= r^2 P(E), & D(rE) &= r^5 D(E). \end{aligned} \quad (15.7)$$

Proof. The perimeter statements are Lemma 4.6(ii),(iii). The Coulomb scaling follows from the change of variables $x = rx'$, $y = ry'$, which produces r^6 from the measures and r^{-1} from the kernel. $\square$

Chapter 16

Balls, the threshold, and exact algebra

Lean module: NoCompromise.Ball.Defs, NoCompromise.Ball.Perimeter,
NoCompromise.Ball.Potential, NoCompromise.Threshold.Algebra,
NoCompromise.Threshold.Comparison, NoCompromise.Threshold.Defs,
NoCompromise.Threshold.Ledger

16.1 The ball

Notation 16.1 (Balls by volume). B_V denotes any ball of volume $V > 0$; its radius is $R = (3V/4\pi)^{1/3}$ and $P_B = P_B(V) := P(B_V)$.

Lemma 16.2 (Volume and perimeter of a ball).

$$P_B = 4\pi R^2 = (36\pi)^{1/3} V^{2/3}, \quad RP_B = 3V. \quad (16.1)$$

Proof. $P(B_R) = \mathcal{H}^2(\partial B_R)$ by Corollary 8.14, and $\mathcal{H}^2(\partial B_R) = 4\pi R^2$ by Lemma 6.15; substitute $V = \frac{4}{3}\pi R^3$. $\square$

Lemma 16.3 (Potential and energy of a ball). For $|x| = r \leq R$,

$$v_{B_R}(x) = 2\pi \left(R^2 - \frac{r^2}{3} \right), \quad (16.2)$$

and consequently

$$D(B_R) = \frac{16\pi^2}{15} R^5 = \frac{V}{5} P_B. \quad (16.3)$$

Proof. The potential is radial and, by Proposition 10.36, satisfies $v'' + \frac{2}{r}v' = -4\pi$ on $(0, R)$. Regularity at the origin gives $v(r) = C - \frac{2\pi}{3}r^2$. Outside the ball the radial harmonic equation and decay give $v(r) = V/r$. The potential is continuous: split its defining integral into a small ball about the moving point, whose contribution is uniformly $O(\rho^2)$, and its complement, where the kernel is uniformly continuous. Matching at $r = R$ therefore yields $C = 2\pi R^2$, which is (16.2). Integrating,

$$D(B_R) = \frac{1}{2} \cdot 4\pi \int_0^R 2\pi \left(R^2 - \frac{r^2}{3} \right) r^2 dr = \frac{16\pi^2}{15} R^5,$$

and the identity with $VP_B/5$ follows from $V = 4\pi R^3/3$. $\square$

Corollary 16.4 (Ball energy).

$$\mathcal{E}(B_V) = \frac{V+5}{5}P_B(V) = (36\pi)^{1/3} \frac{V+5}{5}V^{2/3}. \quad (16.4)$$

Proof. By (16.3), $\mathcal{E}(B_V) = P_B(V) + D(B_V) = (1+V/5)P_B(V)$. Substitute $P_B(V) = (36\pi)^{1/3}V^{2/3}$ from (16.1). $\square$

16.2 The cube-root parameter and the threshold

Definition 16.5 (q and V_*).

$$q := 2^{1/3}, \quad V_* := 5 \frac{2-q^2}{q^2-1}.$$

Lemma 16.6 (Convenient form of V_*). $q > 1$, $q^3 = 2$, $2^{2/3} = q^2$, and

$$V_* = \frac{5q^2}{q+1}. \quad (16.5)$$

Proof. $2 - q^2 = q^3 - q^2 = q^2(q-1)$ and $q^2 - 1 = (q-1)(q+1)$; cancel $q-1 > 0$. $\square$

Lemma 16.7 (The key threshold identity).

$$q^{-2}(V_* + 10) = V_* + 5. \quad (16.6)$$

Proof. Using (16.5), $q^{-2}(V_* + 10) - (V_* + 5) = -5(q^3 - 2)/q^2 = 0$. $\square$

Lemma 16.8 (Numerical bounds on V_*). $V_* > 0$ and

$$\frac{5}{2} < V_* < 4 < 6. \quad (16.7)$$

Proof. For $V_* > 5/2$: by (16.5) this is $2q^2 > q+1$, i.e. $2q^2 - q - 1 = (2q+1)(q-1) > 0$. For $V_* < 4$: first $q < 4/3$, since $q^3 = 2 < 64/27$. The polynomial $5x^2 - 4x - 4$ is increasing for $x \geq 1$ and $5(4/3)^2 - 4(4/3) - 4 = -4/9 < 0$, so $5q^2 < 4(q+1)$, which by (16.5) is $V_* < 4$. $\square$

Proposition 16.9 (One ball versus two equal balls). For $V > 0$, with $L := q^{-2}(V+10)$,

$$2\mathcal{E}(B_{V/2}) = \left(q + \frac{V}{5q^2}\right)P_B(V) = \frac{L}{5}P_B(V), \quad (16.8)$$

and $\mathcal{E}(B_V) = 2\mathcal{E}(B_{V/2})$ holds if and only if $V = V_*$. Moreover $\mathcal{E}(B_V) < 2\mathcal{E}(B_{V/2})$ for $V < V_*$ and $>$ for $V > V_*$.

Proof. A ball of volume $V/2$ has radius R/q , so $2P(B_{V/2}) = 2q^{-2}P_B = qP_B$ (using $2/q^2 = q$), and by (16.3), $2D(B_{V/2}) = 2 \cdot \frac{V/2}{5} \cdot q^{-2}P_B = \frac{V}{5q^2}P_B$. This is (16.8). By (16.4) the equation $\mathcal{E}(B_V) = 2\mathcal{E}(B_{V/2})$ is $V+5 = q^{-2}(V+10)$, an affine equation in V with coefficient $1 - q^{-2} > 0$ on the left, hence with the unique solution $V = V_*$ by (16.6); the sign statements follow from the same monotonicity. $\square$

16.3 The algebraic ledger

The following exact identities are used at the sharp points of the argument. Each is a polynomial identity in q (with $q^3 = 2$) or in a real variable, and each should be a standalone lemma discharged by an ordered-field normalisation tactic after adjoining the positive real cube root q .

Lemma 16.10 (Splitting-function derivative numerator). *For $p > 1$,*

$$\frac{d}{dp} \left[\frac{3(p+2)}{(p-1)(p^3+3p^2+6p+5)} \right] = -\frac{9(p^2+p+1)(p^2+3p+1)}{(p-1)^2(p^3+3p^2+6p+5)^2} < 0. \quad (16.9)$$

Proof. Differentiate by the quotient rule, put the two terms over the positive denominator $(p-1)^2(p^3+3p^2+6p+5)^2$, and expand the numerator. It factors as $-9(p^2+p+1)(p^2+3p+1)$. Every factor other than the leading minus sign is positive for $p > 1$. $\square$

Lemma 16.11 (One-ball contradiction factor). *For $\delta \geq 0$ and $V \in \mathbb{R}$,*

$$V+2 - (V+5-3(1+\delta)^2)(1+\delta) = \delta(4-V+9\delta+3\delta^2). \quad (16.10)$$

Proof. Expand the left side and collect powers of δ : $4\delta - V\delta + 9\delta^2 + 3\delta^3$, which is the displayed right side. $\square$

Lemma 16.12 (The $V = 6$ discriminant). $(V+2)^2 - 16(V-2) = (V-6)^2$.

Proof. Both sides expand to $V^2 - 12V + 36$. $\square$

Lemma 16.13 (Cubic majorant). *For $L > 0$, $x \geq 0$ and $t := 3x/\sqrt{L}$,*

$$\frac{2L^{3/2}}{9} - (Lx - 3x^3) = \frac{L^{3/2}}{9}(t-1)^2(t+2) \geq 0, \quad (16.11)$$

so $Lx - 3x^3 \leq \frac{2}{9}L^{3/2}$.

Proof. Since $L > 0$, substitution of $x = t\sqrt{L}/3$ gives

$$\frac{2L^{3/2}}{9} - (Lx - 3x^3) = \frac{L^{3/2}}{9}(t^3 - 3t + 2) = \frac{L^{3/2}}{9}(t-1)^2(t+2).$$

Here $t \geq 0$, so the last expression is nonnegative. $\square$

Lemma 16.14 (Binding-ratio factorisation). *For $x > 0$ and $a := (5/2)^{1/3}$ (so $a^3 = 5/2$),*

$$\frac{1}{x} + \frac{x^2}{5} - \frac{3}{2a} = \frac{(x-a)^2(x+2a)}{2a^3x} \geq 0, \quad (16.12)$$

with equality exactly at $x = a$.

Proof. The denominator $2a^3x$ is positive. Multiplication by it reduces the identity to

$$2a^3 + \frac{2a^3x^3}{5} - 3a^2x = (x-a)^2(x+2a).$$

Using $2a^3 = 5$, both sides expand to $x^3 - 3a^2x + 2a^3$. The factored right side is nonnegative, and it vanishes exactly when $x = a$. $\square$

Lemma 16.15 (Two-ball smallness at $V \leq 8$). *If $0 < V \leq 8$ and $L = q^{-2}(V + 10)$ then $L/5 \leq 18/(5q^2) < q^4$.*

Proof. The first inequality is $V \leq 8$. For the second, multiply by $5q^2$: the claim is $18 < 5q^6 = 5 \cdot 4 = 20$. $\square$

Lemma 16.16 (Monotonicity of h). *For $h(V) := (V + 10)^3/(V - 2)$ on $[6, 8]$,*

$$h'(V) = \frac{2(V + 10)^2(V - 8)}{(V - 2)^2} \leq 0, \quad h(V) \leq h(6) = 1024 < 1296. \quad (16.13)$$

Proof. The quotient rule gives

$$h'(V) = \frac{3(V + 10)^2(V - 2) - (V + 10)^3}{(V - 2)^2} = \frac{2(V + 10)^2(V - 8)}{(V - 2)^2}.$$

This is nonpositive on $[6, 8]$, so $h(V) \leq h(6) = 16^3/4 = 1024 < 1296$. $\square$

Chapter 17

The value function

Lean module: NoCompromise.Value.BoundedApprox, NoCompromise.Value.Continuity,
NoCompromise.Value.Defs

17.1 Bounded approximation at fixed volume

Lemma 17.1 (Bounded approximation). *Let E have finite positive volume and finite perimeter. There are bounded finite-perimeter sets E_n with*

$$|E_n| = |E|, \quad P(E_n) \rightarrow P(E), \quad D(E_n) \rightarrow D(E).$$

Proof. Steps 1–2: truncate. Lemma 5.13 supplies good radii $R_n \uparrow \infty$ with $|E \cap B_{R_n}| \rightarrow |E|$ and $P(E \cap B_{R_n}) \rightarrow P(E)$.

Step 3: Coulomb converges. Monotone convergence applied to the positive kernel, or (15.4), gives $D(E \cap B_{R_n}) \rightarrow D(E)$.

Step 4: correct the volume. Put $b_n := (|E|/|E \cap B_{R_n}|)^{1/3} \rightarrow 1$ and $E_n := b_n(E \cap B_{R_n})$. Then $|E_n| = |E|$ and, by (15.7), $P(E_n) = b_n^2 P(E \cap B_{R_n}) \rightarrow P(E)$ and $D(E_n) = b_n^5 D(E \cap B_{R_n}) \rightarrow D(E)$. $\square$

17.2 The value function and its properties

Definition 17.2 (Value function). For $V > 0$ put $m(V) := \inf\{\mathcal{E}(E) : |E| = V\}$.

Lemma 17.3 (m is finite and positive). $0 < m(V) \leq \mathcal{E}(B_V) < \infty$ for every $V > 0$.

Proof. The ball gives the upper bound (Corollary 16.4); the isoperimetric inequality (13.3) gives $m(V) \geq c_I V^{2/3} > 0$. $\square$

Lemma 17.4 (Subadditivity). For $V_1, V_2 > 0$, $m(V_1 + V_2) \leq m(V_1) + m(V_2)$.

Proof. Given $\varepsilon > 0$, use Lemma 17.1 to choose *bounded* exact-volume competitors E_i with $\mathcal{E}(E_i) \leq m(V_i) + \varepsilon$. Translate E_2 by te for a unit vector e . For large t the two sets are disjoint at positive distance, so Lemma 4.8 gives $P(E_1 \cup (E_2 + te)) = P(E_1) + P(E_2)$, while (15.5) and (15.6) give $0 \leq I(E_1, E_2 + te) \rightarrow 0$. Hence $m(V_1 + V_2) \leq m(V_1) + m(V_2) + 2\varepsilon$; let $\varepsilon \downarrow 0$. $\square$

Formalisation note 17.5 (Off the critical path). Lemma 17.4 is *not used* anywhere below: continuity of m comes from concavity (Lemma 17.6), and existence comes from strict binding (Proposition 18.9). It is recorded because it is the standard companion to Lemma 17.6, and because its proof is the template for the disjoint-translation argument reused in Lemma 36.5 and Proposition 28.13.

Consequently Lemma 17.1, whose only consumer is Lemma 17.4, is off the critical path as well. This is *not* true of the truncation step inside it: that step is needed by Theorem 12.1 and has been factored out as Lemma 5.13, which *is* on the critical path. What Lemma 17.1 adds beyond Lemma 5.13 — exact volume correction and convergence of the Coulomb term — is what nothing needs. See Formalisation note 40.2.

Lemma 17.6 (Normalised concavity). *The function*

$$V \mapsto V^{-2/3}m(V) = \inf_{|U|=1} (P(U) + V D(U)) \quad (17.1)$$

is concave and finite on $(0, \infty)$.

Proof. Rescale a set of volume V to volume one using (15.7): this gives the displayed identity. The right side is an infimum of affine functions of V , hence concave; it is finite by Lemma 17.3. $\square$

Proposition 17.7 (Continuity of m). *$m : (0, \infty) \rightarrow (0, \infty)$ is continuous.*

Proof. A concave function finite on an open interval is continuous there; multiply (17.1) by $V^{2/3}$. $\square$

Formalisation note 17.8. Proposition 17.7 is what lets Chapter 35 obtain existence at $V = V_*$ without formalising the separate theorem that the set of admissible masses is closed. It is also used in Proposition 19.10.

Chapter 18

The splitting inequality and strict binding

Lean module: NoCompromise.Binding.Splitting, NoCompromise.Binding.Strict

18.1 A scaling lower bound

Lemma 18.1 (Scaling lower bound). *For $V > 0$ and $s \in (0, 1)$,*

$$m(sV) \geq s^{5/3}m(V) + (1-s)s^{2/3}P_B(V). \quad (18.1)$$

Proof. Let $|E| = sV$. Then $|s^{-1/3}E| = V$, so by (15.7)

$$m(V) \leq s^{-2/3}P(E) + s^{-5/3}D(E) = s^{-5/3}\mathcal{E}(E) - (s^{-5/3} - s^{-2/3})P(E).$$

Multiplying by $s^{5/3}$ gives $\mathcal{E}(E) \geq s^{5/3}m(V) + (1-s)P(E)$, and (13.3) gives $P(E) \geq s^{2/3}P_B(V)$. Take the infimum over E . $\square$

Definition 18.2 (The splitting functions). For $s \in (0, 1)$ put

$$N(s) := s^{2/3} + (1-s)^{2/3} - 1, \quad D_0(s) := 1 - s^{5/3} - (1-s)^{5/3}, \quad f(s) := \frac{N(s)}{D_0(s)}.$$

Lemma 18.3 (Positivity). $N(s) > 0$ and $D_0(s) > 0$ for every $s \in (0, 1)$.

Proof. $N > 0$ is Lemma 13.5 with $a = s$, $b = 1-s$. $D_0 > 0$ because $s^{5/3} + (1-s)^{5/3} < s + (1-s) = 1$, using $t^{5/3} < t$ for $t \in (0, 1)$. $\square$

Lemma 18.4 (Two-piece scaling bound). *For $V > 0$ and $s \in (0, 1)$,*

$$m(sV) + m((1-s)V) - m(V) \geq P_B(V) D_0(s) \left(f(s) - \frac{V}{5} \right). \quad (18.2)$$

Proof. Apply (18.1) to s and to $1-s$ and add:

$$\begin{aligned} m(sV) + m((1-s)V) - m(V) &\geq (s^{5/3} + (1-s)^{5/3} - 1)m(V) \\ &\quad + ((1-s)s^{2/3} + s(1-s)^{2/3})P_B(V). \end{aligned} \quad (18.3)$$

The coefficient of $m(V)$ is $-D_0(s) < 0$, so substituting the upper bound $m(V) \leq \mathcal{E}(B_V) = (1 + V/5)P_B(V)$ from Corollary 16.4 *decreases* the right side. Doing so and using the identity

$$(1-s)s^{2/3} + s(1-s)^{2/3} - D_0(s) = N(s)$$

— which is the algebraic rearrangement $(1-s)s^{2/3} + s(1-s)^{2/3} = s^{2/3} + (1-s)^{2/3} - s^{5/3} - (1-s)^{5/3}$ — gives

$$m(sV) + m((1-s)V) - m(V) \geq P_B(V) \left[N(s) - \frac{V}{5} D_0(s) \right],$$

which is (18.2). $\square$

18.2 The one-variable splitting inequality

Lemma 18.5 (Change of variable). *For $s \in (0, 1)$ put $x := s^{1/3}$, $y := (1-s)^{1/3}$, $p := x + y$, $r := xy$. Then $x, y > 0$, $x^3 + y^3 = 1$,*

$$1 < p \leq q^2, \quad (18.4)$$

with $p = q^2$ exactly when $s = 1/2$, and

$$r = \frac{p^3 - 1}{3p}, \quad (18.5)$$

$$N(s) = p^2 - 2r - 1 = \frac{(p-1)^2(p+2)}{3p}, \quad (18.6)$$

$$D_0(s) = 1 - p^2 + 2r + r^2p = \frac{(p-1)^3(p^3 + 3p^2 + 6p + 5)}{9p}. \quad (18.7)$$

Proof. $p > 1$ since $x^3 + y^3 = 1$ and $x, y > 0$ force $x + y > \max(x, y) \geq$ the cube root of $\max(x^3, y^3)$, and more simply $p^3 = 1 + 3rp > 1$. For $p \leq q^2$: expanding,

$$4(x^3 + y^3) - (x + y)^3 = 3(x + y)(x - y)^2 \geq 0,$$

so $p^3 \leq 4 = q^6$, with equality exactly when $x = y$, i.e. $s = 1/2$. Equation (18.5) is $p^3 = x^3 + y^3 + 3xy(x+y) = 1 + 3rp$. Then $N(s) = x^2 + y^2 - 1 = p^2 - 2r - 1$, and substituting (18.5) and simplifying gives the closed form in (18.6). Similarly $x^5 + y^5 = (x^2 + y^2)(x^3 + y^3) - x^2y^2(x+y) = p^2 - 2r - r^2p$, so $D_0(s) = 1 - p^2 + 2r + r^2p$, and substituting (18.5) gives (18.7). $\square$

Lemma 18.6 (f as a function of p). *With p as in Lemma 18.5,*

$$f(s) = F(p) := \frac{3(p+2)}{(p-1)(p^3 + 3p^2 + 6p + 5)}. \quad (18.8)$$

Proof. Divide (18.6) by (18.7). $\square$

Theorem 18.7 (Splitting inequality). *For every $s \in (0, 1)$,*

$$f(s) \geq f(1/2) = \frac{q^2}{q+1} = \frac{V_*}{5}. \quad (18.9)$$

Proof. By Lemma 16.10, $F' < 0$ on $(1, \infty)$, and by (18.4), $p \leq q^2$, so $F(p) \geq F(q^2)$. The endpoint $p = q^2$ corresponds to $s = 1/2$, and

$$f(1/2) = \frac{2 \cdot 2^{-2/3} - 1}{1 - 2 \cdot 2^{-5/3}} = \frac{q - 1}{1 - q^{-2}} = \frac{q^2}{q + 1},$$

which equals $V_*/5$ by (16.5). □

Formalisation note 18.8 (Why this proof). This algebraic proof is specialised to the exponents $2/3$ and $5/3$ and is preferable for formalisation to the entropy-interpolation proof in the literature, because (18.6)–(18.8) and (16.9) are pure polynomial algebra.

18.3 Strict binding

Proposition 18.9 (Strict binding below the threshold). *If $0 < V < V_*$ then for every $s \in (0, 1)$,*

$$m(V) < m(sV) + m((1 - s)V). \tag{18.10}$$

Quantitatively, $m(sV) + m((1 - s)V) - m(V) \geq P_B(V)D_0(s)\frac{V_* - V}{5} > 0$.

Proof. Combine (18.2) with (18.9):

$$m(sV) + m((1 - s)V) - m(V) \geq P_B(V)D_0(s)\left(\frac{V_*}{5} - \frac{V}{5}\right) > 0,$$

using $D_0(s) > 0$ (Lemma 18.3) and $P_B(V) > 0$. □

Chapter 19

Concentration compactness and existence

Lean module: NoCompromise.Compactness.Decomposition,
NoCompromise.Compactness.Nonvanishing, NoCompromise.Existence.Subcritical

The package of this chapter is used twice: for fixed-volume existence (Proposition 19.10) and for the binding-energy corollary (Lemma 38.2).

19.1 Non-vanishing and compactness

Lemma 19.1 (Cube non-vanishing). *Let E_n satisfy $0 < c \leq |E_n| \leq C$ and $P(E_n) \leq C$. Choose a translation of the unit-cube grid so that $|D\mathbf{1}_{E_n}|$ assigns zero mass to every grid face for all n ; a countable intersection of full-measure choices suffices. Let Q_k be the open cubes and $m_{n,k} := |E_n \cap Q_k|$. Then there is $c_0 = c_0(c, C) > 0$ with*

$$\sup_k m_{n,k} \geq c_0 \quad \text{for every } n. \quad (19.1)$$

Proof. If some $m_{n,k} \geq 1/2$ then (19.1) holds with $c_0 \leq 1/2$. Otherwise $m_{n,k} \leq |Q_k \setminus E_n|$, so (7.6) gives $m_{n,k}^{2/3} \leq C_{\text{cube}} P(E_n; Q_k)$, whence

$$\begin{aligned} |E_n| &= \sum_k m_{n,k} = \sum_k m_{n,k}^{1/3} m_{n,k}^{2/3} \\ &\leq \left(\sup_k m_{n,k} \right)^{1/3} C_{\text{cube}} \sum_k P(E_n; Q_k) \leq C_{\text{cube}} C \left(\sup_k m_{n,k} \right)^{1/3}. \end{aligned}$$

The lower volume bound $|E_n| \geq c$ gives $\sup_k m_{n,k} \geq (c/(C_{\text{cube}}C))^3$. □

Formalisation note 19.2. The proof chooses a cube with mass at least half the supremum; *no maximising cube is required*, which matters because the supremum need not be attained.

Lemma 19.3 (Coulomb lower semicontinuity). *If $\mathbf{1}_{E_n} \rightarrow \mathbf{1}_E$ in $L^1_{\text{loc}}(\mathbb{R}^3)$ then $D(E) \leq \liminf_n D(E_n)$.*

Proof. If $L := \liminf_n D(E_n) = +\infty$ there is nothing to prove. Otherwise choose a subsequence n_k with $D(E_{n_k}) \rightarrow L$; local L^1 convergence admits a further subsequence whose characteristic

functions converge a.e. on $\mathbb{R}^3$. Apply Fatou's lemma on $\mathbb{R}^3 \times \mathbb{R}^3$ to the nonnegative integrand $\mathbf{1}_{E_{n_k}}(x)\mathbf{1}_{E_{n_k}}(y)/|x-y|$. The diagonal has product measure zero, so no value of the kernel there is needed, and the result is $D(E) \leq L$. $\square$

Theorem 19.4 (Compactness). *Let $E_n \subset \mathbb{R}^3$ satisfy*

$$0 < c \leq |E_n| \leq C, \quad P(E_n) \leq C. \quad (19.2)$$

Then, after translations and passage to a subsequence, there is a finite-perimeter set E with $0 < |E| \leq C$ such that $\mathbf{1}_{E_n} \rightarrow \mathbf{1}_E$ in $L^1_{\text{loc}}(\mathbb{R}^3)$, and

$$P(E) \leq \liminf_n P(E_n), \quad D(E) \leq \liminf_n D(E_n). \quad (19.3)$$

Proof. Apply Lemma 19.1 and choose for each n a cube with mass at least $\frac{1}{2} \sup_k m_{n,k} \geq c_0/2$; translate it to a fixed cube at the origin. Theorem 4.11 on each ball, followed by a diagonal subsequence, gives L^1_{loc} convergence to a set E of locally finite perimeter. Non-vanishing gives $|E \cap B_R| \geq c_0/2$ for one fixed R , so $|E| > 0$; exhaustion by balls and Fatou give $|E| \leq \liminf_n |E_n| \leq C$. Perimeter lower semicontinuity is Lemma 4.7; Coulomb lower semicontinuity is Lemma 19.3. Finally $P(E) \leq C < \infty$. $\square$

19.2 The one-profile decomposition

Theorem 19.5 (Decomposition). *In the situation of Theorem 19.4 there are radii $R_n \rightarrow \infty$, each a good radius for E_n at the origin, such that with*

$$F_n := E_n \cap B_{R_n}, \quad G_n := E_n \setminus B_{R_n},$$

one has

$$\mathbf{1}_{F_n} \rightarrow \mathbf{1}_E \text{ in } L^1(\mathbb{R}^3), \quad \mathbf{1}_{G_n} \rightarrow 0 \text{ in } L^1_{\text{loc}}(\mathbb{R}^3), \quad (19.4)$$

$$|F_n| \rightarrow |E|, \quad |G_n| = |E_n| - |F_n|, \quad (19.5)$$

$$P(E_n) \geq P(E) + P(G_n) + o(1), \quad (19.6)$$

$$D(E_n) = D(E) + D(G_n) + o(1). \quad (19.7)$$

Proof. Lemma 19.6 constructs the radii and gives (19.4)–(19.5); Lemma 19.7 gives (19.6); Lemma 19.8 gives (19.7). $\square$

Lemma 19.6 (Quantified construction of the radii). *In the situation of Theorem 19.4 there are strictly increasing integers N_j with $N_1 = 1$, integers $k_j \uparrow \infty$, and radii R_n such that: $R_n \in (k_j, k_{j+1})$ for $N_j \leq n < N_{j+1}$; each R_n is a good radius for E_n at 0;*

$$a_n := \mathcal{H}^2(E_n^{(1)} \cap \partial B_{R_n}) \leq 2^{1-j} \rightarrow 0; \quad (19.8)$$

and $|F_n \triangle E| \leq 2^{1-j} \rightarrow 0$, so $\mathbf{1}_{F_n} \rightarrow \mathbf{1}_E$ in $L^1(\mathbb{R}^3)$ and $|F_n| \rightarrow |E|$.

Proof. Choose $k_j \uparrow \infty$ with $|E \setminus B_{k_j}| \leq 2^{-j}$. By local convergence choose strictly increasing N_j so that for $n \geq N_j$, $|(E_n \triangle E) \cap B_{k_{j+1}}| \leq 2^{-j}$; after discarding finitely many terms, take $N_1 = 1$. For $N_j \leq n < N_{j+1}$, Corollary 5.6 gives

$$\begin{aligned} \int_{k_j}^{k_{j+1}} \mathcal{H}^2(E_n^{(1)} \cap \partial B_r) dr &= |E_n \cap (B_{k_{j+1}} \setminus B_{k_j})| \\ &\leq |E \setminus B_{k_j}| + |(E_n \triangle E) \cap B_{k_{j+1}}| \leq 2^{1-j}. \end{aligned}$$

By Lemma 5.10, choose $R_n \in (k_j, k_j + 1)$ among the good radii with $a_n \leq 2^{1-j}$. Then $R_n \rightarrow \infty$, $a_n \rightarrow 0$, and

$$|F_n \triangle E| \leq |(E_n \triangle E) \cap B_{k_j+1}| + |E \setminus B_{k_j}| \leq 2^{1-j}.$$

Since $G_n \cap B_R = \emptyset$ for every fixed R and all large n , the remainders converge locally to $\emptyset$. $\square$

Lemma 19.7 (Perimeter splitting). *With the notation of Lemma 19.6,*

$$P(F_n) + P(G_n) = P(E_n) + 2a_n, \quad (19.9)$$

and consequently

$$P(E_n) \geq P(E) + P(G_n) + o(1).$$

Proof. Adding (5.10) and (5.11) at the good radius R_n gives (19.9), since $P(E_n; B_{R_n}) + P(E_n; \mathbb{R}^3 \setminus \overline{B_{R_n}}) = P(E_n)$ by $|D\mathbf{1}_{E_n}|(\partial B_{R_n}) = 0$. Global L^1 convergence $F_n \rightarrow E$ and Lemma 4.7 give $P(F_n) \geq P(E) + o(1)$, whence $P(E_n) = P(F_n) + P(G_n) - 2a_n \geq P(E) + P(G_n) + o(1)$. $\square$

Lemma 19.8 (Coulomb splitting). *With the notation of Lemma 19.6, $I(F_n, G_n) \rightarrow 0$ and $D(F_n) \rightarrow D(E)$, hence*

$$D(E_n) = D(E) + D(G_n) + o(1).$$

Proof. $D(E_n) = D(F_n) + D(G_n) + I(F_n, G_n)$ by (15.5). Global L^1 convergence and (15.4) give $D(F_n) \rightarrow D(E)$ and

$$|I(F_n, G_n) - I(E, G_n)| \leq C_0 C^{2/3} |F_n \triangle E| \rightarrow 0.$$

For fixed M and large n , $G_n \subset \mathbb{R}^3 \setminus B_{R_n}$, so

$$I(E \cap B_M, G_n) \leq \frac{|E \cap B_M| |G_n|}{R_n - M} \rightarrow 0,$$

while for the tail (15.2) gives

$$I(E \setminus B_M, G_n) \leq \|v_{G_n}\|_\infty |E \setminus B_M| \leq C_0 C^{2/3} |E \setminus B_M|.$$

Let $n \rightarrow \infty$ and then $M \rightarrow \infty$. $\square$

Remark 19.9 (One profile only). Theorem 19.5 is a *one-profile* decomposition. No assertion is made that G_n is itself tight; only its volume and a lower bound for its energy are used downstream. In particular no iterated concentration-compactness induction is needed.

19.3 Existence below the threshold

Proposition 19.10 (Existence for $V < V_*$). *For every $0 < V < V_*$ the infimum $m(V)$ is attained.*

Proof. Let E_n be a minimising sequence with $|E_n| = V$. Since $P(E_n) \leq \mathcal{E}(E_n)$, the perimeters are bounded, so (19.2) holds. Apply Theorems 19.4 and 19.5, obtaining a nonzero local limit E of volume $V_0 \in (0, V]$ together with F_n, G_n .

Case $V_0 < V$. From (19.6) and (19.7), $\mathcal{E}(E_n) \geq \mathcal{E}(E) + \mathcal{E}(G_n) + o(1)$. Since $|G_n| = V - |F_n| \rightarrow V - V_0 > 0$, we have $\mathcal{E}(G_n) \geq m(|G_n|)$, so continuity of m (Proposition 17.7) gives

$$m(V) = \lim_n \mathcal{E}(E_n) \geq \mathcal{E}(E) + \liminf_n m(|G_n|) \geq m(V_0) + m(V - V_0),$$

contradicting Proposition 18.9 with $s = V_0/V$.

Case $V_0 = V$. Then $|F_n| \rightarrow V = |E_n|$, so $|G_n| \rightarrow 0$ and the translated sequence converges to E globally in L^1 . Lower semicontinuity (19.3) gives $\mathcal{E}(E) \leq \liminf_n \mathcal{E}(E_n) = m(V)$; the reverse inequality is the definition of $m(V)$, and $|E| = V$. $\square$

Part IV

Regularity of minimisers

Remark 19.11 (The generic-set discipline). From Chapter 20 through Proposition 26.1, *every* statement is made for an arbitrary set satisfying the ω -minimality inequality of Definition 20.3. The liquid-drop minimiser enters only through Lemma 20.4, which supplies one such ω . This is not stylistic: the same generic theory is applied a second time, with the Coulomb term deleted, in the equality case of the isoperimetric inequality (Chapter 27). A formalisation that hard-codes the liquid-drop energy into these chapters will have to be written twice.

Chapter 20

Penalisation and ω -minimality

Lean module: NoCompromise.Regularity.OmegaMinimal, NoCompromise.Regularity.Penalization, NoCompromise.Regularity.PenalizationDilation, NoCompromise.Regularity.PenalizationQuasiminimal

20.1 Global volume penalisation

Lemma 20.1 (Elementary dilation quotients). *On a fixed neighbourhood of $\rho = 1$ in $(0, \infty)$ the functions*

$$\rho \mapsto \frac{|\rho^2 - 1|}{|\rho^3 - 1|}, \quad \rho \mapsto \frac{|\rho^5 - 1|}{|\rho^3 - 1|}$$

are bounded, with removable values $2/3$ and $5/3$ at $\rho = 1$.

Proof. Factor $\rho^k - 1 = (\rho - 1)(\rho^{k-1} + \dots + 1)$ and cancel $\rho - 1$. □

Lemma 20.2 (Global volume penalisation). *Let E be a fixed-volume minimiser at volume $V > 0$. There is $\Lambda = \Lambda(V, \mathcal{E}(E)) < \infty$ such that*

$$\mathcal{E}(E) \leq \mathcal{E}(F) + \Lambda ||F| - V| \tag{20.1}$$

for every finite-perimeter set F of finite measure.

Proof. Correcting the volume on smooth boundary patches would be circular, because smooth patches are obtained only after quasiminimiser regularity. Global minimality permits a direct correction by dilation.

Suppose no such Λ exists. For every integer $h \geq 1$ choose F_h with $|F_h| \neq V$ and

$$\mathcal{E}(F_h) + h ||F_h| - V| < \mathcal{E}(E). \tag{20.2}$$

Nonnegativity of $\mathcal{E}$ gives $||F_h| - V| \leq \mathcal{E}(E)/h$, so $V_h := |F_h| \rightarrow V$ and in particular $V/2 \leq V_h \leq 2V$ for large h . Put $\rho_h := (V/V_h)^{1/3}$ and $G_h := \rho_h F_h$, so $|G_h| = V$. Since both terms of the energy are nonnegative, (20.2) gives uniform bounds for $P(F_h)$ and $D(F_h)$. By (15.7) and Lemma 20.1, with $C = C(V, \mathcal{E}(E))$,

$$\begin{aligned} |\mathcal{E}(G_h) - \mathcal{E}(F_h)| &\leq |\rho_h^2 - 1|P(F_h) + |\rho_h^5 - 1|D(F_h) \\ &\leq C|\rho_h^3 - 1| = \frac{C}{V_h}|V - V_h| \leq C'|V - V_h|. \end{aligned}$$

Combining with (20.2), $\mathcal{E}(G_h) < \mathcal{E}(E) - (h - C')|V_h - V| < \mathcal{E}(E)$ for large h , contradicting fixed-volume minimality of E . □

20.2 ω -minimality

Definition 20.3 (ω -minimal set). Let $n \geq 2$, $\omega \in [0, \infty)$ and $r_0 \in (0, \infty]$. A set $E \subset \mathbb{R}^n$ of locally finite perimeter is ω -minimal at scales below r_0 if for every $x \in \mathbb{R}^n$, every $0 < r \leq r_0$ and every set F of locally finite perimeter with $F \triangle E \Subset B_r(x)$,

$$P_n(E; B_r(x)) \leq P_n(F; B_r(x)) + \omega |E \triangle F|. \quad (20.3)$$

We say simply ω -minimal (or ω -minimal at unit scale) when $r_0 = 1$. A set that is 0-minimal at scales below every $r_0 < \infty$ is called *locally perimeter minimising*. In dimension $n = 3$ we suppress the perimeter subscript as usual, and in particular

$$P(E; B_r(x)) \leq P(F; B_r(x)) + \omega \frac{4\pi}{3} r^3, \quad (20.4)$$

since $|E \triangle F| \leq |B_r|$. A 0-minimal set is called *locally perimeter minimising* only when the inequality is available at every finite scale, as specified above; plain 0-minimality means unit scale.

Lemma 20.4 (Perimeter quasiminimality of a minimiser). *Let E be a fixed-volume minimiser at volume $V > 0$ and let Λ be as in Lemma 20.2. Then E is ω -minimal with*

$$\omega := C_0(V + |B_1|)^{2/3} + \Lambda < \infty.$$

Proof. Let $F \triangle E \Subset B_r(x)$ with $0 < r \leq 1$. The volumes of E and F are bounded by $V + |B_1|$, so (15.4) gives $|D(E) - D(F)| \leq C_0(V + |B_1|)^{2/3} |E \triangle F|$. Apply (20.1), move the nonlocal term to the right, and use $||F| - V| \leq |E \triangle F|$. Since $E = F$ in a neighbourhood of $\mathbb{R}^3 \setminus B_r(x)$, Lemma 4.8 cancels the identical exterior perimeter contributions, leaving (20.3). $\square$

Formalisation note 20.5 (The only variational input). (20.3) is the *only* variational input to Chapters 21–26. Nothing there may cite Definition 15.3, Definition 15.2, or any Coulomb estimate.

Chapter 21

Density, Ahlfors bounds and the canonical representative

Lean module: NoCompromise.Regularity.Density, NoCompromise.Regularity.DensityAhlfors, NoCompromise.Regularity.DensityCutComparison, NoCompromise.Regularity.DensityEstimates, NoCompromise.Regularity.DensityRadial, NoCompromise.Regularity.RepresentativeBoundary, NoCompromise.Regularity.RepresentativeBounded

Notation 21.1. Throughout this chapter E is ω -minimal (Definition 20.3). For $x \in \partial^e E$ put

$$m_x(r) := |E \cap B_r(x)|, \quad n_x(r) := |B_r(x) \setminus E|.$$

Lemma 21.2 (Coarea derivatives). *For a.e. $r > 0$,*

$$m'_x(r) = \mathcal{H}^2(E^{(1)} \cap \partial B_r(x)), \quad n'_x(r) = 4\pi r^2 - m'_x(r), \quad (21.1)$$

and the cut identities (5.10)–(5.11) hold at r .

Proof. Corollary 5.6 and Lemma 5.10. □

Lemma 21.3 (Cut comparison). *For $x \in \partial^e E$ and a.e. $0 < r \leq 1$,*

$$P(E; B_r(x)) \leq m'_x(r) + \omega m_x(r), \quad (21.2)$$

and symmetrically, comparing with $E \cup B_r(x)$,

$$P(E; B_r(x)) \leq n'_x(r) + \omega n_x(r). \quad (21.3)$$

Proof. Apply (20.3) to the competitor $F = E \setminus B_{r-\eta}(x)$, whose symmetric difference from E is compactly contained in $B_r(x)$, and then let $\eta \downarrow 0$. The BV trace formula (5.11) and Lemma 4.7 give the same inequality as the formal comparison with $E \setminus B_r(x)$, namely (21.2). For (21.3) use $F = E \cup B_{r-\eta}(x)$. □

Constants 21.4. In Lemma 21.5 below, set

$$r_d := \begin{cases} 1, & \omega = 0, \\ \min\{1, c_I/(2\omega|B_1|^{1/3})\}, & \omega > 0, \end{cases} \quad c_I = (36\pi)^{1/3}.$$

Thus $0 < r_d \leq 1$ and $\omega|B_1|^{1/3}r_d \leq c_I/2$; then c_d depends only on c_I . The order of choice is $\omega \rightsquigarrow r_d \rightsquigarrow c_d$.

Lemma 21.5 (Two-sided density). *There are $r_d = r_d(\omega) > 0$ and $c_d = c_d(\omega) \in (0, |B_1|/2)$ such that for every $x \in \partial^e E$ and $0 < r < r_d$,*

$$c_d r^3 \leq |E \cap B_r(x)| \leq (|B_1| - c_d) r^3. \quad (21.4)$$

Proof. Apply (13.3) to $E \cap B_r(x)$ and use (5.10):

$$c_I m_x(r)^{2/3} \leq P(E \cap B_r(x)) = P(E; B_r(x)) + m'_x(r).$$

With (21.2),

$$c_I m_x(r)^{2/3} \leq 2m'_x(r) + \omega m_x(r). \quad (21.5)$$

Since $m_x(r) \leq |B_1| r^3$, choosing r_d as in Constants 21.4 absorbs the last term:

$$m'_x(r) \geq c m_x(r)^{2/3} \quad \text{for a.e. } r < r_d. \quad (21.6)$$

At an essential-boundary point $m_x(r) > 0$ for every $r > 0$, so $\frac{d}{dr} m_x(r)^{1/3} = m'_x(r)/(3m_x(r)^{2/3}) \geq c/3$ a.e., and integrating from zero gives $m_x(r) \geq c_d r^3$. For the complement, use (21.3), apply (13.3) to $B_r(x) \setminus E$, and repeat with n_x ; then $n_x(r) \geq c_d r^3$ is the upper bound in (21.4). $\square$

Lemma 21.6 (Ahlfors perimeter bounds). *After reducing r_d if necessary, there are $0 < c_P < C_P < \infty$, depending only on ω , such that for every $x \in \partial^e E$ and $0 < r < r_d/2$,*

$$c_P r^2 \leq P(E; B_r(x)) \leq C_P r^2. \quad (21.7)$$

Proof. Lower bound. (7.7) together with (21.4).

Upper bound. Fix $r < r_d/2$. Averaging (21.1) over $(r, 2r)$, choose a radius $\rho \in (r, 2r)$ at which (21.2) holds and

$$m'_x(\rho) \leq \frac{m_x(2r) - m_x(r)}{r} \leq C r^2.$$

Then (21.2) at ρ gives $P(E; B_\rho(x)) \leq C r^2 + C \omega r^3 \leq C_P r^2$, and monotonicity of the perimeter measure under inclusion gives the same bound on $B_r(x)$. $\square$

21.1 The canonical representative

Convention 21.7 (The representative). For an ω -minimal set E put

$$\Omega := E^{(1)}. \quad (21.8)$$

From this point on, Ω always denotes the density-one representative of whichever ω -minimal set is under discussion. Statements about $\partial\Omega$ are statements about a genuine topological boundary, not about a null-set equivalence class.

Lemma 21.8 (Openness and representative conversion). *The set $\Omega = E^{(1)}$ is open, $\mathbf{1}_\Omega = \mathbf{1}_E$ a.e., $E^{(0)}$ is open, and*

$$\partial\Omega = \partial^e E = \overline{\partial^* E}. \quad (21.9)$$

Consequently every point of $\partial\Omega$ satisfies (21.4) and (21.7).

Proof. Lebesgue's density theorem gives $\mathbf{1}_\Omega = \mathbf{1}_E$ a.e. Suppose $x \in E^{(1)}$ were not interior to $E^{(1)}$. Then every sufficiently small ball about x contains positive measure of both E and its complement, so (7.7) gives positive perimeter in that ball. Since $\mu_E = \mathcal{H}^2 \llcorner \partial^* E$ (Theorem 8.12), the ball contains a reduced-boundary point, and such points $x_j \rightarrow x$ may be chosen. Put $d_j := |x_j - x|$; the complement-density estimate in $B_{d_j/2}(x_j) \subset B_{d_j}(x)$ gives $|B_{d_j}(x) \setminus E| \geq c d_j^3$, contradicting density one at x . So $E^{(1)}$ is open, and the same argument with the phases exchanged shows $E^{(0)}$ is open. Their common complement is therefore both $\partial\Omega$ and, by definition, $\partial^e E$. Finally $\partial^e E = \overline{\partial^* E}$: the lower perimeter-density bound (21.7) makes every ball centred at an essential-boundary point have positive perimeter and hence meet $\partial^* E$, while the reverse inclusion is automatic. $\square$

Lemma 21.9 (Boundedness). *If E is ω -minimal with $|E| < \infty$ then $\Omega = E^{(1)}$ is bounded, so $\partial\Omega$ is compact.*

Proof. Suppose not. Choose $y_j \in \Omega$ tending to infinity and mutually separated by more than $4r_d$. The balls $B_{r_d}(y_j)$ are pairwise disjoint after a harmless thinning, so finite volume implies infinitely many are not contained in Ω ; keep only those indices. Since $y_j \in \Omega$ and the ball is convex, a segment from y_j to a point outside Ω meets $\partial\Omega$; choose $x_j \in \partial\Omega \cap B_{r_d}(y_j)$. Thin once more so that the balls $B_{r_d/4}(x_j)$ are pairwise disjoint. By the lower phase-density bound (21.4),

$$|E| \geq \sum_j |E \cap B_{r_d/4}(x_j)| \geq \sum_j c_d (r_d/4)^3 = \infty,$$

a contradiction. $\square$

Chapter 22

First variation and almost-monotonicity

Lean module: NoCompromise.Regularity.FirstVariation,
 NoCompromise.Regularity.FixedNormalExcess, NoCompromise.Regularity.Monotonicity,
 NoCompromise.Regularity.PerimeterConvergence, NoCompromise.Regularity.RadialDivergence,
 NoCompromise.Regularity.TangentCone, NoCompromise.Regularity.TangentExistence

22.1 Bounded generalised mean curvature

Lemma 22.1 (Bounded generalised mean curvature). *Let E be ω -minimal and $x \in \mathbb{R}^3$. For every $X \in C_c^1(B_{r_d}(x); \mathbb{R}^3)$,*

$$\left| \int_{\partial^* E} \operatorname{div}_\tau X \, d\mathcal{H}^2 \right| \leq \omega \int_{\partial^* E} |X \cdot \nu_E| \, d\mathcal{H}^2. \quad (22.1)$$

Proof. For $|t| \|DX\|_\infty < 1$ the map F_t^X is a compactly supported C^1 diffeomorphism (Lemma 9.5). Choose $s < r_d \leq 1$ with $\operatorname{spt} X \subseteq B_s(x)$. Injectivity and the fact that F_t^X is the identity off $\operatorname{spt} X$ give $F_t^X(E) \triangle E \subseteq B_s(x)$. Apply (20.3) in $B_s(x)$ to $F = F_t^X(E)$ for both signs of t and divide by $|t|$. The local formula (9.6) gives

$$\frac{P(F_t^X(E); B_s(x)) - P(E; B_s(x))}{t} \longrightarrow \int_{\partial^* E} \operatorname{div}_\tau X \, d\mathcal{H}^2,$$

and Theorem 9.12 gives $|E \triangle F_t^X(E)|/|t| \rightarrow \int_{\partial^* E} |X \cdot \nu_E|$. The two signs of t yield (22.1). $\square$

Remark 22.2 (The scalar curvature is optional). (22.1) also implies the existence of $h_E \in L^\infty(\partial^* E, \mathcal{H}^2)$ with $\|h_E\|_\infty \leq \omega$ and $\int_{\partial^* E} \operatorname{div}_\tau X = \int_{\partial^* E} h_E X \cdot \nu_E$: the measure $\mathcal{H}^2 \llcorner \partial^* E$ is σ -finite, the set $\mathcal{N} := \{X \cdot \nu_E : X \in C_c^1(\mathbb{R}^3; \mathbb{R}^3)\} \subset L^1(\partial^* E)$ is a linear subspace, the left side of (22.1) vanishes whenever the normal trace does, and Hahn–Banach with σ -finite L^1 – L^∞ duality produces h_E . *Nothing later in this document requires it*; all critical-path arguments use (22.1) directly. It should therefore *not* be formalised.

22.2 Almost-monotonicity

Lemma 22.3 (Radial-field tangential divergence). *For a two-plane T_y with unit normal ν , a radial field $X(y) = \eta(|y|)y$ satisfies*

$$\operatorname{div}_{T_y} X = 2\eta(|y|) + \frac{\eta'(|y|)}{|y|} |y^T|^2 = 2\eta + \eta'|y| - \frac{\eta'}{|y|} |y^\perp|^2, \quad (22.2)$$

where $y^T, y^\perp$ are the components of y in T_y and ν . Also $|X \cdot \nu| \leq |\eta| |y^\perp|$.

Proof. Direct computation from $DX = \eta I + \eta'|y|^{-1} y \otimes y$ and $\operatorname{div}_T X = \operatorname{tr}(DX) - \nu \cdot DX\nu$. $\square$

Theorem 22.4 (Almost-monotonicity). *There are absolute constants $c, c_0 > 0$ such that for every ω -minimal E and every $x \in \partial\Omega$ the function*

$$r \mapsto e^{c\omega r} \frac{P(E; B_r(x))}{r^2} \quad (22.3)$$

is nondecreasing on $(0, r_d)$. More precisely, for $0 < \sigma < \rho < r_d$ and $x = 0$,

$$e^{c\omega\rho} \frac{P(E; B_\rho)}{\rho^2} - e^{c\omega\sigma} \frac{P(E; B_\sigma)}{\sigma^2} \geq c_0 \int_{(B_\rho \setminus B_\sigma) \cap \partial^* E} e^{c\omega|y|} \frac{|y^\perp|^2}{|y|^4} d\mathcal{H}^2. \quad (22.4)$$

Consequently the density

$$\Theta(E, x) := \lim_{r \downarrow 0} \frac{P(E; B_r(x))}{r^2} \quad (22.5)$$

exists, and by (21.7) it is finite and positive.

Proof. Translate x to zero and write $\mu := P(E; \cdot)$. On $(0, r_d)$ put

$$f(r) := \frac{\mu(B_r)}{r^2}, \quad A(r) := \int_{B_r \cap \partial^* E} \frac{|y^\perp|^2}{|y|^4} d\mathcal{H}^2(y).$$

Both are locally of bounded variation in the radius away from zero: this is immediate from the monotonicity of the two cumulative radial measures (give the integrand defining A the arbitrary value zero at $y = 0$). Insert the radial field of Lemma 22.3 into (22.1), first with a smooth nonnegative radial cutoff and then with piecewise linear cutoffs. Push the measures forward by $y \mapsto |y|$ and integrate by parts in the one-dimensional radius. This gives the following inequality of Radon measures on $(0, r_d)$:

$$df + c\omega f dr \geq c_0 dA. \quad (22.6)$$

Here is the term-by-term calculation behind (22.6). For a sharp cutoff at a radius r not charged by μ , the three terms in (22.2) give

$$r^3(A'(r) - f'(r)).$$

The right side of (22.1) is at most $\omega \int_{B_r} |y^\perp| d\mu \leq \omega r \mu(B_r) = \omega r^3 f(r)$. Mollification in the radius proves the measure inequality, with absolute constants (indeed one may take $c = c_0 = 1$ with these normalisations); atoms are recovered by one-sided approximation.

Multiplying (22.6) by $e^{c\omega r}$ and integrating on $[\sigma, \rho]$ gives

$$e^{c\omega\rho} f(\rho) - e^{c\omega\sigma} f(\sigma) \geq c_0 \int_{[\sigma, \rho]} e^{c\omega r} dA(r),$$

and the last Stieltjes integral is exactly the right side of (22.4). Thus the quantity (22.3) is nondecreasing. Its limit at zero exists; the Ahlfors bounds (21.7) make that limit finite and positive. $\square$

22.3 Tangent cones

Lemma 22.5 (Perimeter-measure convergence for ω_j -minimal sequences). *Let $s_j \in (0, \infty]$, let E_j be ω_j -minimal at scales below s_j , with $\omega_j \rightarrow \omega_\infty \geq 0$, and let $\mathbf{1}_{E_j} \rightarrow \mathbf{1}_F$ in $L^1_{\text{loc}}(U)$. Suppose the $P(E_j; \cdot)$ are uniformly locally bounded and that*

$$B_R(x) \Subset U \implies R \leq s_j \quad \text{for all sufficiently large } j. \quad (22.7)$$

Then:

- (i) F is locally ω_∞ -minimal in U , meaning that for every $B_R(x) \Subset U$ and every locally finite-perimeter G with $G \Delta F \Subset B_R(x)$,

$$P(F; B_R(x)) \leq P(G; B_R(x)) + \omega_\infty |F \Delta G|;$$

- (ii) $P(E_j; \cdot) \rightarrow P(F; \cdot)$ as measures on U ; in particular $P(E_j; A) \rightarrow P(F; A)$ for every open $A \Subset U$ with $P(F; \partial A) = 0$.

Proof. (i): Let $G \Delta F \Subset B_R(x) \Subset U$. Choose $a < b < R$ so that $G = F$ a.e. outside $B_a(x)$. On $B_b(x) \setminus \overline{B_a(x)}$ one has $E_j \rightarrow F = G$ in L^1 . Proposition 5.16 and its averaging identity give radii $t_j \in (a, b)$ and competitors

$$G_j = (G \cap B_{t_j}(x)) \cup (E_j \setminus B_{t_j}(x))$$

whose trace errors tend to zero; choose the radii also to carry no perimeter mass for the finitely many sets involved. By (22.7), $b \leq s_j$ eventually, and $G_j \Delta E_j \Subset B_b(x)$, so (20.3) applies in $B_b(x)$. Cancel the common exterior perimeter and use (5.14); lower semicontinuity on the left gives

$$P(F; B_a(x)) \leq P(G; B_a(x)) + \omega_\infty |F \Delta G|.$$

Since $F = G$ on $B_R(x) \setminus \overline{B_a(x)}$, locality (and a good intermediate radius followed by approximation if necessary) adds the same outer perimeter to both sides and proves the asserted inequality on $B_R(x)$.

(ii): By Theorem 3.8 a subsequence of $P(E_j; \cdot)$ converges weakly-* to some $\lambda \geq P(F; \cdot)$ (the inequality from Lemma 4.7). For the reverse, apply the same construction with $G = F$ inside a ball compactly contained in U . The penalty now tends to zero because ω_j is bounded and $E_j \rightarrow F$ locally in L^1 ; hence $\limsup_j P(E_j; B) \leq P(F; B)$ for every concentric smaller ball whose boundary is null for the relevant measures. Such balls form a basis (only countably many radii are excluded), so outer regularity gives $\lambda \leq P(F; \cdot)$. Thus $\lambda = P(F; \cdot)$. Every subsequential limit is the same, proving convergence of the full sequence; the final assertion is the Portmanteau continuity-set criterion. $\square$

Lemma 22.6 (Fixed-normal excess under strict perimeter convergence). *Let $U \subset \mathbb{R}^n$ be open, let E_j, E have locally finite perimeter in U , and suppose*

$$\mathbf{1}_{E_j} \rightarrow \mathbf{1}_E \quad \text{in } L^1_{\text{loc}}(U), \quad |D\mathbf{1}_{E_j}| \rightarrow |D\mathbf{1}_E| \quad \text{as Radon measures on } U.$$

If $\nu_E = \nu$ for one fixed unit vector ν at $|D\mathbf{1}_E|$ -a.e. point of U , then for every nonnegative $\phi \in C_c(U)$,

$$\int_U \phi |\nu_{E_j} - \nu|^2 d|D\mathbf{1}_{E_j}| \longrightarrow 0. \quad (22.8)$$

Proof. Local L^1 convergence gives $D\mathbf{1}_{E_j} \rightarrow D\mathbf{1}_E$ distributionally. The assumed convergence of the variation measures supplies uniform variation bounds on compact subsets, so approximation of a continuous test field by smooth ones upgrades this to weak convergence of the vector Radon measures. Using $D\mathbf{1}_{E_j} = -\nu_{E_j}|D\mathbf{1}_{E_j}|$ and $|\nu_{E_j}| = |\nu| = 1$,

$$\int \phi |\nu_{E_j} - \nu|^2 d|D\mathbf{1}_{E_j}| = 2 \int \phi d|D\mathbf{1}_{E_j}| + 2 \int \phi \nu \cdot D\mathbf{1}_{E_j}.$$

The limit is $2 \int \phi d|D\mathbf{1}_E| + 2 \int \phi \nu \cdot D\mathbf{1}_E = 0$, because $D\mathbf{1}_E = -\nu|D\mathbf{1}_E|$. $\square$

Formalisation note 22.7 (Extracted from a proof). Lemma 22.5 is a step of the blow-up argument for almost-monotonicity, and is used again inside the proof of the height bound. It is stated separately here precisely because it has two independent consumers (Lemma 22.8 and Lemma 24.8), and because (ii) — convergence of *measures*, not just of characteristic functions — is the part that is easy to use without proving.

Lemma 22.8 (Existence of tangent cones). *Let E be ω -minimal and $x \in \partial\Omega$. Every sequence $r_j \downarrow 0$ has a subsequence along which*

$$E_{x,r_j} := \frac{E - x}{r_j} \quad (22.9)$$

converges in L^1_{loc} to a locally perimeter-minimising set C with $0 \in \partial(C^{(1)})$, with $P(C; \cdot)$ the limit of the rescaled perimeter measures, and $P(C; B_R)/R^2 = \Theta(E, x)$ for every $R > 0$.

Proof. E_{x,r_j} is ωr_j -minimal at scales below $1/r_j$, with $\omega r_j \rightarrow 0$. Thus (22.7) holds on $U = \mathbb{R}^3$. Moreover (21.7) gives uniform local perimeter bounds, so Theorem 4.11 gives a local L^1 limit C along a subsequence and Lemma 22.5 makes C locally perimeter minimising with convergence of the perimeter measures. The density identity first follows at every radius R satisfying $P(C; \partial B_R) = 0$:

$$\frac{P(C; B_R)}{R^2} = \lim_j \frac{P(E; B_{Rr_j}(x))}{(Rr_j)^2} = \Theta(E, x).$$

Only countably many radii are excluded. The function $R \mapsto P(C; B_R)$ is continuous from below because the balls B_s increase to B_R as $s \uparrow R$; approximation of an arbitrary R from below by nonexceptional radii therefore proves the identity for every $R > 0$. Finally, the two phase-density estimates at x , after rescaling and passing to the local L^1 limit, give fixed positive lower density fractions for both C and its complement in every ball about 0. Thus $0 \in \partial^e C$, and Lemma 21.8, applied to the locally minimizing set C , gives $0 \in \partial(C^{(1)})$. $\square$

Proposition 22.9 (Tangent cones are cones). *The set C of Lemma 22.8 satisfies $e^t C = C$ modulo null sets for every $t \in \mathbb{R}$, and its density-one representative is an actual cone with vertex 0.*

Proof. Lemma 22.8 says that C is locally perimeter minimising, that $0 \in \partial(C^{(1)})$, and that all its density ratios are constant. Fix $0 < \sigma < \rho$. After a dilation making $\rho < 1$, apply (22.4) directly to C with $\omega = 0$; local perimeter minimality is preserved by dilation and supplies the unit-scale hypothesis. The left side is zero, so

$$\int_{(B_\rho \setminus B_\sigma) \cap \partial^* C} \frac{|y^\perp|^2}{|y|^4} d\mathcal{H}^2 = 0 \quad (22.10)$$

for all $0 < \sigma < \rho$. This avoids any passage to the limit in an integrand depending on the measure-theoretic normal. Hence the radial vector lies in the approximate tangent plane $\mathcal{H}^2$ -a.e. By the polar identity this says $y \cdot D\mathbf{1}_C = 0$ as a measure. For completeness, localise the field $X(y) = y$ on the support of a test function and let $\Phi_t(y) = e^t y$. Change variables and the BV integration-by-parts identity give

$$\frac{d}{dt} \int_{\mathbb{R}^3} \mathbf{1}_C(\Phi_{-t}(x)) \varphi(x) dx = - \int_{\mathbb{R}^3} \varphi(x) X(x) \cdot D(\mathbf{1}_C \circ \Phi_{-t})(x) = 0;$$

the last equality is the transported form of $X \cdot D\mathbf{1}_C = 0$. At $t = 0$ the integral is $\int \mathbf{1}_C \varphi$, so $\mathbf{1}_C \circ \Phi_{-t} = \mathbf{1}_C$ a.e. for every t . Thus $e^t C = C$ modulo null sets, and Convention 21.7 makes $C^{(1)}$ a cone. $\square$

Chapter 23

The phase-preserving cylindrical deformation

Lean module: NoCompromise.Regularity.CompressionJacobian,
NoCompromise.Regularity.CompressionProfileTheorem, NoCompromise.Regularity.Cylinders,
NoCompromise.Regularity.Deformation, NoCompromise.Regularity.SlabCap

Remark 23.1 (Placement). This construction depends only on Part III's predecessors in Part 1 — specifically on Theorem 9.1, Proposition 5.15 and Theorem 4.11 — and is used only by Lemma 24.11. It is placed here, next to its consumer, rather than in the foundations. Remark 8.15 records that Chapter 8 does not use it, so there is no cycle.

Notation 23.2 (Cylinders). $C_r := B_r^2 \times (-r, r)$, and $C_r^\nu(x)$ denotes its rotation with axis ν centred at x ; $C_r^\nu := C_r^\nu(0)$ and $C_r := C_r^{e_3}$.

Definition 23.3 (Slab-and-cap configuration). Let E have locally finite perimeter, $0 < \eta_0 < 1$, $c \in \mathbb{R}$, $\rho > 0$. Say E is in *slab-and-cap configuration* (ρ, c, η_0) if

$$|c| + \eta_0 \rho < \rho, \quad (23.1)$$

and

$$\partial^* E \cap C_\rho \subset \{|x_3 - c| < \eta_0 \rho\} \quad (23.2)$$

and the two caps carry the opposite constant phases:

$$B_\rho^2 \times \{-\rho\} \subset E^{(1)}, \quad B_\rho^2 \times \{\rho\} \subset E^{(0)} \quad (23.3)$$

up to $\mathcal{H}^2$ -null sets. The *reversed* configuration exchanges the two inclusions in (23.3).

Lemma 23.4 (The compression profile). Let $\frac{1}{2} < \sigma < \tau < 1$ and $d := \tau - \sigma$. The radial profile

$$\alpha(x') := \begin{cases} 0, & |x'| \leq \sigma, \\ \sin^2\left(\frac{\pi(|x'| - \sigma)}{2d}\right), & \sigma < |x'| < \tau, \\ 1, & |x'| \geq \tau \end{cases} \quad (23.4)$$

is C^1 and Lipschitz on $\mathbb{R}^2$ and satisfies, a.e.,

$$|\nabla \alpha|^2 \leq \frac{\pi^2}{d^2} \alpha(1 - \alpha) \leq \frac{\pi^2}{d^2} (1 - \alpha^2). \quad (23.5)$$

For $0 < \varepsilon < 1/2$ put $\beta_\varepsilon := \varepsilon + (1 - \varepsilon)\alpha$. Then $\varepsilon \leq \beta_\varepsilon \leq 1$, $\beta_\varepsilon = 1$ at $|x'| = \tau$, and

$$|\nabla \beta_\varepsilon|^2 \leq \frac{C}{d^2} (1 - \beta_\varepsilon^2). \quad (23.6)$$

Proof. (23.5) is a direct computation from $\alpha' = \frac{\pi}{d} \sin \theta \cos \theta$ with $\theta = \pi(|x'| - \sigma)/(2d)$, using $\alpha = \sin^2 \theta$. For (23.6), note $1 - \beta_\varepsilon^2 = (1 - \varepsilon)(1 - \alpha)(1 + \beta_\varepsilon)$ while $(1 - \varepsilon)^2(1 - \alpha^2) \leq 2(1 - \varepsilon)(1 - \alpha)(1 + \beta_\varepsilon)$. $\square$

Lemma 23.5 (Jacobian of the compression). *Let $T_\varepsilon(x', x_3) := (x', c + \beta_\varepsilon(x')(x_3 - c))$. For a two-plane $S = \nu^\perp$ with $\nu = (\nu', \nu_3)$,*

$$J_2(DT_\varepsilon|_S)^2 = |(\beta_\varepsilon \nu' - (x_3 - c)\nu_3 \nabla \beta_\varepsilon, \nu_3)|^2, \quad (23.7)$$

and consequently

$$J_2(DT_\varepsilon|_S) \leq 1 + \frac{C(x_3 - c)^2}{d^2}. \quad (23.8)$$

T_ε is bi-Lipschitz on the transition annulus, with inverse $T_\varepsilon^{-1}(x', z) = (x', c + (z - c)/\beta_\varepsilon(x'))$.

Proof. (23.7) is (6.8) applied to DT_ε . For (23.8), Young's inequality on the mixed term gives, for $0 < \beta_\varepsilon < 1$,

$$\begin{aligned} 2\beta_\varepsilon |x_3 - c| |\nu'| |\nu_3| |\nabla \beta_\varepsilon| &\leq (1 - \beta_\varepsilon^2) |\nu'|^2 \\ &\quad + \frac{\beta_\varepsilon^2 (x_3 - c)^2 \nu_3^2 |\nabla \beta_\varepsilon|^2}{1 - \beta_\varepsilon^2}, \end{aligned}$$

with the endpoint cases by continuity. Using (23.6) and $|\nu'|^2 + \nu_3^2 = 1$ absorbs the mixed term into $(1 - \beta_\varepsilon^2) |\nu'|^2$ and gives (23.8). $\square$

Theorem 23.6 (Phase-preserving cylindrical deformation). *Let E be in slab-and-cap configuration $(1, c, \eta_0)$ with $\eta_0 < 1/2$. Fix radii*

$$\frac{1}{2} < \sigma < \frac{5}{8} < \frac{7}{8} < \tau < 1.$$

Assume that both are cylindrically regular for E , meaning

$$|D\mathbf{1}_E|((\partial B_s^2) \times (-1, 1)) = 0 \quad (s = \sigma, \tau). \quad (23.9)$$

Such pairs have full measure in the displayed product of intervals: the pushforward of the finite measure $|D\mathbf{1}_E| \llcorner C_1$ by $x \mapsto |x'|$ has at most countably many atoms. Then there is a set F of locally finite perimeter which

- *agrees with E outside $B_\tau^2 \times (-1, 1)$,*
- *agrees with $\{x_3 < c\}$ in $B_\sigma^2 \times (-1, 1)$,*
- *has the same upper and lower cap traces as E ,*

and satisfies

$$\begin{aligned} P(F; B_\tau^2 \times (-1, 1)) &\leq P(E; (B_\tau^2 \setminus \overline{B_\sigma^2}) \times (-1, 1)) + \pi \sigma^2 \\ &\quad + \frac{C}{(\tau - \sigma)^2} \int_{\partial^* E \cap C_1} (x_3 - c)^2 d\mathcal{H}^2. \end{aligned} \quad (23.10)$$

Proof. Condition (23.9) and the local trace theorem on the smooth part of each vertical wall imply that the traces from its two sides agree $\mathcal{H}^2$ -a.e.; the circular edges are $\mathcal{H}^2$ -null. Thus cutting at either radius creates only the trace jump explicitly counted below.

Step 1: extend the slices. For $\sigma < |x'| < \tau$, extend the vertical slice of E from $(-1, 1)$ to all of $\mathbb{R}$ by the value one below -1 and zero above 1 . By (23.3) and the gluing formula (5.13) this extension creates no derivative measure on the planes $x_3 = \pm 1$.

Step 2: define the competitor. Let F_ε equal the image of the extended E under T_ε in the transition annulus, equal E outside $B_\tau^2 \times (-1, 1)$, and equal $\{x_3 < c\}$ in $B_\sigma^2 \times (-1, 1)$. At the outer wall $\beta_\varepsilon = 1$, so the traces match exactly. Clearance (23.1) ensures that both the original and compressed interfaces stay strictly between the caps, so the transformed cap traces are still one below and zero above.

Step 3: inner-wall error. By Theorem 8.12, (23.2) makes $D\mathbf{1}_E$ vanish in each of the connected lower and upper subcylinders outside the slab. A BV function with zero distributional derivative on a connected open set is a.e. constant (mollify on compact subsets), and (23.3) identifies these constants as one below and zero above. Lemma 5.7 therefore says that almost every original vertical slice is one below $c - \eta_0$ and zero above $c + \eta_0$. Its ε -compressed trace can differ from $\mathbf{1}_{\{x_3 < c\}}$ only on $(c - \varepsilon\eta_0, c + \varepsilon\eta_0)$. So (5.13) adds inner-wall perimeter at most

$$(2\pi\sigma)(2\varepsilon\eta_0). \quad (23.11)$$

Step 4: transformed interface. The bi-Lipschitz perimeter transport formula (9.2) applies to T_ε ; combining (23.8), the horizontal disk in the core, and (23.11) gives

$$\begin{aligned} P(F_\varepsilon; B_\tau^2 \times (-1, 1)) &\leq P(E; (B_\tau^2 \setminus \overline{B_\sigma^2}) \times (-1, 1)) + \pi\sigma^2 \\ &\quad + \frac{C}{d^2} \int_{\partial^* E \cap C_1} (x_3 - c)^2 d\mathcal{H}^2 + C\varepsilon. \end{aligned} \quad (23.12)$$

Step 5: remove ε . The right side of (23.12) is uniform in ε , so Theorem 4.11 gives an L^1 limit F along a sequence $\varepsilon \downarrow 0$. The definitions on the core and exterior pass directly to the limit. More strongly than trace convergence, clearance (23.1) leaves fixed collars of both caps on which every F_ε is identically one below and zero above; local L^1 convergence preserves those collars and hence the asserted cap traces. Lemma 4.7 removes the $C\varepsilon$ term. $\square$

Formalisation note 23.7 (What is *not* assumed). The proof does not assume that transformed affine sheets remain affine, and it uses no collision, current, or cancellation theorem. Every trace, gluing, Jacobian and limiting assertion is one of Propositions 5.15, Theorem 9.1, Lemma 23.5 or Theorem 4.11.

Chapter 24

The ε -regularity theorem

Lean module: NoCompromise.Regularity.ApproxHarmonic, NoCompromise.Regularity.EpsReg,
 NoCompromise.Regularity.Excess, NoCompromise.Regularity.FluxDefect,
 NoCompromise.Regularity.GeometricRecurrence, NoCompromise.Regularity.GraphApprox,
 NoCompromise.Regularity.GraphBadBase, NoCompromise.Regularity.GraphGoodBase,
 NoCompromise.Regularity.GraphProjectedExcess, NoCompromise.Regularity.GraphSingleValued,
 NoCompromise.Regularity.GraphTwoPointCaps, NoCompromise.Regularity.HarmonicAffine,
 NoCompromise.Regularity.HarmonicBlowup, NoCompromise.Regularity.HarmonicBlowupUniform,
 NoCompromise.Regularity.HeightBound, NoCompromise.Regularity.ReversePoincare,
 NoCompromise.Regularity.TiltImprovement, NoCompromise.Regularity.ZeroExcess

24.1 Excess and its elementary comparisons

Definition 24.1 (Cylindrical and spherical excess). For E of locally finite perimeter, $x \in \mathbb{R}^3$, $r > 0$ and $\nu \in \mathbb{S}^2$,

$$\mathbf{E}(E, x, r, \nu) := \frac{1}{r^2} \int_{\partial^* E \cap C_r^\nu(x)} |\nu_E - \nu|^2 d\mathcal{H}^2, \quad (24.1)$$

$$\mathbf{E}^b(E, x, r, \nu) := \frac{1}{r^2} \int_{\partial^* E \cap B_r(x)} |\nu_E - \nu|^2 d\mathcal{H}^2. \quad (24.2)$$

Lemma 24.2 (Ball–cylinder comparison). *From $B_r(x) \subset C_r^\nu(x) \subset B_{\sqrt{2}r}(x)$,*

$$\mathbf{E}^b(E, x, r, \nu) \leq \mathbf{E}(E, x, r, \nu) \leq 2 \mathbf{E}^b(E, x, \sqrt{2}r, \nu). \quad (24.3)$$

Proof. Use $B_r(x) \subset C_r^\nu(x) \subset B_{\sqrt{2}r}(x)$. The second inclusion changes the normalization from r^{-2} to $(\sqrt{2}r)^{-2}$ and therefore produces the factor 2. $\square$

Lemma 24.3 (Change of centre). *If $z \in C_{r/8}^\nu(x)$ then $C_{r/8}^\nu(z) \subset C_r^\nu(x)$, hence*

$$\mathbf{E}(E, z, r/8, \nu) \leq 64 \mathbf{E}(E, x, r, \nu). \quad (24.4)$$

Proof. The inclusion $C_{r/8}^\nu(z) \subset C_r^\nu(x)$ bounds the numerator by that defining $\mathbf{E}(E, x, r, \nu)$, while $(r/8)^{-2} = 64r^{-2}$. $\square$

Formalisation note 24.4 (These are the only two). Lemmas 24.2 and 24.3 are the only ball–cylinder and change-of-centre comparisons used in this chapter. In particular (24.4) is what makes the iteration of §24.9 available at *every* nearby boundary point, not just at the original centre.

24.2 Zero-excess classification

Lemma 24.5 (Zero-excess classification). *Let F have finite perimeter in a connected open cylinder C and suppose*

$$\nu_F = e_3 \quad |D\mathbf{1}_F| \text{-a.e. in } C. \quad (24.5)$$

Then, after changing F on a null set, exactly one of the following holds in C : $F \cap C = \emptyset$; $F \cap C = C$; or $F \cap C = C \cap \{x_3 < c\}$ for some $c \in \mathbb{R}$, where the cutting plane is allowed to lie outside C . In the nonconstant case only the lower halfspace occurs.

Proof. The polar decomposition (4.2) and (24.5) give

$$D_i \mathbf{1}_F = 0 \quad (i = 1, 2), \quad D_3 \mathbf{1}_F = -|D\mathbf{1}_F|. \quad (24.6)$$

Testing the first two identities against tensor-product test functions and using Fubini shows $\mathbf{1}_F(x', x_3) = g(x_3)$ a.e. in every smaller cylinder. The last identity says the distributional derivative of the $\{0, 1\}$ -valued g is nonpositive, so g is non-increasing; a non-increasing function taking only the values 0, 1 a.e. is identically 0, identically 1, or $\mathbf{1}_{(-\infty, c)}$. Compatibility on overlapping smaller cylinders gives the stated form on C , including the two degenerate cases when the cutting plane lies outside the cylinder. $\square$

Formalisation note 24.6 (Both degenerate cases are needed). The two constant cases are not decoration: they occur in the compactness argument of Lemma 24.8, where the limit may a priori be a full or empty cylinder, and they are excluded there by the density estimate at the origin rather than by hypothesis. Similarly the specification *lower* halfspace, rather than “a halfspace”, is what makes the orientation in (23.3) determinate.

24.3 The height bound

Constants 24.7. In Lemma 24.8 below, ε_h depends only on η ; the order of choice is $\eta \rightsquigarrow \varepsilon_h$.

Lemma 24.8 (Height bound). *For every $\eta \in (0, 1/8)$ there is $\varepsilon_h(\eta) > 0$ such that: if E is ω -minimal, $0 \in \partial\Omega$, $0 < r \leq 1$, and*

$$\mathbf{E}(E, 0, r, e_3) + \omega r \leq \varepsilon_h(\eta), \quad (24.7)$$

then

$$\partial\Omega \cap C_{3r/4} \subset \{|x_3| < \eta r\}. \quad (24.8)$$

Proof. Scale to $r = 1$; the rescaled set is ωr -minimal at scales below $1/r \geq 1$. If the assertion failed there would be sets E_j with quasiminimality constants $\omega_j \rightarrow 0$, admissible comparison scales at least one, excess tending to zero, and points

$$x_j \in \partial\Omega_j \cap C_{3/4}, \quad |(x_j)_3| \geq \eta.$$

The Ahlfors bounds (21.7) and Theorem 4.11 give, after a subsequence, $\mathbf{1}_{E_j} \rightarrow \mathbf{1}_F$ in $L^1(C_{7/8})$, and every ball compactly contained in $C_{7/8}$ has radius less than one, so (22.7) holds there. Lemma 22.5(ii) gives convergence of the perimeter measures on every smaller cylinder whose boundary carries no limiting perimeter. Since $\int_{\partial^* E_j \cap C_{7/8}} |\nu_{E_j} - e_3|^2 \rightarrow 0$, fix $0 \leq \phi \in C_c(C_{7/8})$. The polar identity gives

$$\int \phi |\nu_{E_j} - e_3|^2 d|D\mathbf{1}_{E_j}| = 2 \int \phi d|D\mathbf{1}_{E_j}| + 2 \int \phi e_3 \cdot D\mathbf{1}_{E_j}.$$

Perimeter-measure convergence handles the first term, while local L^1 convergence gives weak convergence of the distributional derivatives and handles the second. The left side tends to zero, hence $D\mathbf{1}_F = -e_3|D\mathbf{1}_F|$ on $C_{7/8}$, which is exactly (24.5). Thus Lemma 24.5 makes F a horizontal halfspace. Because $0 \in \partial\Omega_j$ and both phases have a uniform density at the origin by (21.4), the constant cases are excluded and the limiting plane passes through zero.

After a further subsequence $x_j \rightarrow x_\infty$ with $|(x_\infty)_3| \geq \eta$. A ball of radius $c\eta$ centred at x_j lies strictly in one phase of the limiting halfspace, yet (21.4) at the boundary point x_j puts a fixed positive amount of *each* phase in that ball. This contradicts the local L^1 convergence. $\square$

24.4 Phase separation and the reverse Poincaré inequality

Lemma 24.9 (Exact flux-defect identity for the excess). *Let E be in slab-and-cap configuration $(1, c, \eta_0)$ with $\eta_0 < 1$. For a.e. $s \in (1/2, 1)$,*

$$\int_{\partial^* E \cap (B_s^2 \times (-1, 1))} \nu_E \cdot e_3 d\mathcal{H}^2 = \pi s^2, \quad (24.9)$$

and consequently

$$\frac{1}{2} \int_{\partial^* E \cap (B_s^2 \times (-1, 1))} |\nu_E - e_3|^2 d\mathcal{H}^2 = P(E; B_s^2 \times (-1, 1)) - \pi s^2. \quad (24.10)$$

Proof. For a.e. s , slicing by $x' \mapsto |x'|$ gives no perimeter mass on the vertical wall and the interior and exterior traces there agree. Apply the trace product formulas of Theorem 5.8 chartwise to the two caps and the smooth vertical wall of $D_s := B_s^2 \times (-1, 1)$; the two circular edges are $\mathcal{H}^2$ -null. This identifies the boundary measure of $E \cap D_s$. Gauss–Green (8.20) for the constant field e_3 now has zero lateral flux, zero upper-cap contribution, and lower-cap contribution $-\pi s^2$ because the trace of E is respectively unrestricted, zero and one on those three pieces. Since the outward normal of the lower cap is $-e_3$, cancellation of the total flux gives (24.9) with the displayed positive sign. Expanding $|\nu_E - e_3|^2 = 2 - 2\nu_E \cdot e_3$ and using $\mathcal{H}^2(\partial^* E \cap \cdot) = P(E; \cdot)$ (Theorem 8.12) gives (24.10). $\square$

Formalisation note 24.10 (Where orientation enters). (24.10) is *the* point at which orientation enters the ε -regularity proof. An unoriented area Jacobian must not be used to estimate $|\nu_E - e_3|^2$ pointwise: the flux identity (24.9) depends on the *signed* cap traces (23.3), and replacing it by an unoriented bound loses exactly the cancellation that makes the excess a *defect*.

Lemma 24.11 (Reverse Poincaré). *There is $\eta_0 \in (0, 1/2)$, absolute, with the following property. Let E be ω -minimal, let $0 < \rho \leq 2^{-1/2}$, and suppose that after a rigid motion E is in slab-and-cap configuration (ρ, c, η_0) . Then*

$$\mathbf{E}(E, 0, \rho/2, e_3) \leq \frac{C}{\rho^4} \int_{\partial^* E \cap C_\rho} (x_3 - c)^2 d\mathcal{H}^2 + C\omega\rho. \quad (24.11)$$

If the two phases in (23.3) are reversed, the same holds with $-e_3$ in place of e_3 .

Proof. Put

$$\tilde{E} := \{y : \rho y \in E\}, \quad \tilde{c} := c/\rho, \quad \tilde{\omega} := \rho\omega.$$

Then $\tilde{E}$ has slab-and-cap configuration $(1, \tilde{c}, \eta_0)$. Scaling (20.3) directly shows that $\tilde{E}$ satisfies the same comparison with coefficient $\tilde{\omega}$ for every competitor supported compactly in a ball of radius at most $1/\rho$; this larger admissible radius must be retained after normalising the geometric scale.

Choose $\sigma \in (1/2, 5/8)$ and $\tau \in (7/8, 1)$ so that (23.9) holds for $\tilde{E}$ and (24.10) holds at $s = \sigma$. The excluded sets of radii are null, so such a pair exists. Apply Theorem 23.6 to obtain F . Since

$$\sqrt{\tau^2 + 1} < \sqrt{2} \leq 1/\rho,$$

one can choose R with $\sqrt{\tau^2 + 1} < R \leq 1/\rho$. The support of $F \triangle \tilde{E}$ lies in the closed cylinder $\overline{B_\tau^2} \times [-1, 1] \Subset B_R$. Hence the scaled quasiminimality inequality is applicable. The construction gives matching traces on the outer wall and caps, while cylindrical regularity removes perimeter mass from the outer wall; locality therefore cancels the identical part outside the cylinder and yields

$$P(\tilde{E}; B_\tau^2 \times (-1, 1)) \leq P(F; B_\tau^2 \times (-1, 1)) + \tilde{\omega}|F \triangle \tilde{E}|.$$

Now $|F \triangle \tilde{E}| \leq 2\pi\tau^2$, and cylindrical regularity at σ decomposes the first perimeter into its core and annular parts. Combining this with (23.10), and using $\tau - \sigma > 1/4$, gives

$$P(\tilde{E}; B_\sigma^2 \times (-1, 1)) - \pi\sigma^2 \leq C \int_{\partial^* \tilde{E} \cap C_1} (y_3 - \tilde{c})^2 d\mathcal{H}^2(y) + C\tilde{\omega}. \quad (24.12)$$

By (24.10) at $s = \sigma$ and $C_{1/2} \subset B_\sigma^2 \times (-1, 1)$,

$$\mathbf{E}(\tilde{E}, 0, 1/2, e_3) \leq C \int_{\partial^* \tilde{E} \cap C_1} (y_3 - \tilde{c})^2 d\mathcal{H}^2(y) + C\tilde{\omega}.$$

Finally, cylindrical excess is scale invariant, the displayed height integral equals $\rho^{-4} \int_{\partial^* E \cap C_\rho} (x_3 - c)^2 d\mathcal{H}^2(x)$, and $\tilde{\omega} = \rho\omega$. This is (24.11). For the reversed configuration apply the proved case to E^c , whose outward normal is $-\nu_E$ and whose quasiminimality inequality is identical. $\square$

Remark 24.12 (The slab hypothesis is supplied, not assumed). Under the hypotheses of Theorem 24.31, (23.2) follows from Lemma 24.8, and (23.3) follows from the fact that the limiting halfspace in Lemma 24.8 has outward normal e_3 . When Lemma 24.11 is applied at the *new* affine plane in Lemma 24.26, verifying (23.2)–(23.3) requires a separate argument; see (24.46). Skipping that verification is the most common way to produce a false proof of (24.42).

24.5 Maximal-function graph approximation

Definition 24.13 (Projected excess measure and its maximal density). For E of locally finite perimeter let $\pi(x', x_3) := x'$ and put, for Borel $A \subset \mathbb{R}^2$,

$$\mu(A) := \int_{\partial^* E \cap \pi^{-1}(A) \cap C_{3/4}} |\nu_E - e_3|^2 d\mathcal{H}^2, \quad (24.13)$$

and $M_{1/8}\mu(x') := \sup_{0 < \rho < 1/8} \mu(B_\rho^2(x'))/(\pi\rho^2)$.

Lemma 24.14 (Good base set). *Let E have locally finite perimeter, let μ be its projected excess measure from Definition 24.13, and let $\gamma, c > 0$. Put $c_\gamma := c\gamma^2$. Then*

$$|\{M_{1/8}\mu > c_\gamma\}| \leq Cc_\gamma^{-1}\mu(B_{3/4}^2), \quad (24.14)$$

so with $G_0 := \{M_{1/8}\mu \leq c\gamma^2\} \cap B_{1/2}^2$,

$$|B_{1/2}^2 \setminus G_0| \leq Cc^{-1}\gamma^{-2} \mathbf{E}(E, 0, 1, e_3). \quad (24.15)$$

Proof. (24.14) is Lemma 7.12; then $\mu(B_{3/4}^2) \leq \mathbf{E}(E, 0, 1, e_3)$ by Definitions 24.1 and 24.13. $\square$

Constants 24.15. In Lemma 24.16 below, The order of choice is $\gamma \rightsquigarrow \tau \rightsquigarrow \eta \rightsquigarrow (c, \varepsilon_g)$: first the target Lipschitz constant γ ; then a slab width $\tau < \min\{1/64, \gamma/256\}$; then a height parameter $\eta := \min\{\gamma/4, 1/4\}$ for Lemma 24.8; then c and ε_g small enough that $Cc\gamma^2 \leq \varepsilon_h(\eta)$ and $\varepsilon_g \leq \varepsilon_h(\eta)$, and also $\varepsilon_g|B_1|^{1/3} \leq c_I/2$ so that Constants 21.4 gives $r_d = 1$ throughout this unit-scale argument.

Lemma 24.16 (Two-point Lipschitz estimate on the good base). *Let $\gamma \in (0, 1/8)$. There are $c = c(\gamma) > 0$ and $\varepsilon_g(\gamma) > 0$, chosen in the order of Constants 24.15, such that, if E is ω -minimal with $0 \in \partial\Omega$ and*

$$\mathbf{E}(E, 0, 1, e_3) + \omega \leq \varepsilon_g(\gamma), \quad (24.16)$$

then any two reduced-boundary points $p = (p', p_3)$, $q = (q', q_3)$ with $p, q \in C_{1/2}$ and $p', q' \in G_0$ satisfy

$$|p_3 - q_3| \leq \gamma|p' - q'|. \quad (24.17)$$

Moreover the phases on the two horizontal faces are fixed, in the trace sense:

$$B_{1/2}^2 \times \{-1/2\} \subset E^{(1)}, \quad B_{1/2}^2 \times \{1/2\} \subset E^{(0)} \quad \text{up to } \mathcal{H}^2\text{-null sets.} \quad (24.18)$$

Proof. First apply Lemma 24.8 at $(0, 1)$ with a slab width

$$\tau < \min\{1/64, \gamma/256\}, \quad (24.19)$$

so that $\partial\Omega \cap C_{3/4} \subset \{|x_3| < \tau\}$. Since $D\mathbf{1}_E$ vanishes in each of the connected regions $B_{3/4}^2 \times (-3/4, -\tau)$ and $B_{3/4}^2 \times (\tau, 3/4)$, the two corresponding phase values are constants $\ell, u \in \{0, 1\}$. For a.e. $s \in (1/2, 5/8)$, Gauss–Green in $B_s^2 \times (-1/2, 1/2)$ gives

$$\int_{\partial^* E \cap (B_s^2 \times (-1/2, 1/2))} \nu_E \cdot e_3 d\mathcal{H}^2 = \pi s^2(\ell - u).$$

On the other hand the left side is the perimeter in that cylinder minus one half of its excess numerator. The cylinder contains $B_{1/4}$, so the Ahlfors lower bound makes its perimeter at least a fixed positive constant, whereas (24.16) makes the excess numerator arbitrarily small. Thus the left side is positive, forcing $(\ell, u) = (1, 0)$. This proves (24.18); in particular no unrecorded replacement of E by its complement is being made.

Set $d := |p' - q'|$ and $h := |p_3 - q_3|$. If $d \geq 1/32$ then the global slab bound and (24.19) give $h \leq 2\tau < \gamma d$. Suppose $d < 1/32$ and, towards a contradiction, $h > \gamma d$. Put $\rho := 2 \max\{d, h\}$; the global slab bound gives $\rho < 1/8$ and $q \in C_{3\rho/4}(p)$. Since $p' \in G_0$, Definition 24.13 gives

$$\mathbf{E}(E, p, \rho, e_3) \leq Cc\gamma^2. \quad (24.20)$$

By Constants 24.15, Lemma 24.8 applies at (p, ρ) with height parameter $\eta = \min\{\gamma/4, 1/4\}$ and gives $h < \eta\rho$. If $d \geq h$ this says $h < 2\eta d \leq \frac{\gamma}{2}d$, contradicting $h > \gamma d$. If $h > d$ it says $h < 2\eta h \leq h/2$, again a contradiction. The case $d = 0$ is included: it forces $h = 0$. $\square$

Lemma 24.17 (Single-valuedness from slicing). *In the situation of Lemma 24.16, for a.e. $x' \in B_{1/2}^2$ the restricted vertical slice $t \mapsto \mathbf{1}_E(x', t)$ on $(-1/2, 1/2)$ is a one-dimensional BV function whose oriented jump number between its lower and upper traces is exactly one. If $x' \in G_0$, (24.17) permits at most one reduced-boundary point in $\{x'\} \times (-1/2, 1/2)$. Hence, outside a planar null set, $\partial^* E \cap C_{1/2} \cap \pi^{-1}(G_0)$ is a single-valued γ -Lipschitz graph over a Borel set $G \subset G_0$.*

Proof. Lemma 5.7 gives the slice BV property. The slab conclusion places no boundary on the portions with $|x_3|$ between the thin slab and $1/2$; equation (24.18) says that their traces are the lower-one and upper-zero phases. Thus the one-dimensional fundamental theorem for BV on $(-1/2, 1/2)$ makes the oriented jump number equal to one. The rest is (24.17). $\square$

Lemma 24.18 (Bad-base area bound). *With $B := B_{1/2}^2 \setminus G$,*

$$\begin{aligned} \mathcal{H}^2(\partial^* E \cap C_{1/2} \cap \pi^{-1}(B)) &\leq |B| + 3 \int_{\partial^* E \cap C_{1/2} \cap \pi^{-1}(B)} (1 - \nu_3) d\mathcal{H}^2 \\ &\leq |B| + \frac{3}{2} \mathbf{E}(E, 0, 1, e_3). \end{aligned} \quad (24.21)$$

Proof. Write $1 = (1 - |\nu_3|) + |\nu_3|$ and integrate over $S_B := \partial^* E \cap C_{1/2} \cap \pi^{-1}(B)$. The localized signed slicing identity from Lemma 24.17 says that the signed sum of jump orientations in $(-1/2, 1/2)$ is one, and therefore bounds $\int_{S_B} |\nu_3| d\mathcal{H}^2$ by $|B| + \int_{S_B} (|\nu_3| - \nu_3) d\mathcal{H}^2$. Now use the elementary inequalities

$$1 - |t| \leq 1 - t, \quad |t| - t \leq 2(1 - t) \quad (-1 \leq t \leq 1), \quad (24.22)$$

and $|\nu_E - e_3|^2 = 2(1 - \nu_3)$. $\square$

Theorem 24.19 (Graph approximation). *For every $\gamma \in (0, 1/8)$ there are $C = C(\gamma) < \infty$ and $\varepsilon_g(\gamma) > 0$ with the following property; the constant C below depends only on γ , and every later use fixes γ before any other constant (Lemmas 24.21 and 24.23, Constants 24.25), so no uniformity in γ is asserted or needed. Let E be ω -minimal with $0 \in \partial\Omega$ and (24.16). Then there are a Borel set $G \subset B_{1/2}^2$ and a function*

$$f : B_{1/2}^2 \rightarrow \mathbb{R}, \quad \text{Lip}(f) \leq \gamma, \quad (24.23)$$

such that

$$\partial^* E \cap C_{1/2} \cap \pi^{-1}(G) = \{(x', f(x')) : x' \in G\} \quad (24.24)$$

up to $\mathcal{H}^2$ -null sets,

$$|B_{1/2}^2 \setminus G| + \mathcal{H}^2((\partial^* E \cap C_{1/2}) \setminus \text{graph } f) \leq C\gamma^{-2} \mathbf{E}(E, 0, 1, e_3), \quad (24.25)$$

and

$$\int_{B_{1/2}^2} |\nabla f|^2 \leq C \mathbf{E}(E, 0, 1, e_3). \quad (24.26)$$

Proof. Lemma 24.17 produces the single-valued γ -Lipschitz graph over $G \subset G_0$; extend its height from G to all of $B_{1/2}^2$ by the McShane formula, which preserves the Lipschitz constant. If the height bound used in Lemma 24.16 is $|x_3| < \tau$, compose this extension with the nearest-point projection onto $[-\tau, \tau]$. This projection is 1-Lipschitz and fixes the values on G , so the resulting extension still has Lipschitz constant at most γ , has the same good graph, and also satisfies

$$\|f\|_{L^\infty(B_{1/2}^2)} \leq \tau. \quad (24.27)$$

This gives (24.23)–(24.24).

Equation (24.25) combines (24.15) with (24.21).

For (24.26): on the good graph,

$$|\nu_E - e_3|^2 = 2 \left(1 - \frac{1}{\sqrt{1 + |\nabla f|^2}} \right) \geq c |\nabla f|^2 \quad (24.28)$$

because $|\nabla f| \leq \gamma \leq 1/8$. Here the signed slice identity and the lower-one/upper-zero cap separation orient every good jump upward, so the outward normal is $(-\nabla f, 1)/\sqrt{1+|\nabla f|^2}$; without this sign information the first equality would be false. On the bad planar set the McShane extension has $|\nabla f| \leq \gamma$ a.e., so its Dirichlet energy there is at most $\gamma^2|B| \leq C \mathbf{E}(E, 0, 1, e_3)$ by (24.15). Corollary 6.13 on the good graph converts (24.28) into (24.26). $\square$

24.6 Approximate harmonicity

Lemma 24.20 (Approximate harmonicity). *Fix $\gamma \in (0, 1/8)$. There is $C_\gamma < \infty$, depending only on γ and chosen before the clamping height in (24.27), such that the graph approximation f of Theorem 24.19 may be taken to satisfy in addition, for every $\zeta \in C_c^1(B_{1/2}^2)$,*

$$\left| \int_{B_{1/2}^2} \nabla f \cdot \nabla \zeta \right| \leq C_\gamma (\mathbf{E}(E, 0, 1, e_3) + \omega) \|\nabla \zeta\|_\infty. \quad (24.29)$$

At a general scale $r \in (0, 1]$, for the graph approximation of the set rescaled to unit scale,

$$\left| \int_{B_{r/2}^2} \nabla f \cdot \nabla \zeta \right| \leq C_\gamma r^2 (\mathbf{E}(E, 0, r, e_3) + \omega r) \|\nabla \zeta\|_\infty. \quad (24.30)$$

Proof. Let $\eta \in C_c^1((-3/4, 3/4))$ be identically one on the slab containing $\partial\Omega \cap C_{3/4}$ (available by Lemma 24.8), and use in (22.1) the vertical field $X(x', x_3) := \zeta(x')\eta(x_3)e_3$; the derivatives of η vanish on the boundary portion under consideration. Since $\text{spt } \zeta \subset B_{1/2}^2$, this field is compactly supported in B_1 , as required by Lemma 22.1. On the good graph, with $Q := \sqrt{1+|\nabla f|^2}$,

$$\text{div}_\tau X = \frac{\nabla f \cdot \nabla \zeta}{Q^2}, \quad d\mathcal{H}^2 = Q \, dx', \quad (24.31)$$

so the graph contribution is $\int_G \nabla f \cdot \nabla \zeta / Q$. Since $|\nabla f| \leq \gamma$,

$$\left| \int_G \left(1 - \frac{1}{Q}\right) \nabla f \cdot \nabla \zeta \right| \leq C_\gamma \|\nabla \zeta\|_\infty \int_G |\nabla f|^2 \leq C_\gamma \mathbf{E} \|\nabla \zeta\|_\infty \quad (24.32)$$

by (24.26). The boundary omitted by the graph has area bounded by (24.25), and on it $|\text{div}_\tau X| \leq C \|\nabla \zeta\|_\infty$, so its contribution is at most

$$C\gamma^{-2} \mathbf{E} \|\nabla \zeta\|_\infty. \quad (24.33)$$

The McShane extension contributes on the bad planar set at most

$$\int_{B_{1/2}^2 \setminus G} |\nabla f| |\nabla \zeta| \leq \gamma |B_{1/2}^2 \setminus G| \|\nabla \zeta\|_\infty \leq C\gamma^{-1} \mathbf{E} \|\nabla \zeta\|_\infty. \quad (24.34)$$

Finally the right side of (22.1) is at most $\omega \int_{\partial^* E} |X \cdot \nu_E| \leq C\omega \|\zeta\|_\infty$. Indeed, cover the portion of $\partial^* E \cap B_1$ meeting the support of X by a fixed dimension-dependent number of balls of radius $1/4$ centred on the boundary, and apply the Ahlfors upper bound (21.7) in each ball; the smallness convention in Constants 24.15 makes these constants absolute. Combining (24.32)–(24.34) gives the two-term estimate

$$\left| \int_{B_{1/2}^2} \nabla f \cdot \nabla \zeta \right| \leq C_\gamma \mathbf{E}(E, 0, 1, e_3) \|\nabla \zeta\|_\infty + C\omega \|\zeta\|_\infty, \quad (24.35)$$

with $C_\gamma \leq C(1 + \gamma^{-2})$. Since $\|\zeta\|_\infty \leq C\|\nabla\zeta\|_\infty$ for compactly supported ζ on $B_{1/2}^2$, this yields (24.29). Only the clean form is used later, and only the fact that C_γ depends on γ alone matters; the two-term form and the explicit growth in γ^{-2} are not part of the statement.

Scaling. Write $x = r\xi$, $f(x') = r\tilde{f}(\xi')$, $\tilde{\zeta}(\xi') := \zeta(r\xi')$. The physical left side is r times the unit-scale one, while $\|\nabla_\xi \zeta\|_\infty = r\|\nabla_x \zeta\|_\infty$; this gives the factor r^2 in (24.30). $\square$

24.7 Harmonic blow-up

Lemma 24.21 (Compactness of the normalised graphs). *Fix $\gamma \in (0, 1/8)$, and use this same graph parameter for every j . Let E_j be ω_j -minimal at unit scale with $0 \in \partial\Omega_j$, satisfying (24.16), and put*

$$a_j := (\mathbf{E}(E_j, 0, 1, e_3) + \omega_j)^{1/2} > 0, \quad a_j \longrightarrow 0. \quad (24.36)$$

Let f_j be the graph approximations of Theorem 24.19 and

$$\bar{f}_j := \int_{B_{1/2}^2} f_j, \quad g_j := \frac{f_j - \bar{f}_j}{a_j}. \quad (24.37)$$

Then, after a subsequence,

$$g_j \rightharpoonup h \text{ in } H^1(B_{1/2}^2), \quad g_j \rightarrow h \text{ in } L^2(B_{1/2}^2), \quad (24.38)$$

and h is harmonic (hence smooth).

Proof. (24.26) and Poincaré give a uniform H^1 bound for g_j ; Theorem 4.16 gives (24.38) after a subsequence. Dividing (24.29) by a_j makes the right side $O(a_j)$, so $\int \nabla h \cdot \nabla \zeta = 0$ for every test function. Weakly harmonic implies smooth by Lemma 10.18. $\square$

Formalisation note 24.22 (Strong L^2 , not strong H^1). Only strong L^2 convergence is claimed. Strong H^1 would require an additional energy-convergence argument and is *unnecessary*, because Lemma 24.11 converts height control into tilt control. Do not attempt to prove the stronger statement.

Lemma 24.23 (Uniform harmonic approximation). *Fix $\gamma \in (0, 1/8)$. There is $C_b(\gamma) < \infty$ such that for every $\tau_0 \in (0, 1]$ there is $0 < \varepsilon_b(\tau_0, \gamma) \leq \varepsilon_g(\gamma)$ with the following property. If E is ω -minimal at unit scale with $0 \in \partial\Omega$ and $a := (\mathbf{E}(E, 0, 1, e_3) + \omega)^{1/2}$ satisfies $0 < a^2 \leq \varepsilon_b$, then, for every graph approximation f furnished by Theorem 24.19, with $\bar{f} := \int_{B_{1/2}^2} f$, there is a harmonic h on $B_{1/4}^2$ with*

$$\int_{B_{1/4}^2} \left| \frac{f - \bar{f}}{a} - h \right|^2 \leq \tau_0, \quad \int_{B_{1/4}^2} (|h|^2 + |\nabla h|^2) \leq C_b(\gamma). \quad (24.39)$$

The same assertion includes the clamped extension in (24.27). The bound $C_b(\gamma)$ is independent of τ_0 and of the chosen slab width.

Proof. The gradient estimate (24.26) and the planar Poincaré inequality give a uniform H^1 bound, depending only on the fixed γ , for $(f - \bar{f})/a$. Choose $C_b(\gamma)$ larger than the square of that bound. If the approximation failed for some τ_0 , choose sets with $a_j \downarrow 0$ and a failing graph approximation for each set. Lemma 24.21 gives a harmonic limit h and strong L^2 convergence on $B_{1/4}^2$; weak lower semicontinuity gives its stated H^1 bound. This contradicts the failure. Clamping preserves the graph, Lipschitz and Dirichlet estimates used here, so the bound is uniform over those extensions as well. $\square$

24.8 Improvement of tilt

Lemma 24.24 (Affine approximation of a harmonic function). *For h harmonic on $B_{1/4}^2$ and $\theta \in (0, 1/32)$,*

$$\int_{B_{4\theta}^2} |h - h(0) - \nabla h(0) \cdot x'|^2 \leq C\theta^6 \int_{B_{1/4}^2} |\nabla h|^2. \quad (24.40)$$

Proof. Interior derivative estimates (10.10) bound $\|D^2 h\|_\infty$ on $B_{1/8}^2$ by $C\|\nabla h\|_{L^2(B_{1/4}^2)}$; Taylor's theorem then gives a pointwise bound $C\theta^2\|D^2 h\|_\infty$ on $B_{4\theta}^2$, and squaring and integrating over a set of area $C\theta^2$ gives θ^6 . $\square$

Constants 24.25. Fix the graph parameter once and for all at $\gamma_0 := 1/16$. The order of choice in Lemma 24.26 is

$$\gamma_0 \rightsquigarrow C \rightsquigarrow \theta \rightsquigarrow \tau_t \rightsquigarrow \tau_0 \rightsquigarrow \varepsilon_t(\theta).$$

The graph, harmonic and reverse-Poincaré bounds at this fixed γ_0 give C before the scale θ is chosen. For each $\theta \in (0, 1/32)$ choose a slab width τ_t , put $\tau_0 = \theta^6$, and then shrink $\varepsilon_t(\theta)$ to meet the explicitly listed smallness conditions in the proof. Corollary 24.27 chooses one scale after this uniform C is known.

Lemma 24.26 (Improvement of tilt). *There is an absolute $C \geq 1$ such that for every $\theta \in (0, 1/32)$ there is $\varepsilon_t(\theta) > 0$ with the following property. If E is ω -minimal at unit scale with $0 \in \partial\Omega$, put*

$$a := (\mathbf{E}(E, 0, 1, e_3) + \omega)^{1/2} \quad \text{and assume} \quad a^2 \leq \varepsilon_t(\theta). \quad (24.41)$$

Then there is a unit vector ν' with $|\nu' - e_3| \leq Ca$ such that

$$\mathbf{E}(E, 0, \theta, \nu') \leq C\theta^2 \mathbf{E}(E, 0, 1, e_3) + C\theta\omega. \quad (24.42)$$

In particular the same C works for all these scales; only the admissible initial excess $\varepsilon_t(\theta)$ changes with θ .

Proof. Fix $\gamma_0 = 1/16$. In this proof every generic constant C depends only on the estimates for that fixed graph parameter, the harmonic derivative and affine-approximation estimates, and the absolute constants of Lemma 24.11. None depends on θ . The H^1 bound in (24.39) and Lemma 10.14 give a fixed $C_h \geq 1$ such that every harmonic approximant used below satisfies $|h(0)| + |\nabla h(0)| \leq C_h$.

Now let $\theta \in (0, 1/32)$. Choose

$$0 < \tau_t < \min\{\theta^3, \eta_0\theta/32, 1/64, \gamma_0/256\}, \quad \tau_0 := \theta^6, \quad (24.43)$$

where η_0 is the fixed slab constant of Lemma 24.11. Choose the positive $\varepsilon_t(\theta)$ so that

$$\varepsilon_t(\theta) \leq \min \left\{ \varepsilon_g(\gamma_0), \varepsilon_h(\tau_t), \varepsilon_b(\theta^6, \gamma_0), \theta^6, \left(\frac{\eta_0\theta}{64C_h} \right)^2, \left(\frac{1}{16C_h} \right)^2 \right\}.$$

These choices affect the hypotheses, not the constants in the estimates. If $a = 0$, take $\nu' = e_3$; zero excess at unit scale gives zero excess in the smaller cylinder. Assume $a > 0$.

The height bound gives $\partial\Omega \cap C_{3/4} \subset \{|x_3| < \tau_t\}$. Choose the graph approximation with its extension clamped to $[-\tau_t, \tau_t]$, so $|\bar{f}| \leq \tau_t$. The phase-separation argument of Lemma 24.16 gives

the lower-one and upper-zero regions outside this slab in $C_{3/4}$: each of the two boundary-free regions is connected, so its phase is constant, and the cap traces in that lemma identify the two constants. Shrinking the slab is permitted by (24.43). Apply Lemma 24.23 to this graph with tolerance θ^6 , and define

$$\nu' := \frac{(-a\nabla h(0), 1)}{\sqrt{1 + a^2|\nabla h(0)|^2}}. \quad (24.44)$$

Then $|\nu' - e_3| \leq 2C_h a \leq 1/8$. In particular the rotated cylinder $C_{2\theta}^{\nu'}$ lies inside $C_{3\theta} \subset C_{3/4}$ and its horizontal projection lies inside $B_{3\theta}^2 \subset B_{4\theta}^2$.

Step 1: height above the new affine plane. Let the plane be

$$x_3 = \bar{f} + a(h(0) + \nabla h(0) \cdot x'). \quad (24.45)$$

On the portion of the boundary contained in graph f , its squared vertical discrepancy is at most

$$2a^2 \left(\int_{B_{1/4}^2} \left| \frac{f - \bar{f}}{a} - h \right|^2 + \int_{B_{4\theta}^2} |h - h(0) - \nabla h(0) \cdot x'|^2 \right) \leq Ca^2\theta^6$$

after integration over the base. Here (24.39) and (24.40) apply, and the area element of the graph is bounded by $\sqrt{1 + \gamma_0^2}$. Distance to the affine plane is no larger than this vertical discrepancy.

For a boundary point in $C_{2\theta}^{\nu'}$ outside graph f , $|x_3| \leq \tau_t$, $|\bar{f}| \leq \tau_t$ and the bound on $h(0), \nabla h(0)$ make its squared distance at most $C(\tau_t^2 + a^2)$. By (24.25), the area of such points is bounded by $C\gamma_0^{-2} \mathbf{E}(E, 0, 1, e_3)$. Since $\tau_t^2 \leq \theta^6$, $a^2 \leq \theta^6$ and $\mathbf{E}(E, 0, 1, e_3) \leq a^2$, their contribution is at most $Ca^2\theta^6$. Thus the full boundary height-square in the rotated cylinder is at most $Ca^2\theta^6$, with C independent of θ .

Step 2: slab and caps relative to the new plane. Write $b := \bar{f} + ah(0)$ and $c := b/\sqrt{1 + a^2|\nabla h(0)|^2}$, the signed distance of the plane from zero. The choices above give

$$|b| + 3\theta a|\nabla h(0)| \leq \tau_t + (1 + 3\theta)C_h a < \eta_0\theta/8. \quad (24.46)$$

Consequently every boundary point in $C_{2\theta}^{\nu'}$ has distance less than $\tau_t + \eta_0\theta/8 < 2\eta_0\theta$ from the new plane. Also $|c| \leq |b| < \eta_0\theta/8$, so $|c| + 2\eta_0\theta < 2\theta$ because $\eta_0 < 1/2$. This verifies both the slab and clearance conditions of Definition 23.3 at radius 2θ .

A point on either cap has the form $x = \pm 2\theta\nu' + y$, with $y \cdot \nu' = 0$ and $|y| < 2\theta$. Since $|\nu' - e_3| \leq 1/8$, its third coordinate is less than $-\theta$ on the lower cap and greater than θ on the upper cap. Both caps lie in $C_{3/4}$ and outside the original slab, so their traces are respectively one and zero by the phase separation already established. The complete slab-and-cap configuration therefore holds after rotating ν' to e_3 .

Step 3: apply reverse Poincaré. Since $2\theta < 2^{-1/2}$, Lemma 24.11 applies and gives

$$\mathbf{E}(E, 0, \theta, \nu') \leq C\theta^{-4}(a^2\theta^6) + C\omega\theta.$$

Substitute $a^2 = \mathbf{E}(E, 0, 1, e_3) + \omega$ and use $\theta < 1$ to absorb $C\theta^2\omega$ into $C\theta\omega$. Every constant in these three steps was independent of θ ; choose one common $C \geq 1$ larger than those constants and $2C_h$. This establishes the quantified statement. $\square$

Corollary 24.27 (Excess decay). *Let $C \geq 1$ be the uniform constant in Lemma 24.26, and choose once and for all*

$$\theta := \min\{1/64, 1/(4C)\} \in (0, 1/32), \quad C\theta^2 \leq \theta/4 \leq \theta/2. \quad (24.47)$$

Then there is $\varepsilon_0 > 0$, absolute, such that: if E is ω -minimal, $0 \in \partial\Omega$, $0 < r \leq 1$, and

$$\mathbf{E}(E, 0, r, e_3) + \omega r \leq \varepsilon_0, \quad (24.48)$$

then there is a unit vector ν' with

$$\mathbf{E}(E, 0, \theta r, \nu') \leq \frac{\theta}{2} \mathbf{E}(E, 0, r, e_3) + C\theta\omega r. \quad (24.49)$$

Proof. Obtain C first from Lemma 24.26, then choose θ by (24.47), then set $\varepsilon_0 := \varepsilon_t(\theta)$. Under $x = r\xi$ the excess is invariant and the quasiminimality coefficient becomes ωr , with admissible comparison radii at least one since $r \leq 1$. Apply the lemma at this scale and use $C\theta^2 \leq \theta/2$. $\square$

24.9 Iteration and completion

Lemma 24.28 (Geometric recurrence). *Let $E_0 \geq 0$, $\theta \in (0, 1)$, $\omega, r \geq 0$, and let $E_j \geq 0$ satisfy*

$$E_{j+1} \leq \frac{\theta}{2} E_j + C\theta\omega r_j, \quad r_j := \theta^j r. \quad (24.50)$$

Then $E_j \leq C'\theta^j(E_0 + \omega r)$ for every j , with C' depending only on C .

Proof. Divide by θ^{j+1} : the recurrence for $\tilde{E}_j := E_j/\theta^j$ becomes $\tilde{E}_{j+1} \leq \frac{1}{2}\tilde{E}_j + C\omega r$, whose solution is bounded by $\tilde{E}_0 + 2C\omega r$. $\square$

Lemma 24.29 (The optimal normals are Cauchy). *In the iteration of Corollary 24.27 at scales $r_j = \theta^j r$, with E_j the excess relative to the normal ν_j selected at step j ,*

$$|\nu_{j+1} - \nu_j|^2 \leq C(E_j + \omega r_j), \quad (24.51)$$

and consequently $\nu_j \rightarrow \nu_\infty$ with

$$|\nu_j - \nu_\infty| \leq C\theta^{j/2}(E_0 + \omega r)^{1/2}. \quad (24.52)$$

Proof. On a cylinder of radius comparable to r_{j+1} , expand $|\nu_{j+1} - \nu_j|^2$ through ν_E , bound the two error integrals by the excesses E_j and E_{j+1} , and use the Ahlfors lower bound (21.7) to divide by a positive multiple of r_{j+1}^2 . Because θ is fixed, this gives $|\nu_{j+1} - \nu_j|^2 \leq C(E_j + E_{j+1})$; the recurrence (24.50) gives (24.51). Then Lemma 24.28 gives $E_j + \omega r_j \leq C\theta^j(E_0 + \omega r)$, and summing (24.51) gives (24.52). $\square$

Lemma 24.30 (Uniform iteration at nearby centres). *There is $\varepsilon'_0 > 0$, absolute, such that if E is ω -minimal, $0 \in \partial\Omega$, $0 < r \leq 1$, and $\mathbf{E}(E, 0, r, e_3) + \omega r \leq \varepsilon'_0$, then (24.48) holds at every $z \in \partial\Omega \cap C_{r/8}$ at scale $r/8$, and hence the iteration of Corollary 24.27 and Lemma 24.29 runs at every such centre.*

Proof. $C_{r/8}(z) \subset C_{r/2}$, so (24.4) gives $\mathbf{E}(E, z, r/8, e_3) \leq 64 \mathbf{E}(E, 0, r, e_3)$; take $\varepsilon'_0 := \varepsilon_0/64$ (also shrinking to absorb the $\omega r/8$ term). $\square$

Theorem 24.31 (ε -regularity). *There are absolute constants $\varepsilon_0 > 0$, $\theta \in (0, 1/16)$ and $C < \infty$ such that the following holds. Let E be ω -minimal with $0 \in \partial\Omega$, let $0 < r \leq 1$, and assume*

$$\mathbf{E}(E, 0, r, e_3) + \omega r \leq \varepsilon_0. \quad (24.53)$$

Then $\partial\Omega \cap C_{r/4}$ is, after rotation, the graph of a $C^{1,1/2}$ function, and

$$|\nu_\Omega(x) - \nu_\Omega(y)| \leq Cr^{-1/2}|x - y|^{1/2}(\mathbf{E}(E, 0, r, e_3) + \omega r)^{1/2} \quad (24.54)$$

for $x, y \in \partial\Omega \cap C_{r/8}$.

Proof. Step 1: iterate. Apply (24.49) recursively at the scales $r_j = \theta^j r$. Lemma 24.28 gives $E_j \leq C\theta^j(E_0 + \omega r)$ and Lemma 24.29 gives the limiting normal ν_∞ with the rate (24.52).

Step 2: run the iteration at every nearby centre. Lemma 24.30.

Step 3: one graph. The limiting normal at every reduced-boundary point in a sufficiently small common cylinder makes an angle less than $\pi/4$ with e_3 . Indeed, at a reduced-boundary point, differentiation of the excess measure and (24.52) identify the limiting normal with the measure-theoretic normal ν_E there. Hence $D_3 \mathbf{1}_E = -\nu_E \cdot e_3 |D\mathbf{1}_E|$ has one sign in the common cylinder. By BV slicing (Lemma 5.7), for a.e. vertical line the one-dimensional slice of $\mathbf{1}_E$ is therefore monotone and has at most one jump. After rescaling, the phase-separation conclusion (24.18) of Lemma 24.16 gives uniform lower-one and upper-zero regions, and hence exactly one jump for a.e. base point. Let $f(x')$ be that jump height on this full-measure set.

The preceding a.e. graph has a unique continuous extension. To see this without making an exceptional-slice assumption, take two sequences of reduced-boundary points converging to topological-boundary points. Apply the height bound at the first point at the smallest dyadic iteration scale which contains the second. Because all limiting normals make angle less than $\pi/4$ with e_3 , the resulting cone estimate says that two sufficiently near boundary points with the same horizontal projection must coincide, and that the vertical separation tends to zero with the horizontal separation. The reduced boundary is dense in $\partial\Omega$ by Lemma 21.8, so this estimate passes to the limiting topological-boundary points. The lower-one and upper-zero regions show that every vertical segment over the smaller base disk meets the boundary. Consequently the extended f is continuous and its graph is the entire topological boundary in the smaller cylinder, not merely an a.e. subset. Local graph extensions agree on overlaps because they represent the same density-one open set.

Step 4: Hölder estimate. Let $d := |x - y|$ and choose the largest iteration scale s with $s \asymp d$. Take the fixed comparison factor in $s \asymp d$ large enough that one ball of radius cs centred at a boundary point lies in both scale- s cylinders. Expanding the squared difference of the two scale- s normals through ν_E on that ball, using the two excess bounds and dividing by its Ahlfors lower bound, gives

$$|\nu_{x,s} - \nu_{y,s}| \leq C(d/r)^{1/2}(\mathbf{E}(E, 0, r, e_3) + \omega r)^{1/2}.$$

The two tails from these scale- s normals to their limiting normals have the same bound by (24.52). This proves (24.54) first on the reduced boundary and then everywhere by continuity. The continuous graph has this field as its normal; solving the normal–gradient relation in the cone $\nu \cdot e_3 > 1/2$ gives $f \in C^{1,1/2}$. $\square$

Chapter 25

Classification of minimising cones

Lean module: `NoCompromise.Cones.ThreeDim`, `NoCompromise.Cones.TwoDim`,
`NoCompromise.Tests.Constants`

25.1 The two-dimensional case

Lemma 25.1 (Structure of a minimising cone in the plane). *Let $C \subset \mathbb{R}^2$ be a nontrivial locally perimeter-minimising cone. Then the link $\Gamma := \partial C \cap S^1$ is a finite set, ∂C in the punctured plane is exactly the union of the corresponding rays, and $\#\Gamma$ is even.*

Proof. Replace C by its density-one representative. Cover the annulus $B_2 \setminus \overline{B_{1/2}}$ by finitely many polar-coordinate sectors on which $(r, \theta) \mapsto r(\cos \theta, \sin \theta)$ and its inverse are bi-Lipschitz. Lemma 7.4 puts the pullback of $\mathbf{1}_C$ in BV on every such rectangle. Cone invariance makes this pullback independent of r a.e.; Lemma 5.7 therefore shows that its angular factor is a $\{0, 1\}$ -valued BV function on every coordinate arc. Compatibility on overlaps gives a BV function on the compact circle S^1 .

A one-dimensional $\{0, 1\}$ -valued BV function has only finitely many jumps: each jump has variation one, while its total variation is finite. Its jump set is precisely $\Gamma = \partial C \cap S^1$. Indeed, away from a jump the angular factor is a.e. constant on an arc, and cone invariance makes the corresponding open sector one phase; at a jump both phases occupy sectors of positive density. Dilation then shows that the punctured boundary is exactly the union of the rays through these jump points. Traversing the circle, the two values alternate at successive jumps and return to the starting value, so the number of jumps is even. No rectifiable-set coarea theorem or curvature argument is used. $\square$

Lemma 25.2 (Two-dimensional minimising cones are halfplanes). *A nontrivial locally perimeter-minimising cone in $\mathbb{R}^2$ is a halfplane.*

Proof. *Two rays.* If there are exactly two rays with angle $\alpha \neq \pi$, replace the two radial segments in B_1 by the chord joining their endpoints. The competitor has the same trace on ∂B_1 and its added interface has length $2 \sin(\alpha/2) < 2$, contradicting minimality.

Four or more rays. If there are $2m \geq 4$ rays, enumerate their endpoints cyclically and pair 1 with 2, 3 with 4, and so on. The corresponding chords are disjoint, each lying in the closed sector between consecutive rays. Fill the alternating caps so that the trace on ∂B_1 is unchanged. No chord is longer than the two radial segments it replaces, and at least one is strictly shorter:

equality would require a paired arc of angle π , and not all paired arcs can have angle π when there are at least four rays. Hence the total perimeter strictly decreases, a contradiction.

In both comparisons the competitor is understood to agree with C outside B_1 . Its symmetric difference is compactly contained in, say, B_2 ; apply local minimality there and use locality to cancel the unchanged annulus. Thus exactly two antipodal rays occur. $\square$

25.2 The three-dimensional case

Lemma 25.3 (Tangent cones of a cone are cylinders). *Let $C \subset \mathbb{R}^3$ be a nontrivial locally perimeter-minimising cone and $p \in \partial C \setminus \{0\}$. Then every tangent limit T of C at p is invariant under translations in the direction p , and there is a measurable $L \subset p^\perp$ with*

$$T = L \times \mathbb{R}p \quad \text{modulo null sets.} \quad (25.1)$$

Moreover L is a cone in $p^\perp$ centred at the origin.

Proof. Blow up at p along $r_j \downarrow 0$. Cone invariance gives, whenever $1 + r_j t > 0$,

$$\mathbf{1}_C(p + r_j(z + tp)) = \mathbf{1}_C\left(p + \frac{r_j}{1 + r_j t}z\right), \quad (25.2)$$

and $(1 + r_j t)^{-1} \rightarrow 1$. Translation and dilation continuity in L^1_{loc} therefore show T is invariant under translations in $\mathbb{R}p$. Choosing coordinates with $p = e_3$, invariance means $\mathbf{1}_T(x', x_3 + t) = \mathbf{1}_T(x', x_3)$ for every t ; Fubini and the distributional identity $D_3 \mathbf{1}_T = 0$ give (25.1). That T is itself a cone centred at 0 is Proposition 22.9 applied at p , whose almost-monotonicity error vanishes because C is exactly minimising; hence L is a cone in $p^\perp$. $\square$

Lemma 25.4 (Product competitor and descent of minimality). *Let $T = L \times \mathbb{R}$ be locally perimeter minimising in $\mathbb{R}^3$. Then L is locally perimeter minimising in $\mathbb{R}^2$.*

Proof. Testing $D\mathbf{1}_T$ with product test fields on a unit vertical interval first shows that L has locally finite perimeter; equivalently this is the product case of the BV slicing identity. Let $A \subset \mathbb{R}^2$ have locally finite perimeter and $A \triangle L \Subset B_R^2$. For $T_0 > 1$ define the explicit competitor

$$F_{T_0} := (A \times (-T_0, T_0)) \cup (L \times ((-\infty, -T_0] \cup [T_0, \infty))), \quad (25.3)$$

so $F_{T_0} \triangle T \Subset B_R^2 \times (-T_0 - 1, T_0 + 1)$. Put $Q_{T_0} := B_{R+1}^2 \times (-T_0 - 1, T_0 + 1)$. The BV product and gluing formulas give the finite local identity

$$P(F_{T_0}; Q_{T_0}) - P(T; Q_{T_0}) = 2T_0(P_2(A; B_{R+1}^2) - P_2(L; B_{R+1}^2)) + 2|A \triangle L|. \quad (25.4)$$

The last term is the pair of horizontal jump interfaces at heights $\pm T_0$; it is *independent* of T_0 . Since the symmetric difference is compactly contained in Q_{T_0} , apply local minimality in an enclosing ball and use locality to cancel the region where the two sets agree. This gives the corresponding inequality on Q_{T_0} ; division by $2T_0$ and $T_0 \rightarrow \infty$ give $P_2(L; B_{R+1}^2) \leq P_2(A; B_{R+1}^2)$. Since $A = L$ on the annulus $B_{R+1}^2 \setminus B_R^2$, this is exactly $P_2(L; B_R^2) \leq P_2(A; B_R^2)$. $\square$

Regression test 25.5 (Horizontal cost). (25.4) must have horizontal cost exactly $2|A \triangle L|$. Getting this factor wrong (e.g. $|A \triangle L|$ or $4|A \triangle L|$) still gives a T_0 -independent term, so the limit argument still works and the error is invisible. A regression test on (25.4) is therefore worth writing.

Lemma 25.6 (Smoothness of the punctured cone). *Let $C \subset \mathbb{R}^3$ be a nontrivial locally perimeter-minimising cone. Then $\partial C \setminus \{0\}$ is a smooth embedded surface with zero mean curvature.*

Proof. By Lemmas 25.3, 25.4 and 25.2, every tangent cone of C at every nonzero boundary point is a halfspace; the empty and full planar factors are excluded by the two-sided density estimate at the tangent-cone vertex. Let C_j denote the rescalings along a subsequence converging to such a halfspace, and let ν be its normal. Perimeter-measure convergence from Lemma 22.8 and Lemma 22.6, applied to a continuous cutoff equal to one on a smaller cylinder, show that $\int |\nu_{C_j} - \nu|^2 d|D\mathbf{1}_{C_j}| \rightarrow 0$ there. Rescaling back, the cylindrical excess of C is therefore arbitrarily small at a sufficiently small scale around any $p \neq 0$. Theorem 24.31 (with $\omega = 0$) then makes $\partial C \setminus \{0\}$ a $C^{1,1/2}$ embedded surface. Its weak mean curvature vanishes by (22.1) with $\omega = 0$. The Campanato/Schauder package of Chapter 10 (Theorems 10.27 and 10.29), which is logically prior to this chapter, makes it $C^{2,\alpha}$ with zero right side, and the mean curvature then vanishes pointwise. Smoothness follows by iterating the Schauder estimate to all orders. $\square$

Formalisation note. The formalisation proves the $C^{2,\alpha}$ regularity and the pointwise zero-mean-curvature equation here, proves Theorem 25.8 from that alone (dilation invariance forces the graph to be affine), and then obtains the smoothness asserted in this lemma a posteriori: ∂C is a plane.

Lemma 25.7 (The link is a single great circle). *With C as in Lemma 25.6, the link $\Gamma := \partial C \cap \mathbb{S}^2$ is a single great circle.*

Proof. By Lemma 25.6, Γ is a smooth compact embedded one-manifold, hence a finite disjoint union of circles. To obtain its Euler equation, let Y be a smooth tangent vector field on $\mathbb{S}^2$, let $\eta \in C_c^\infty((1/2, 2))$, and use the ambient variation $X(r\theta) := \eta(r)Y(\theta)$ in (22.1) with $\omega = 0$. The first variation of the conical surface factors as

$$0 = \int_{1/2}^2 \eta(r) dr \int_{\Gamma} \operatorname{div}_{\Gamma} Y ds. \quad (25.5)$$

Choosing η with nonzero integral shows the geodesic-curvature vector of each component vanishes. If a component is parametrised by unit speed γ , its acceleration in $\mathbb{R}^3$ is normal to the sphere, so $\gamma'' = -\gamma$ and hence $\gamma(t) = a \cos t + b \sin t$ for orthonormal a, b : its image is a great circle. Two distinct great circles intersect, whereas distinct components of the embedded link are disjoint. So Γ is one great circle. $\square$

Formalisation note. The formalisation proves Theorem 25.8 directly from the $C^{2,\alpha}$ regularity of Lemma 25.6 and derives this lemma as a corollary (local flatness of ∂C plus a local-to-global great-circle argument); the endgame of the argument above, that a nonempty family of pairwise disjoint great circles is a single great circle, is formalised as well.

Theorem 25.8 (Three-dimensional minimising cones are halfspaces). *Every nontrivial locally perimeter-minimising cone in $\mathbb{R}^3$ is a halfspace.*

Proof. ∂C is the plane spanned by the great circle of Lemma 25.7, so C is one of the two halfspaces. $\square$

Chapter 26

Elimination of the singular set

Lean module: NoCompromise.Cones.NoSingular

Proposition 26.1 (Every boundary point is regular). *Let E be ω -minimal with $|E| < \infty$. Then every point of $\partial\Omega$ is regular; $\partial\Omega$ is a compact embedded $C^{1,1/2}$ surface with finitely many connected components; and*

$$\Omega = \text{int } \overline{\Omega}, \quad P(\Omega) = \mathcal{H}^2(\partial\Omega). \quad (26.1)$$

Proof. Fix $x \in \partial\Omega$. Lemma 22.8 and Proposition 22.9 supply a locally perimeter-minimising tangent cone; Theorem 25.8 says it is a halfspace. Perimeter-measure convergence along the tangent sequence and Lemma 22.6 imply that the cylindrical excess relative to the limiting halfspace tends to zero, so at some scale (24.53) holds and Theorem 24.31 gives x a $C^{1,1/2}$ graph neighbourhood. Since x was arbitrary there are no singular points.

Compactness of $\partial\Omega$ (Lemma 21.9) gives a finite graph cover, whence (26.1) by Corollary 8.14. Finally $\partial\Omega$ has finitely many components: the components of a manifold are open in the manifold, and compactness extracts a finite subcover from the component cover. $\square$

Chapter 27

The isoperimetric equality case

Lean module: `NoCompromise.Isoperimetric.Rigidity`

Theorem 27.1 (Isoperimetric rigidity). *Let $E \subset \mathbb{R}^3$ have $0 < |E| < \infty$, finite perimeter, and $P(E) = c_I |E|^{2/3}$. Then E is, up to a null set, a translate of a ball of volume $|E|$. Conversely balls give equality.*

Proof. Step 1: E is ω -minimal. Equality makes E a perimeter minimiser among sets of its volume. Apply the proof of Lemma 20.2 with the Coulomb term deleted — i.e. with $\mathcal{E}$ replaced by P — which gives a finite volume-penalty constant, and then the proof of Lemma 20.4 with the Coulomb Lipschitz term deleted. Thus the generic ω -minimal regularity theory applies with no liquid-drop term built into it.

Step 2: a bounded smooth representative. Chapters 21–26 apply verbatim and give a bounded representative Ω with compact $C^{1,1/2}$ boundary; the bootstrap of §28.3 with v_Ω replaced by 0 makes it $C^{3,\beta}$, which is all that Steps 3–5 need: the ABP chain of Proposition 13.4 and Corollary 13.6 uses only a C^3 boundary (see the formalisation note after Proposition 13.4). *These chapters use only the inequality (13.3), never this theorem.*

Step 3: one component. Apply Corollary 13.6; by Lemma 13.5 equality leaves exactly one component.

Step 4: equality in the ABP chain. In the chain of Proposition 13.4 every integrand inequality is nonnegative, so equality of the endpoints implies

- $|\nabla z(\Gamma) \setminus B_1| = 0$;
- $|G \setminus \Gamma| = 0$, because $\Delta z/3 > 0$;
- $D^2 z = (\Delta z/3)I$ a.e. on G , by equality in the arithmetic–geometric mean inequality for a positive semidefinite matrix.

Step 5: conclude. Smoothness extends the Hessian identity everywhere, so $\nabla z(x) = \alpha x + b$ with $\alpha = P(E)/(3|E|) > 0$. The affine image $\nabla z(G)$ is open, contains B_1 , and has the same volume as B_1 ; an open proper superset would have strictly larger volume, so $\nabla z(G) = B_1$ and $G = \alpha^{-1}(B_1 - b)$ is a ball. The converse is Lemma 16.2 and a direct computation. $\square$

Chapter 28

The Euler–Lagrange equation and the smooth bootstrap

Lean module: `NoCompromise.Energy.PotentialHolder`, `NoCompromise.Stationary.Bootstrap`,
`NoCompromise.Stationary.Connected`, `NoCompromise.Stationary.Defs`,
`NoCompromise.Stationary.EulerLagrange`, `NoCompromise.Stationary.MinimizerContext`,
`NoCompromise.Stationary.ScalingIdentity`, `NoCompromise.Variation.CoulombBoundary`

Notation 28.1. For the rest of Part IV, E is a fixed-volume minimiser at volume $V > 0$ and $\Omega := E^{(1)}$ is its bounded open representative with compact $C^{1,1/2}$ boundary, as produced by Lemmas 20.4, 21.9 and Proposition 26.1. ν denotes the continuous outward normal (Convention 2.3).

28.1 The nonlocal first variation

Lemma 28.2 (First variation of the Coulomb energy). *Let Ω be a bounded C^1 domain and $X \in C_c^\infty(\mathbb{R}^3; \mathbb{R}^3)$. Then*

$$\left. \frac{d}{dt} \right|_{t=0} D(F_t^X(\Omega)) = \int_{\partial\Omega} v_\Omega X \cdot \nu \, d\mathcal{H}^2. \quad (28.1)$$

Proof. Pull both integrations back to Ω :

$$D(F_t^X(\Omega)) = \frac{1}{2} \iint_{\Omega \times \Omega} \frac{JF_t^X(x) JF_t^X(y)}{|F_t^X(x) - F_t^X(y)|} \, dx \, dy.$$

Differentiating the Jacobians and the kernel gives an integrand bounded by

$$C \left(\frac{1}{|x - y|} + \frac{|X(x) - X(y)|}{|x - y|^2} \right) \leq \frac{C'}{|x - y|},$$

since X is Lipschitz; this majorant is integrable on the bounded product $\Omega \times \Omega$. Dominated convergence and exchange of x, y yield

$$\begin{aligned} \left. \frac{d}{dt} \right|_0 D(F_t^X(\Omega)) &= \iint_{\Omega \times \Omega} \left(\frac{\operatorname{div} X(x)}{|x - y|} - \frac{(x - y) \cdot X(x)}{|x - y|^3} \right) dx \, dy \\ &= \iint_{\Omega \times \Omega} \operatorname{div}_x \left(\frac{X(x)}{|x - y|} \right) dx \, dy. \end{aligned} \quad (28.2)$$

For the last step, fix $y \in \Omega$ outside a null set and apply Corollary 10.6(ii) to $W_y(x) := X(x)/|x - y|$, obtaining (10.5). Both sides of (10.5) are integrable in y : the volume integrand is bounded by

$C(|x-y|^{-1} + |x-y|^{-2})$ and the boundary kernel is integrable on $\partial\Omega \times \Omega$ in local graph coordinates. Fubini gives (28.1). $\square$

Formalisation note 28.3. The boundary is only $C^{1,1/2}$ at this stage, so the proof must not use a smooth tubular normal graph. Straight ambient perturbations (Definition 9.4) suffice, and Corollary 10.6 is stated for C^1 domains for exactly this reason.

Theorem 28.4 (Constrained first variation). *There is $\lambda \in \mathbb{R}$ such that for every $X \in C_c^\infty(\mathbb{R}^3; \mathbb{R}^3)$,*

$$\int_{\partial\Omega} \operatorname{div}_\tau X \, d\mathcal{H}^2 = \int_{\partial\Omega} (\lambda - v_\Omega) X \cdot \nu \, d\mathcal{H}^2. \quad (28.3)$$

Explicitly, with

$$A(X) := \int_{\partial\Omega} (\operatorname{div}_\tau X + v_\Omega X \cdot \nu), \quad B(X) := \int_{\partial\Omega} X \cdot \nu,$$

one has $A(X) = \lambda B(X)$ with $\lambda = A(Y)/B(Y)$ for any Y as below.

Proof. Choose a compactly supported smooth field Y equal to x on a neighbourhood of $\bar{\Omega}$. By Corollary 10.6(i),

$$B(Y) = \int_{\partial\Omega} Y \cdot \nu = 3V \neq 0. \quad (28.4)$$

For a given X consider $E_{t,s} := F_s^Y(F_t^X(\Omega))$. By Theorem 9.8 its volume is C^1 in (t, s) with nonzero s -derivative at the origin, so the implicit function theorem (B6) gives a C^1 function $s = s(t)$ with $s(0) = 0$ and $|E_{t,s(t)}| = V$. Differentiating minimality at $t = 0$ and eliminating $s'(0)$ gives $A(X) - \lambda B(X) = 0$ with $\lambda := A(Y)/B(Y)$; using Theorems 9.7, 9.8 and Lemma 28.2 to identify A and B gives (28.3). $\square$

Remark 28.5. In a graph chart, vertical test fields reduce (28.3) to the weak prescribed-mean-curvature equation (28.7) below. No smooth collar is required at any point.

28.2 Regularity of the Newtonian potential

Lemma 28.6 ($C^{1,\beta}$ regularity of v_Ω). *For bounded measurable $\Omega \subset \mathbb{R}^3$,*

$$v_\Omega \in C^{1,\beta}(\mathbb{R}^3) \quad \text{for every } \beta < 1. \quad (28.5)$$

Proof. Differentiation under the integral is valid because $|x-y|^{-2}$ is locally integrable:

$$\nabla v_\Omega(x) = - \int_{\Omega} \frac{x-y}{|x-y|^3} \, dy.$$

Continuity follows by splitting the integral into a small ball and its complement. For the Hölder estimate let $h := |x-z| < 1/4$. On $B_{2h}(x) \cup B_{2h}(z)$ the integral of either gradient kernel is $O(h)$. On the complement, the mean value theorem for $K(\xi) := \xi/|\xi|^3$ gives $|K(x-y) - K(z-y)| \leq Ch|x-y|^{-3}$, and since Ω is bounded, integrating over dyadic annuli gives

$$|\nabla v_\Omega(x) - \nabla v_\Omega(z)| \leq Ch(1 + |\log h|). \quad (28.6)$$

For every $\beta < 1$, $h(1 + |\log h|) \leq C_\beta h^\beta$. $\square$

Formalisation note 28.7 (No Calderón–Zygmund). No second-derivative L^p theorem is needed anywhere. Lemma 28.6 is proved directly, and §32.1 uses tubular cutoffs rather than a Calderón–Zygmund estimate. Note also the logarithm in (28.6): v_Ω is *not* claimed to be $C^{1,1}$, and no later step requires it.

28.3 The smooth bootstrap

Lemma 28.8 (The graph equation). *In a graph chart $x_3 = f(x')$ over a planar disk, (28.3) implies the weak equation*

$$\operatorname{div}\left(\frac{\nabla f}{\sqrt{1+|\nabla f|^2}}\right) = \pm(\lambda - v_\Omega(x', f(x'))), \quad (28.7)$$

where the sign depends only on which side of the chart is Ω . Only this direction is used later.

Proof. Use a vertical field $X = \zeta(x')\eta(x_3)e_3$ with $\eta \equiv 1$ near the graph. Writing $Q := \sqrt{1+|\nabla f|^2}$, direct differentiation gives $\operatorname{div}_\tau X = (\nabla f \cdot \nabla \zeta)/Q^2$, while $d\mathcal{H}^2 = Q dx'$ and $X \cdot \nu d\mathcal{H}^2 = \pm \zeta dx'$. Substitution in (28.3) and planar integration by parts give (28.7). $\square$

Lemma 28.9 (Uniform ellipticity of the mean-curvature operator). *For*

$$a_{ij}(p) := \frac{\delta_{ij}}{\sqrt{1+|p|^2}} - \frac{p_i p_j}{(1+|p|^2)^{3/2}}$$

and $|p| \leq M$, the matrix $(a_{ij}(p))$ is uniformly elliptic with constants depending only on M , and $p \mapsto a_{ij}(p)$ is smooth.

Proof. Diagonalise in the frame $\{p/|p|\} \cup p^\perp$: the eigenvalues are $(1+|p|^2)^{-3/2}$ and $(1+|p|^2)^{-1/2}$. $\square$

Proposition 28.10 ($C^{2,\beta}$ from the quasilinear Schauder lemma). *With f as in Lemma 28.8 and $f \in C^{1,1/2}$, one has $f \in C^{2,1/2}$; and then $f \in C^{2,\beta}$ for every $\beta < 1$.*

Proof. The $C^{1,1/2}$ regularity from Theorem 24.31 makes $a_{ij}(\nabla f)$ uniformly elliptic with $C^{0,1/2}$ coefficients (Lemma 28.9), and the right side of (28.7) is $C^{0,1/2}$ by Lemma 28.6. Apply Theorem 10.29 with $\alpha = 1/2$ to get $f \in C^{2,1/2}$. For its full-norm estimate, $\|f\|_\infty$ is finite on the fixed smaller graph chart; it is retained as an input, while the derivative estimate is independent of an additive height constant. Then ∇f is Lipschitz and $v_\Omega \in C^{1,\beta}$, so the coefficients and the right side are $C^{0,\beta}$ for every $\beta < 1$; reapplying Theorem 10.29 gives $f \in C^{2,\beta}$. $\square$

Proposition 28.11 (The last derivative). *With f as in Proposition 28.10, $f \in C^{3,\beta}$ for every $\beta < 1$. Consequently*

$$\partial\Omega \in C^{3,\beta} \quad \text{for every } \beta < 1. \quad (28.8)$$

Proof. One more derivative requires a separate line of argument: the divergence-form estimate alone does not automatically add it. Put

$$a_{ij}(x) := D_j A_i(\nabla f(x)), \quad g(x) := \pm(\lambda - v_\Omega(x, f(x))),$$

where $A_i(p) := p_i/\sqrt{1+|p|^2}$. Then $a_{ij} \in C^{1,\beta}$ and $g \in C^{1,\beta}$ by Propositions 28.10 and Lemma 28.6, and for $w_k := \partial_k f$, differentiating the weak equation gives

$$\partial_i(a_{ij}\partial_j w_k) = \partial_k g. \quad (28.9)$$

Expanding the divergence — legitimate now because $a \in C^{1,\beta}$ and $w_k \in C^{1,\beta}$ — yields the nondivergence equation

$$a_{ij}\partial_{ij} w_k = \partial_k g - (\partial_i a_{ij})\partial_j w_k, \quad (28.10)$$

whose right side is $C^{0,\beta}$. Theorem 10.21, including its harmless bounded first-order term, gives $w_k \in C^{2,\beta}$. $\square$

Corollary 28.12 (Pointwise Euler–Lagrange equation). *On $\partial\Omega$,*

$$H + v_\Omega = \lambda. \quad (28.11)$$

Proof. By (28.8) the weak equation (28.7) may be expanded pointwise, and the left side is H in the sign convention of Convention 2.4. $\square$

28.4 Connectedness and the scaling identity

Proposition 28.13 (Connectedness). *Ω is connected.*

Proof. The compact smooth boundary has finitely many components (Proposition 26.1), so the bounded open set Ω has finitely many connected components. If Ω were disconnected, write $\Omega = A \dot{\cup} B$ with A, B nonempty unions of components. Their closures are disjoint: a common boundary point would have a graph neighbourhood in which the interior side is connected, forcing the two components to coincide. Compactness then gives positive separation.

Translate B by te . For all large t , Lemma 4.8 gives $P(A \cup (B + te)) = P(A) + P(B) = P(\Omega)$, and the two self-interactions are unchanged. The original energy contains the strictly positive cross-term $I(A, B)$ (Lemma 15.10), while $I(A, B + te) \rightarrow 0$ (Lemma 15.11). A large translation is therefore a same-volume competitor of strictly lower energy, contradicting Definition 15.3. $\square$

Proposition 28.14 (Scaling identity for the multiplier).

$$3V\lambda = 2P(\Omega) + 5D(\Omega) = 5\mathcal{E}(\Omega) - 3P(\Omega). \quad (28.12)$$

Proof. Choose a compactly supported smooth field X equal to x on a neighbourhood of the bounded set $\bar{\Omega}$; the compact cutoff is needed because x itself is not compactly supported. Its straight perturbation $x \mapsto x + tX(x)$ agrees there exactly with dilation by $1 + t$. By (15.7), $P(r\Omega) = r^2P(\Omega)$, $D(r\Omega) = r^5D(\Omega)$ and $|r\Omega| = r^3V$, so stationarity of $\mathcal{E} - \lambda|\cdot|$ at $r = 1$ — i.e. (28.3) with this X — gives $2P(\Omega) + 5D(\Omega) = 3V\lambda$. The second form follows from $\mathcal{E} = P + D$. $\square$

Definition 28.15 (Stationary domain). A *stationary domain of volume V with multiplier λ* is a bounded connected open $\Omega \subset \mathbb{R}^3$ with $|\Omega| = V$, boundary of class C^3 , satisfying (28.11) pointwise on $\partial\Omega$.

Corollary 28.16 (Minimisers are stationary). *Every fixed-volume minimiser at volume $V > 0$ has a representative which is a stationary domain of volume V , and it satisfies (28.12).*

Proof. Collect Lemma 21.9, Propositions 26.1, 28.11, 28.13, Corollary 28.12 and Proposition 28.14. $\square$

Part V

The hull, the exterior potential, and the capacitary inequality

Chapter 29

The filled hull

Lean module: NoCompromise.Hull.Defs, NoCompromise.Hull.Properties,
NoCompromise.Stationary.Defs

Notation 29.1. Throughout Part V, Ω is a bounded connected $C^{3,\beta}$ stationary domain of volume V with multiplier λ (Definition 28.15), as produced by Corollary 28.16.

Definition 29.2 (Filled hull). Let U_∞ be the unbounded connected component of $\mathbb{R}^3 \setminus \overline{\Omega}$ and put

$$K := \mathbb{R}^3 \setminus U_\infty. \quad (29.1)$$

Thus K fills every bounded complementary component of Ω .

Lemma 29.3 (Hull properties). K is compact and regular closed, $K = \overline{\text{int } K}$; the sets $\text{int } K$ and $\mathbb{R}^3 \setminus K$ are connected; ∂K is connected; and

$$\overline{\Omega} \subset K, \quad \partial K \subset \partial\Omega. \quad (29.2)$$

Moreover ∂K is $C^{3,\beta}$, the outward normals and curvatures of K and Ω agree on ∂K , and

$$P(K) \leq P(\Omega). \quad (29.3)$$

Proof. Write $\partial\Omega = \Sigma_1 \dot{\cup} \dots \dot{\cup} \Sigma_k$ (Proposition 26.1). Theorem 14.14 says $\mathbb{R}^3 \setminus \partial\Omega$ has $k+1$ components. One of them is the connected set Ω (Proposition 28.13), so there are exactly k components of $\mathbb{R}^3 \setminus \overline{\Omega}$. Every Σ_i has one collar side in Ω and one in a complementary component (Theorem 14.12); every complementary component has nonempty boundary contained in $\partial\Omega$, so it meets at least one Σ_i . Hence the map sending Σ_i to the component on its non- Ω side is a surjection between finite sets of size k , therefore a bijection. Exactly one boundary component faces U_∞ ; call it Σ_∞ .

By construction $\partial K = \partial U_\infty = \Sigma_\infty$; all other boundary components become interior interfaces once their adjacent cavities are filled. The union of Ω , all bounded complementary components, and collars crossing their common interfaces is connected and equals $\text{int } K$, and the outer collar gives $K = \overline{\text{int } K}$. Normal agreement holds because both outward normals point into U_∞ . Finally $P(K) = \mathcal{H}^2(\Sigma_\infty) \leq \sum_i \mathcal{H}^2(\Sigma_i) = P(\Omega)$. $\square$

Formalisation note 29.4 (Representative bookkeeping). Five distinct objects are now in play: the measurable set E , its density-one open representative Ω , its closure $\overline{\Omega}$, the filled hull K , and the exterior $\mathbb{R}^3 \setminus K$. Confusing any two of them produces a false statement. A Lean development should give each a distinct name and prove the inclusions (29.2) once, as a reusable lemma, rather than re-deriving them in each consumer.

Chapter 30

The exterior capacitary potential

Lean module: NoCompromise.Capacity.Flux, NoCompromise.Capacity.Kelvin,
NoCompromise.Capacity.Levels, NoCompromise.Capacity.Potential

30.1 Existence, uniqueness and the Hopf sign

Theorem 30.1 (Capacitary potential). *There is a unique*

$$u \in C^\infty(\mathbb{R}^3 \setminus K) \cap C_{\text{loc}}^{2,\alpha}(\overline{\mathbb{R}^3 \setminus K}) \quad (0 < \alpha < 1)$$

with

$$\Delta u = 0 \text{ on } \mathbb{R}^3 \setminus K, \quad u = 1 \text{ on } \partial K, \quad u(x) \rightarrow 0 \text{ } (|x| \rightarrow \infty). \quad (30.1)$$

It satisfies

$$0 < u < 1 \text{ outside } K, \quad \partial_{\nu_K} u < 0 \text{ on } \partial K. \quad (30.2)$$

Extending u by the constant 1 on K gives a continuous, locally H^1 function on $\mathbb{R}^3$.

Proof. Translate so that $0 \in \text{int } K$ and choose

$$0 < a < R_0, \quad B_a \subset \text{int } K \subset K \subset B_{R_0}. \quad (30.3)$$

Step 1: annular minimisation. For $R > R_0$ let $A_R := B_R \setminus K$. In the affine Sobolev class with traces 1 on ∂K and 0 on ∂B_R , minimise $J(v) := \frac{1}{2} \int_{A_R} |\nabla v|^2$. A fixed smooth interpolant makes the class nonempty; (10.1) on the zero-trace difference space gives coercivity, and the direct method (Proposition 10.8) gives a unique minimiser u_R , weakly harmonic. Testing with $(u_R - 1)_+$ and $(u_R)_-$ gives $0 \leq u_R \leq 1$; Lemma 10.18 and Proposition 10.13 give strict inequalities inside. Boundary flattening and Theorem 10.32 give $C^{2,\alpha}$ regularity up to either boundary. Here A_R is connected: an exterior path may have every portion outside B_R replaced by a path on a sphere of radius in (R_0, R) , which is disjoint from K .

For all sufficiently large R , a single smooth function equal to one near K and zero outside B_{2R_0} is an admissible competitor. Consequently $\int_{A_R} |\nabla u_R|^2 \leq C(K)$ uniformly.

Step 2: monotone limit. If $R_2 > R_1$, comparison on A_{R_1} gives $u_{R_2} \geq u_{R_1}$, since the larger-domain solution is nonnegative on ∂B_{R_1} . The family is monotone and bounded by one; local elliptic estimates and monotone convergence give a harmonic limit u with *no subsequence needed*,

and the fixed inner-boundary estimates preserve the trace $u = 1$. Local weak convergence of the gradients and exhaustion pass the uniform energy bound to the limit:

$$\int_{\mathbb{R}^3 \setminus K} |\nabla u|^2 < \infty. \quad (30.4)$$

Step 3: barriers. The function $R_0/|x|$ is harmonic on the exterior (its pole lies in K), is ≥ 1 on ∂K , and tends to zero, so comparison gives

$$u(x) \leq \frac{R_0}{|x|}. \quad (30.5)$$

For a lower barrier on the finite annulus use

$$b_R(r) := \frac{a(R/r - 1)}{R - a}, \quad (30.6)$$

harmonic for $a < r < R$, equal to 1 at $r = a$ and 0 at $r = R$. Since $|x| \geq a$ on ∂K , $b_R(|x|) \leq 1 = u_R$ there, and both vanish on ∂B_R ; comparison gives $u_R \geq b_R$, and letting $R \rightarrow \infty$,

$$\frac{a}{|x|} \leq u(x) \quad (|x| \geq R_0). \quad (30.7)$$

So $u \rightarrow 0$ and $u \not\equiv 0$.

Step 4: uniqueness. For two solutions apply Lemma 10.11 to their difference on $B_R \setminus K$; the boundary difference on ∂B_R tends uniformly to zero.

Step 5: the Hopf sign. Corollary 10.16, using an exterior tangent ball at $p \in \partial K$ and the radial barrier (10.12), gives $\partial_{\nu_K}(1 - u) > 0$. $\square$

Notation 30.2 (w). $w := |\nabla u|$. On ∂K ,

$$\nabla u = -w \nu_K, \quad w > 0. \quad (30.8)$$

30.2 Capacity, flux and Kelvin asymptotics

Definition 30.3 (Capacity).

$$\text{Cap}(K) := \frac{1}{4\pi} \int_{\mathbb{R}^3 \setminus K} |\nabla u|^2, \quad \mathbf{C} := \text{Cap}(K). \quad (30.9)$$

Lemma 30.4 (Decay). $u(x) = O(|x|^{-1})$ and $|\nabla u(x)| = O(|x|^{-2})$.

Proof. (30.5) rescaled on dyadic annuli combined with the interior gradient estimate (10.10). $\square$

Lemma 30.5 (Flux identity).

$$4\pi \mathbf{C} = \int_{\partial K} w \, d\mathcal{H}^2, \quad (30.10)$$

and for every regular value $s \in (0, 1)$,

$$\int_{\{u=s\}} w \, d\mathcal{H}^2 = 4\pi \mathbf{C}. \quad (30.11)$$

Proof. Integrate by parts on $B_R \setminus K$ using (10.4) and let $R \rightarrow \infty$, using Lemma 30.4: this gives (30.10). Applying (10.4) on $\{s < u < 1\}$ instead gives (30.11); regular values are dense by Theorem 11.5. $\square$

Lemma 30.6 (Kelvin transform and the multipole expansion). *The Kelvin transform*

$$\tilde{u}(z) := |z|^{-1}u\left(\frac{z}{|z|^2}\right) \quad (30.12)$$

extends smoothly across the origin, so $\tilde{u}(z) = C_\infty + a \cdot z + O(|z|^2)$ and, transforming back with differentiated remainders,

$$u(x) = \frac{C_\infty}{|x|} + \frac{a \cdot x}{|x|^3} + O(|x|^{-3}), \quad (30.13)$$

$$\nabla u(x) = -C_\infty \frac{x}{|x|^3} + O(|x|^{-3}), \quad (30.14)$$

$$D^2 u(x) = C_\infty \left(3 \frac{x \otimes x}{|x|^5} - \frac{I}{|x|^3} \right) + O(|x|^{-4}). \quad (30.15)$$

Moreover $C_\infty = C$.

Proof. $\tilde{u}$ is bounded and harmonic on a punctured ball by Lemma 30.4; the removability argument of Proposition 10.37 — test against a cutoff vanishing on B_ε , bound the boundary error by $\sup \tilde{u}$ times the capacity of B_ε — makes it distributionally harmonic, and difference-quotient interior regularity makes it smooth. Taylor's theorem then gives the expansion, and differentiating the explicit Kelvin transformation gives (30.13)–(30.15). The inward flux through a large sphere is $4\pi C_\infty + o(1)$; comparing with (30.10) gives $C_\infty = C$. $\square$

Lemma 30.7 (Asymptotic level geometry). *For small regular $\varepsilon > 0$ the level $\{u = \varepsilon\}$ is a radial graph $r_\varepsilon(\theta) = C/\varepsilon + O(1)$ with two angular derivatives controlled uniformly, and with the normal toward infinity,*

$$w = \frac{\varepsilon^2}{C} + O(\varepsilon^3), \quad H = \frac{2\varepsilon}{C} + O(\varepsilon^2), \quad d\mathcal{H}^2 = \frac{C^2}{\varepsilon^2}(1 + O(\varepsilon))d\omega. \quad (30.16)$$

Consequently

$$\int_{\{u=\varepsilon\}} (Hw - 4\varepsilon^{-1}w^2) = -8\pi\varepsilon + O(\varepsilon^2). \quad (30.17)$$

Proof. The implicit function theorem applied to $u(r\theta) = \varepsilon$ using (30.13)–(30.15). $\square$

Formalisation note 30.8 (Off the critical path). Lemma 30.7 is *not used* below. Chapter 31 needs the far-field normalisation in the translated coordinates of Lemma 31.2 and re-derives exactly what it needs as Lemma 31.5; in particular (30.17) is the untranslated precursor of (31.6). Lemma 30.7 is retained because it is the natural place to record the level geometry, and because comparing it with Lemma 31.5 is a useful consistency check (Regression test 39.5). A formalisation may skip it.

30.3 Connectedness of regular levels

Lemma 30.9 (Regular levels are connected). *Every regular level $\{u = s\}$, $0 < s < 1$, is connected.*

Proof. Extend u by 1 on K . The superlevel $\{u > s\}$ is bounded by (30.5). Every one of its components meets K : otherwise u would be harmonic on a bounded component, equal to s on its boundary and $> s$ inside, contradicting Proposition 10.13. Since K is connected (Lemma 29.3) it lies in one component, so the whole superlevel is connected.

The sublevel $\{u < s\}$ contains the complement of a large ball by (30.5). A bounded component would have boundary value s and an interior minimum below s , contradicting the minimum principle. So the sublevel is connected. These are exactly the two complementary components of the regular level. That level is a compact smooth surface with finitely many, say k , connected components; Theorem 14.14 says its complement has $k + 1$ components. Hence $k + 1 = 2$ and $k = 1$. $\square$

Chapter 31

Euclidean closure of the capacity level-set argument

Lean module: `NoCompromise.CapacitaryK.Asymptotics`, `NoCompromise.CapacitaryK.Inequality`, `NoCompromise.CapacitaryK.PositiveMeasure`, `NoCompromise.CapacitaryK.Representatives`, `NoCompromise.CapacitaryK.SlabIdentities`

This chapter replaces any informal instruction to “pass through the critical levels”. The argument is entirely Euclidean: it uses the positive measure $\Delta|\nabla u|$, the weighted coarea formula (5.5), and a scalar measure inequality, and it retains every critical-level contribution as a nonnegative singular measure. *No Riemannian or conformal geometry appears anywhere.*

Convention 31.1 (Level-set orientation). The outward unit normal of the superlevel set $\{u \geq t\}$ is $\nu := -\nabla u/|\nabla u|$, and $H := \operatorname{div}_{\Sigma_t} \nu$, so a round sphere has $H > 0$. This is consistent with Convention 2.4.

31.1 Far-field normalisation

Lemma 31.2 (Normalised expansion at infinity). *After translating the origin by a/C (with a from Lemma 30.6), one has*

$$u(r, \theta) = \frac{C}{r} + \frac{Q(\theta)}{r^3} + O(r^{-4}), \quad \int_{\mathbb{S}^2} Q \, d\theta = 0, \quad (31.1)$$

where Q is the restriction to $\mathbb{S}^2$ of a homogeneous harmonic polynomial of degree two.

Proof. Expand the Kelvin transform in its convergent Taylor series; harmonicity makes every homogeneous Taylor term harmonic, and a homogeneous harmonic polynomial of positive degree has zero spherical mean. Translating the origin removes the degree-one term, which is the only term that could produce a relative correction of order t . $\square$

Lemma 31.3 (Level asymptotics from (31.1)). *For small $t > 0$ the level $\{u = t\}$ is the radial graph*

$$r_t(\theta) = \frac{C}{t} \left(1 + \frac{Q(\theta)}{C^3} t^2 + O(t^3) \right), \quad (31.2)$$

and uniformly in θ ,

$$|\nabla u| = \frac{t^2}{C} \left(1 + \frac{Q(\theta)}{C^3} t^2 + O(t^3) \right), \quad (31.3)$$

$$d\mathcal{H}_t^2 = \frac{C^2}{t^2} \left(1 + 2 \frac{Q(\theta)}{C^3} t^2 + O(t^3) \right) d\theta. \quad (31.4)$$

The remainders $O(t^3)$ are uniform in θ together with their first and second θ -derivatives. Consequently, with H the mean curvature of the level and $p(t) := \int_{\{u=t\}} H |\nabla u| d\mathcal{H}^2$,

$$p(t) = 8\pi t + O(t^4) \quad (t \downarrow 0), \quad (31.5)$$

obtained by integrating the expansions; the curvature of the radial graph involves only θ -derivatives of r_t at fixed t .

Proof. The implicit function theorem for large r gives (31.2); differentiating (31.1), substituting (31.2), and using that tangential derivatives first enter quadratically give (31.3)–(31.4). $\square$

Definition 31.4 (The level functionals). For a regular value $t \in (0, 1)$ put

$$F(t) := \int_{\{u=t\}} w^2 d\mathcal{H}^2.$$

(Definition 31.14 extends F to all t .)

Lemma 31.5 (Far-field normalisation of F). $F(t) = 4\pi t^2 + O(t^5)$ and $F'(t) = 8\pi t + O(t^4)$ for small t ; in particular

$$\lim_{t \downarrow 0} \frac{F(t) - 4\pi t^2}{t^4} = 0, \quad \lim_{t \downarrow 0} \left(F'(t) - \frac{4F(t)}{t} + 8\pi t \right) = 0. \quad (31.6)$$

Proof. Multiply (31.3) squared by (31.4) and integrate: the first correction has zero spherical mean by (31.1), so it cancels. All sufficiently small levels are regular, so $F' = p$ there by the slab identity, and the derivative estimate is (31.5); the $O(t^5)$ bound on $F - 4\pi t^2$ alone would not control the derivative. $\square$

Formalisation note. The formalisation establishes (31.5) by the volume route $p(t) = \mu\{0 < u \leq t\} = \int_{\{0 < u \leq t\}} \Delta |\nabla u| dx$, with Bochner's identity $\Delta |\nabla u| = (|D^2 u|^2 - |\nabla |\nabla u||^2) / |\nabla u|$ for harmonic u , the second-order expansion (31.1) with its gradient and Hessian remainders, and the zero spherical mean of Q ; the level expansions of Lemma 31.3 are proved as stated but are not needed for it.

31.2 The positive measure $\Delta |\nabla u|$

Lemma 31.6 (Bochner and positivity). Let $w_\varepsilon := (w^2 + \varepsilon^2)^{1/2}$. Then, where u is harmonic,

$$\Delta w_\varepsilon = \frac{|D^2 u|^2}{w_\varepsilon} - \frac{|D^2 u \nabla u|^2}{w_\varepsilon^3} \geq \frac{\varepsilon^2 |D^2 u|^2}{w_\varepsilon^3} \geq 0. \quad (31.7)$$

Proof. Direct computation using $\Delta u = 0$ and Cauchy–Schwarz $|D^2 u \nabla u|^2 \leq |D^2 u|^2 |\nabla u|^2$. $\square$

Proposition 31.7 ($\mu := \Delta w$ is a positive Radon measure). Δw is a positive distribution on $\mathbb{R}^3 \setminus K$ and hence a positive Radon measure μ there.

Proof. $w_\varepsilon \rightarrow w$ locally uniformly, so (31.7) makes Δw a positive distribution. A positive distribution has order zero: if ϕ is supported in a fixed compact set with $|\phi| \leq 1$, choose a cutoff $\eta \geq 1$ there and use $-\eta \leq \phi \leq \eta$ to bound $|\langle \Delta w, \phi \rangle|$ by $\langle \Delta w, \eta \rangle$. Then Theorem 3.4 (in its positive case) produces μ . $\square$

Lemma 31.8 (Level-set identities on the regular set). *At a point where $w > 0$, choose an orthonormal frame e_1, e_2, ν tangent-normal to the level surface. Since $\nabla u = -w\nu$ and $\Delta u = 0$,*

$$D^2 u(e_i, e_j) = -w A_{ij}, \quad D^2 u(e_i, \nu) = -e_i w, \quad D^2 u(\nu, \nu) = Hw, \quad \partial_\nu w = -Hw, \quad (31.8)$$

where A is the second fundamental form of the level set. Consequently the absolutely continuous density of μ on the regular set is

$$\Delta w = \frac{|D^2 u|^2 - |\nabla w|^2}{w} = w|A|^2 + \frac{|\nabla_\Sigma w|^2}{w}. \quad (31.9)$$

Proof. Differentiate $\nabla u = -w\nu$ and use $\Delta u = 0$ to eliminate the trace. $\square$

31.3 Two exact slab identities

Lemma 31.9 (Curvature slab identity). *Let $0 < a < b < 1$ be regular values. Then*

$$\mu(\{a < u < b\}) = \int_{\{u=b\}} Hw \, d\mathcal{H}^2 - \int_{\{u=a\}} Hw \, d\mathcal{H}^2. \quad (31.10)$$

Proof. Approximate w by w_ε in the smooth slab $\{a < u < b\}$, apply (10.4), and let $\varepsilon \downarrow 0$; formula (31.8) fixes the boundary signs via $\partial_\nu w = -Hw$. No mass is lost in the limit: the left side converges weakly as measures; because w is bounded away from zero in neighbourhoods of both regular boundary levels, w and $\mu = \Delta w$ are smooth there, so μ gives those two surfaces zero mass. The slab is therefore a continuity set for the limiting measure, while the boundary fluxes converge smoothly. $\square$

Lemma 31.10 (Flux slab identity). *Let $0 < a < b < 1$ be regular values. Then*

$$F(b) - F(a) = \int_{\{a < u < b\}} \nabla w \cdot \nabla u \, dx = \int_a^b \int_{\{u=t\}} Hw \, d\mathcal{H}^2 \, dt. \quad (31.11)$$

Proof. Apply (10.4) to $w\nabla u$: its normal flux is w^2 on the upper level and $-w^2$ on the lower level, and $\Delta u = 0$, giving the first equality. For the second use (5.5) on $\{w > 0\}$ together with $\partial_\nu w = -Hw$ from (31.8), and Lemma 31.11 to discard $\{w = 0\}$. $\square$

Lemma 31.11 (The gradient vanishes a.e. on the zero set). *Let $g \geq 0$ be locally Lipschitz on an open subset of $\mathbb{R}^n$. Then $\nabla g = 0$ a.e. on $\{g = 0\}$.*

Proof. Restrict to almost every coordinate line and apply the one-dimensional statement that a differentiable nonnegative function has derivative zero at a.e. one of its zeros. $\square$

31.4 Canonical BV representatives through critical values

Definition 31.12 (The BV representative p). Fix a small regular value t_0 . Define the right-continuous BV function

$$p(t) := p(t_0) + \begin{cases} \mu(\{t_0 < u \leq t\}), & t \geq t_0, \\ -\mu(\{t < u \leq t_0\}), & t < t_0, \end{cases} \quad p(t_0) := \int_{\{u=t_0\}} Hw d\mathcal{H}^2. \quad (31.12)$$

Lemma 31.13 (p agrees with the geometric integral). *At every regular value t ,*

$$p(t) = \int_{\{u=t\}} Hw d\mathcal{H}^2. \quad (31.13)$$

Proof. (31.10). □

Definition 31.14 (The canonical F). Put

$$\widehat{F}(t) := F(t_0) + \int_{t_0}^t p(s) ds. \quad (31.14)$$

By Lemma 31.10, $\widehat{F}(t) = \int_{\{u=t\}} w^2$ at every regular value. We henceforth write F for $\widehat{F}$, thereby defining F at critical values.

Proposition 31.15 (Distributional structure).

$$F \in W_{\text{loc}}^{1,1}(0,1), \quad F' = p \text{ a.e.}, \quad Dp = u_{\#}\mu, \quad (31.15)$$

where $u_{\#}\mu$ is the pushforward of μ under u .

Proof. The first two assertions are (31.14). For $Dp = u_{\#}\mu$: the identity holds on intervals with regular endpoints by (31.10), and such intervals generate the Borel sets because regular values are dense (Theorem 11.5); monotone convergence gives it for all Borel sets. □

Lemma 31.16 (Density of $u_{\#}\mu$ on the regular set). *On the open set of regular values,*

$$\frac{d(u_{\#}\mu)}{dt} = \int_{\{u=t\}} (|A|^2 + |\nabla_{\Sigma} \log w|^2) d\mathcal{H}^2, \quad (31.16)$$

and the part of $u_{\#}\mu$ carried by critical values is a nonnegative singular measure.

Proof. Apply (5.5) to the density (31.9), dividing by $|\nabla u| = w$. The remaining measure is positive by Proposition 31.7 and is carried by the critical values; those values are Lebesgue-null by Theorem 11.5, so this remainder is singular. □

31.5 The differential inequality as a measure inequality

Lemma 31.17 (The total-curvature input). *For every regular value $t \in (0,1)$,*

$$\int_{\{u=t\}} \kappa d\mathcal{H}^2 \leq 4\pi, \quad (31.17)$$

where κ is the Gauss curvature of the level surface.

Proof. Every regular level is a compact connected smooth embedded closed surface by Lemma 30.9; apply Theorem 14.43. $\square$

Proposition 31.18 (Measure differential inequality). *For a.e. regular t ,*

$$\frac{d(u_{\#}\mu)}{dt} \geq \int_{\{u=t\}} H^2 - 8\pi \geq \frac{(\int_{\{u=t\}} Hw)^2}{\int_{\{u=t\}} w^2} - 8\pi = \frac{p(t)^2}{F(t)} - 8\pi, \quad (31.18)$$

and consequently, as distributions on $(0, 1)$,

$$Dp \geq \left(\frac{p^2}{F} - 8\pi \right) dt. \quad (31.19)$$

Proof. By Convention 2.4, $|A|^2 = H^2 - 2\kappa$, so (31.16) and (31.17) give $\frac{d(u_{\#}\mu)}{dt} \geq \int H^2 - 8\pi$. Cauchy–Schwarz with weight w and $F = \int w^2$ give the middle inequality, and Lemma 31.13 identifies the numerator. At a regular value $F(t) > 0$. The first inequality also shows that p^2/F is locally integrable: its positive part is bounded by the locally finite absolutely continuous density of $u_{\#}\mu$ plus 8π . Define the quotient arbitrarily on the null set of critical values. Since the singular part of $u_{\#}\mu$ is nonnegative, (31.18) upgrades to the distributional inequality (31.19) via $Dp = u_{\#}\mu$. $\square$

Lemma 31.19 (The monotone quantity). *Put*

$$h(t) := p(t) - \frac{4F(t)}{t} + 8\pi t. \quad (31.20)$$

Then, using $DF = p dt$ and (31.19),

$$Dh \geq \left(\frac{p}{\sqrt{F}} - \frac{2\sqrt{F}}{t} \right)^2 dt \geq 0, \quad (31.21)$$

so the right-continuous representative of h is nondecreasing.

Proof. $D(-4F/t) = (-4p/t + 4F/t^2)dt$, and adding (31.19) and $8\pi dt$ gives $Dh \geq (p^2/F - 4p/t + 4F/t^2)dt$, which is the displayed square. $\square$

Proposition 31.20 (The two Euclidean inequalities on levels). *For every $t \in (0, 1)$,*

$$p(t) \geq \frac{4F(t)}{t} - 8\pi t, \quad (31.22)$$

and

$$F(t) \geq 4\pi t^2. \quad (31.23)$$

Proof. By (31.6), $\lim_{t \downarrow 0} h(t) = 0$; monotonicity (Lemma 31.19) gives $h \geq 0$, which is (31.22). Put $z(t) := F(t) - 4\pi t^2$; (31.22) gives

$$z'(t) \geq \frac{4z(t)}{t} \text{ a.e.,} \quad \text{equivalently} \quad \left(\frac{z(t)}{t^4} \right)' \geq 0. \quad (31.24)$$

The first limit in (31.6) says $z(t)/t^4 \rightarrow 0$ as $t \downarrow 0$, so $z \geq 0$, which is (31.23). $\square$

Theorem 31.21 (Euclidean capacity inequalities).

$$\int_{\partial K} |\nabla u|^2 d\mathcal{H}^2 \geq 4\pi, \quad \int_{\partial K} H|\nabla u| d\mathcal{H}^2 \geq 4 \int_{\partial K} |\nabla u|^2 d\mathcal{H}^2 - 8\pi. \quad (31.25)$$

Proof. The boundary level $t = 1$ is regular by the Hopf sign (30.2). Taking the left limit in (31.22) and (31.23), and using smooth convergence of the levels to ∂K , defines the endpoint values $F(1) := \lim_{t \uparrow 1} F(t)$ and $p(1) := \lim_{t \uparrow 1} p(t)$ and gives

$$\int_{\partial K} |\nabla u|^2 = F(1) \geq 4\pi, \quad \int_{\partial K} H|\nabla u| = p(1) \geq 4F(1) - 8\pi.$$

□

Remark 31.22 (What was retained). All possible critical-level contributions have been retained as the nonnegative singular part of $u_{\#}\mu$ (Lemma 31.16). Nothing was discarded at a critical value, and no Sard theorem stronger than Theorem 11.5 was used.

Part VI

The threshold theorems

Chapter 32

The Coulomb–capacity identity

Lean module: `NoCompromise.Green.Identity`

Remark 32.1 (Only one case is needed). The headline theorem uses Green’s identity only for the charge density $\mathbf{1}_\Omega$. The stronger measurable-set version is therefore *not* stated and must not be formalised.

Notation 32.2. $v := v_\Omega$. Since $\Omega \subset K$ (Lemma 29.3), v is harmonic on $\mathbb{R}^3 \setminus K$; Lemma 28.6 gives $v \in C^{1,\beta}(\mathbb{R}^3)$, and Proposition 10.36 gives

$$-\Delta v = 4\pi \mathbf{1}_\Omega \quad (32.1)$$

distributionally.

32.1 Interior flux via a tubular cutoff

Lemma 32.3 (Interior flux).

$$\int_{\partial K} \partial_{\nu_K} v \, d\mathcal{H}^2 = -4\pi V. \quad (32.2)$$

Proof. Let d be the signed distance to ∂K , negative in K , in a fixed tubular neighbourhood (Theorem 14.12). Choose $\eta \in C^\infty(\mathbb{R})$ with $\eta = 1$ on $(-\infty, -1]$, $\eta = 0$ on $[0, \infty)$, $\eta' \leq 0$, and set

$$\chi_\varepsilon(x) := \begin{cases} 1, & d(x) \leq -\varepsilon, \\ \eta(d(x)/\varepsilon), & -\varepsilon < d(x) < 0, \\ 0, & d(x) \geq 0. \end{cases} \quad (32.3)$$

After smoothing the two flat joins, $\chi_\varepsilon \in C_c^\infty$, $0 \leq \chi_\varepsilon \leq 1$, and $\chi_\varepsilon \rightarrow \mathbf{1}_K$ pointwise. Testing (32.1) with χ_ε ,

$$-4\pi \int_\Omega \chi_\varepsilon = \langle \Delta v, \chi_\varepsilon \rangle = - \int \nabla v \cdot \nabla \chi_\varepsilon. \quad (32.4)$$

In tubular coordinates $x = p + t\nu_K(p)$ the right side is

$$- \int_{\partial K} \int_{-\varepsilon}^0 \partial_{\nu_K} v(p + t\nu_K(p)) \frac{1}{\varepsilon} \eta'(t/\varepsilon) J(p, t) \, dt \, d\mathcal{H}^2(p),$$

with $J(p, t) \rightarrow 1$ uniformly. Since ∇v is continuous (Lemma 28.6) and $-\int_{-1}^0 \eta' = 1$, this tends to $\int_{\partial K} \partial_{\nu_K} v$. Dominated convergence on the left of (32.4), together with $\Omega \subset K$, gives (32.2). $\square$

32.2 The outer Wronskian cancellation

Lemma 32.4 (Multipole expansion of v). *Since Ω is bounded, for $|x| > 2 \sup_{y \in \Omega} |y|$*

$$\frac{1}{|x - y|} = \frac{1}{|x|} + \frac{x \cdot y}{|x|^3} + O\left(\frac{|y|^2}{|x|^3}\right), \quad (32.5)$$

with a uniform $O(|x|^{-4})$ remainder after differentiating in x . Integrating over $y \in \Omega$,

$$v(x) = \frac{V}{|x|} + \frac{b \cdot x}{|x|^3} + O(|x|^{-3}) \quad (32.6)$$

with differentiated remainder.

Proof. Taylor expansion of the kernel and integration. □

Lemma 32.5 (The outer integral vanishes). *With u from Theorem 30.1,*

$$\int_{\partial B_R} (v \partial_r u - u \partial_r v) d\mathcal{H}^2 = O(R^{-2}) \longrightarrow 0. \quad (32.7)$$

Proof. By (30.13) and (32.6), the monopole terms cancel exactly:

$$\frac{V}{R} \left(-\frac{C}{R^2} \right) - \frac{C}{R} \left(-\frac{V}{R^2} \right) = 0.$$

Every remaining term is $O(R^{-4})$ pointwise, so its integral over ∂B_R , of area $4\pi R^2$, is $O(R^{-2})$. □

Theorem 32.6 (Coulomb–capacity identity for Ω).

$$\frac{1}{4\pi} \int_{\partial K} v_\Omega |\nabla u| d\mathcal{H}^2 = |\Omega| = V. \quad (32.8)$$

Proof. Apply Green's second identity to u and v on $A_R := B_R \setminus K$, where both are smooth. The outward normal of A_R on its inner boundary is $-\nu_K$; since $u = 1$ and $\partial_{-\nu_K} u = w$ on ∂K (Notation 30.2),

$$\int_{\partial K} (vw + \partial_{\nu_K} v) d\mathcal{H}^2 + \int_{\partial B_R} (v \partial_r u - u \partial_r v) d\mathcal{H}^2 = 0. \quad (32.9)$$

Lemma 32.5 kills the outer integral as $R \rightarrow \infty$, giving $\int_{\partial K} vw = -\int_{\partial K} \partial_{\nu_K} v$, and Lemma 32.3 evaluates the right side as $4\pi V$. □

Chapter 33

The capacitary estimate for stationary domains

Lean module: NoCompromise.Stationary.CapacitaryEstimate

Notation 33.1. $C := \text{Cap}(K) > 0$, $I := \int_{\partial K} |\nabla u|^2 d\mathcal{H}^2$, and $P := P(\Omega)$.

Lemma 33.2 (Integrating the Euler–Lagrange equation).

$$\lambda C - V = \frac{1}{4\pi} \int_{\partial K} H |\nabla u| d\mathcal{H}^2, \quad (33.1)$$

and consequently, by (31.25),

$$\lambda C - V \geq \frac{I}{\pi} - 2. \quad (33.2)$$

Proof. Since $\partial K \subset \partial\Omega$ and the outward normals agree (Lemma 29.3), (28.11) gives $H + v_\Omega = \lambda$ on ∂K . Multiply by $|\nabla u|/(4\pi)$ and integrate: the flux formula (30.10) evaluates $\lambda \int_{\partial K} |\nabla u|/(4\pi) = \lambda C$, and Theorem 32.6 evaluates the v_Ω term as V . This is (33.1); the second inequality of (31.25) gives (33.2). $\square$

Lemma 33.3 (Two lower bounds for I).

$$I \geq 4\pi, \quad \text{hence} \quad \lambda C - V \geq 2, \quad (33.3)$$

and

$$I \geq \frac{(\int_{\partial K} |\nabla u|)^2}{P(K)} = \frac{16\pi^2 C^2}{P(K)} \geq \frac{16\pi^2 C^2}{P}, \quad \text{hence} \quad \lambda C - V \geq \frac{16\pi C^2}{P} - 2. \quad (33.4)$$

Combining,

$$\lambda C - V \geq \max\left\{2, \frac{16\pi C^2}{P} - 2\right\}. \quad (33.5)$$

Proof. (33.3) is the first inequality of (31.25) fed into (33.2). For (33.4), Cauchy–Schwarz on ∂K and the flux identity (30.10) give the displayed lower bound for I , and (29.3) replaces $P(K)$ by P ; feed into (33.2). $\square$

Lemma 33.4 (Elimination of the capacity). *Put*

$$s := C\sqrt{\frac{4\pi}{P}} > 0, \quad y := \lambda\sqrt{\frac{P}{4\pi}}, \quad (33.6)$$

so $\lambda C = sy$. Then (33.5) becomes

$$y \geq \frac{V+2}{s}, \quad y \geq 4s + \frac{V-2}{s}, \quad (33.7)$$

and consequently

$$y \geq \begin{cases} V+2, & 0 < V \leq 6, \\ 4\sqrt{V-2}, & V \geq 6. \end{cases} \quad (33.8)$$

Proof. Substituting (33.6) into (33.5) gives (33.7), so $y \geq \max\{(V+2)/s, 4s + (V-2)/s\}$.

If $s \leq 1$: the first term gives $y \geq V+2$. For $V \geq 6$ this already gives $y \geq 4\sqrt{V-2}$, because $(V+2)^2 - 16(V-2) = (V-6)^2 \geq 0$ (Lemma 16.12).

If $s \geq 1$: consider $g_V(s) := 4s + (V-2)/s$. For $0 < V \leq 6$ its minimum on $[1, \infty)$ is at $s = 1$, giving $V+2$. For $V \geq 6$ its minimum is at $s = \sqrt{V-2}/2 \geq 1$ and equals $4\sqrt{V-2}$. $\square$

Proposition 33.5 (The capacity estimate). *Every bounded connected C^3 stationary domain Ω of volume $V > 0$ with multiplier λ satisfies*

$$\lambda\sqrt{\frac{P(\Omega)}{4\pi}} \geq \begin{cases} V+2, & 0 < V \leq 6, \\ 4\sqrt{V-2}, & V \geq 6. \end{cases} \quad (33.9)$$

Proof. (33.8) with $y = \lambda\sqrt{P(\Omega)/(4\pi)}$. $\square$

Formalisation note 33.6 (The interface). Proposition 33.5 is the *only* statement from Parts IV–V that Chapters 34–36 use, apart from Corollary 28.16 and Proposition 28.14. Everything after it is elementary one-variable algebra. A formalisation can therefore develop Chapters 34–38 against (33.9) as a hypothesis, in parallel with the heavy machinery.

Chapter 34

Classification at volumes at most V_*

Lean module: NoCompromise.Classification.Subcritical

Notation 34.1 (The isoperimetric deficit parameter). For a fixed-volume minimiser Ω at volume V put

$$\delta := \sqrt{\frac{P(\Omega)}{P_B(V)}} - 1 \geq 0, \quad z := 1 + \delta, \quad (34.1)$$

the inequality by Theorem 13.7. Then

$$P(\Omega) = P_B z^2, \quad \sqrt{\frac{P(\Omega)}{4\pi}} = Rz. \quad (34.2)$$

Lemma 34.2 (Upper bound from the one-ball comparison). *Let Ω be a fixed-volume minimiser at volume $V > 0$ with multiplier λ . Then*

$$\lambda \sqrt{\frac{P(\Omega)}{4\pi}} \leq (V + 5 - 3z^2)z. \quad (34.3)$$

Proof. Comparison with the ball gives $\mathcal{E}(\Omega) \leq \mathcal{E}(B_V) = \frac{V+5}{5}P_B$ (Corollary 16.4). By the scaling identity (28.12) and $RP_B = 3V$ (Lemma 16.2),

$$\lambda \sqrt{\frac{P(\Omega)}{4\pi}} = \frac{Rz}{3V} (5\mathcal{E}(\Omega) - 3P(\Omega)) \leq \frac{Rz}{3V} ((V+5)P_B - 3P_B z^2) = (V+5-3z^2)z.$$

□

Proposition 34.3 (Classification at $V \leq V_*$). *At every volume $0 < V \leq V_*$, every fixed-volume minimiser is a ball, uniquely up to translation and null sets.*

Proof. Let Ω be a minimiser of volume $V \leq V_*$; by Corollary 28.16 take the bounded connected smooth representative and its multiplier λ . Since $V \leq V_* < 4 < 6$ (Lemma 16.8), Proposition 33.5 gives

$$\lambda \sqrt{\frac{P(\Omega)}{4\pi}} \geq V + 2, \quad (34.4)$$

while Lemma 34.2 gives the opposite bound (34.3). By Lemma 16.11,

$$V + 2 - (V + 5 - 3(1 + \delta)^2)(1 + \delta) = \delta(4 - V + 9\delta + 3\delta^2),$$

which is *strictly positive* for $\delta > 0$ because $V \leq V_* < 4$. That contradicts (34.4) together with (34.3). Hence $\delta = 0$, i.e. equality in the isoperimetric inequality, and Theorem 27.1 makes Ω a translate of B_V up to a null set. $\square$

Chapter 35

Existence and uniqueness at the endpoint

Lean module: NoCompromise.Classification.Endpoint

Proposition 35.1 (Endpoint existence and uniqueness). *$m(V_*) = \mathcal{E}(B_{V_*})$, the ball B_{V_*} is a fixed-volume minimiser at volume V_* , and it is the unique one up to translation and null sets.*

Proof. Proposition 19.10 gives existence for $V < V_*$ and Proposition 34.3 then gives $m(V) = \mathcal{E}(B_V)$ for $0 < V < V_*$. Choose $V_n \uparrow V_*$. Continuity of m (Proposition 17.7) and continuity of the ball-energy formula (16.4) give

$$m(V_*) = \lim_n m(V_n) = \lim_n \mathcal{E}(B_{V_n}) = \mathcal{E}(B_{V_*}),$$

so B_{V_*} is a minimiser. Uniqueness is Proposition 34.3 at $V = V_*$. □

Remark 35.2 (Why this route). This avoids formalising the separate theorem that the set of masses admitting minimisers is closed. It is logically valid because classification *below* the endpoint (Proposition 34.3 for $V < V_*$) is established before endpoint existence is invoked. Note the order: Proposition 34.3 is stated for $V \leq V_*$ but its proof does not use existence at any volume, so there is no circularity.

Chapter 36

Nonexistence above the threshold

Lean module: NoCompromise.Nonexistence.Sharp, NoCompromise.Nonexistence.Slicing

36.1 The elementary bound $V \leq 8$

Lemma 36.1 (Slicing inequality). *Let Ω be a fixed-volume minimiser at volume $V > 0$ with the bounded representative of Corollary 28.16. For $\nu \in \mathbb{S}^2$ and $\ell \in \mathbb{R}$ put*

$$\Omega_{\nu,\ell}^{\pm} := \Omega \cap \{x : \pm(\nu \cdot x - \ell) > 0\}.$$

Then for a.e. ℓ ,

$$P(\Omega_{\nu,\ell}^+) + P(\Omega_{\nu,\ell}^-) - P(\Omega) = 2\mathcal{H}^2(\Omega \cap \{\nu \cdot x = \ell\}) \quad (36.1)$$

and

$$\iint_{\Omega_{\nu,\ell}^+ \times \Omega_{\nu,\ell}^-} \frac{dx dy}{|x - y|} \leq 2\mathcal{H}^2(\Omega \cap \{\nu \cdot x = \ell\}). \quad (36.2)$$

Proof. For a.e. ℓ the plane is transverse to $\partial\Omega$, the two pieces have finite perimeter, the omitted planar slice has volume zero, and (36.1) is the planar case of Proposition 5.11.

Translate the two pieces far apart. Their perimeters and self-interactions are unchanged, while their mutual interaction tends to zero (Lemma 15.11). Minimality and passage to the translation limit give $\mathcal{E}(\Omega) \leq \mathcal{E}(\Omega_{\nu,\ell}^+) + \mathcal{E}(\Omega_{\nu,\ell}^-)$; combining with (36.1) and (15.5) gives (36.2). $\square$

Lemma 36.2 (The two integral identities). *For $x, y \in \mathbb{R}^3$ and $\nu \in \mathbb{S}^2$,*

$$\int_{\mathbb{R}} \mathbf{1}_{\{\nu \cdot x > \ell > \nu \cdot y\}} d\ell = (\nu \cdot (x - y))_+, \quad (36.3)$$

and for $a \neq 0$,

$$\frac{1}{4\pi} \int_{\mathbb{S}^2} \frac{(\nu \cdot a)_+}{|a|} d\nu = \frac{1}{4\pi} \int_0^{2\pi} \int_0^{\pi/2} \cos \theta \sin \theta d\theta d\varphi = \frac{1}{4}. \quad (36.4)$$

Proof. (36.3) is direct. For (36.4) use rotational invariance to take $a = |a|e_3$ and compute in spherical coordinates. $\square$

Proposition 36.3 (The bound $V \leq 8$). *If the fixed-volume problem has a minimiser at volume $V > 0$, then $V \leq 8$. In particular no minimiser exists for $V > 8$.*

Proof. Integrate (36.2) in ℓ . Tonelli applies because all integrands are nonnegative, and the value on the diagonal $x = y$ may be defined to be zero. By (36.3) on the left and Cavalieri on the right,

$$2V \geq \iint_{\Omega \times \Omega} \frac{(\nu \cdot (x - y))_+}{|x - y|} dx dy. \quad (36.5)$$

Average over $\nu \in \mathbb{S}^2$ with normalised surface measure; a second application of Tonelli and (36.4) give $2V \geq \frac{1}{4}|\Omega|^2 = \frac{1}{4}V^2$. Since $V > 0$, $V \leq 8$. $\square$

36.2 Sharp nonexistence for $V_* < V \leq 8$

Notation 36.4. $L := q^{-2}(V + 10)$, so that $2\mathcal{E}(B_{V/2}) = \frac{L}{5}P_B(V)$ by Proposition 16.9.

Lemma 36.5 (Two-ball comparison). *Let Ω be a fixed-volume minimiser at volume $0 < V \leq 8$, with z, δ as in Notation 34.1 and multiplier λ . Then*

$$\lambda \sqrt{\frac{P(\Omega)}{4\pi}} \leq (L - 3z^2)z, \quad (36.6)$$

and

$$z^2 \leq \frac{L}{5} < q^4, \quad \text{hence} \quad z < q^2 \quad \text{and} \quad q^{-2}z < 1. \quad (36.7)$$

Proof. Place two balls of volume $V/2$ at distance tending to infinity. Their cross-interaction tends to zero (Lemma 15.11) and their perimeters add (Lemma 4.8), so minimality gives $\mathcal{E}(\Omega) \leq 2\mathcal{E}(B_{V/2}) = \frac{L}{5}P_B$ (Proposition 16.9). Substituting into the scaling identity exactly as in Lemma 34.2 gives (36.6). Also

$$z^2 = \frac{P(\Omega)}{P_B} \leq \frac{\mathcal{E}(\Omega)}{P_B} \leq \frac{L}{5},$$

and $L/5 < q^4$ is Lemma 16.15. $\square$

Proposition 36.6 (Nonexistence for $V_* < V \leq 6$). *No fixed-volume minimiser exists at a volume V with $V_* < V \leq 6$.*

Proof. Suppose Ω minimises at such a V . Using (16.6) in the form $q^{-2}(V_* + 10) = V_* + 5$, the algebraic identity

$$V + 2 - (L - 3z^2)z = \delta(4 - V_* + 9\delta + 3\delta^2) + (V - V_*)(1 - q^{-2}z) \quad (36.8)$$

holds. Its first term is nonnegative because $V_* < 4$ (Lemma 16.8) and $\delta \geq 0$; its second is strictly positive because $V > V_*$ and $q^{-2}z < 1$ by (36.7). Hence $(L - 3z^2)z < V + 2$, so (36.6) gives $\lambda \sqrt{P(\Omega)/4\pi} < V + 2$. But $V \leq 6$ and Proposition 33.5 give $\lambda \sqrt{P(\Omega)/4\pi} \geq V + 2$, a contradiction. $\square$

Formalisation note 36.7 (Verify (36.8) symbolically). (36.8) is the sharp point of the whole argument: it is where V_* is used as the exact threshold rather than as an approximate one. Expand both sides in δ and V , substituting $L = q^{-2}(V + 10)$ and using $q^3 = 2$ only through (16.6). It should be discharged by normalisation, not by a chain of inequalities. Compare Lemma 16.11, which is its $V = V_*$ specialisation.

Proposition 36.8 (Nonexistence for $6 < V \leq 8$). *No fixed-volume minimiser exists at a volume V with $6 < V \leq 8$.*

Proof. Suppose Ω minimises at such a V . By Lemma 16.13 with $x = z$, $Lz - 3z^3 \leq \frac{2}{9}L^{3/2}$, and since $L = (V + 10)/q^2$ with $q^3 = 2$,

$$\frac{2L^{3/2}}{9} = \frac{(V + 10)^{3/2}}{9}. \quad (36.9)$$

So (36.6) gives

$$\lambda \sqrt{\frac{P(\Omega)}{4\pi}} \leq \frac{(V + 10)^{3/2}}{9}. \quad (36.10)$$

We claim $\frac{(V+10)^{3/2}}{9} < 4\sqrt{V-2}$ for $6 < V \leq 8$. Both sides are positive, so it suffices to prove $(V + 10)^3/(V - 2) < 1296$, which is Lemma 16.16. But Proposition 33.5 with $V \geq 6$ gives the contradictory lower bound $\lambda\sqrt{P(\Omega)/4\pi} \geq 4\sqrt{V-2}$. $\square$

Corollary 36.9 (Nonexistence above the threshold). *No fixed-volume minimiser exists at any volume $V > V_*$.*

Proof. Propositions 36.6 and 36.8 cover $V_* < V \leq 8$; Proposition 36.3 covers $V > 8$. $\square$

Chapter 37

Assembly of the main theorem

Lean module: NoCompromise.Main

Theorem 37.1 (Main theorem). *Let $\mathcal{E}$ be as in Definition 15.2, V_* as in Definition 16.5, and let “minimiser” mean fixed-volume minimiser in the sense of Definition 15.3. Then for every $V > 0$:*

- (i) if $0 < V \leq V_*$, the minimisers at volume V are exactly the translates of a round ball of volume V , up to Lebesgue-null sets;*
- (ii) if $V > V_*$, there is no minimiser at volume V .*

Proof. For $0 < V < V_*$: Proposition 19.10 gives existence and Proposition 34.3 gives that every minimiser is a ball, unique up to translation and null sets.

At $V = V_*$: Proposition 35.1 gives existence, and Proposition 34.3 gives uniqueness.

For $V > V_*$: Corollary 36.9 gives nonexistence.

Conversely a translate of B_V has energy $\mathcal{E}(B_V) = m(V)$ for $V \leq V_*$, so every such translate is a minimiser. $\square$

Chapter 38

The minimal binding energy

Lean module: NoCompromise.Binding.BallRatio, NoCompromise.Binding.Main,
NoCompromise.Binding.Ratio

38.1 Relaxed ratio minimisation

Definition 38.1 (Relaxed ratio infimum). For $A > 0$ put

$$e_{\leq}(A) := \inf \left\{ \frac{\mathcal{E}(E)}{|E|} : 0 < |E| \leq A \right\}. \quad (38.1)$$

Lemma 38.2 (The relaxed infimum is attained). *For every $A > 0$ the infimum (38.1) is attained.*

Proof. Let E_n be a minimising sequence, $V_n := |E_n|$, $e := e_{\leq}(A)$, so $\mathcal{E}(E_n)/V_n \rightarrow e$. The isoperimetric inequality gives $\mathcal{E}(E_n)/V_n \geq c_I V_n^{-1/3}$, so $V_n \geq c_A > 0$; since $V_n \leq A$ and the ratios are bounded, $P(E_n) \leq \mathcal{E}(E_n) \leq C_A$. Hence (19.2) holds; apply Theorems 19.4 and 19.5 after translations, obtaining a nonzero local limit E , a tight part $F_n \rightarrow E$ globally, and a remainder G_n . Local convergence and exhaustion give $0 < |E| \leq \liminf_n V_n \leq A$.

If $|G_n| > 0$ then $\mathcal{E}(G_n) \geq e|G_n|$ by Definition 38.1 (note $|G_n| \leq A$); if $|G_n| = 0$ the same inequality is trivial. Energy splitting (19.6)–(19.7) yields

$$\mathcal{E}(E_n) \geq \mathcal{E}(E) + e|G_n| + o(1).$$

Because the ratios converge and $V_n \leq A$, $\mathcal{E}(E_n) = eV_n + o(1)$. Also $V_n = |F_n| + |G_n|$ with $|F_n| \rightarrow |E|$. Subtracting $e|G_n|$ and passing to the limit gives $e|E| \geq \mathcal{E}(E)$, and the reverse inequality follows from the definition of e since $0 < |E| \leq A$. Hence $\mathcal{E}(E) = e|E|$. $\square$

Formalisation note 38.3 (No convergence of V_n is needed). Boundedness by A is enough to turn ratio convergence into the additive error $o(1)$. Do not add a hypothesis that V_n converges.

Lemma 38.4 (A ratio optimiser is a fixed-volume minimiser). *Any minimiser of (38.1), of volume M , is a fixed-volume minimiser at volume M .*

Proof. A same-volume competitor of lower energy would have a lower ratio. $\square$

38.2 Stabilisation and reduction to balls

Lemma 38.5 (Stabilisation after volume eight). *$A \mapsto e_{\leq}(A)$ is nonincreasing, and*

$$e_{\leq}(A) = e_{\leq}(8) \quad \text{for every } A \geq 8. \quad (38.2)$$

Consequently

$$\inf_{0 < |E| < \infty} \frac{\mathcal{E}(E)}{|E|} = e_{\leq}(8), \quad (38.3)$$

and the global infimum is attained.

Proof. Monotonicity is clear from (38.1). Let $A \geq 8$ and let E_A attain $e_{\leq}(A)$ (Lemma 38.2). Its volume M_A is the volume of a fixed-volume minimiser (Lemma 38.4), so $M_A \leq 8$ by Proposition 36.3. Therefore $e_{\leq}(A) = \mathcal{E}(E_A)/M_A \geq e_{\leq}(8)$, and monotonicity gives the reverse. $\square$

Lemma 38.6 (Optimal ball volume). *For $V > 0$,*

$$\frac{\mathcal{E}(B_V)}{V} = (36\pi)^{1/3} \left(V^{-1/3} + \frac{1}{5} V^{2/3} \right), \quad (38.4)$$

this is minimised exactly at $V = 5/2$, and

$$\frac{\mathcal{E}(B_{5/2})}{5/2} = (36\pi)^{1/3} \frac{3}{2(5/2)^{1/3}} = 3 \left(\frac{9\pi}{5} \right)^{1/3}. \quad (38.5)$$

Proof. (38.4) is (16.4) divided by V . Put $x := V^{1/3} > 0$ and $a := (5/2)^{1/3}$; Lemma 16.14 gives

$$\frac{1}{x} + \frac{x^2}{5} - \frac{3}{2a} = \frac{(x-a)^2(x+2a)}{2a^3x} \geq 0,$$

with equality exactly at $x = a$, i.e. $V = 5/2$. Evaluating at $V = 5/2$ gives (38.5). $\square$

Theorem 38.7 (Binding-energy corollary). *The infimum*

$$e_* := \inf \left\{ \frac{\mathcal{E}(E)}{|E|} : E \subset \mathbb{R}^3 \text{ measurable, } 0 < |E| < \infty \right\}$$

equals $3(9\pi/5)^{1/3}$, is attained, and every optimiser is a translate of a ball of volume $5/2$, up to a Lebesgue-null set.

Proof. By Lemma 38.5 the global infimum e_* is attained; let E_* be an optimiser and $M := |E_*|$. By Lemma 38.4 it is a fixed-volume minimiser at volume M , so Theorem 37.1 gives $M \leq V_*$ and makes E_* a ball up to translation and a null set. Among balls, Lemma 38.6 makes $M = 5/2$ the unique optimal volume, and $5/2 < V_*$ by Lemma 16.8, so this volume lies in the existence range. Finally (38.5) evaluates $e_* = 3(9\pi/5)^{1/3}$. $\square$

Part VII

Formalisation apparatus

Chapter 39

Regression tests

Lean module: `NoCompromise.Tests.Constants`, `NoCompromise.Tests.Curvature`,
`NoCompromise.Tests.Iteration`, `NoCompromise.Tests.LevelFunctionals`,
`NoCompromise.Tests.Signs`

The tests below exist to catch sign, orientation and normalisation errors — the failure mode most likely to produce a Lean development that compiles and proves something false. Each should be an actual theorem in the Lean development, not a comment. They are grouped by which convention they probe.

39.1 Conventions 2.3 and 2.4: normals and curvature

Regression test 39.1 (The model exterior potential). For $u = C/|x|$ on $\{|x| > r\}$, with the normal toward infinity,

$$H = \frac{2}{r}, \quad w = \frac{C}{r^2}, \quad \langle \nabla w, \nabla u \rangle = Hw^2.$$

A sign error in Convention 2.4 flips the third identity.

Regression test 39.2 (Stationarity of the round ball). For $\Omega = B_R$, (28.11) holds with $H = 2/R$ and $v_{B_R} = 2\pi(R^2 - r^2/3)$ from (16.2); solving gives an explicit λ , which must agree with (28.12), i.e. $3V\lambda = 2P_B + 5D(B_V)$. *Checking both routes agree is the test.*

Regression test 39.3 (The inner normal in Green's identity). In (32.9) the outward normal of the exterior on its inner boundary is $-\nu_K$, so

$$\int_{\partial K} v_\Omega w \, d\mathcal{H}^2 = - \int_{\partial K} \partial_{\nu_K} v_\Omega \, d\mathcal{H}^2.$$

Dropping the minus sign leaves (32.8) looking plausible but gives $-V$.

Regression test 39.4 (The tubular flux). The tubular-cutoff computation of Lemma 32.3 must return

$$\int_{\partial K} \partial_{\nu_K} v_\Omega \, d\mathcal{H}^2 = -4\pi V,$$

not $+4\pi V$ and not $-V$. Test it on $\Omega = K = B_R$, where $v_{B_R}(x) = V/|x|$ outside and $\partial_{\nu_K} v = -V/R^2$, giving $-V/R^2 \cdot 4\pi R^2 = -4\pi V$.

39.2 Chapter 31: the level functionals

Regression test 39.5 (The model level functionals). For $u = C/r$ the Euclidean level functions of Chapter 31 satisfy

$$F(t) = 4\pi t^2, \quad p(t) = 8\pi t, \quad h(t) = 0.$$

The third is the equality case of Lemma 31.19 and is the sharpest available check on (31.20).

Regression test 39.6 (The Bochner density). For $u = C/r$, $\Delta|\nabla u| = 2C/r^4$, matching (31.9) since a sphere of radius r has $|A|^2 = 2/r^2$ and $w = C/r^2$, hence $w|A|^2 = 2C/r^4$ and $\nabla_\Sigma w = 0$.

39.3 Chapter 24: the iteration

Regression test 39.7 (Normalisation of the recurrence). The recurrence

$$E_{j+1} \leq (\theta/2)E_j + C\theta^{j+1}\omega r$$

must normalise, after division by θ^{j+1} , to a geometric sum with ratio $1/2$ (Lemma 24.28). If the ratio comes out as $\theta/2$ or as 1, the choice (24.47) has been applied in the wrong place.

Regression test 39.8 (Excess of an exact halfspace and of a tilted plane). $\mathbf{E}(H, 0, r, \nu) = 0$ for the halfspace H with outward normal ν , and for a halfspace with outward normal ν' at angle ϑ to ν , $\mathbf{E}(H', 0, r, \nu)$ is a positive absolute multiple of ϑ^2 for small ϑ . This pins the normalisation r^{-2} in Definition 24.1.

39.4 Chapter 14: the total curvature bound

Regression test 39.9 (Sphere and torus). For $\Sigma = \partial B_R$ with the outward normal, $\kappa \equiv 1/R^2$, the Gauss map is a diffeomorphism, $\iota_y(n) = 1$ for every y , and $\int_\Sigma \kappa d\mathcal{H}^2 = 4\pi$ — equality in (14.28). For a standard torus of revolution, $\int_\Sigma \kappa d\mathcal{H}^2 = 0$, and a height function in a generic direction has $c_0 = c_2 = 1$, $c_1 = 2$, so $c_0 - c_1 + c_2 = 0$: this exercises Proposition 14.42 and Proposition 14.35 in a case where the inequality is slack and κ changes sign.

Regression test 39.10 (The antipodal averaging is not vacuous). On the round sphere, $\iota_y(n) = \iota_{-y}(n) = 1$, so $\iota_y + \iota_{-y} = 2$ and Proposition 14.35 is tight. A version of (14.27) that dropped the antipodal term — bounding $\iota_y(n)$ alone by 2 rather than $\iota_y + \iota_{-y}$ — would still pass on the sphere but would give only $\int_\Sigma \kappa \leq 8\pi$, which is too weak for Chapter 31. Test the factor $\frac{1}{2}$ explicitly.

Regression test 39.11 (Curvature sign convention). For $\Sigma = \partial B_R$ with the *outward* normal, the shape operator of Definition 14.15 is $A = \frac{1}{R}I$, so $H = 2/R > 0$ and $\kappa = 1/R^2 > 0$, matching Convention 2.4. With the inward normal both signs of H flip while κ does not — which is why (14.12) is orientation-independent and (14.9) is not.

39.5 Chapter 25: the product competitor

Regression test 25.5 (horizontal cost exactly $2|A\Delta L|$) belongs here.

39.6 Chapter 16: the constants

Regression test 39.12 (Headline constants). The following must be provable numerically (or as exact algebraic identities):

$$V_* = \frac{5 \cdot 2^{2/3}}{2^{1/3} + 1} \approx 3.51207, \quad \frac{5}{2} < V_* < 4,$$

the optimal ball volume is $5/2$, and $e_* = 3(9\pi/5)^{1/3}$. The ratio of the two ways of writing V_* in Definition 16.5 and (16.5) must be exactly 1.

Chapter 40

Dependency overview

Lean module: (none)

The `\uses` graph is acyclic, but the expository document order is not itself a topological order: a small number of theorem wrappers precede their technical lemmas, and a few low-cost modules are grouped thematically. A formalisation must topologically sort the labeled nodes. Remarks 10.1 and 13.1 record the two most important architectural constraints. The table below records a coarse topological layering and the size class of each layer; “L” marks a dominant cost.

	Layer	Depends on	Size
1	Signed measures (Ch. 3)	base only	L
2	BV core (Ch. 4)	1	L
3	Coarea, slicing, traces (Ch. 5)	2	medium
4	Area formula (Ch. 6)	base only	L
5	Sobolev, Poincaré, relative isoperimetry, maximal (Ch. 7)	2, 3	medium
6	De Giorgi structure (Ch. 8)	2–5	L
7	Transport, first variation (Ch. 9)	4, 6	medium
8	Elliptic estimates (Ch. 10)	2, 5	L
9	Sard (Ch. 11)	base only	small
10	Smooth approximation (Ch. 12)	3, 9	small
11	Sharp isoperimetric inequality (Ch. 13)	8, 10	small
12	Flows, transversality, Morse count, total curvature (Ch. 14)	4, 9, base	L
13	The functional (Ch. 15–16)	2, 11	small
14	Value function, splitting, existence (Ch. 17–19)	3, 5, 11, 13	medium
15	Penalisation, density, representative (Ch. 20–21)	6, 11, 13	medium
16	Monotonicity, tangent cones (Ch. 22)	7, 15	medium
17	Deformation (Ch. 23)	3, 7	medium
18	ε -regularity (Ch. 24)	2, 5, 16, 17	L
19	Cones, no singular set (Ch. 25–26)	8, 18	medium
20	Isoperimetric rigidity (Ch. 27)	11, 19	small
21	Euler–Lagrange, bootstrap (Ch. 28)	7, 8, 19	medium

	Layer	Depends on	Size
22	Hull (Ch. 29)	12, 21	small
23	Exterior potential (Ch. 30)	8, 9, 22	medium
24	Appendix K closure (Ch. 31)	1, 3, 12, 23	medium
25	Green identity (Ch. 32)	8, 12, 21, 23	small
26	Capacitary estimate (Ch. 33)	13, 24, 25	small
27	Classification, endpoint, nonexistence, assembly (Ch. 34–37)	13, 14, 20, 21, 26	small
28	Binding energy (Ch. 38)	13, 14, 27	small

Formalisation note 40.1 (The three dominant costs). Layers 1, 12 and 18 — signed Riesz representation, the flow/degree package, and the ε -regularity iteration — together with the De Giorgi structure theorem (layer 6) and the area formula (layer 4) account for most of the work. Within layer 12 the dominant item is §14.1 alone (Formalisation note 14.8). Layers 26–28, which contain the sharp mathematics of the paper, are comparatively small. Do not schedule the project as if the difficulty were distributed evenly along the document.

Formalisation note 40.2 (Items off the critical path). Exactly six numbered results in this document are stated but never used by Theorem 37.1 or Theorem 38.7:

- Lemma 17.4 (subadditivity of m);
- Lemma 17.1 (bounded approximation; its only consumer is the previous item);
- Corollary 12.2 (energy convergence of the smooth approximants);
- Lemma 30.7 (level asymptotics before renormalisation);
- Proposition 7.10 (relative isoperimetry on a planar disk).
- Theorem 10.33 (a standalone boundary regularity theorem for uniformly elliptic nondivergence-form equations).

Each carries an inline note saying so (Formalisation notes 17.5 and 30.8, Remark 7.7, Formalisation note 10.34, and the remark inside Corollary 12.2). They are retained because each is the natural companion of a neighbouring result, but a formalisation may skip all six at no cost. Everything else in the document is reachable from the two main theorems by following `\uses` edges; if a tool finds a seventh unreachable item, that is a defect in this document.

Note also Remark 22.2, which describes a construction that is deliberately *not* given the status of a numbered result precisely so that it is not formalised.

Formalisation note 40.3 (Two independent tracks). Layers 13–14 and 26–28 depend on the heavy layers only through Theorem 13.7, and Proposition 33.5 respectively. A team can therefore build the “liquid-drop” half against those two statements as hypotheses while the “geometric measure theory” half is being built, and discharge the hypotheses at the end. Formalisation note 33.6 makes the interface explicit.

Chapter 41

Complete blueprint-to-module map

Lean module: (this chapter is the answer)

This chapter records the assignment of all 378 target and context items to 207 Lean modules. The module root is `NoCompromise`; mathematical declarations retain the `LiquidDrop` namespace. All 302 mathematical results have an owner, and all 59 remarks and formalisation notes are attached separately as guidance. No range or wildcard stands for an unassigned item.

The tree below lists each module with the items it owns. The assignment was checked mechanically for coverage and acyclicity using every declared `\uses` edge, shared conventions, and documented context prerequisites (including the regression tests' references).

The formalisation is complete: all 302 mathematical results are proved in the Lean library, which has 1,206 files. The tree records the planned home of each item. Some items were implemented in other files, and most supporting lemmas live in files that the tree does not list. The design rules and notes below describe the plan the formalisation started from.

41.1 Design rules used

- (R1) One mathematical concept per file. A file that needs a section heading to be navigable should be split.
- (R2) Planned imports form a DAG. Build in a topological order of the module imports; the thematic layers of Chapter 40 are only an overview.
- (R3) Each layer has a single `...Basic` or `...Defs` file holding its definitions, so that downstream files import definitions without importing proofs.
- (R4) The regularity theory is stated for `OmegaMinimal` (Definition 20.3) and never for the liquid-drop energy, per Remark 19.11. The energy appears only in `Energy/`, `Regularity/Penalization.lean`, and `Nonexistence/`, `Classification/`, `Binding/`.
- (R5) Each convention has one owner at the layer where its objects are available. `Conventions.lean` holds the primitive ambient setting, null-set convention, test classes and constant discipline. The polar sign belongs to `BV/Basic.lean`; curvature belongs to `Surface/Geometry.lean`; the first-variation sign belongs with its theorems; the density representative belongs to the regularity layer. Mathematical claims inside a convention must still be proved.
- (R6) Every quantitative statement takes its constants as explicit hypotheses or returns them existentially in the declared order of Convention 2.7. Constants are never section variables.

(R7) Files in the `Sard/`, `Flow/`, `Topology/` and `Surface/` trees import nothing from `BV/` in the planned graph, so they can be built independently of the whole measure-theoretic half. `Surface/` does import `Area/` (for `thm:area-formula-rect`, `lem:sphere-area` and `lem:smooth-surface-rectifiable`), which itself depends only on the base.

41.2 The tree

```
NoCompromise.lean -- root: imports Main and Binding.Main
```

```
NoCompromise/
```

```
Area/
```

```
AlmostLinear.lean
  -- const:almost-linear
  -- lem:almost-linear
Cofactor.lean
  -- lem:cofactor
Formula.lean
  -- thm:area-formula-2d
GoodPieces.lean
  -- def:good-pieces
GoodPiecesCover.lean
  -- lem:good-pieces-exhaust
Graph.lean
  -- cor:graph-area
Linear.lean
  -- def:jacobian
  -- lem:linear-distortion
PieceFormula.lean
  -- lem:area-one-piece
RankDeficient.lean
  -- lem:rank-deficient
Rectifiable.lean
  -- def:rectifiable
  -- thm:area-formula-rect
  -- lem:smooth-surface-rectifiable
Sphere.lean
  -- lem:sphere-area
```

```
BV/
```

```
AnnularGluingFull.lean
  -- prop:annular-gluing
Basic.lean
  -- conv:outward
  -- lem:test-class
  -- thm:polar
  -- lem:perimeter-invariances
  -- lem:lsc
  -- lem:locality
BoundaryTraces.lean
  -- thm:traces
Coarea.lean
```

```

    -- thm:bv-coarea
CoareaCritical.lean
    -- lem:coarea-critical-part
CoareaL1Lebesgue.lean
    -- cor:coarea-L1
CoareaRegular.lean
    -- lem:coarea-regular-part
Compactness.lean
    -- lem:translation-estimate
    -- lem:grid-approx
    -- thm:bv-compactness
Defs.lean
    -- def:density
    -- def:variation
    -- def:perimeter
ExactCuts.lean
    -- prop:cut-identities
Gluing.lean
    -- prop:gluing
GoodRadii.lean
    -- def:good-radius
    -- lem:good-radius-ae
GoodTruncation.lean
    -- lem:good-truncation
JumpDisintegration.lean
    -- lem:bv-slices
Rellich.lean
    -- lem:L2-interp
Slicing.lean
    -- cor:slicing
SmoothApprox.lean
    -- thm:smooth-approx
    -- cor:smooth-approx-energy
StrictApprox.lean
    -- lem:bv-product-smooth
    -- lem:strict-approx
WeightedCoarea.lean
    -- thm:weighted-coarea

Ball/
  Defs.lean
    -- not:BV
  Perimeter.lean
    -- lem:ball-perimeter
  Potential.lean
    -- lem:ball-coulomb
    -- cor:ball-energy

Binding/
  BallRatio.lean
    -- lem:optimal-ball-volume
  Main.lean
    -- thm:main-binding

```

```

Ratio.lean
  -- def:relaxed-ratio
  -- lem:relaxed-attained
  -- lem:ratio-is-fixed
  -- lem:stabilization
Splitting.lean
  -- lem:scaling-lower
  -- def:splitting-functions
  -- lem:ND-positive
  -- lem:two-piece
  -- lem:splitting-substitution
  -- lem:splitting-F
  -- thm:splitting
Strict.lean
  -- prop:strict-binding

CapacitaryK/
  Asymptotics.lean
    -- conv:level-orientation
    -- lem:K-normalization
    -- lem:K-level-asymptotics
    -- def:K-F
    -- lem:K-far-field
  Inequality.lean
    -- lem:K-gauss-bonnet-input
    -- prop:K-measure-inequality
    -- lem:K-h-monotone
    -- prop:K-two-ineq
    -- thm:capacitary-inequalities
  PositiveMeasure.lean
    -- lem:K-bochner
    -- prop:K-mu
    -- lem:K-level-identities
  Representatives.lean
    -- def:K-p
    -- lem:K-p-geometric
    -- def:K-Fhat
    -- prop:K-structure
    -- lem:K-pushforward-density
  SlabIdentities.lean
    -- lem:K-slab-H
    -- lem:K-slab-F
    -- lem:gradient-vanishes-on-zero-set

Capacity/
  Flux.lean
    -- def:capacity
    -- lem:potential-decay
    -- lem:flux-identity
  Kelvin.lean
    -- lem:kelvin
    -- lem:level-asymptotics
  Levels.lean

```

```

    -- lem:level-connected
Potential.lean
    -- thm:capacitary-potential
    -- not:w

Classification/
Endpoint.lean
    -- prop:endpoint
Subcritical.lean
    -- not:z-delta
    -- lem:one-ball-upper
    -- prop:classification

Compactness/
Decomposition.lean
    -- thm:decomposition
    -- lem:radii-construction
    -- lem:perimeter-splitting
    -- lem:coulomb-splitting
Nonvanishing.lean
    -- lem:cube-nonvanishing
    -- lem:coulomb-lsc
    -- thm:compactness

Cones/
NoSingular.lean
    -- prop:no-singular-points
ThreeDim.lean
    -- lem:cone-tangent-cylinder
    -- lem:cone-descent
    -- lem:cone-smooth
    -- lem:cone-link-great-circle
    -- thm:cone-3d
TwoDim.lean
    -- lem:cone-2d-link
    -- lem:cone-2d
Conventions.lean
    -- conv:ambient
    -- conv:null
    -- conv:test-fields
    -- conv:constants

DeGiorgi/
Blowup.lean
    -- thm:halfspace-blowup
ConeConcentration.lean
    -- lem:cone-concentration
ExactDensity.lean
    -- thm:exact-density
NormalFlux.lean
    -- lem:normal-flux
Rectifiability.lean
    -- lem:rectifiability-criterion

```

```

Reduced.lean
  -- not:degiorgi
  -- def:reduced-boundary
  -- lem:polar-differentiation
ReducedDensity.lean
  -- lem:reduced-density
ReducedPerimeter.lean
  -- lem:reduced-perimeter
SmoothBoundary.lean
  -- cor:smooth-perimeter
Structure.lean
  -- thm:degiorgi-structure
  -- thm:gauss-green

Elliptic/
Boundary.lean
  -- not:half-ball
  -- thm:boundary-C1a
  -- thm:boundary-C2a
  -- thm:boundary-nondiv
  -- thm:boundary-neumann
Caccioppoli.lean
  -- lem:caccioppoli
CampanatoComparison.lean
  -- lem:campanato-compare
CampanatoGrowth.lean
  -- lem:campanato-energy-growth
CampanatoHolder.lean
  -- thm:campanato
CampanatoHolderDatum.lean
  -- lem:Gh
CampanatoIteration.lean
  -- lem:campanato-iteration
ClassicalAdmissible.lean
  -- thm:W11-gauss-green
ClassicalKernelFlux.lean
  -- cor:W11-GG-uses
FrozenDecay.lean
  -- lem:frozen-decay
HarmonicDerivative.lean
  -- lem:harmonic-deriv
HarmonicMeanValue.lean
  -- lem:mean-value
HolderInterpolation.lean
  -- lem:holder-interp
Hopf.lean
  -- prop:hopf
HopfExterior.lean
  -- cor:hopf-exterior
InteriorH2.lean
  -- lem:interior-H2
KelvinRemovable.lean
  -- prop:kelvin-removable

```

```

Newtonian.lean
  -- prop:newtonian
NewtonianSchauder.lean
  -- thm:newtonian-schauder
NondivSchauder.lean
  -- thm:nondiv-schauder
Quasilinear.lean
  -- thm:quasilinear-schauder
Schauder.lean
  -- const:nondiv-order
SobolevChain.lean
  -- lem:sobolev-Ck
StrongMaximumHarmonic.lean
  -- prop:strong-max
TraceL1.lean
  -- conv:domain-classes
  -- prop:trace-L1
WeakDirichlet.lean
  -- prop:weak-dirichlet
WeakMaximum.lean
  -- lem:weak-max
WeakSolutions.lean
  -- prop:weak-neumann

Energy/
Coulomb.lean
  -- not:C0
  -- lem:potential-bound
  -- lem:coulomb-bilinear
  -- lem:coulomb-lipschitz
  -- lem:coulomb-disjoint
  -- lem:cross-decay
CoulombDefs.lean
  -- def:coulomb
Defs.lean
  -- def:energy
  -- def:minimizer
NullInvariance.lean
  -- lem:null-invariance
PotentialHolder.lean
  -- lem:potential-C1beta
Scaling.lean
  -- lem:coulomb-scaling

Existence/
Subcritical.lean
  -- prop:existence-subcritical

Flow/
FlowBox.lean
  -- thm:flow-box
  -- cor:flow-manifold
ODE.lean

```

```

-- thm:global-flow
-- lem:flow-group
-- lem:flow-gronwall
-- lem:flow-variational
-- thm:flow-Ck

Green/
  Identity.lean
    -- not:v-omega
    -- lem:interior-flux
    -- lem:v-expansion
    -- lem:wronskian
    -- thm:green-identity

Hull/
  Defs.lean
    -- def:hull
  Properties.lean
    -- lem:hull-properties

Isoperimetric/
  ABP.lean
    -- lem:abp-neumann
    -- lem:abp-contact
    -- prop:iso-smooth
    -- lem:concavity-23
    -- cor:iso-smooth-components
    -- thm:sharp-isoperimetric
  Rigidity.lean
    -- thm:isoperimetric-rigidity
Main.lean
  -- thm:main

Measure/
  SignedRiesz.lean
    -- def:locally-bounded
    -- def:upper-variation
    -- lem:upper-variation-additive
    -- thm:signed-riesz
    -- lem:riesz-exhaustion
  WeakStar.lean
    -- lem:C0-separable
    -- def:exhaustion-cutoffs
    -- thm:weak-star-compactness

Nonexistence/
  Sharp.lean
    -- not:L
    -- lem:two-ball-comparison
    -- prop:nonexistence-6
    -- prop:nonexistence-8
    -- cor:nonexistence
  Slicing.lean

```

```

-- lem:slicing-inequality
-- lem:slicing-integrals
-- prop:V-1e-8

Regularity/
ApproxHarmonic.lean
  -- lem:approx-harmonic
CompressionJacobian.lean
  -- lem:compression-jacobian
CompressionProfileTheorem.lean
  -- lem:compression-profile
Cylinders.lean
  -- not:cylinders
Deformation.lean
  -- thm:deformation
Density.lean
  -- const:rd
DensityAhlfors.lean
  -- lem:ahlfors
DensityCutComparison.lean
  -- lem:cut-comparison
DensityEstimates.lean
  -- lem:two-sided-density
DensityRadial.lean
  -- not:mx-nx
  -- lem:mx-derivative
EpsReg.lean
  -- lem:normals-cauchy
  -- lem:uniform-centers
  -- thm:eps-regularity
Excess.lean
  -- def:excess
  -- lem:excess-ball-cyl
  -- lem:excess-center
FirstVariation.lean
  -- lem:bounded-H
FixedNormalExcess.lean
  -- lem:fixed-normal-excess-convergence
FluxDefect.lean
  -- lem:flux-defect
GeometricRecurrence.lean
  -- lem:geometric-recurrence
GraphApprox.lean
  -- thm:graph-approx
GraphBadBase.lean
  -- lem:bad-base-area
GraphGoodBase.lean
  -- lem:good-base
GraphProjectedExcess.lean
  -- def:projected-excess
GraphSingleValued.lean
  -- lem:single-valued
GraphTwoPointCaps.lean

```

```

-- const:graph
-- lem:two-point-lipschitz
HarmonicAffine.lean
-- lem:harmonic-affine
HarmonicBlowup.lean
-- lem:blowup-compactness
HarmonicBlowupUniform.lean
-- lem:harmonic-approx-uniform
HeightBound.lean
-- const:height
-- lem:height-bound
Monotonicity.lean
-- thm:monotonicity
OmegaMinimal.lean
-- def:omega-minimal
Penalization.lean
-- lem:penalization
PenalizationDilation.lean
-- lem:dilation-quotients
PenalizationQuasiminimal.lean
-- lem:quasiminimal
PerimeterConvergence.lean
-- lem:perimeter-measure-convergence
RadialDivergence.lean
-- lem:radial-divergence
RepresentativeBoundary.lean
-- conv:representative
-- lem:open-representative
RepresentativeBounded.lean
-- lem:bounded-representative
ReversePoincare.lean
-- lem:reverse-poincare
SlabCap.lean
-- def:slab-cap
TangentCone.lean
-- prop:tangent-is-cone
TangentExistence.lean
-- lem:tangent-cone-minimizing
TiltImprovement.lean
-- const:tilt
-- lem:tilt-improvement
-- cor:excess-decay
ZeroExcess.lean
-- lem:zero-excess

Sard/
Equidimensional.lean
-- lem:sard-equidim
-- cor:sard-charts
FourToThree.lean
-- thm:sard-4to3
OneDimensional.lean
-- lem:sard-1d

```

```

Scalar.lean
  -- lem:sard-2d
ThreeDimensional.lean
  -- thm:sard-3d

Sobolev/
BV.lean
  -- lem:GN-smooth
  -- thm:bv-sobolev
  -- lem:planar-sobolev-L2
Extension.lean
  -- lem:bilip-chain
  -- lem:bv-extension
H1Extension.lean
  -- lem:H1-extension-disk
Maximal.lean
  -- def:maximal
  -- lem:maximal
PlanarDomain.lean
  -- lem:planar-GN
Poincare.lean
  -- thm:bv-poincare
PoincareTrace.lean
  -- prop:poincare-trace
RelativeCubes.lean
  -- prop:rel-iso-cube
RelativeScaling.lean
  -- prop:rel-iso-ball
  -- prop:rel-iso-disk
Rellich.lean
  -- thm:rellich

Stationary/
Bootstrap.lean
  -- lem:mc-elliptic
  -- prop:bootstrap-C2
  -- prop:bootstrap-C3
  -- cor:EL-pointwise
CapacitaryEstimate.lean
  -- not:cap-estimate
  -- lem:integrate-EL
  -- lem:two-bounds-I
  -- lem:eliminate-capacity
  -- prop:cap-estimate
Connected.lean
  -- prop:connected
Defs.lean
  -- def:stationary
  -- not:stationary
EulerLagrange.lean
  -- thm:constrained-first-var
  -- lem:graph-PMC
MinimizerContext.lean

```

```

-- not:minimizer-rep
ScalingIdentity.lean
-- prop:scaling-identity
-- cor:minimizer-stationary

Surface/
Geometry.lean
-- conv:curvature
-- def:frame-free
Morse.lean
-- lem:morse-height
-- lem:morse-distinct
-- lem:morse-coords
-- lem:one-manifold
-- lem:morse-band
-- cor:sublevel-stable
-- lem:local-sectors
MorseCount.lean
-- def:merging-saddle
-- lem:level-adjacency
-- lem:sublevel-closure
-- lem:count-submerging
-- lem:merge-disjoint
-- prop:morse-count
Shape.lean
-- lem:hess-height
-- lem:gauss-jacobian
-- conv:gauss-orientation
-- lem:sign-kappa-index
-- lem:critical-normal
TotalCurvature.lean
-- lem:regular-fiber-finite
-- def:index-sum
-- lem:gauss-map-extension
-- thm:total-curvature-index
-- prop:index-sum-antipodal
-- thm:total-curvature-bound

Tests/
Constants.lean
-- test:product-cost
-- test:constants
Curvature.lean
-- test:total-curvature
-- test:antipodal
Iteration.lean
-- test:recurrence
-- test:excess-values
LevelFunctionals.lean
-- test:K-model
-- test:K-bochner
Signs.lean
-- test:model-potential

```

```

-- test:ball-stationary
-- test:green-normal
-- test:tubular-flux
-- test:kappa-sign

Threshold/
Algebra.lean
  -- lem:Vstar-form
  -- lem:Vstar-identity
  -- lem:Vstar-bounds
Comparison.lean
  -- prop:two-ball
Defs.lean
  -- def:threshold
Ledger.lean
  -- lem:ledger-Fprime
  -- lem:ledger-oneball
  -- lem:ledger-six
  -- lem:ledger-cubic
  -- lem:ledger-binding
  -- lem:ledger-L5
  -- lem:ledger-h

Topology/
Parity.lean
  -- prop:orientation-parity
  -- thm:component-count
Transversality.lean
  -- thm:transversality
  -- cor:transv-preimage
  -- lem:handshake
Tubular.lean
  -- thm:tubular

Value/
BoundedApprox.lean
  -- lem:bounded-approx
Continuity.lean
  -- lem:m-concave
  -- prop:m-continuous
Defs.lean
  -- def:value
  -- lem:m-finite
  -- lem:subadditivity

Variation/
ConstantTranslation.lean
  -- lem:sym-diff-constant
CoulombBoundary.lean
  -- lem:coulomb-first-var
FieldStability.lean
  -- lem:sym-diff-stability
OneDimensionalTranslation.lean

```

```

-- lem:1d-translation
Perimeter.lean
-- thm:first-var-perimeter
Piola.lean
-- lem:piola
Straight.lean
-- conv:first-variation
StraightCofactor.lean
-- lem:straight-cofactor
StraightDiffeo.lean
-- def:straight
-- lem:straight-diffeo
SymmetricDifferenceFull.lean
-- thm:sym-diff
TransportC1.lean
-- thm:transport-perimeter
VolumeBoundary.lean
-- thm:first-var-volume

```

41.3 Notes on the tree

1. **Elliptic/ClassicalCalculus.lean must be built early.** Despite living under Elliptic/, it is imported by DeGiorgi/Structure.lean (for Corollary 8.14). It depends only on BV/Rellich.lean and chart calculus. See Remark 10.1. If a strict layering is preferred, move it to Calculus/ChartGaussGreen.lean at the top of the tree; the labels are unchanged.
2. **Isoperimetric/Rigidity.lean is deliberately far from Isoperimetric/ABP.lean.** ABP.lean depends on Elliptic/ only; Rigidity.lean depends on the whole Regularity/tree. Keeping them in one file would create a false impression of a cycle and would force the whole regularity development to be imported by every consumer of (13.3). See Remark 13.1.
3. **Regularity/OmegaMinimal.lean holds only the definition.** This lets Isoperimetric/Rigidity.lean and Regularity/Penalization.lean both supply instances without either importing the other, per (R(R4)).
4. **The threshold algebra is the first pilot.** Start with Conventions.lean and Threshold/Defs.lean, then Threshold/Algebra.lean and Threshold/Ledger.lean. The ledger is pure algebra, and its statements (Lemmas 16.10–16.16, 16.15) are exactly the ones a normalisation tactic can discharge. Doing it first de-risks the sharp constants and gives an early win. Note that $q = 2^{1/3}$ should be introduced once, as a real cube root with $q^3 = 2$ and $q > 1$, and never as a Real.rpow expression to be re-derived. Prove agreement with the showcase’s LiquidDrop.criticalVolume. The two-ball comparison is assigned to Threshold/Comparison.lean, so the pilot does not import ball energy or its geometric prerequisites.
5. **Flow/ plus Surface/Morse.lean is the optional-cost centre.** Flow/ exists only for lem:morse-band and lem:one-manifold, which exist only for Surface/MorseCount.lean. Formalisation note 14.46 gives the two ways this block might shrink, neither of them verified. Surface/Shape.lean and the proof of the index-integral formula are independent of Flow/. The complete Surface/TotalCurvature.lean file, however, contains Theorem 14.43, which imports Proposition 14.35 from Surface/MorseCount.lean and therefore depends transitively

on `Flow/`. Build the complete file after the Morse count; deferring `Flow` would require a further file split or an explicit conditional interface for that count.

6. **Four files were deleted relative to the first draft of this document:** `Topology/Triangulation.lean`, `Surface/Frame.lean`, `Surface/Stokes.lean` and `Surface/GaussBonnet.lean`, replaced by `Surface/Shape.lean`, `Surface/Morse.lean`, `Surface/MorseCount.lean` and `Surface/TotalCurvature.lean`. The reason is Formalisation note 14.45: the deleted route needed abstract simplicial complexes with geometric realisation, which the pinned `mathlib` does not provide in any usable form (`AlgebraicTopology/Simplicial*` is category-theoretic simplicial sets; `Analysis/Convex/SimplicialComplex` is convex geometry). Note that no `Topology/Degree.lean` is needed either: Formalisation note 14.39 explains why the index sum is never shown to be a degree.
7. **Regularity/ has seventeen planned files.** The ε -regularity argument is where a monolithic file would be least maintainable: the smallness constants of Constants 24.7, 24.15 and 24.25 are chosen in three different files and must be threaded in the declared order. Making each choice a separate file's *output* — an existential statement, per (R(R6)) — is what keeps the order honest.
8. **Tests/ should be built incrementally, not at the end.** Add each test once its actual prerequisites are available. The completed map records these dependencies explicitly: for example, `test:constants` also checks the optimal ball volume and binding energy, so its complete implementation comes after those results. Its elementary threshold checks may be extracted earlier during the pilot.
9. **Thematic overview of the build.** Following Chapter 40: Conventions $\rightarrow$ Threshold/Ledger $\rightarrow$ Measure $\rightarrow$ BV $\rightarrow$ Area $\rightarrow$ Elliptic/ClassicalCalculus $\rightarrow$ Sobolev $\rightarrow$ DeGiorgi $\rightarrow$ Variation $\rightarrow$ Elliptic (rest) $\rightarrow$ Sard $\rightarrow$ BV/SmoothApprox $\rightarrow$ Isoperimetric/ABP $\rightarrow$ Energy, Ball, Threshold $\rightarrow$ Value, Binding/Splitting $\rightarrow$ Compactness, Existence $\rightarrow$ Regularity $\rightarrow$ Cones $\rightarrow$ Isoperimetric/Rigidity $\rightarrow$ Stationary $\rightarrow$ Flow, Topology, Surface $\rightarrow$ Hull, Capacity $\rightarrow$ CapacitaryK $\rightarrow$ Green $\rightarrow$ Stationary/CapacitaryEstimate $\rightarrow$ Classification, Nonexistence $\rightarrow$ Main $\rightarrow$ Binding/Main.
The `Flow/Topology/Surface` block is placed late because nothing before `Hull` needs the complete block; its elementary surface definitions are available earlier. For actual scheduling, use a topological order of the module imports, which interleaves these thematic groups as required.
10. **Blueprint node names.** The $\text{\LaTeX}$ labels in this document are intended to be the blueprint node names verbatim. When a Lean declaration is written for an item, add `\lean{Decl.Name}` to that item and `\leanok` when it compiles without `sorry`. Do not rename labels: Chapter 40 and this chapter both reference them.
11. **What may never appear.** No axiom, no constant, no `sorry` in a finished file, and no theorem whose statement cites a published paper. If an item cannot be proved from its declared `\uses` set plus Chapter 1, that is a defect in this document and should be reported rather than patched with a hypothesis.

41.4 Provenance

The mathematical target is O. Chodosh and M. Gianocca, *No compromise in the liquid drop model*, arXiv:2608.11517. The source paper uses Bonacini–Cristoferi for regularity and Agostiniani–Mazzieri for capacitary monotonicity; *neither is assumed here* — they are replaced by Chapters 20–26 and Chapter 31 respectively, and are used only as provenance and as checks on constants. The slicing idea of Chapter 36 originates with Frank–Killip–Nam; the compactness and existence context of Chapters 19 and 38 is related to work of Frank–Lieb and Frank–Nam. The arguments needed here are written out in full.